- B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic
Pool Function Selected	B = Delta value setting from remocon

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool "B" value)

** Pool function also can be select at Zone 2 when optional PCB is connected and Zone 2 system is select.

- Maximum/minimum regulation of Target Water Temperature.

	Heating	**Cooling**
Max	55°C / 60°C*	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C

- Target Zone 2 Water Temperature (Heat mode) during SG ready control
  - If buffer tank selection is "NO" then following shift is carried out.
    - While digital input is "10" or "11" then,
      Final Target Zone 2 water temperature
      = Target Zone 2 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is "YES"
    - No shift of Target Zone 2 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to "Buffer tank temperature control"

- Target Zone 2 Water Temperature (Cool mode) during SG ready control
  - Final Target Zone 2 water temperature = Target Zone 2 water temperature + SG Cool Setting

16.1.6.3 Zone Temperature Control Contents

- During Standard System (Optional PCB not connected)
  - Only 1 zone temperature control is available
  - This zone room temperature is control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temperature)
  - Target Zone Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor. Target Water Temperature will set same as Target Zone Water Temperature
  - Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
  - Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- During Extension System (Optional PCB connected)

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- During Extension System (Optional PCB connected)

- Buffer Tank connection select "NO" &
  - One zone system is select

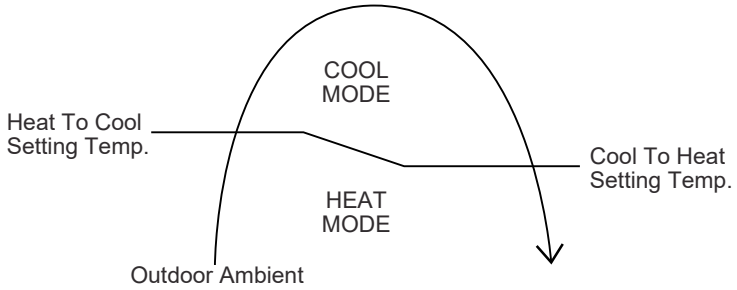
- This zone room temperature control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temp.)
- Target Zone Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor.
- Target Water Temperature will set same as Target Zone Water Temperature
- Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
- Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- 2 zone system select
  - Each zone room temperature is control by each sensor which select from either one of the 4 room sensor
  - Target Zone 1 & 2 Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor.
    * Zone Mixing Valve & Zone pump will operate to achieve Target Zone Water Temperature which refer to zone sensor.
    * Zone Sensor will detect if zone sensor is open or short.
  - Target Water Temperature will set same as the active & higher zone water temperature setting. (When cooling mode, lower zone water temp setting)
  - Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
  - Heat pump and water pump OFF when ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat).

- Buffer Tank Connection select "YES" &
  - 1 zone system or 2 zone system select
    - Each zone control by each sensor which select from either one of the 4 room sensor
    - Target Zone 1 & 2 Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor. Each zone have their own Target Zone Water Temperature.
      * Zone Mixing Valve & Zone pump will operate to achieve each Target Zone Water Temperature which refer to zone sensor
      * Zone Sensor will detect if zone sensor is open or short.
    - Target Buffer Tank Temperature will be set as active & higher zone water temperature setting + Buffer Delta T. (Heating) Target Buffer Tank Temperature will be set as active & lower zone water temperature setting (Cooling).
    - Target Water Temperature will set as Target Buffer Tank Temperature + [2°C] (Heating) Target Water Temperature will set as Lower or Active Target Zone Water Temp. + [-3°C] (Cooling)
    - Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM thermo OFF
      - Heat mode: ROOM thermo OFF (Buffer Tank Temperature > Target Buffer Tank + [0°C]
      - Cool mode: ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat)

16.1.7 Auto Mode Operation

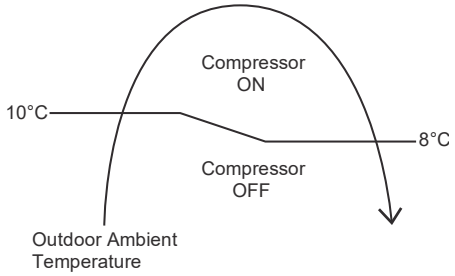


- Control details:
  - To enable the unit to operate either heat or cool mode automatically, heat to cool set temperature and cool to heat set temperature can be set by control panel.
  - Automatic operation is judged based on control panel setting temperature and outdoor ambient temperature.

  * Minimum setting of heat to cool set temperature is 1°C higher than cool to heat set temperature.

- Judgement control:
  - If outdoor ambient temperature < Heat to Cool Set Temperature, unit will operate in Heat Mode or else the unit will operate in Cool Mode.
  - If current operation is Cool mode, outdoor ambient temperature > Cool to Heat Temperature, unit will maintain Cool mode operation or else the unit will operate Heat mode.
  - If current operation is Heat mode, outdoor ambient temperature >Heat to Cool Temperature, unit will maintain Heat mode operation or else the unit will operate Cool mode.
  - Every 60 minutes the outdoor ambient temperature is judged.
  - When Auto + Tank mode is selected, operation mode switching is judged by both outdoor ambient temperature and indoor air temperature.

16.1.8 Auto Cooling Mode Operation Limit



- Auto Mode Cooling Only operation will start once the outdoor ambient temperature reaches 10°C and compressor will continue to run until the outdoor ambient temperature drops to 8°C.
- Due to this limitation,
  If Heat to Cool temperature is set lower than 10°C, the compressor will not operates until the outdoor ambient temperature reaches 10°C or higher.

16.1.9 Tank Mode Operation

- 3 ways valve direction
  - 3 ways valve switch to tank side during Tank Thermo ON condition. Switch 3 ways valve to room side when tank achieve Tank Thermo OFF temperature.

- Tank Thermo ON/OFF Characteristic

  Tank Thermo OFF
  Case 1: Internal Tank Heater is select and Tank Heater ON
  - Tank temperature > Tank Set Temperature continuously for 15 seconds. **OR**
  - Water outlet >75°C
- Case 2: Tank Heater OFF OR External Heater is select
  - When heat pump OFF due to water thermos & Tank temperature > Tank water set temperature for continuously 20 seconds. **OR**
  - Tank temperature > Tank set temperature + 1°C for continuously 20 seconds.

  Tank Thermo ON
  Case 1: Tank Heater ON (Internal Tank Heater)
  - Tank temperature < Tank set temperature + R/C (Tank re-heat temperature)

  Case 2: Tank Heater OFF (Internal Tank Heater)
  - Tank temperature < Tank water set temperature + R/C (Tank re-heat temperature)

  * When tank thermo ON, water pump will ON for 3 minutes then only heat pump turn ON.
  * Tank water set temperature = tank set temperature or 52°C whichever lower.

- 2 ways valve close
- Heat pump Thermostat Characteristic
  - Heat pump Water Outlet set temperature is set to below table:

Outdoor ambient temperature	Heat pump target water outlet temperature
< -10°C	56°C
> -10°C	59°C

Characteristic of heat pump thermos ON/OFF under tank mode condition:

Water Outlet Thermo Condition

- Heat pump thermos OFF temperature:
  1 Heat pump thermo OFF temperature = Target Water outlet temperature + (1°C)
  2 Water outlet temperature > heat pump thermo OFF temperature for continuously 3 minutes, heat pump OFF but water pump continue ON.

- Heat pump thermo ON temperature
  1 Heat pump thermo ON temperature = water inlet during thermo OFF time + [-3°C]
  2 When water outlet temperature < heat pump thermo ON temperature, heat pump ON.

Water inlet thermo protection condition

- Heat pump thermo OFF temperature:
  1 Water inlet temperature > [60°C/55°C] for continuously 30 seconds, heat pump OFF, water pump continue ON.

- Heat pump thermo ON temperature:
  1 Heat pump thermos ON temp = water inlet temperature < [60°C/55°C].

Outdoor ambient temperature	Water inlet temperature
< -10°C	55°C
> -10°C	60°C

Thermo ON/OFF for Heat Pump in Tank Operation:

When tank temperature achieve heat pump OFF condition, refer below condition:
Conditon 1 : Tank Heater ON (Internal Tank Heater)

- Heat pump will turn OFF, water pump continue ON and room heater will continue ON if tank temperature below tank heater thermo ON condition. 3 ways valve will only switch to room side after tank temperature reach tank heater thermo OFF condition.

Conditon 2 : Tank Heater OFF (Internal Tank Heater)

- If tank temperature achieve tank thermo OFF, heat pump turn OFF, water pump turn OFF, room heater OFF and 3 ways valve switch to room side.

When tank temperature achieve heat pump ON condition, water pump ON, heat pump ON and room heater turn OFF.

Heat pump OFF condition at Tank Mode

- Tank temperature > tank water set temperature continuously for 20 seconds after heat pump thermos OFF due to water thermo. (Heat pump turn OFF but water pump continue ON and room heater turn ON to achieve tank set temperature) **OR**
- Tank temperature > tank set temperature + [1°C] for continuously 20 seconds. (Heat pump OFF, water pump OFF, room heater OFF and 3 ways valve switch to room side)

Heat pump ON condition at Tank Mode

- Tank temperature < tank water set temperature + R/C setting (Tank re-heat temp) (Water pump turn ON OR continue ON, heat pump ON and 3 ways valve switch to tank side or maintain at tank side)

Tank heater control

- Internal heater only operates to tank side if Tank heater ON and backup heater is enable.

Internal heater turn ON condition:

- Tank temperature < tank set temperature **AND**
- Heat pump thermos OFF **AND**
- 20 minutes from previous heater off **AND**
- Internal tank heater selects USE from control panel.

Internal heater turn OFF condition:

- Tank temperature > tank set temperature for continuously 15 seconds **OR**
- Heat pump thermo ON **OR**
- Mode change or operation is off by control panel.

16.1.10 Heat + Tank Mode Operation

1 3 ways valve control:
   - 3 ways valve switch to room side during room heat-up interval and switch to tank side during tank heat-up interval. Both modes will switch alternately. Tank mode is the initial running mode of Heat + Tank mode.

2 Heat pump operation control:
   - During room heat-up interval
     - Follow normal heating operation.
       Switching to tank side depends to below cases:
       Case 1:
       [Previous switch from tank interval to room interval due to thermo OFF]
     - Switch to tank heat-up interval when Tank temp < Tank thermos ON temp (Room heat-up interval ends)
       Case 2:
       [If heating operation at room side is less than 30 minutes and switch to tank side 3 times consecutively]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
       Case 3:
       [Previous switch from tank interval to room interval due to tank interval timer is complete]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.
   - During Tank heat-up interval
     - Tank interval is the first mode running when heat + tank mode is select.
     - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
     - Heat pump operates according to normal tank mode operation.

3 Room heater control:
   - During heating heat-up interval
     - Follow normal room heater control operation.

4 Tank heater control:
   - During heating heat-up interval
     - Internal tank heater will not function under heating heat-up interval.
   - During tank heat-up interval
     - Internal tank heater will turn ON after heat pump thermo off to boil tank temperature to tank set temperature.
   - 2 ways valve control is open
   - Indoor water pump control:
     - Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

16.1.11 Cool + Tank Mode Operation

1 3 ways valve control:
   - 3 ways valve switch to room side during room cooling interval and switch to tank side during tank heat-up interval. Both mode will switch alternately. Tank mode is the initial mode of cool + tank mode.

2 Heat pump operation control:
   - During room heat-up interval
     - Follow normal cooling operation.
       Switching to tank side depends to below cases:
       Case 1:
       [Previous switch from tank interval to room interval due to thermo OFF]
     - Switch to tank heat-up interval when Tank temperature < Tank Thermo ON temperature (Room interval will ends)
       Case 2:
       [If cooling operation at room side is less than 30 minutes and switch to tank side for 3 times consecutively]

- Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
  Case 3:
  [Previous switch from tank interval to room interval is due to tank interval timer is complete]
- Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.

  *Tank Thermo ON temperature:

Internal Tank Heater select USE	<Tank set temperature + R/C setting (Tank re-heat temperature)
Others	<Tank water set temperature + R/C setting (Tank re-heat temperature)

- During Tank heat-up interval
  - Tank interval is the first mode running when the cool + tank mode is select.
  - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
  - Heat pump operates according to normal tank mode operation.

3 Room heater control:
- During room cooling interval
  - Room heater is OFF and not operates.

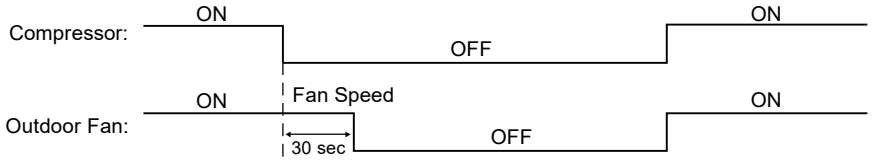
4 Tank heater control:
- During room cooling interval
  - Internal tank heater will not function under room cooling interval.
- During tank heat-up interval
  - Internal tank heater will turn ON after heat pump thermos off to boil tank temperature to tank set temperature.

5 2 ways valve is close.

6 Indoor water pump control:
- Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

16.1.12 Outdoor Fan Motor Operation

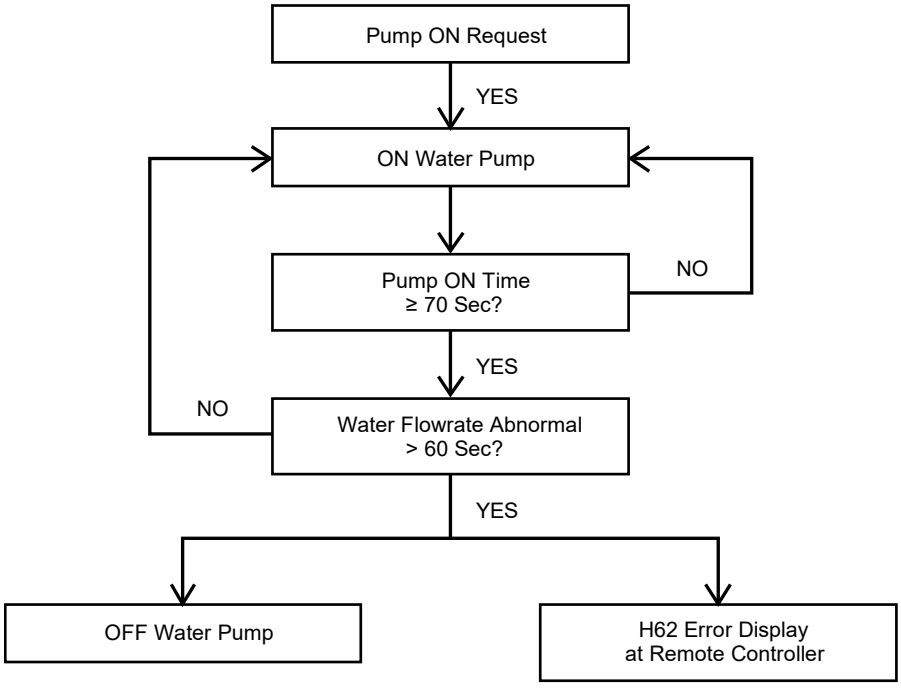
Outdoor fan motor is adjusted according to operation condition. It starts when compressor starts operation and it stops 30 seconds after compressor stops operation.



16.2 Water Pump

16.2.1 Water Pump Control

- Once the indoor unit is ON, the water pump will be ON immediately and no error judgment for 70 seconds. However, during this 70 seconds operation, if there is any abnormality cause at outdoor or malfunction, the compressor should be OFF immediately and restart delay after 3 minutes.
- The system will start checking on the water flow level after operation start for 70 seconds. If water flow level is detected low continuously 60 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H62 error occurs).
- When error happens, the power has to be reset to clear the error.
- If there is no error indication, the water pump shall be continuously running.
- The water pump will remain ON when compressor OFF due to thermostat OFF setting is reached.
- Water pump will OFF when room thermo **OR** tank thermo **OR** buffer tank thermo OFF.
- Water pump will delay 15 seconds to turn OFF when request to OFF except during anti-freeze deice activate or air purge mode.



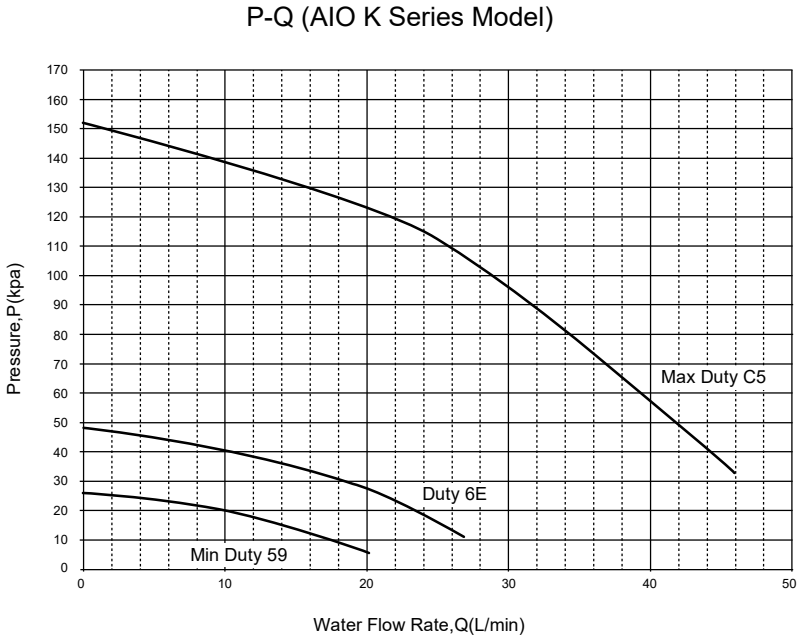
Maximum pump speed setting on remote control

1) When Pump flowrate setting is ΔT
   Standard pump speed is automatically controlled to get the designed water temperature different between water inlet and outlet (ΔT). The maximum pump speed setting limitation can be adjusted by the installer according to water circuit pressure drop condition.

2) When pump flowrate setting is Max. Duty
   Indoor water pump speed will operate at the maximum pump speed setting at room side operation. The maximum pump speed setting can be adjusted by the installer according to water circuit pressure drop condition.

However, the following sequences do not follow maximum pump duty setting by remote control.
- Pump down mode
- Air purge mode
- Normal deice

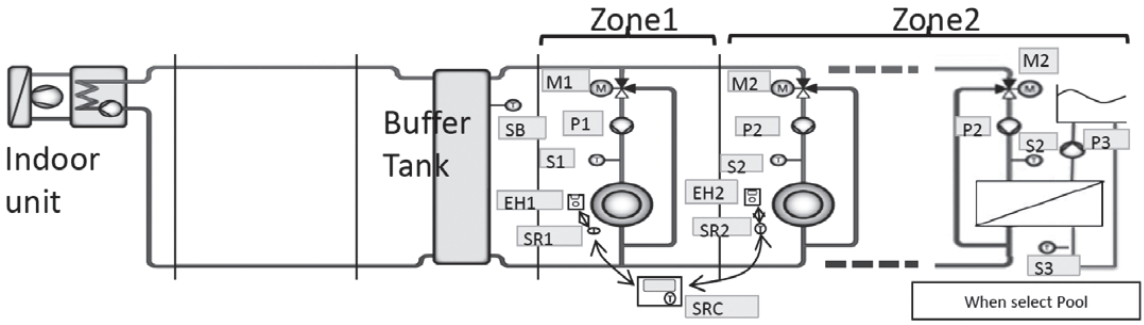
1) P-Q graph for different pump HEX duty



16.2.2 Zone Water Pump Control

Purpose:

- Water pump install at each zone to circulate the water inside each zone during buffer tank connection selected "YES" or 2 zone systems.



Content:

- AC type water pump install for this zone water pump control. When optional PCB connected, 230V output will drive this zone pump.
- There are three pump can be connected through Optional PCB. (Zone 1 Pump, Zone 2 Pump, & Pool Pump)

  * Zone 1 pump [P1] use to circulate zone 1 water circuit & Zone 1 mixing valve [M1] adjust to control the Zone 1 target water temperature.

  * Zone 2 pump [P2] use to circulate zone 2 water circuit & Zone 2 mixing valve [M2] adjust to control the Zone 2 target water temperature.

  * When Pool Function select as Zone 2 circuit, [P2] use to circulate water to heat exchanger which use to transfer heat to pool water.

  * Pool pump [P3] circulates the pool water through the heat exchanger to get warm water.

- Zone 1 and Zone 2 water pump start condition:
  - Zone room request ON
    (eg. Zone 1 thermo ON, only zone 1 pump will turn ON)
- Zone 1 and Zone 2 water pump stop condition:
  - Zone room request OFF
- Pool water pump start condition:
  - Pool Zone request ON **AND**
  - Pool function is selected

- Pool water pump stop condition
  - Pool zone: Zone room request OFF **OR**
  - Pool function is cancel

    * Zone 1 & Zone 2 water pump need to turn OFF when antifreeze deice pump stop control activate and turn ON back after the antifreeze deice pump stop control end under setting of "NO" buffer tank connection.

Zone Pump Prohibit ON control:

- Start condition:
  Zone 1 water temperature ≥ 75°C continuously for 5 minutes
  *stop zone water pump operates if the zone water fulfilled.
- Cancel condition:
  After 30 minutes from start condition fulfilled.
  *zone water pump operates according to normal condition.

Zone Pump Control during Anti-Freeze

Zone pump control during Zone Anti-Freeze Control:

- When Zone Anti-Freeze Flag=1, Zone Pump Turn ON.
- When Zone Anti-Freeze Flag=0, Zone Pump Turn OFF.

Zone pump control during Indoor Anti-Freeze Control:

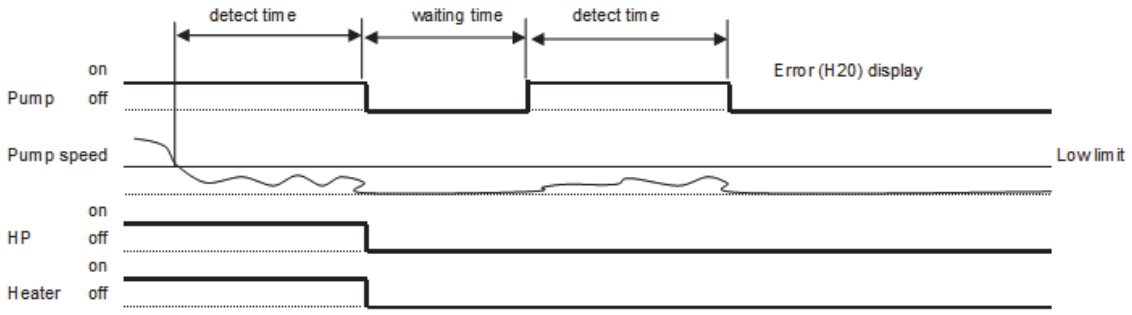
- Zone pump only ON/OFF if the Extension PCB connected and Buffer Tank select "NO" condition

  When Indoor Anti-Freeze flag=1, Zone Pump Turn ON
  When Indoor Anti-Freeze flag=0, Zone Pump Turn OFF

* Pool Water Pump will not affected by both Indoor anti-freeze control or zone anti-freeze control.

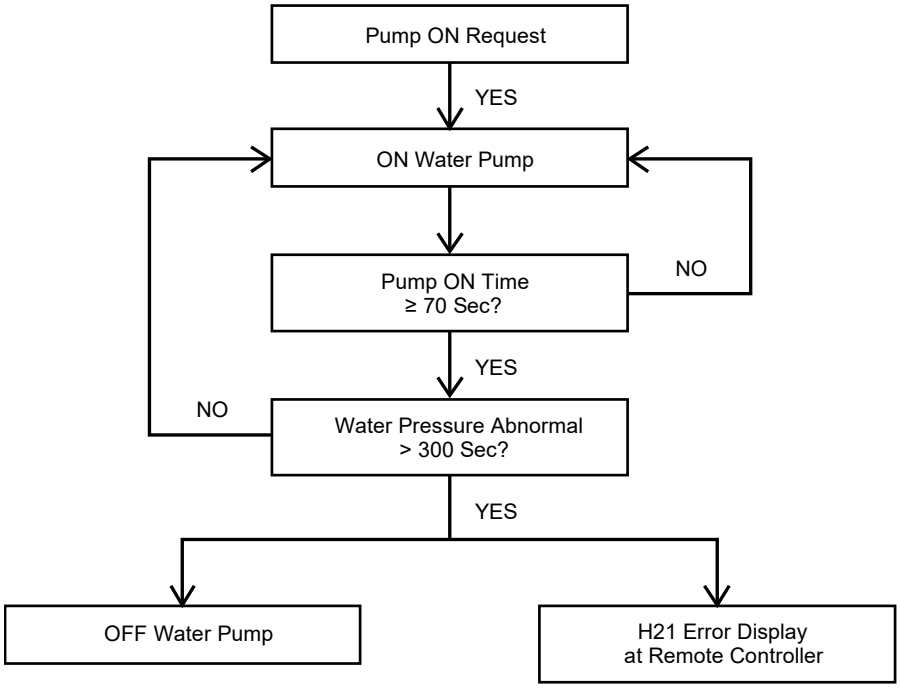
16.2.3 Water Pump Speed Feedback Error

- Basically pump speed feedback is control by micon.
- When pump speed is below low limit or over high limit for a few seconds, micon detect pump error and system is stopped.
- Error detection conditions:
  - Detect abnormal water pump speed for continuous 10 secs.
  - Current pump speed < 700 rpm or
  - Current pump speed > 6000 rpm for 10 seconds.
- Control contents:
  - When error occurs, water pump, heating and heater is stopped for 30 seconds then restart again (Retry control).
  - When micon detect error again, system is stopped and error code [H20] is displayed at control panel.



16.3 Water Pressure Control

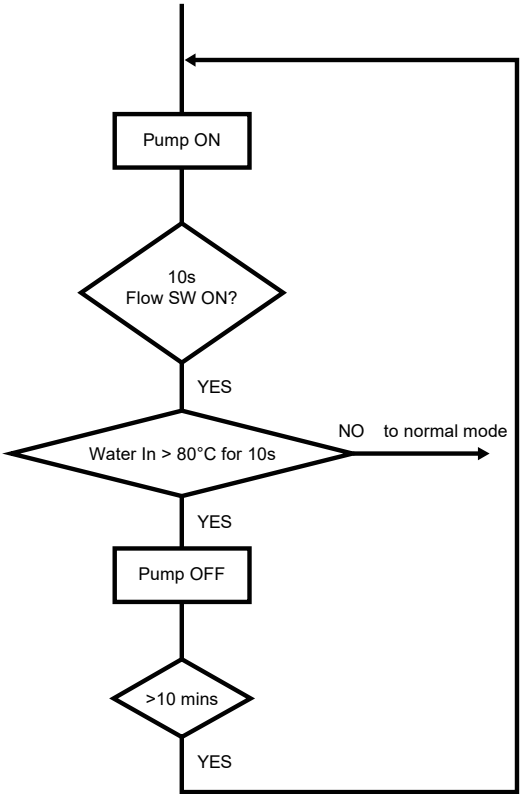
- Once the indoor unit is ON, the water pump will be ON immediately and no error judgment for 70 seconds. However, during this 70 seconds operation, if there is any abnormality cause at outdoor or malfunction, the compressor should be OFF immediately and restart delay after 3 minutes.
- The system will start checking on the water pressure after operation start for 70 seconds. If water pressure is detected low continuously 300 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H21 error occurs).
- When error happens, the power has to be reset to clear the error.
- If there is no error indication, the water pump shall be continuously running.



16.4 Indoor Unit Safety

16.4.1 Indoor Unit Safety Control

1 When water pump is ON, the system will start checking flow switch status (ON/OFF).
2 If the flow switch ON for 10 seconds, the system will check on the water inlet temperature for 10 seconds.
If the water inlet temperature not exceeds 80°C, the water pump shall be continuously running with normal mode.
If the water inlet temperature exceeds 80°C for continuously 10 seconds, the water pump will be OFF immediately.
3 After water pump OFF for more than 10 minutes, it will be ON back and the indoor unit safety control checking is restarted.



16.5 Auto Restart Control

1 When the power supply is cut off during the operation of Air-to-Water Heatpump, the compressor will reoperate after power supply resumes.

16.6 Indication Panel

LED	Operation
Color	Green
Light ON	Operation ON
Light OFF	Operation OFF

Note:
- If Operation LED is blinking, there is an abnormality operation occurs.

16.7 Indoor Back-Up Heater Control

16.7.1 Indoor Electric Heater Control

1 Normal Heating Mode
- Heater On condition:
  a. Heater switch is ON
  b. After Heatpump thermo ON for Remote Control Set Delay Time mins
  c. After water pump operate **[3]** mins
  d. Outdoor air temperature < Outdoor set temperature for heater
  e. When water outlet temperature < Water set temperature + Remote Control Heater ON Setting
  f. **[20]** minutes since previous Backup heater Off
     * When heatpump cannot operate due to error happens during normal operation, heater will go into force mode automatic
     * Heater need to operate during deice operation

- Heater Stop Condition:
  a. When outdoor set temperature > outdoor set temperature + **[+2°C]** for continuous 15 secs OR
  b. When water out temp > water set temperature + Remote Control Heater OFF Setting for continuous 15 secs OR
  c. Heater switch is Off OR
  d. Heatpump thermo-off or OFF condition

2 Force Heater Mode
- Heater On condition:
  a. After water pump operate **[3]** mins
  b. When water outlet temperature < water set temperature + Remote Control Heater ON Setting
  c. **[20]** minutes since previous Backup heater Off

- Heater Stop condition
  a. Force mode off **OR**
  b. When water outlet temperature > water set temperature + Remote Control Heater OFF Setting for continuous 15 secs

* Do not operate heater at the following situation
1 Water outlet temperature sensor, and water inlet sensor abnormal
2 Flow switch abnormal
3 Circulation pump stop condition

16.7.2 Room Heater Operation during Deice

Purpose:

- To protect the indoor Heat Exchanger from ice forming and prevent heat exchanger plate breakage.

Control content:

- This Heater protection control will activate only if the backup heater is enable at custom setup by remote controller. Once fulfil the start condition, room heater will turn ON together (base on max heater capacity selection) and stop together if stop condition is fulfilled.
  * If the heater is request to turn ON OLP feedback will be detected.

Starting conditions:

1. During normal deice operation 4~9
2. Water outlet temperature < 10°C or
3. Outdoor air temperature < -10°C or
4. Water inlet temperature < 27°C

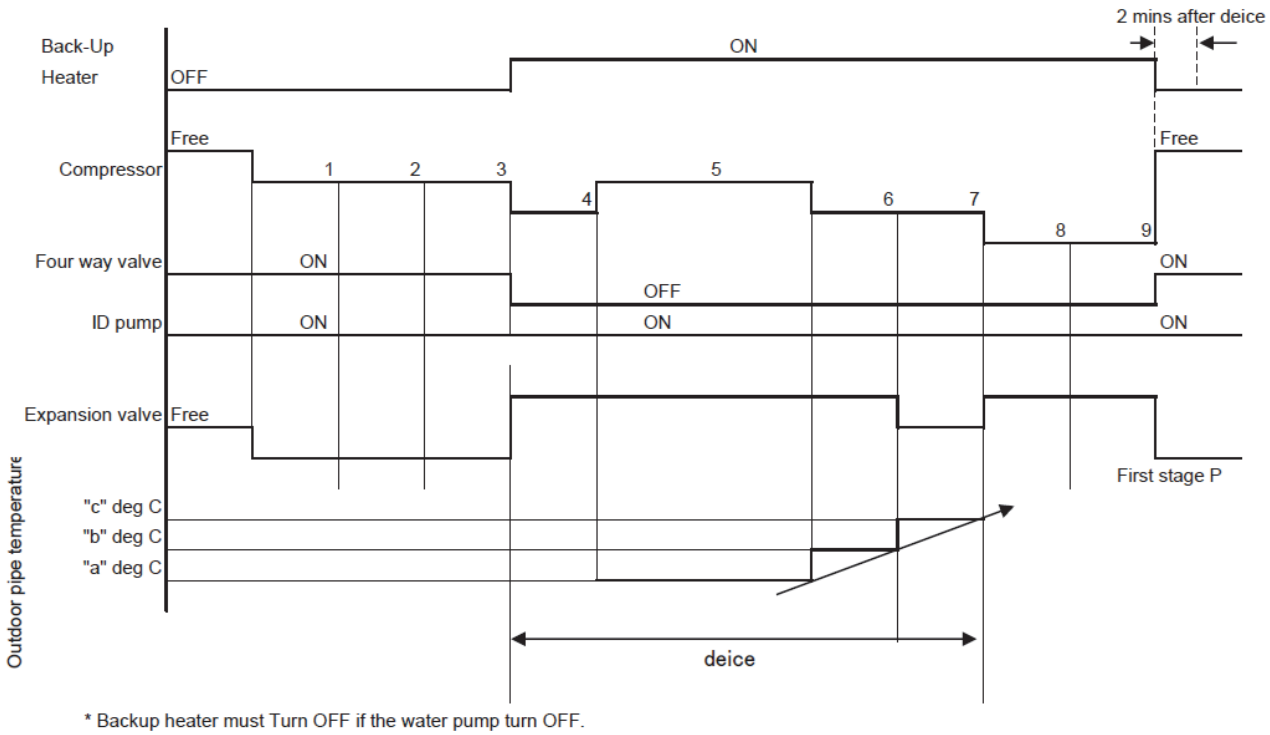
Heater operates when 1 ~2 fulfilled **OR** When 1 and 3 ~ 4 is fulfilled.
However, this control does not relate to Heater ON/OFF button on remote control.

Stop condition:

- When normal deice end or
- Water outlet temperature > 45°C

However, room heater keeps ON if indoor electric heater control activate.

< Deice operation time chart >



* Backup heater must Turn OFF if the water pump turn OFF.

16.8 Tank Heater Control

16.8.1 Tank Heater Remote Control Setting

1 Tank heater selection:
External: - Booster Heater use to heat up tank when select external
Internal: - Backup Heater use to heat up tank when select internal
* When select External Tank Heater, Heater Delay ON Timer need to set. (range 20 min ~ 3 hrs)

2 Tank Heater ON/OFF selection by user.

The remote control Tank set Temperature range will change according to the External and Internal Tank Heater use.

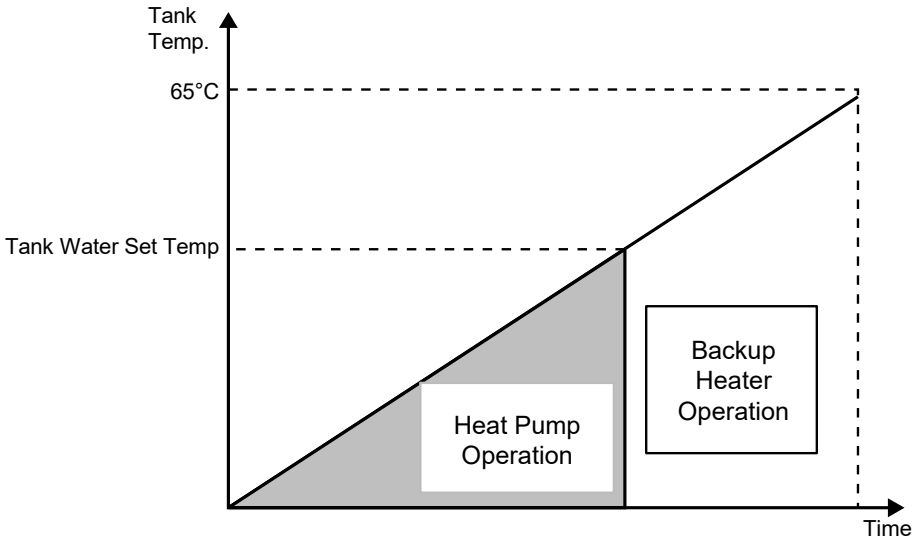
Tank Heater Selection	Range
External	40 ~ 75°C
Internal	40 ~ 65°C

16.8.2 Internal Heater Control at Tank Mode

- Internal heater turn ON condition:
  1 Internal Heater select for Tank heater by remote controller
  2 Tank Heater select ON by user.
  3 Backup Heater Enable
  4 Tank Temperature < Tank Set Temperature
  5 Heat Pump Thermo OFF
  6 20 minutes from previous heater off.

- Room heater turn OFF condition:
  1 Tank Temperature > Tank Set Temperature + [0°C] for continuous 15 seconds. **OR**
  2 Heat Pump Thermo ON **OR**
  3 Mode Change or Operation OFF by remote controller **OR**

  * Backup Heater Turn ON/OFF all together according to the selected heater capacity.



16.9 Base Pan Heater Control (Optional)

- To enable the base pan heater function, control panel initial setting has to be manually adjusted by activating Base Pan Heater menu.
- There are 2 optional start condition can be selected, Type A or Type B.
- Control details:
  1 Type A: (Default Auto Mode)
  Start conditions:
    - When outdoor air temperature ≤ 3°C during heating and deice operation is ON.

  Control contents:
    - Base pan heater is ON during deice operation and continues ON for 10 minutes after deice operation ends.

  Cancel condition:
    - When outdoor temperature > 6°C after deice end or
    - When operation is not at heating mode or
    - Base pan heater ON timer count is completed.

  2 Type B: (ON Mode)
  Start conditions:
    - When outdoor air temperature is ≤ 5°C and operates in heating mode, base pan heater is ON.

  Cancel conditions:
    - When outdoor air temperature is > 7°C or
    - When operation is not at heating mode.

16.10 Force Heater Mode

Purpose of Force Heater Mode:

- As a backup heat source when heat pump error. Force heater Mode only control backup heater to heat up the room circuit, and turn ON back up heater or booster heater to boil up tank water base on the tank heater selection (internal or external).

Force Heater Control start condition:

- Force heater request ON by user during error OR auto turn ON by remote controller during error **AND**
  (Force Heater mode can be operate regardless of mode selection, remocon will send the latest mode selection force bit by bit to indoor. Indoor will judge to turn ON heater to room side if it is heat mode selected, and turn ON heater to heat tank water base on tank heater selection)
- During Error Happen (exclude the error list below)

Error List which not allow Force Heater operation

H12	Capacity Mismatch	H76	Indoor-Remote Controller Communication Error
H20	Abnormal Water Pump	H95	Abnormal Voltage Connection
H62	Abnormal Water Flow	F37	Abnormal Water Inlet sensor
H70	Abnormal Back-up Heater OLP	H45	Abnormal Water Outlet sensor
H74	PCB Communication Error	H21	Water Pressure Error
[When tank mode operate with external heater selected & tank heater select ON]			
H72	Abnormal tank sensor	H91	Abnormal tank heater OLP

Force Heater Control Stop Condition:

- Force Heater request OFF **OR**
- Operation OFF request **OR**
- Power reset **OR**
- Error of above list happens during force heater operation.

Control contents:

- After fulfill start condition, indoor will operate the force heater operation according to below mode condition
  Heat mode Only: Turn ON backup heater to achieve room heat pump target water temperature.
  Heat + Tank mode: Turn ON backup heater to heat up room **OR** Turn ON Heater to Boil up tank water.
  Cool mode Only: Water pump and backup heater will OFF in force heater mode.
  Cool + Tank mode: Operate pump and internal Heater OR External heater to Boil up tank water.
  Tank mode Only: Operate pump and internal Heater OR External heater to Boil up tank water.
  * For heat mode condition, backup heater will only turn ON if the backup heater is enable regardless of Room Heater Selection.
  * For tank mode condition, If internal heater selected backup heater will turn ON to boil up tank water.
  If external heater selected , booster heater will turn ON to boil up tank water regardless of tank heater selection.

Room Side: (Heat Mode):

- When force heater mode start condition fulfilled, turn ON water pump and turn ON backup heater follow below control.
- Operate the 3 ways valve at room side only and turn ON 2 ways valve as heat mode operation.
- Turn ON the zone pump and mixing valve if system select 2 zone system or Buffer tank connect YES, control according to normal zone pump and mixing valve control.
- When Force heater mode stop condition fulfilled, turn OFF heater as below condition and turn OFF water pump after pump delay time.

Backup Heater On Condition:

- When Force Heater Control start condition fulfill **AND**
- After water pump operate 2 minutes **AND**
- When water outlet temperature < water set temperature + Remote Control Heater ON Setting **AND**
- 20 minutes since previous Backup heater Off **AND**
- Backup Heater Enable

Backup Heater Stop condition:

- Force mode off **OR**
- Operation off **OR**
- When water outlet temperature > water set temperature + Remote Control Heater OFF Setting for continuous 15 secs **OR**
  * ON/OFF follow normal heater sequence.

Tank side (Tank mode):

- When tank mode select and force heater bit received, turn ON backup heater (INTERNAL) or Booster Heater (External) depend on the tank heater selection.
- If tank heater selection is INTERNAL, follow normal thermo judgement to switch 3 ways valve to tank side and room side.
- If tank heater selection is EXTERNAL, only turn ON booster heater according to tank thermo.

<u>Tank Heater selection is INTERNAL:</u>
Backup Heater ON Condition:

- After water pump operate 2 mins **AND**
- When tank temperature < Tank set temperature - [Remocon Set Tank Re-heat Temp] **AND**
- 20 minutes since previous Backup heater OFF **AND**
- Backup Heater Enable

Backup Heater OFF condition:

- Force mode off **OR**
- When tank temperature > Tank set temperature for continuous 15 secs **OR**
- Tank Operation OFF

<u>Tank Heater selection is EXTERNAL:</u>
Booster Heater ON condition:

- Force Heater mode ON **AND**
- Tank temperature < tank set temperature + [Remocon Set Tank Re-heat Temp] - 1°C, **AND**
- 20 minutes since previous heater off.

Booster Heater OFF condition:

- Tank temperature > tank set temperature for continuous 15 secs.
- Force mode OFF
- Tank Mode Operation OFF
  (During tank interval or tank mode condition, water pump and 3 ways valve will OFF)

16.11 Powerful Operation

Powerful mode is use to increase the capacity of heat pump to achieve higher target temperature. Powerful mode is applicable when heat mode is operating.

Remote control setting:
On quick menu of remote control, there is 4 options of powerful mode can be select.

- OFF : Cancel powerful mode
- 30 minutes : Set powerful for 30 minutes
- 60 minutes : Set powerful for 60 minutes
- 90 minutes : Set powerful for 90 minutes

Control contents:
During the time set by remote control, powerful will activate according to 2 shift up controls. However, this function is applicable only for heating. Remote control will transmit the signal to indoor unit once this function is select then transmit OFF signal to indoor when the timer is complete. Indoor will transmit signal to outdoor for frequency control.

Indoor setting temperature shift

- If system is standard system (Optional PCB is not connected)
  - Target water temperature will shift up to Wlo or Whi whichever higher.
- If system is extension system (Optional PCB is connected)
  - Target water Zone 1 and Zone 2 temperature will shift up to Wlo or Whi whichever higher.

  * If “Direct Type” temperature control is select, this powerful shift up setting is not effective.

- **Start condition**
  - Powerful function is select by remote control.

- **End Condition**
  - OFF/ON button is pressed.
  - Powerful function is OFF by remote control.

16.12 Quiet Operation

Quiet mode is use to reduce the noise of outdoor unit by reducing the frequency or fan speed.

Quiet level
There are 3 level (Level 1, Level 2, Level 3) to set by quick menu function on remote control.

Control content
Once the quiet function is select, the remote control will transmit the signal to indoor and outdoor unit.

Start condition

1) In case of “Sound” is set for “Quiet priority” on the remote control
   If any of below condition is achieved
   - Quiet mode is set on remote control.
   - Quiet mode is request ON by weekly timer.

2) In case of “Capacity” is set for “Quiet priority” on the remote control
   If condition { a) or b) } and { c) or d) or e) } is achieved
   a) Quiet mode is set on remote control.
   b) Quiet mode is request ON by weekly timer.
   c) During heating : Water outlet temperature > Target Water Temperature + [-3°C]
   d) During Cooling : Water outlet temperature < Target Water Temperature + [3°C]
   e) During tank heat up : Tank temperature > Tank set Temperature + [-3°C]

Stop condition

1) In case of “Sound” is set for “Quiet priority” on the remote control
   If any of below condition is achieved
   - OFF/ON button is pressed.
   - Quiet mode is OFF by remote control.
   - Quiet mode is request OFF by weekly timer.

2) In case of “Capacity” is set for “Quiet priority” on the remote control
   If any of below condition is achieved
   - OFF/ON button is pressed.
   - Quiet mode is OFF by remote control.
   - Quiet mode is request OFF by weekly timer.
   - During heating : Water outlet temperature ≤ Target Water Temperature + [-5°C] for continuously [30] minutes.
   - During Cooling : Water outlet temperature ≥ Target Water Temperature + [5°C] for continuously [30] minutes.
   - During tank heat up : Tank temperature ≤ Tank set temperature + [-5°C] for continuously [30] minutes.

16.13 Sterilization Mode

- Purpose:
  - To sterilize water tank by setting the required boiling temperature.

- Remote control setting
  - Days for sterilization function to start can be select.
  - Time of selected day to start sterilization function.
  - Boiling temperature (Internal heater is 55°C ~ 65°C)
  - Maximum operation time is 5 minutes to 1 hour.

- Start condition
  - Tank connection set to “YES” by remote control
  - Sterilization function selects “YES”.
  - Sterilization signal received from remote controller by timer.
  - Tank mode request ON.

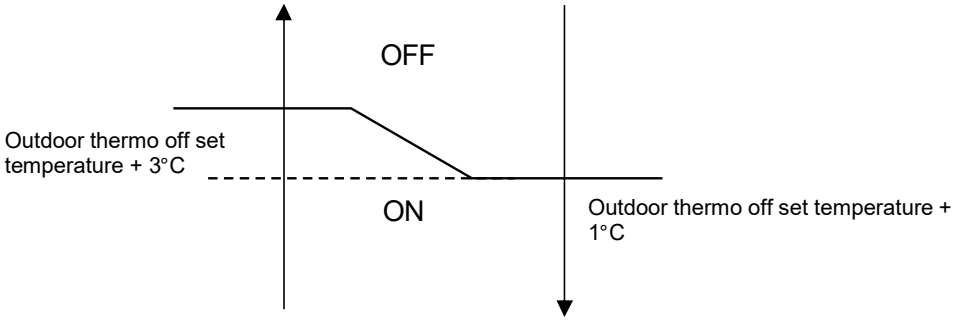
- Stop condition
  - When boiling timer is completed. Boiling timer (Remote control set maximum operation time) start counting once tank achieve boiling set temperature **OR**
  - After 8 hours of operation since sterilization start.
  - Tank mode request OFF.

- Control content:
  - During sterilization function activation time, target tank set temperature will internally change to boiling set temperature.
  - During sterilization activates, heat pump and heater (external or internal) will operate as normal tank mode to achieve the boiling set temperature.
  - Sterilization operation will end when stop condition is fulfill.
  - After sterilization is complete, tank set temperature will resume to normal operation.

* Tank temperature may not achieve boiling set temperature if tank heater is select OFF **OR** external compressor switch.

16.14 Outdoor Ambient Thermo OFF Control

- Purpose:
  - Stop provides heating to room side during high outdoor ambient condition.



- Control content:
  - Heating outdoor ambient thermos OFF control only applicable when heat pump operate in heat mode. (This control will not activate when running in tank side)
  - Heat pump and water pump will turn OFF when outdoor ambient is higher than outdoor thermo OFF set temperature.
  - Heat pump thermos ON when outdoor ambient < outdoor thermos OFF set temperature + 1°C.

16.15 Alternative Outdoor Ambient Sensor Control

Purpose of the Alternative Outdoor Ambient Sensor:

- It is some possibility that the air to water heat pump unit will install at a location where the original ambient sensor is expose to direct sunlight. Therefore, another optional ambient sensor can be connect to indoor PCB and locate at new and better reading location to improve the heat pump performance.

Control Detail:

- Remocon can select either the extra outdoor ambient sensor is connected or not. (YES/NO)
- The alternative outdoor ambient sensor will connect to indoor unit main PCB terminal.

  - when alternative sensor select NO
    - Original Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
    - Data communication direction : OUTDOOR send outdoor temperature reading to INDOOR.
    - Error judge : OUTDOOR will judge the original outdoor sensor error (F36 display if error detect). No judge error on alternative outdoor sensor

  - when alternative sensor select YES
    - Alternative Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
    - Data communication direction : INDOOR send outdoor temperature reading to OUTDOOR.
    - Error judge : INDOOR will judge the Extra outdoor sensor error only after operation ON request received from remocon.
      (F36 display if error detect). No judge error on original outdoor sensor.

16.16 Force DHW mode

Purpose:
When user want to use hot water now, user can press this force DWH mode under the quick menu to operate tank only mode to boil up the tank temperature.

Remocon setting:
Force DHW function can be activate under quick menu.

Control Content:

- when press the Force DHW function during operation OFF condition:
  - When receive this Force DHW bit from remocon, indoor will run tank only mode regardless of the mode selection.
  - After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to operation OFF with previous mode selection.
- When press the Force DHW function during operation ON condition:

- When receive this Force DHW bit from remocon, indoor will memories the running mode and run tank only mode regardless of the mode selection.
- After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to previous memories running mode.

  * when operation OFF or mode change request from remocon during force DHW mode operation, End force DHW mode and follow the new request operation.

  * Once receive force DHW mode from remocon, indoor direct start tank mode and consider tank thermo ON. Thermo OFF only when achieve tank thermo OFF depend on the Tank System Setting.

16.17 SMART DHW mode

Panasonic All In One model provide the option to choose STANDARD DHW Mode or SMART DHW Mode for Tank Heat Up according to requirement. SMART DHW mode comparatively consume lower tank heat up power but longer re-heat time than STANDARD DHW Mode.

SMART DHW control

- During SMART DHW start time 20:00 (Default Setting) to SMART DHW stop time 05:00 (Default setting)
  Heat pump re-heat the tank water only when tank temperature drop below 20°C (Default setting)

- Time between 05:00 to 20:00
  Heat pump reheat the tank water when tank temperature as below condition

Condition 1: Tank Heater ON
Reheat when tank temperature below tank set temperature + R/C (Tank re-heat Temperature) - 3°C

Condition 2: Tank Heater OFF
Reheat when tank temperature below Tank set temperature or 52°C (Whichever lower) + R/C (Tank re-heat Temperature) -3°C

* SMART DHW start time, stop time and SMART ON Temperature can change in CUSTOM menu.

STANDARD DHW Mode

- Heat pump always reheat the tank water when tank temperature as below condition

Condition 1: Tank Heater ON
Reheat when tank temperature below tank set temperature + R/C (Tank re-heat temperature)

Condition 2: Tank Heater OFF
Reheat when tank temperature below Tank set temperature or 52°C (Whichever lower) + R/C (Tank re-heat temperature)

16.18 DHW Capacity Setting

DHW Capacity is heat pump heating capacity output control during tank boiling operation. There are two DHW capacity setting (VARIABLE & STANDARD) which can be set in remote control.

VARIABLE DHW Capacity:
- Heat pump operate with efficient (Low) Capacity to boil tank temperature during re-heat process. And heat pump regulated to operate with high capacity to boil tank temperature when tank temperature drop below 25°C.

STANDARD DHW Capacity:
- Heat pump operate according to outdoor rated heating capacity during tank boiling process.

16.19 Anti Freeze Control

- Anti freeze protection control menu can be set YES or NO by control panel.
- In heatpump system, there are 3 types of anti freeze control:
  1. Expansion tank anti-freeze control
     - Expansion tank anti freeze heater ON condition:
       - Outdoor ambient temp. < 3°C
     - Expansion tank anti freeze heater OFF condition:
       - Outdoor ambient temp. > 4°C

  2. Water pump circulation anti freeze control
     - Water pump turns ON when **ALL** below conditions are fulfilled:
       - Heatpump OFF (stand by) OR error occurs.
       - Water flowing flag is ON.
       - Water flow switch is not abnormal.
       - Outdoor ambient temp. < 3°C OR outdoor ambient temp. sensor is abnormal.
       - Water inlet/outlet temp. < 6°C.
       - After 5 minutes from previous water pump OFF.
     - Water pump turn OFF when **ANY** of below conditions is fulfilled:
       - Outdoor ambient temperature ≥ 4°C.
       - During -5°C < outdoor ambient temp. < 4°C
         - After water pump ON for 4 minutes, and water inlet temp. ≥ 8°C.
         - Else, shift to back up heater anti freeze control.
       - During outdoor ambient temp. < -5°C
         - After water pump ON for 4 minutes, and water inlet/outlet ≥ 20°C.
         - Else, shift to back up heater anti freeze control.
       - When operation ON and once compressor starts
     - However, if flow switch is abnormal (H62), then water pump circulation anti freeze control will not activate.

  3. Back up heater anti freeze control:
     - Back up heater turn ON when **ALL** below conditions is fulfilled:
       - Water inlet/outlet temp. < 6°C.
       - Water pump circulation anti freeze control activated and water pump ON for 4 minutes.
     - Back up heater turns OFF when ANY of below conditions is fulfilled:
       - Water inlet/outlet temp. > 28°C.
       - Water pump circulation anti freeze control deactivated/water pump OFF.
     - However, if back up heater is abnormal (H70) then back up heater anti freeze control will not activate.

16.19.1 Zone Anti-Freeze Control

- If buffer tank selection is “NO” and Anti- Freeze function select “NO” from remote control, this control cannot activate.

Start condition:

- After [5] min from previous Zone pump off. **AND**
- Outdoor air temp < [3] °C **OR** Outdoor sensor is abnormal. **AND**
- Zone water temperature < [6]˚C **OR** Zone Sensor Short or Open

Cancel condition:

- After water Zone pump ON [4] min **AND**
- Outdoor air temp ≥ [4]°C **OR**
- During -5 °C ≤ Outdoor air temp < [4] °C **OR**
  Zone water temperature sensor > [8] °C
- During Outdoor air temp < [-5] °C
  Zone water temperature sensor > [20] °C
  *However, Zone water temperature sensor is Open or Short, Condition C and D is ignored.

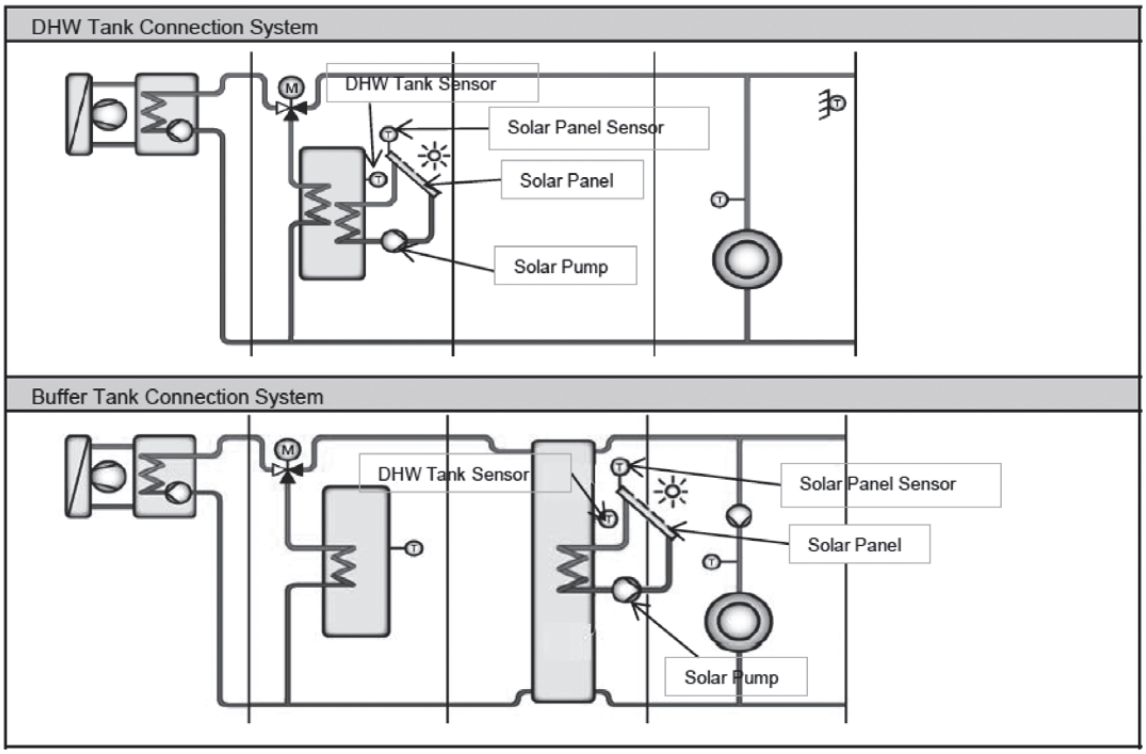
16.20 Solar Operation (Optional)

16.20.1 Solar Operation:

Solar function:

- This function allow user to control the solar pump to operate depend on the solar sensor reading compare to the tank installed. Solar pump will circulate the hot water energy store at solar panel to heat up the DHW Tank or Buffer Tank.

Solar Connection Diagram:



- Solar function can only enable when the Optional PCB is connected.
- Few part as below need to install to control the solar operation:
  - Solar Panel
  - Solar Pump
  - Solar Panel Sensor
  - Tank Sensor (Buffer tank sensor OR DHW Tank sensor depend on the connection direction)
    * During Solar Connection to the system, installer need to alert on the high water temperature may flow to the zone circuit or DHW piping circuit. Therefore pipe which withstand higher water temperature need to be installed.
- Solar remote control setting
  1 Solar Setting can only be set when the optional PCB connection is select "YES"
  2 By remote controller, Setting as below list can be set for solar function operation (Installer Menu)
    - Solar Function ("YES" or "NO)
    - Tank Connection Direction ("DHW TANK" or "BUFFER TANK")
    - Delta T turns ON: Difference temperature setting between solar panel sensor and Tank to turn on solar pump. (Range :5 ~ 15°C)
    - Delta T turns OFF: Difference temperature setting between solar panel thermistor and Tank to turn off solar pump. (Range :2 ~ 10°C)
    - Outdoor temp for Anti-Freeze : Outdoor temp to start Anti-Freeze control for solar circuit. (Range : -20 ~ 10°C)
    - Tank Temperature HI Limit Set (Range : 70 ~ 90°C)

16.20.2 Solar Operation Control

- Solar function can only be activate if the solar function selection “YES” from remote control. To achieve hot water from solar panel, indoor need to control the solar pump and circulate hot water from solar panel.

  Under normal case:
  - Solar pump start condition:
    - Solar panel temperature > Delta T turn on setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **AND**
    - Tank temperature (DHW or Buffer) < Solar HI Limit Temp (R/C) **AND**
    - Operation ON with heat mode (apply to solar connect to "Buffer Tank" case)

    * Condition c) ignore if the solar system is connect to DHW tank (control active under operation OFF time for Tank connection case)

  - Solar pump stop condition:
    - Solar panel temperature < Delta T turn OFF setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **OR**
    - Tank hot water temp >= Solar HI Limit Temp (R/C) + [2]°C

  Under solar Anti-freeze protection control:
  - Solar pump start condition:
    - Outdoor temp < Outdoor temp setting for Anti-Freeze (R/C)
  - Solar pump stop condition:
    - Outdoor temp > Outdoor temp setting for Anti-Freeze + [2]°C

**However, During Cool mode this function cannot activate if Tank selection is “Buffer Tank".
**Solar pump can operate even if Heat pump is under error stop.

- Solar operation during error:
  - During Tank sensor (DHW or Buffer depend on selection) abnormal, Solar operation will not able to function.
  - During Solar Panel sensor detect OPEN (not include SHORT), Solar operation will not able to function too.

16.21 Boiler Bivalent Control

- Boiler is an additional or alternative heat source to heat up the room when necessary.
- Purpose of this control is to turn ON and turn OFF the Boiler output signal when boiler heating capacity needed in the system.
- Boiler is possible to connect to DHW Tank and Buffer Tank depends on the installer.
- Boiler operation parameter need to be set on Boiler itself, indoor do not control the boiler operation direction and operation.
- There is two option of control pattern can be set by remote controller: AUTO OR SG ready Mode

  Auto Control Mode:
  1 There are Alternative mode, Parallel mode, & Advance Parallel mode available to select by installer to fit to the total system.

  Remote control setting value:
  1 Outdoor Ambient Set = (Range: -15°C ~ 35°C)

- **Alternative Mode**
  - Only one heat source operates at one time, either heat pump or boiler depends on condition.

  * External pump will turn ON when the external pump selection is ON when boiler is ON even heat pump is OFF.

  Control detail:
  During Operation ON at Heat mode or Tank mode or Heat + Tank Mode
  - Boiler signal turn ON and heat pump and water pump turn OFF when:
    - Outdoor ambient < Outdoor Ambient Set **AND**
    - Boiler prohibit flag = 0

** However indoor water pump can operate when Anti-freeze control condition fulfilled.

  - Boiler signal turn OFF and heat pump and water pump turn ON when:
    - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
    - Boiler prohibit flag = 1

- **Parallel Mode**
  - Parallel mode allows heat pump and boiler ON at the same time. Boiler operates as an additional heating capacity when low heat pump capacity at low ambient condition.

Control detail:
During operation ON at Heat mode or Tank mode or Heat + Tank mode
- Boiler signal turns ON when:
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when:
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Boiler prohibit flag = 1

- **Advance Parallel Mode**
  - Advance parallel mode allow heat pump to operate and turn ON boiler only when ambient and temperature condition is fulfilled.

Remote control setting value:
1 Outdoor Ambient Set = (Range : -15°C ~ 35°C)
2 Selection of boiler connection direction. (Heat only, DHW only, Heat & DHW)
3 Setting data under Heat Direction
   - Start Temperature | START_TEMP |
   - Start Delay Timer | START_TIMER |
   - Stop Temperature | STOP_TEMP |
   - Stop Delay Timer | STOP_TIMER |
4 Setting data under DHW Direction
   - Delay Timer | DELAY_TIMER |

- **SG ready Control Mode**
  - Using same SG ready from Sub Board input to control boiler ON/OFF output.

* When this SG ready is select for bivalent control, default SG ready function will change to control bivalent output
- Remote controller can set the External Pump ON/OFF like bivalent alternative mode

Control Content
Indoor will follow the SG ready bit input to control ON/OFF heat pump and boiler
00 : Heat pump OFF, Boiler OFF 10 : Heat pump OFF, Boiler ON
01 : Heat pump ON, Boiler OFF 11 : Heat pump ON, Boiler
* External pump will turn ON when the external pump selection is ON when boiler is ON even heat pump is OFF.

Control detail:
During operation ON at Heat Mode
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Buffer tank temperature < Target Buffer Tank Temperature + [START_TEMP] for [START_TIMER] **AND**
  - Heat pump operate at room side AND
  - Connection of Boiler to Heating Select "YES" From installer menu **AND**
  - Buffer Tank connection select "YES" **AND**
  - Boiler prohibit flag = 0

- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [-2°C] **OR**
  - Buffer Tank temperature > Target Buffer Tank temperature + [STOP_TEMP] for [STOP_TIMER] **OR**
  - Heat pump not at room side. **OR]**
  - Boiler prohibit flag = 1

During operation ON at Tank Mode
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Heat pump operate at tank side for continuous | DELAY_TIMER | **AND**
  - Connection of Boiler to DWH Tank select "YES" from installer menu. **AND**
  - Boiler prohibit flag = 0

- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Heat pump not operates at tank side. **OR**
  - Boiler prohibit flag = 1

Boiler prohibit flag control

Purpose:
- For product safety. Boiler signal is OFF when water temperature is too high.

Start condition:
- Water outlet ≥ 85°C continues for 5 minutes.
- Water inlet ≥ 85°C continues for 5 minutes.
- Zone1 water temp ≥ 75°C continues for 5 minutes.
- Zone2 water temp ≥ 75°C continues for 5 minutes.

Contents:
- After start condition fulfilled, set boiler prohibit flag = 1

Cancel condition:
- After 30 minutes from start condition fulfilled.

Contents:
- Set boiler prohibit flag = 0

Smart Bivalent Control Mode:

1 When this mode is selected by installer, it will decide whether to use heat pump or boiler to heat up room and sanitary tank.

Remote control setting value:

1 Energy price → Electricity price 1 ~ 10 = (Range : 0 ~ 999.9 */kWh)
Energy price → Boiler price = (Range : 0 ~ 999.9 */kWh) → Boiler efficiency = (Range : 0 ~ 99%)
2 Season setting → Season 1 ~ 4 = (Range : Jan ~ Dec)
3 Schedule setting → Season 1 ~ 4 → Pattern 1 ~ 4 → Start time = (Range : 12am ~ 11pm) → Price = (Range : 0 ~ 10)

* Installer is to input all the remote control setting as shown above based on regional electricity price rate and various season.

Control detail:
- Usage of heat pump or boiler to heat up is decided based on which heat source is cheaper as a result of the calculation from these remote control setting.
- It varies depends on heat pump's efficiency at various outdoor ambient condition and the boiler's efficiency.

16.22 External Room Thermostat Control (Optional)

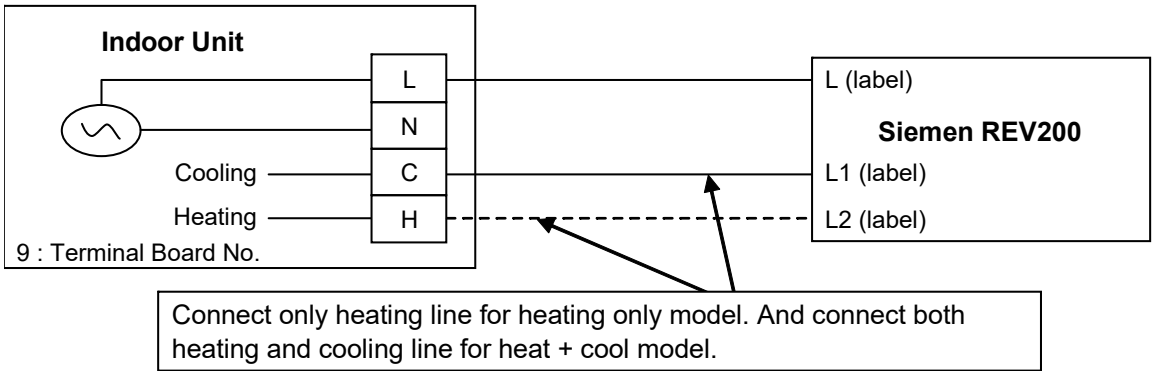
Purpose:

1 Better room temperature control to fulfill different temperature request by external room thermostat.
Recommended external room thermostat:

Maker	Characteristic
Siemen (REV200)	Touch panel
Siemen (RAA20)	Analog

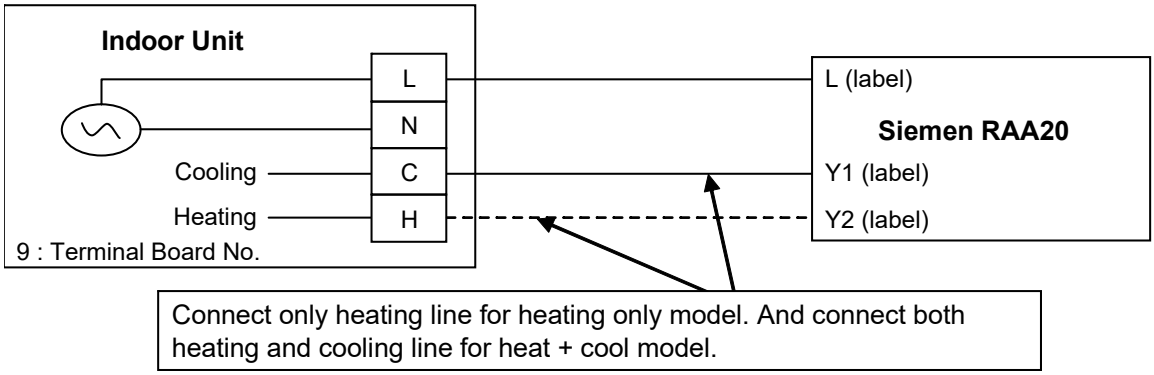
Connection of external room thermostat:
Wire Connection and thermo characteristic of Siemen REV200:

Setting	L/L1 (H)	Heat Thermo	L/L2 (C)	Cool Thermo
Set Temp < Actual Temp	Open Circuit	OFF	Short Circuit	ON
Set Temp > Actual Temp	Short Circuit	ON	Open Circuit	OFF



Wire Connection and thermo characteristic of Siemen RAA20:

Setting	L/Y1 (H)	Heat Thermo	L/Y2 (C)	Cool Thermo
Set Temp < Actual Temp	Open Circuit	OFF	Short Circuit	ON
Set Temp > Actual Temp	Short Circuit	ON	Open Circuit	OFF



Control Content:

- External room thermostat control activate only when remote thermostat connection select YES by Indoor control panel.
- When indoor running heat mode, refer thermo On/Off from heating line feedback. And when indoor running cool mode, refer thermo On/Off from cooling line feedback.
- Heat pump Off immediately when receive thermo off feedback.

16.23 Three Ways Valve Control

Purpose:

- 3 ways valve is used to change flow direction of hot water from heat pump between heating side and tank side.

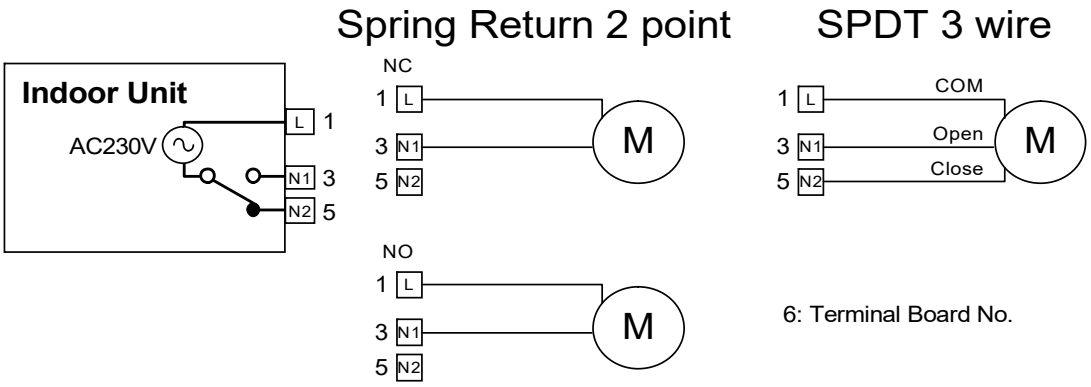
Control contents:

1 3 ways valve switch Off:
   - During 3 ways valve switch Off time, the hot water will provide heat capacity to heating side.

2 3 ways valve switch On:
   - During 3 ways valve switch On time, the hot water will provide heat capacity to tank side.

3 Stop condition:
   - During stop mode, 3 ways valve will be in switch off position.

4 When deice flag received from outdoor
   - During Heat mode, 3 ways valve maintain at room side.
   - During Tank mode, 3 ways valve switch ON to tank side regardless of tank thermo conditions.
   - During Heat + Tank mode, 3 ways valve direction is as below
     a) When receive deice bit from outdoor, if 3 ways valve is ON to tank side, maintain at tank side.
     b) When receive deice bit from outdoor, if 3 ways valve is OFF at room side, maintain at room side when any of below condition achieved :-
     i) Tank temperature < [40°C]
     ii) Remote controller 'DHW Defrost' select 'NO'
     Else, switch ON to Tank side.
     * Resume back to room interval after deice operation complete.



* During pump down and force mode, fix 3 ways valve in close condition.

* Recommended Parts : SFA 21/18 (Siemens)

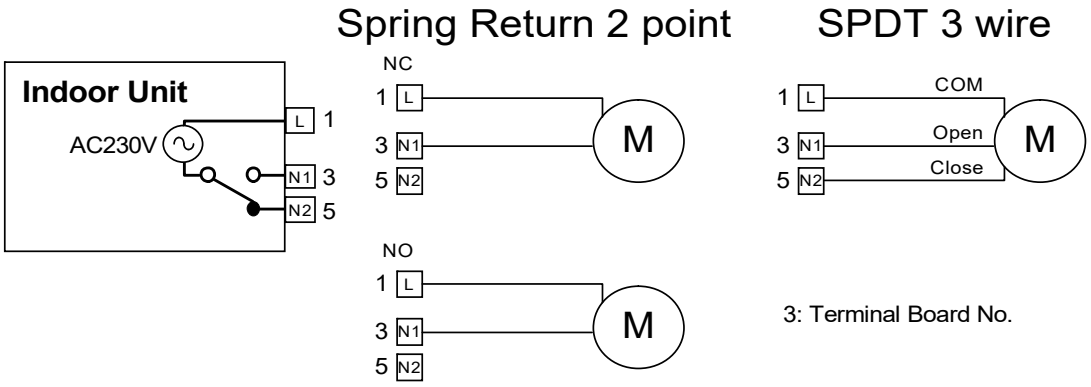
16.24 Two Ways Valve Control

Functionality of 2 ways valve:

- Use to allow hot water to floor heating panel or block cold water to floor heating panel.

Control contents:

1 When indoor running in heat mode, OPEN the 2 ways valve.
2 When indoor running in cool mode, CLOSE the 2 ways valve.
3 Stop condition:
   a. During stop mode, fix 2 ways valve in close condition.

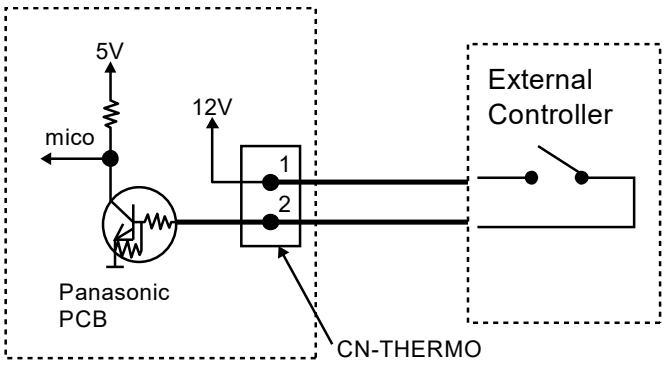


* During pump down mode, fix 2 ways valve in close condition.
* During force mode, open 2 ways valve.

* Recommended Parts : SFA 21/18 (Siemens)

16.25 External OFF/ON Control

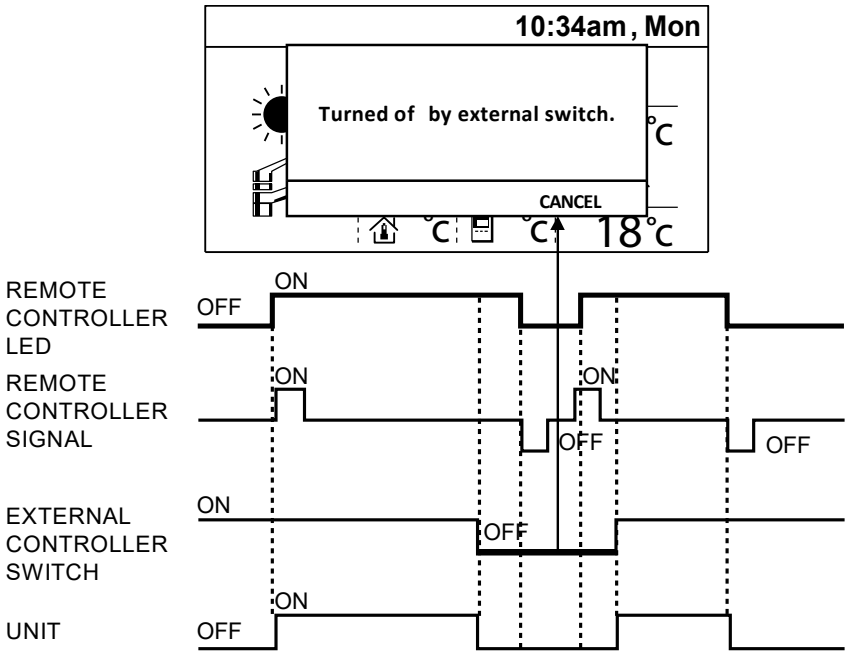
- Communication circuit between indoor unit and external controller is as per below.



- Maximum length of communication cable is 50 meter.
- Control content:

External Control Switch	Control Panel OFF/ON	Control Panel Power LED	System Status
ON	ON	ON	ON
ON	OFF	OFF	OFF
OFF	ON	ON	OFF
OFF	OFF	OFF	OFF

Remocon Screen Display and Control Detail:



When External SW connection select "YES" from remocon installer menu:

- Heating or Cooling system will operate normally if the External Switch signal is ON.
- Once the External Switch turn OFF, System Turn OFF (Heat pump, water pump, heater etc…)
- Remocon LED remain ON or OFF according to the current operation request.
- Pop up menu at remocon main screen as above screen to inform customer system stop by External Switch.
- It is possible to press cancel and return to main screen to do change of operation setting while waiting the External Switch turn ON back.
- Remocon LED will always follow the latest changes from remocon.
- If no action on remocon for continuous 5 minutes, the pop up screen will show again on the screen.
- But once the External Switch Turn ON back, pop up screen will disappear and system can operate normally according to the latest operation setting and request.

16.26 External Compressor Switch (Optional PCB)

External compressor switch port can have two purpose of control as below:

- Heat source ON/OFF function (Dip switch Pin 3 on PCB "OFF")
- Heater ON/OFF function (Dip switch Pin 3 on PCB "ON")
- Heat source ON/OFF function

Purpose:

- Heat pump ON/OFF function is use to turn OFF the high power consumption device (Heat pump, & Heater) when there is energy or electric current limitation. Other optional function still can be operate under heat pump and heater OFF condition.

Control Detail:

- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "OFF" This heat pump ON/OFF function will activate
- The ON/OFF signal of this External Compressor Switch is same as External Switch.
- When the External Compressor Switch is ON:
  - Heat pump system operate normally
- When the External Compressor Switch is OFF:
  - Heat pump, Indoor water pump & Heater (Booster heater & Backup Heater) need to turn OFF
  - Solar, Boiler and zone control can be operate follow normal control condition.
    * pump delay OFF also included in this control

  (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

- Heater ON/OFF function

Purpose:

- Heater ON/OFF function is use to turn OFF the heater (backup heater & booster heater) when there is energy or electric current limitation. Heat pump and other optional function still can operate.

Control detail:

- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "ON" This heater ON/OFF function will activate
- When the External Compressor Switch is ON:
  - Heat pump and heater operate normally
- When the External Compressor Switch is OFF:
  - Backup heater and booster heater cannot operate even heater request is ON.
  - Heat pump and option function (Solar, Boiler and zone control) can be operate follow normal control condition.

  (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

16.27 Heat/Cool Switch (Optional PCB)

Purpose:

- User can switch the running mode from heat to cool or cool to heat through external installed Heat/Cool switch. This kind of heat / cool switch may built in inside the field supply room remocon as well.

Control contents:

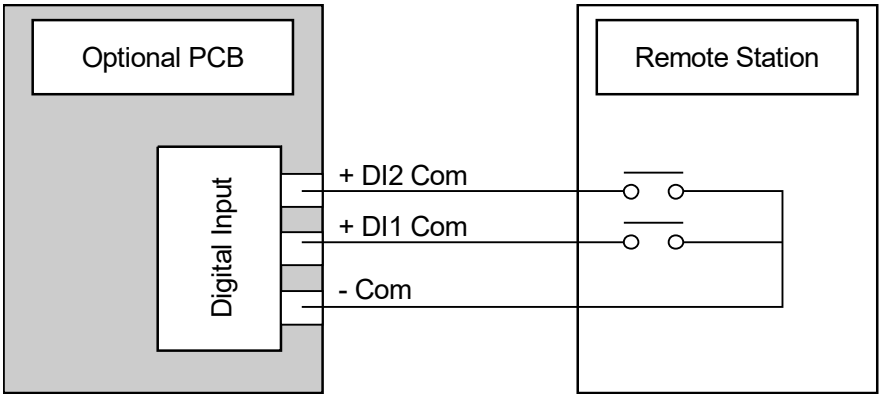
- Heat/Cool Switch can only be set when Cool Function is "enable" at custom menu setting, & Extension PCB select "YES" & Zone 1 not set "Pool" condition.
- This heat/cool switch control will be activate only when installer set the Heat/Cool Switch "USE" through remocon.
- Once the Heat/Cool Switch Set "USE", remocon will check indoor send Signal to judge the option of mode select.
  - When Heat/Cool Switch Contact Open : Remocon only can select Heat Mode, or Heat + Tank Mode, or Tank Mode
  - When Heat/Cool Switch Contact Close : Remocon only can select Cool Mode, or Cool + Tank Mode, or Tank Mode
- Operation ON/OFF will depend on remocon request.
- When Heat Mode is running with Contact Open, user change this setting to contact close, indoor will this signal to remocon judge and change mode to cool and send back to indoor. And it is same as from cool mode change to heat mode.
  * This switch have higher priority, remocon follow indoor send signal when control activated.
  * There is no effect to the operation when the mode running is only Tank Mode.

  (Weekly Timer are ignored and cannot be set during Heat / Cool Switch is "Enable" Condition.)

16.28 SG Ready Control (Optional PCB)

Purpose:

- To set ON/OFF of heat pump and target temperature by digital input of third party device if necessary in field.



Remote control setting

For this function, following items need to be set on Remote Control (installer menu) –

- SG control = YES or NO
- Capacity up setting 1
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]
  - Cooling capacity [-15 ~ 0 %]
- Capacity up setting 2
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]
  - Cooling capacity [-15 ~ 0 %]

Control contents:

If SG control on remote control = "Yes", then following control only activate by digital input.

- **While Digital input is " 00 " (Normal operation)**
  - Normal operation. Once detect '00' system will operate back to normal condition.
    (All the target set temperature for heating side and DHW side will return back to previous set temperature when digital signal change from "10' or "11" back to "00".)
- **While digital input is detected " 01 " (HP stop)**
  - Heat pump & room heater & tank heater cannot operate.
    (Solar control and Boiler back up and 2 Zone control can activate.)
- **While digital input is detected " 10 " (Capacity 1)**
  - Target temperature for heating and DHW Tank is changed according to the percentage set by Remote control setting. However, which setting temperature is change depend on system setting.
  - Target water temperature of cooling is changed according to the adjustment value set by remocon setting.
- **While digital input is detected " 11 " (Capacity 2)**
  - Target temperature for heating and DHW Tank is changed according to the percentage set by Remote control setting. However, which setting temperature is change depend on system setting.
  - Target water temperature of cooling is changed according to the adjustment value set by remocon setting.
- **While digital input is detected " 10 " (Capacity 1)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

If Buffer selection is "YES"
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 1) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) % * (Max regulation depend on the tank max setting limit)

If Buffer selection is "NO"
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 1) %
(Zone 1 and Zone 2 will change according to its own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) % * (Max regulation depends on the tank max setting limit)

Setting temperature for cooling is changed
New Target water temperature = target water temperature + Remote Control setting for cool (*Capacity 1) * (Min/Max regulation of cooling water set apply)

- **While digital input is detected " 11 " (Capacity 2)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

If Buffer selection is "YES"
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 2) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) % * (Max regulation depends on the tank max setting limit)

If Buffer selection is "NO"
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 2) %
(Zone 1 and Zone 2 will change according to it's own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) % * (Max regulation depends on the tank max setting limit)
** This function is not applicable for Cooling mode.

Setting temperature for cooling is changed
New Target water temperature = target water temperature + Remote Control setting for cool (*Capacity 2) * (Min/Max regulation of cooling water set apply)

16.29 Demand Control (Optional PCB)

Remote control setting:

- When Optional PCB connection select 'YES", Demand Control function can select "YES" or "NO".

Purpose:

- After the demand control select YES, below control will activated.
  - 0-10V Demand control

0-10V Demand control

- Demand control is use to reduce the current usage of heat pump unit by third party device.

Control start condition:

- Select “YES” at Demand control at installer menu.
- 0-10V input for this electrical current control is detected.

Control content:

- If start condition is fulfilled, indoor will receive the voltage signal from optional PCB. Indoor will send the rate value to outdoor unit.
- Outdoor will change the current limit according to the percentage receive from indoor unit.

16.30 Holiday Mode

- Purpose:
  Promotes energy saving by allowing the user to stop the system during holiday and enables the system to resume at the preset temperature after holiday.

- Control details:
  - Indoor operate the unit according running mode request. Target temperature will follow holiday setting temperature.
    - If heat mode request is receive, Target Water Out Temperature will change according to holiday shift temperature set.
      [If heat is set OFF at holiday, unit, water pump and zone control will OFF]
    - If tank mode request is receive, Target Tank Set Temperature will change according to the holiday tank shift temperature set.
      [If tank is set OFF at holiday, heat pump and tank heater will OFF]
  - After days of holiday have been set, heat pump will stop and only resume operation at the end of holiday countdown.

- Start condition:
  - Holiday timer set and the holiday timer start
    * The day holiday mode was set is counted as day 1.

- Stop condition:
  - OFF/ON button is pressed.
  - Holiday timer is reached.

16.31 Dry Concrete

- Purpose
  Provide heat to floor heating panel and dry the wet concrete during installation.

- Setting condition:
  - Dry concrete parameter can be set through remote control under system setup.
  - Parameters are possible to set up to 99 days with different target set temperature

- Control details:
  - Dry concrete mode will be activates when select ON from service setup.
  - Once start dry concrete function, remote control will send step 1 setting temperature to indoor unit.
    * This temperature is set at zone temperature. If system is 2 zones, both zone target temperature is set as same temperature.
  - Heat pump will start heat mode operation to room side with received target water outlet temperature.
    * Heat pump will operate according to Heat pump Target Water Temperature.
  - After complete day 1 setup operation, day 2 data will be send to indoor at 12.00am on the second day.
  - Each preset data will be send every day until dry concrete mode is complete, unit will turns OFF and exit dry concrete function.
  - 3 ways valve and booster heater will turn OFF and 2 ways valve will turns ON.

- Cancel condition:
  - Dry concrete mode is complete and OFF signal is received.
  - OFF signal is received by pressing OFF/ON button.

16.32 Flow Sensor

- The water flow sensor serves as an overload protector that shuts down the unit when the water level is detected to be low.
- Abnormal flow detection:

Sequence	Abnormal flow	Normal flow
Normal case	Flow rate < 7 l/min or ≥ 69 l/min	≥ 7 l/min
During status 2~6 on Anti-freeze deice	Flow rate ≥ 7 l/min	< 7 l/min

17. Protection Control (WH-UDZ03KE5)

17.1 Protection Control for All Operations

17.1.1 Time Delay Safety Control

1 The compressor will not start for three minutes after stop of operation.

17.1.2 30 Seconds Forced Operation

1 Once the compressor starts operation, it will not stop its operation for 30 seconds.
2 However, it can be stopped using control panel at indoor unit.

17.1.3 Total Running Current Control

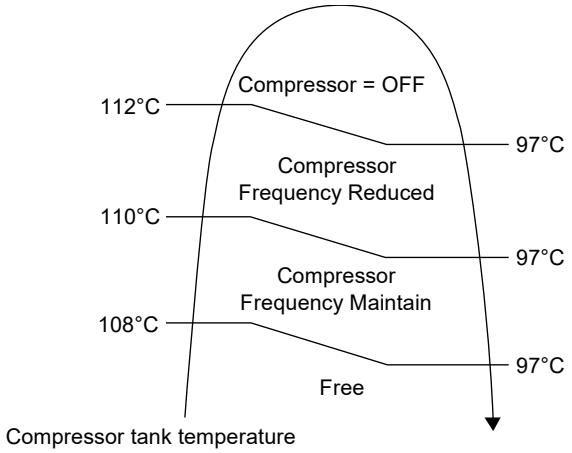
1 When the outdoor running current exceeds X value, the compressor frequency will decrease.
2 If the outdoor running current does not exceed X value, the compressor frequency will return to normal operating frequency.
3 If the outdoor running current continue to increase till exceed Y value, compressor will stop, and if this occurs 3 times within 20 minutes, system will stop operation and OFF/ON control panel LED will blink (F16 error occurs).

	WH-UDZ03KE5	
Operation Mode	X (A)	Y (A)
Heating	11.0	14.0
Cooling	7.0	14.0

A. DC Peak Current Control
1 When the current to IPM exceeds set value, compressor will stop. Compressor will restart after three minutes.
2 If the set value exceeds again for more than 30 seconds after the compressor restarts, operation will restart after two minutes.
3 If the set value exceeds again for within 30 seconds after the compressor restarts, operation will restart after one minute. If this condition repeats continuously for seven times, system will stop operation and OFF/ON control panel LED will blink (F23 error occurs).

17.1.4 Compressor Overheating Prevention Control

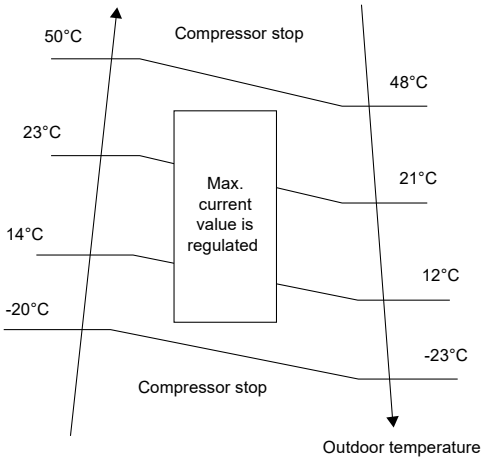
- The compressor operating frequency is regulated in accordance to compressor tank temperature as shown in below figures. When the compressor tank temperature exceeds 112°C, compressor will stop, and if this occurs 4 times within 30 minutes, system will stop operation and OFF/ON control panel LED will blink (F20 error occurs).



17.1.5 High Pressure Sensor Control

- Purpose:
  - To protect the system operation.
- Detection period:
  - After compressor on for 1 minute.
- Detection conditions:
  - When abnormal high voltage detection, 5 V or when open circuit detection 0V for 5 seconds continuously.
- After detection:
  - When abnormality is detected continue 5 seconds, unit stop operation.
  - OFF/ON control panel LED will blink (H64 error occurs).

17.1.6 Outside Temperature Current Control



17.2 Protection Control for Heating Operation

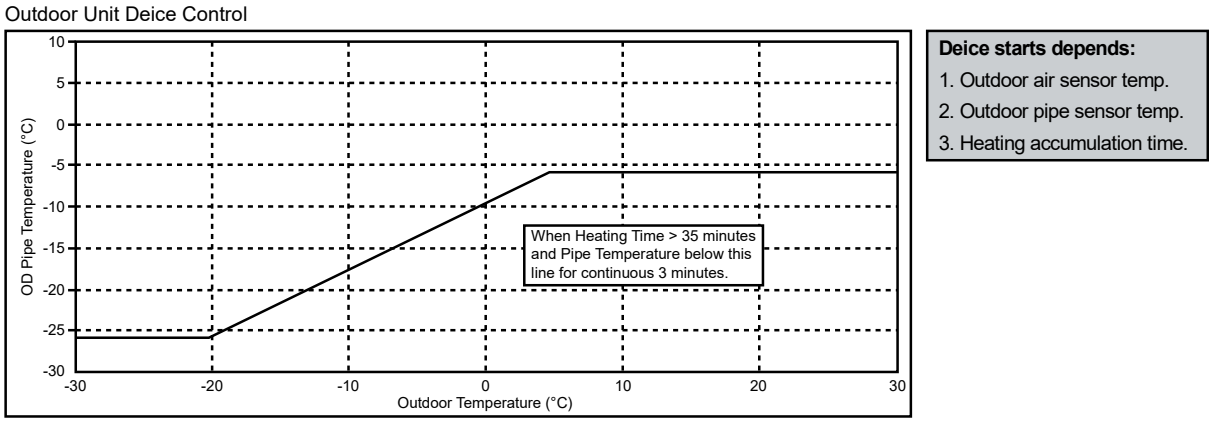
17.2.1 Outdoor Air Temperature Control

The maximum current value is regulated when the outdoor air temperature rises above 14°C in order to avoid compressor overloading.

17.2.2 Deice Operation

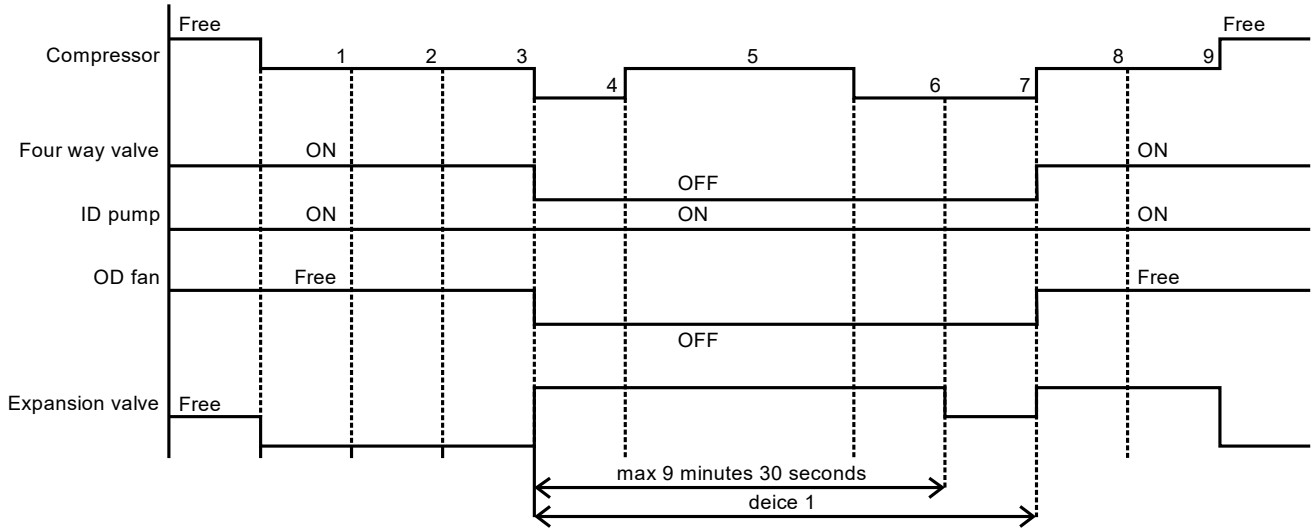
When outdoor pipe temperature and outdoor air temperature is low, deice operation start where outdoor fan motor stop.
Normally, deice start if pipe sensor temperature fulfill deice condition. If remote controller set to AUTO force defrost setting, unit will start force deice after heat pump operate for 3 hours without deice at below outdoor temperature 5°C or outdoor pipe sensor temperature fulfill deice condition and Deice accumulation timer reach 2 hours.

- Deice judging condition



Deice operation time diagram

a. Deice mode 1 control:



17.2.3 Force Defrost Operation

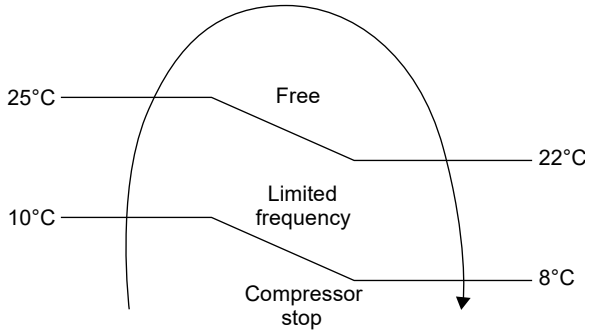
Force defrost can be set through remote control with two selection (Manual OR Auto).

- If Manual defrost set, heat pump only run force defrost at heat mode when force defrost request from quick menu remote control.
- If Auto defrost set, heat pump automatically run force defrost operation when any of below conditions is achieved :
  i) after 3 hours heating accumulation time without defrost when ambient below 5°C.
  ii) after outdoor pipe sensor temperature fulfill deice condition and Deice accumulation timer reach 2 hours.
  * Deice accumulation timer will not reset when remote control is pressed OFF.

17.3 Protection Control for Cooling Operation

17.3.1 Outdoor Air Temperature Control

- The Compressor operating frequency is regulated in accordance to the outdoor air temperature as shown in the diagram below.
- This control will begin 1 minute after the compressor starts.
- Compressor frequency will adjust base on outdoor air temperature.



17.3.2 Freeze Prevention Control 1

1 When indoor heat exchanger temperature is lower than 0°C continuously for 10 seconds, compressor will stop operating.
2 Compressor will resume its operation three minutes after the indoor heat exchanger is higher than 1°C.
3 Indoor heat exchanger freeze prevention (H99) will memory in error history.

18. Protection Control (WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5)

18.1 Protection Control for All Operations

18.1.1 Time Delay Safety Control

1 The compressor will not start for three minutes after stop of operation.

18.1.2 30 Seconds Forced Operation

1 Once the compressor starts operation, it will not stop its operation for 30 seconds.
2 However, it can be stopped using control panel at indoor unit.

18.1.3 Total Running Current Control

1 When the outdoor running current exceeds X value, the compressor frequency will decrease.
2 If the outdoor running current does not exceed X value, the compressor frequency will return to normal operating frequency.
3 If the outdoor running current continue to increase till exceed Y value, compressor will stop, and if this occurs 3 times within 20 minutes, system will stop operation and OFF/ON control panel LED will blink (F16 error occurs).

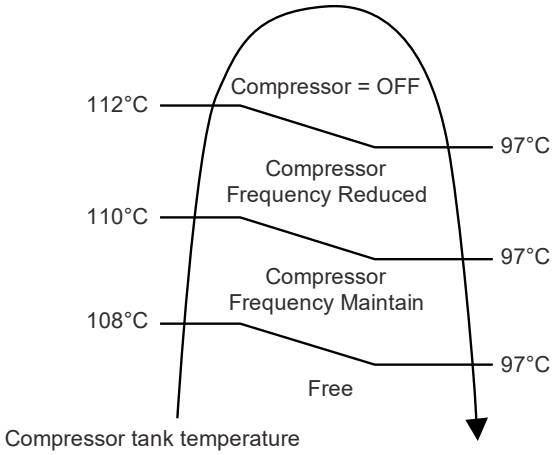
	UDZ05KE5		UDZ07KE5		UDZ09KE5	
Operation Mode	X (A)	Y (A)	Y (A)	Y (A)	X (A)	Y (A)
Heating	12.5	14.6	14.9	16.9	14.9	16.9
Cooling	8.5	14.6	14.1	16.9	14.1	16.9

A. DC Peak Current Control

1 When the current to IPM exceeds set value, compressor will stop. Compressor will restart after three minutes.
2 If the set value exceeds again for more than 30 seconds after the compressor restarts, operation will restart after two minutes.
3 If the set value exceeds again for within 30 seconds after the compressor restarts, operation will restart after one minute. If this condition repeats continuously for seven times, system will stop operation and OFF/ON control panel LED will blink (F23 error occurs).

18.1.4 Compressor Overheating Prevention Control

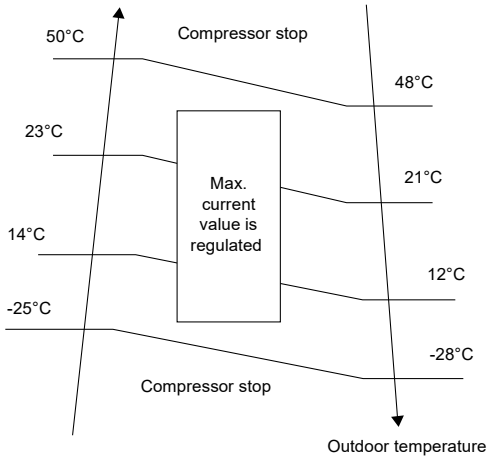
- The compressor operating frequency is regulated in accordance to compressor tank temperature as shown in below figures. When the compressor tank temperature exceeds 112°C, compressor will stop, and if this occurs 4 times within 30 minutes, system will stop operation and OFF/ON control panel LED will blink (F20 error occurs).



18.1.5 High Pressure Sensor Control

- Purpose:
  - To protect the system operation.
- Detection period:
  - After compressor on for 1 minute.
- Detection conditions:
  - When abnormal high voltage detection, 5 V or when open circuit detection 0V for 5 seconds continuously.
- After detection:
  - When abnormality is detected 4 times within 120 minutes, unit stop operation.
  - OFF/ON control panel LED will blink (H64 error occurs).

18.1.6 Outside Temperature Current Control



18.2 Protection Control for Heating Operation

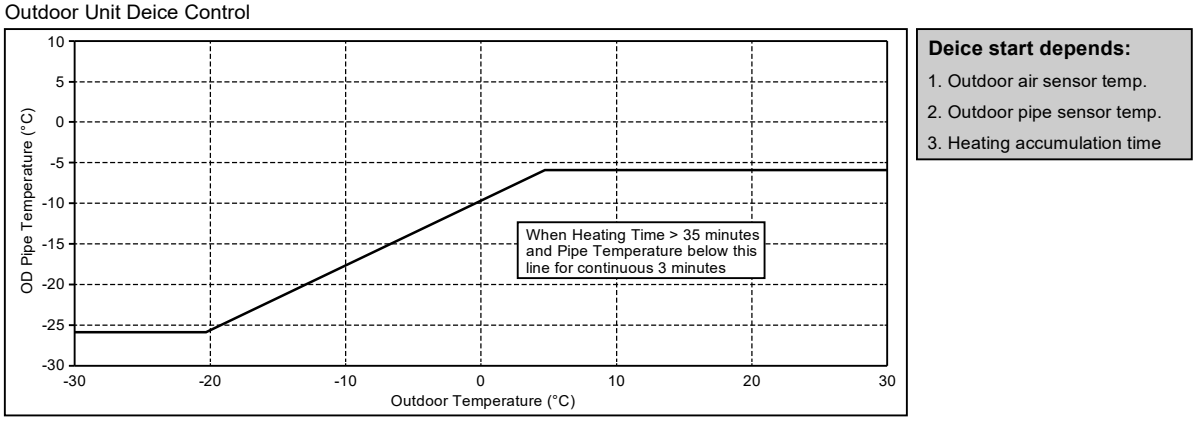
18.2.1 Outdoor Air Temperature Control

The maximum current value is regulated when the outdoor air temperature rises above 14°C in order to avoid compressor overloading.

18.2.2 Deice Operation

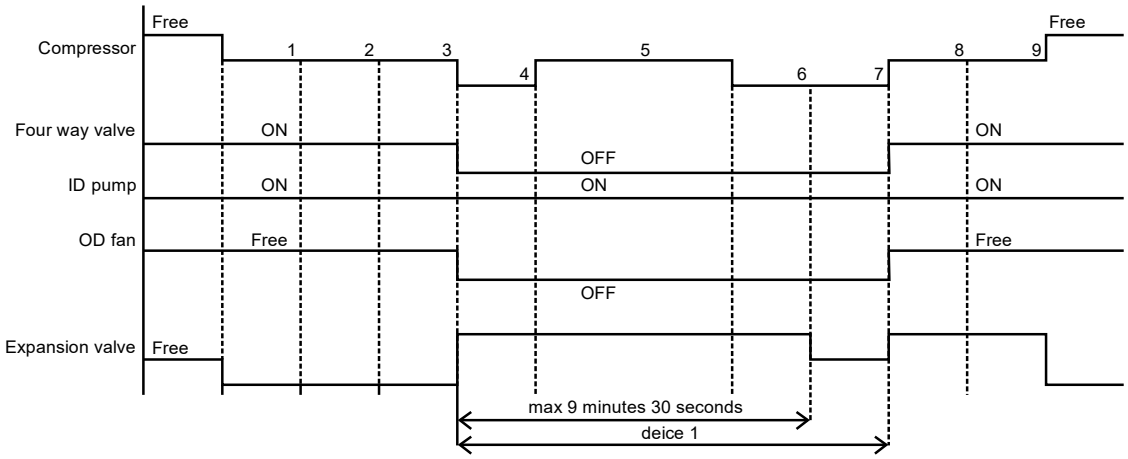
When outdoor pipe temperature and outdoor air temperature is low, deice operation start where outdoor fan motor stop.
Normally, deice start if pipe sensor temperature fulfill deice condition. If remote controller set to AUTO force defrost setting, unit will start force deice after heat pump operate for 3 hours without deice at below outdoor temperature 5°C or outdoor pipe sensor temperature fulfill deice condition and Deice accumulation timer reach 2 hours.

- Deice judging condition



Deice operation time diagram

a. Deice mode 1 control:



18.2.3 Force Defrost Operation

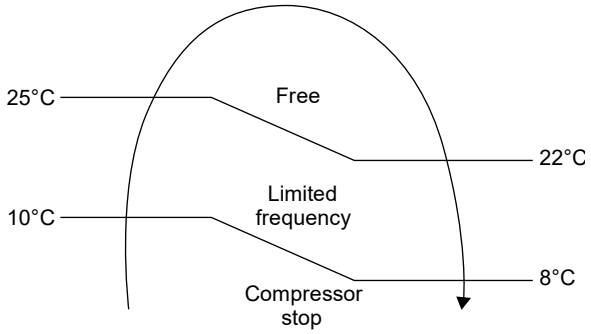
Force defrost can be set through remote control with two selection (Manual OR Auto).

- If Manual defrost set, heat pump only run force defrost at heat mode when force defrost request from quick menu remote control.
- If Auto defrost set, heat pump automatically run force defrost operation when any of below conditions is achieved :
  i) after 3 hours heating accumulation time without defrost when ambient below 5°C.
  ii) after outdoor pipe sensor temperature fulfill deice condition and Deice accumulation timer reach 2 hours.
  * Deice accumulation timer will not reset when remote control is pressed OFF.

18.3 Protection Control for Cooling Operation

18.3.1 Outdoor Air Temperature Control

- The Compressor operating frequency is regulated in accordance to the outdoor air temperature as shown in the diagram below.
- This control will begin 1 minute after the compressor starts.
- Compressor frequency will adjust base on outdoor air temperature.



18.3.2 Freeze Prevention Control 1

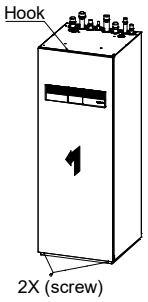
1 When indoor heat exchanger temperature is lower than 0°C continuously for 10 seconds, compressor will stop operating.
2 Compressor will resume its operation three minutes after the indoor heat exchanger is higher than 1°C.
3 Indoor heat exchanger freeze prevention (H99) will memory in error history.

19. Servicing Guide

19.1 How to take out Front Plate

Open and Close Front Plate

1 Remove the 2 mounting screws of Bottom Front Plate.
2 Slide it upwards to unhook the Bottom Front Plate hook.
3 Reverse above steps 1~2 for close it.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

19.2 Test Run

1 Before test run, make sure below items have been checked:-
   a) Pipework are properly done.
   b) Electric cable connecting work are properly done.
   c) Tank Unit is filled up with water and trapped air is released.
   d) Please turn on the power supply after filling the tank until full.
2 Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run. Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

3 For normal operation, Water Pressure reading should be in between 0.05 MPa and 0.3 MPa (0.5 bar and 3 bar). If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4 After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

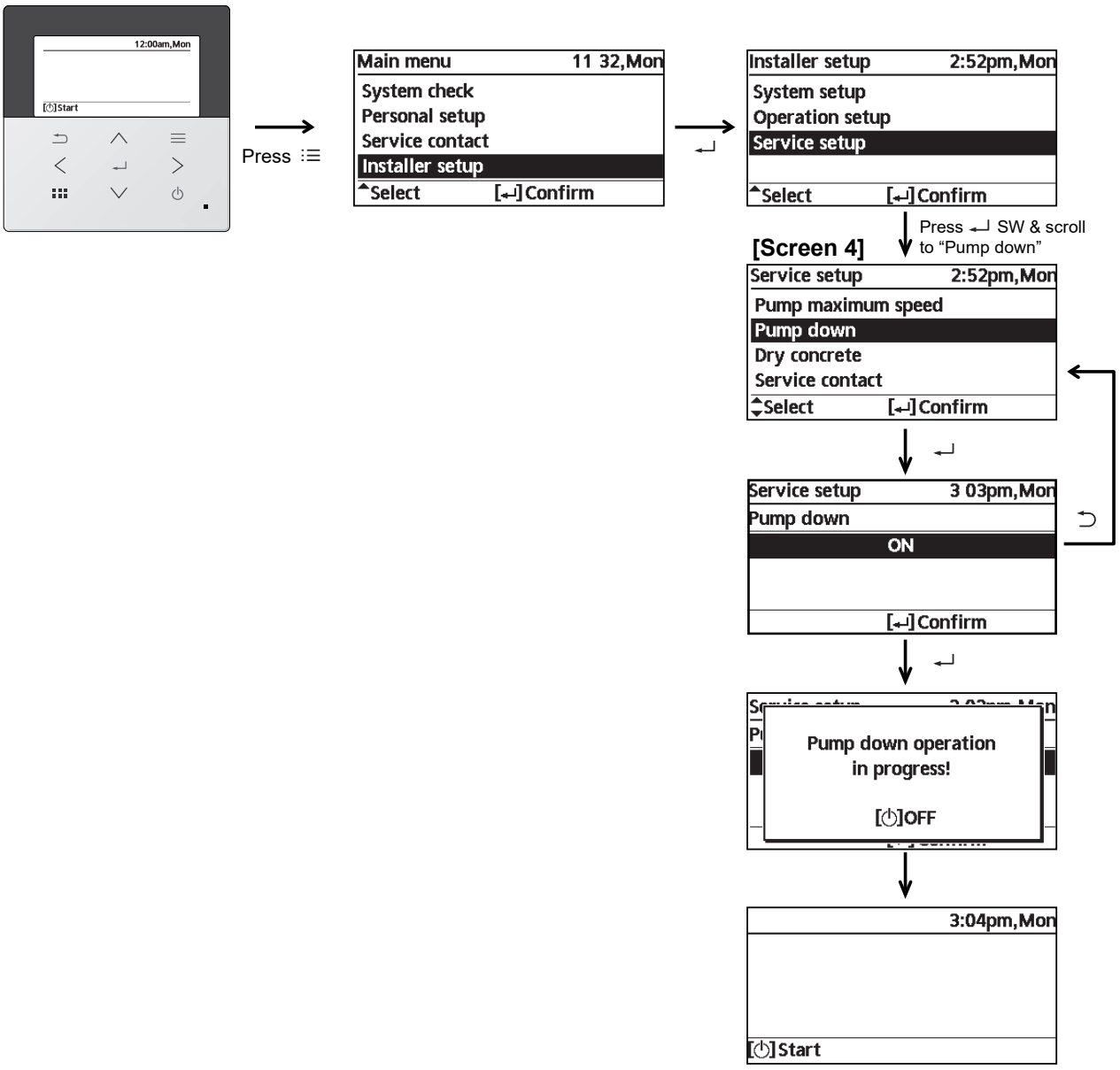
19.3 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

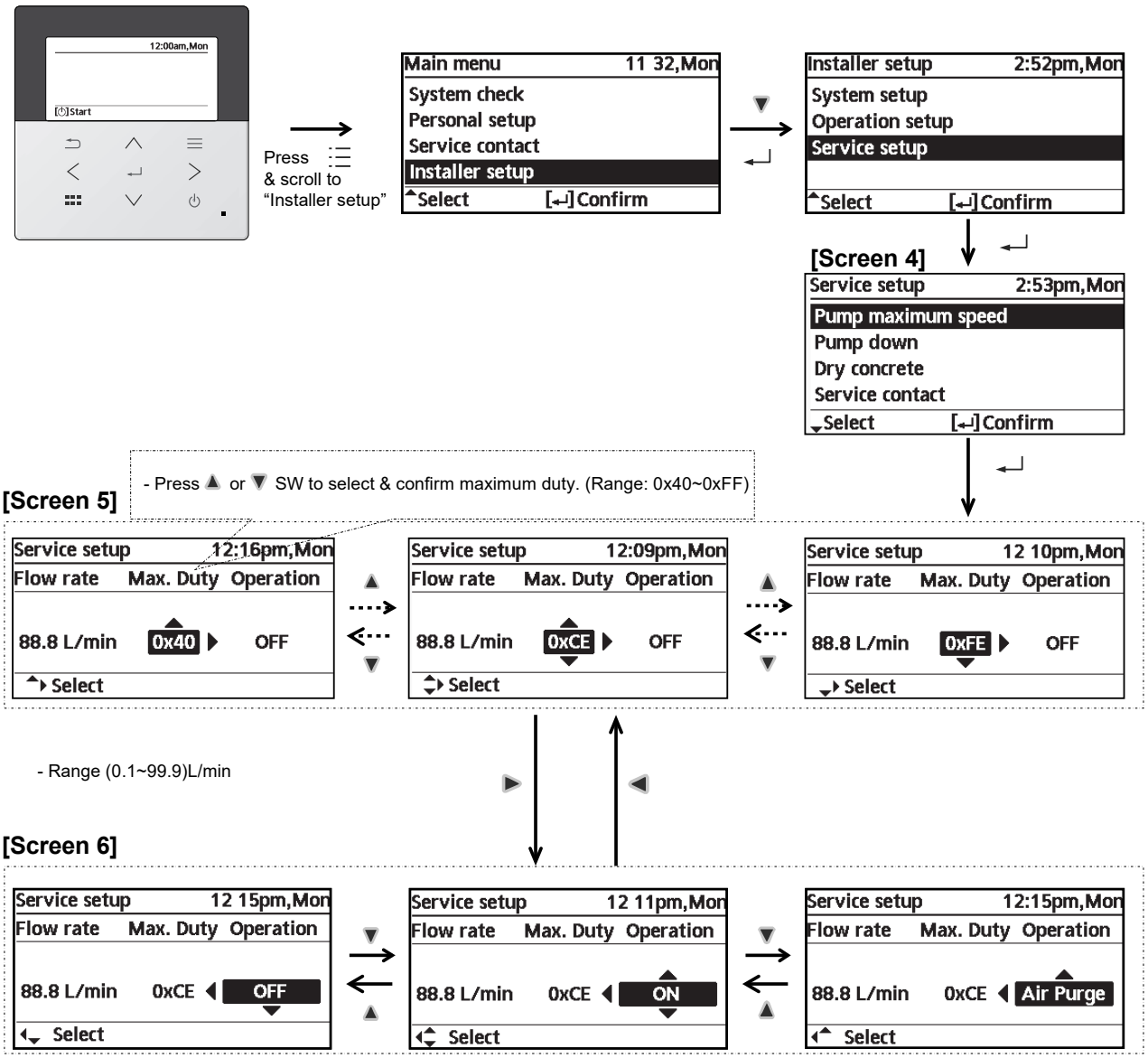
- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit’s piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m.

19.4 Pump Down Procedures

Refer below steps for proper pump down procedure.



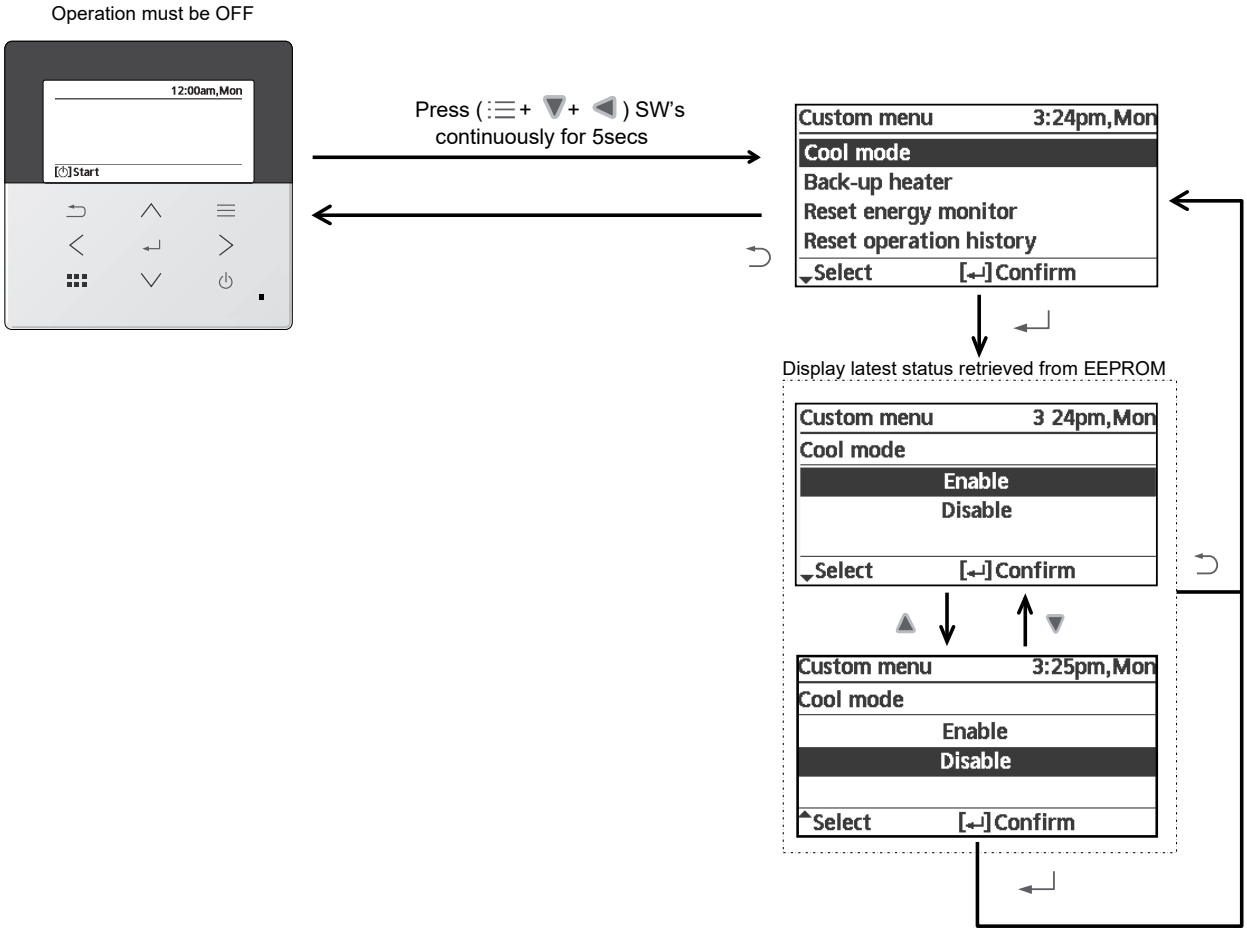
19.5 How To Adjust Pump Speed



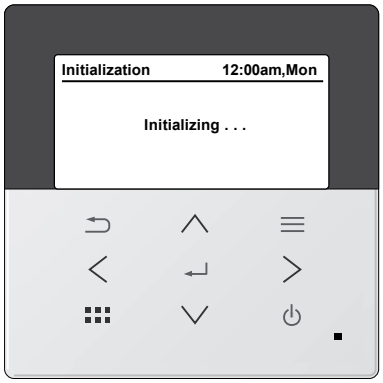
NOTE:

1. Whenever at [Screen 5], if press ⏻ SW to OFF, pump operation should be turned OFF.
2. Whenever at [Screen 6], if press ⏻ SW to OFF, pump operation should be turned OFF.

19.6 How To Unlock Cool Mode

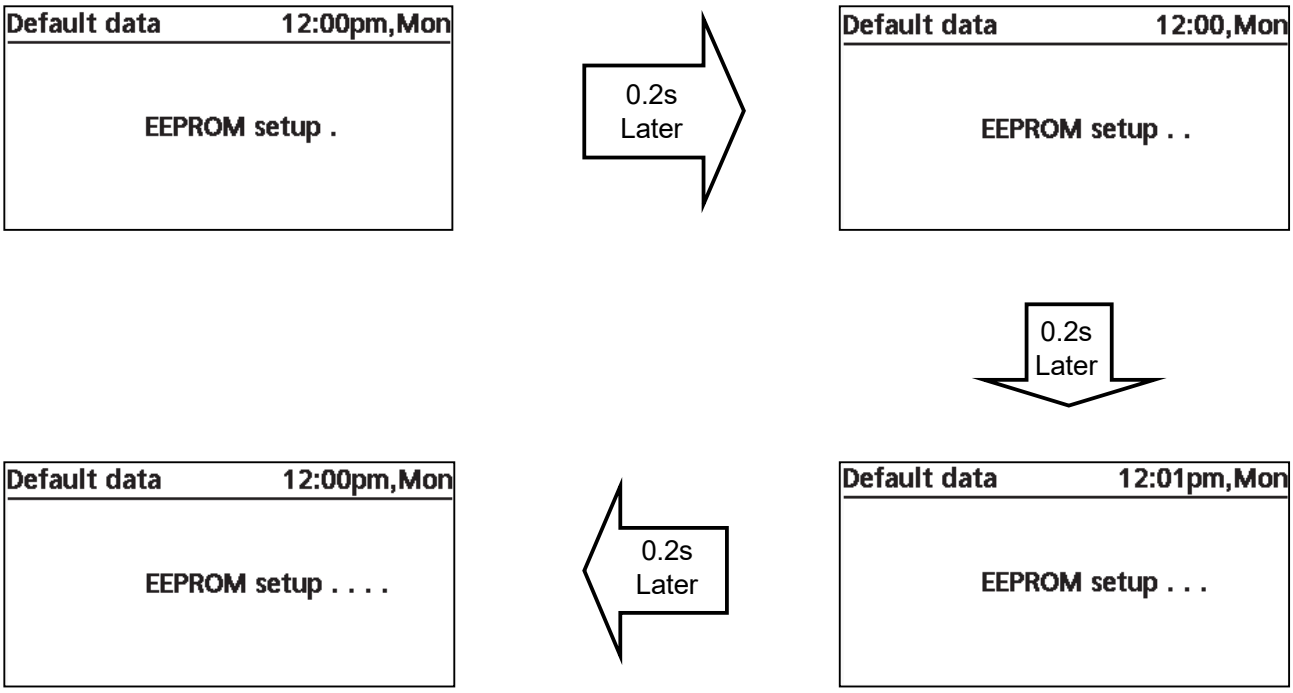


19.7 EEPROM Factory Default Data Setup Procedure

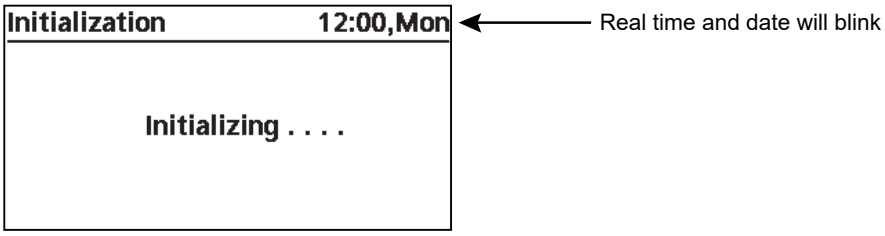


- EEPROM default data setup is only possible during initialization process.
- Press (▲, ▼, ◀, ▶) simultaneously for 5secs continuously, initialization process will stop & EEPROM default data setup process will start.

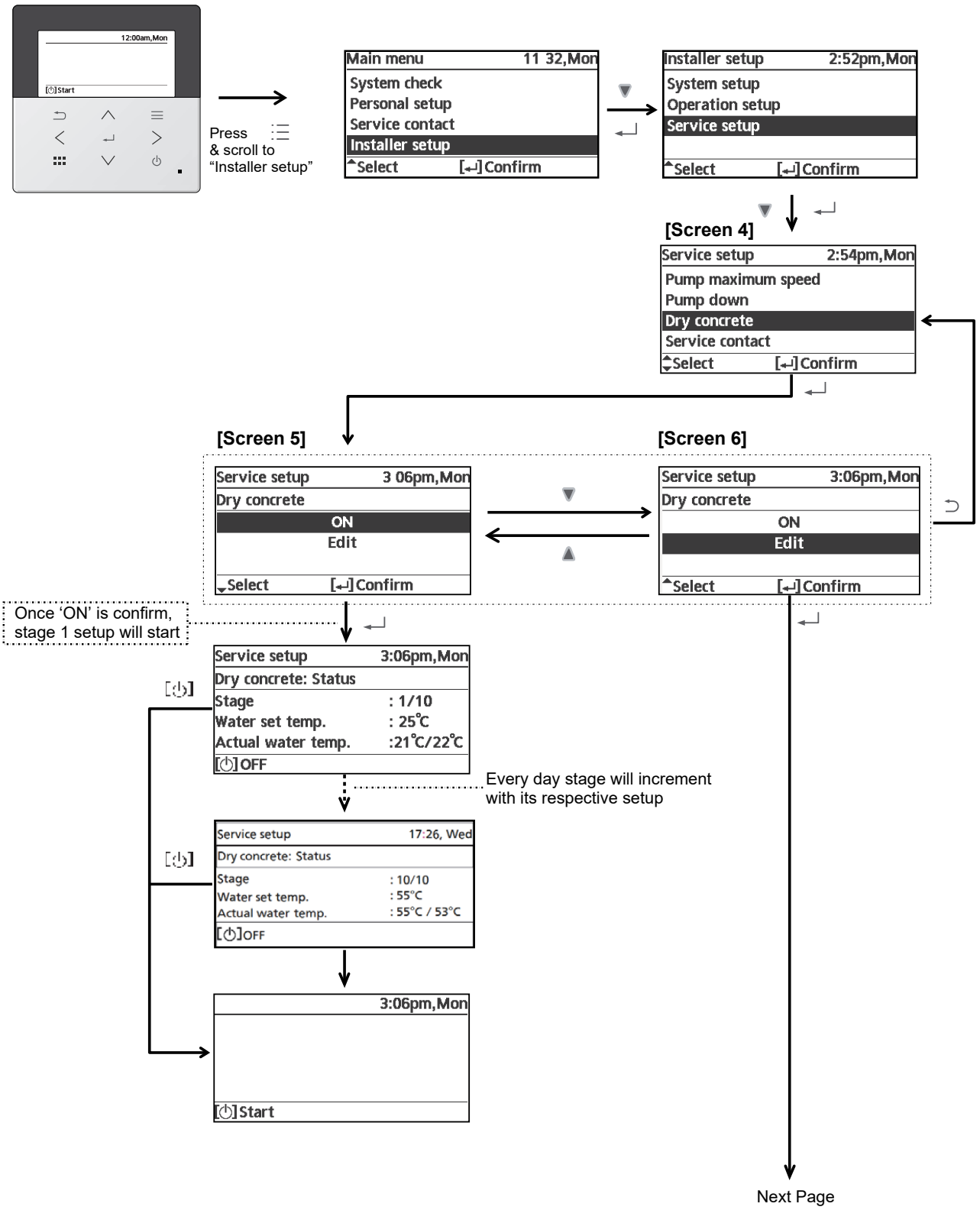
During EEPROM default data setup process, display should be as shown below.

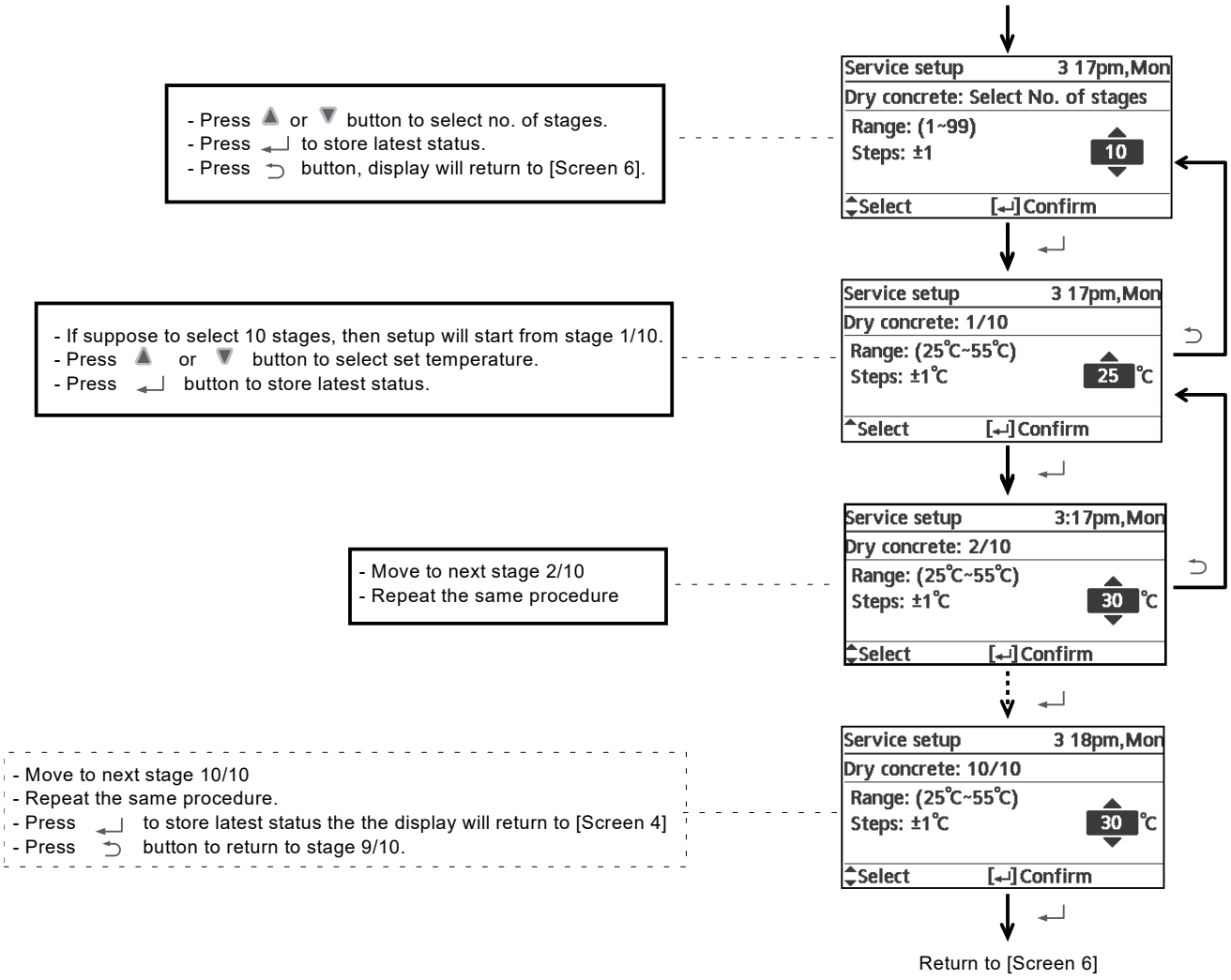


- Once EEPROM default data setup process is complete, initialization process will re-start from beginning.



19.8 Dry Concrete Setup




- Press ▲ or ▼ button to select no. of stages.
- Press ↵ to store latest status.
- Press ↩ button, display will return to [Screen 6].
Service setup 3 17pm,Mon
Dry concrete: Select No. of stages
Range: (1~99)
Steps: ±1
10
Select [↵]Confirm
- If suppose to select 10 stages, then setup will start from stage 1/10.
- Press ▲ or ▼ button to select set temperature.
- Press ↵ button to store latest status.
Service setup 3 17pm,Mon
Dry concrete: 1/10
Range: (25℃~55℃)
Steps: ±1℃
25 ℃
Select [↵]Confirm
- Move to next stage 2/10
- Repeat the same procedure
Service setup 3:17pm,Mon
Dry concrete: 2/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
- Move to next stage 10/10
- Repeat the same procedure.
- Press ↵ to store latest status the the display will return to [Screen 4]
- Press ↩ button to return to stage 9/10.
Service setup 3 18pm,Mon
Dry concrete: 10/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
Return to [Screen 6]

20. Maintenance Guide

In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

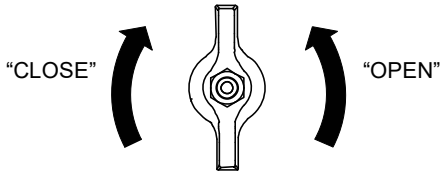
1 Charging and Discharging the Water

Make sure all the piping installations are properly done before carry out below steps.

Charge the Water

- For domestic hot water tank

a. Set the Domestic Hot Water Tank Discharge (Drain Tap) to “CLOSE”.



Domestic Hot Water Tank Discharge (Drain Tap)

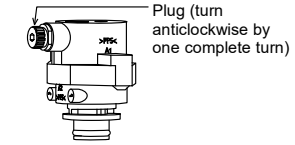
b. Set all Tap / Shower “OPEN”.

c. Start filling water to the Domestic Hot Water Tank via Tube Connector. After 20~40min, water should flow out from Tap / Shower. Else, please contact your local authorized dealer.

d. Check and make sure no water leaking at the tube connecting points.

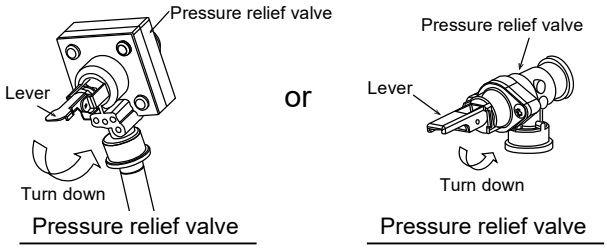
- For Space Heating / Cooling

a. Turn the plug on the Air Purge Valve outlet anticlockwise by one complete turn from fully closed position.



Air purge valve

b. Set the Pressure Relief Valve lever “DOWN”.



Pressure relief valve　　Pressure relief valve

c. Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Space Heating / Cooling circuit via Tube Connector. Stop filling water if the free water flow through Pressure Relief Valve Drainage.

d. Turn ON the Tank Unit and make sure Water Pump is running.

e. Check and make sure no water leaking at the tube connecting points.

f. The water may drip from this discharge hose. Therefore must guide the hose without close or block the outlet of the hose.

Discharge the Water

- For domestic hot water tank

a. Turn OFF power supply.

b. Set the Domestic Hot Water Tank Discharge (Drain Tap) to “OPEN”.

c. Open Tap / Shower to allow air inlet.

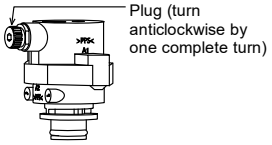
d. Turn the Safety Relief Valve ㉒ knob counterclockwise slightly and hold it until all air is released from this pipeline. Then recover the knob to original position after ensured the pipeline is emptied.

e. After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

2 Check Water Pressure *(0.1 MPa = 1 bar)
Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure Gauge). If necessary add water into Tank Unit (via Tube Connector).

3 Check Pressure Relief Valve
- Check for correct operation of Pressure Relief Valve by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keep on draining out from the Tank Unit, switch off the system, and then contact your local authorized dealer.

4 Air Purge Valve
Air purge valve must be installed at all high points in a closed water circuit system.
An automatic air purge valve is provided inside the indoor unit. To automatically purge the air from the system, turn the plug on the air purge valve outlet anticlockwise by one complete turn from fully closed position.
Excessive air is automatically purged if the plug is kept in this position.



Air purge valve

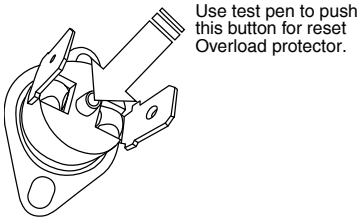
5 Indoor Unit Control Board Area
Thorough visual inspection of the control board and look for defects, i.e. loose connection, melting of wire insulator and etc.

6 RCCB/ELCB
Ensure the RCCB/ELCB set to “ON” condition before check RCCB/ELCB.
Turn on the power supply to the tank unit.
This testing could only be done when power is supplied to the tank unit.

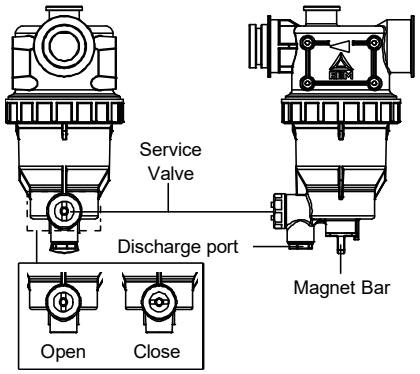
⚠ WARNING
Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the “TEST” button on the RCCB/ELCB. The lever would turn down and indicate “0”, if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the tank unit.
- If RCCB/ELCB functions normal, set the lever to “ON” again after testing finish.

7 Reset Overload Protector
Overload Protector serves the safety purpose to prevent the water over heating. When the Overload Protector trip at high water temperature, take below steps to reset it.
a. Take out the cover.
b. Use a test pen to push the centre button gently in order to reset the Overload Protector.
c. Fix the cover to the original fixing condition.



8 Maintenance for Magnetic Water Filter Set
a. Turn OFF power supply.
b. Place a container below Magnetic Water Filter Set.
c. Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set.
d. By using Allen key (8mm), remove the Cap of Discharge Port.
e. By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
f. Reinstall the Cap of Discharge Port and Magnet Bar.
g. Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 5 for details.)
h. Turn ON power supply.



9 Maintenance for Safety Relief Valve (25)
It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

20.1 Maintenance for Magnetic Water Filter Set

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

20.1.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

20.1.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it. In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer. When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

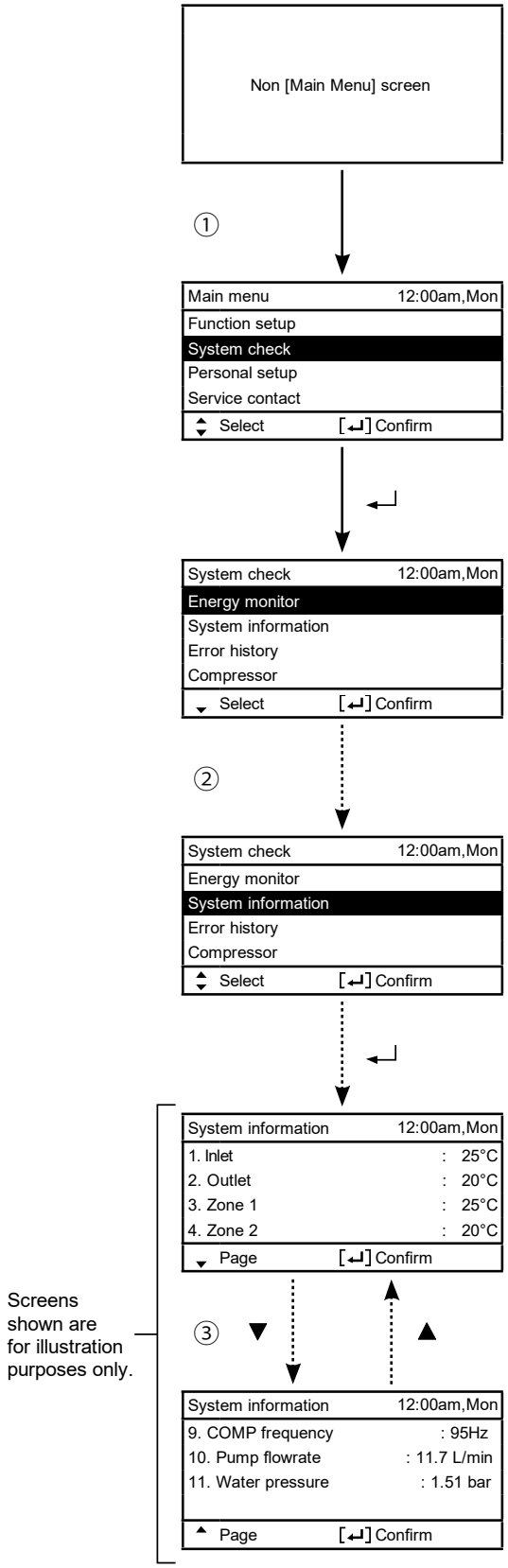
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

20.1.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to "System check".
2. Press ↵ and scroll to "System information".
3. Press ↵ and search for "Water pressure".



20.1.4 Specifications

20.1.4.1 Specifications of fresh water was heat transfer medium in brazed heat exchanger

Parameter	Quality Limits for Tap Water on the Secondary Side
Temperature	Below 60°C
pH	7 to 9
Alkalinity	60mg/l < HCO_3 < 300mg/l
Conductivity	< 500µS/cm
Hardness	[Ca^+, Mg^+] / [HCO_3^-] > 0.5
Chloride	< 200mg/l at 60°C
Sulphate	[SO_4^{2-}] < 100mg/l and [HCO_3^-] / [SO_4^{2-}] > 1
Nitrate	NO_3 < 100mg/l
Chlorine	< 0.5mg/l

20.1.4.2 External filter

Solids in the water must be filtered.
Minimum filter mesh size required for the field supply external filter in the water inlet is 20 mesh.

21. Troubleshooting Guide

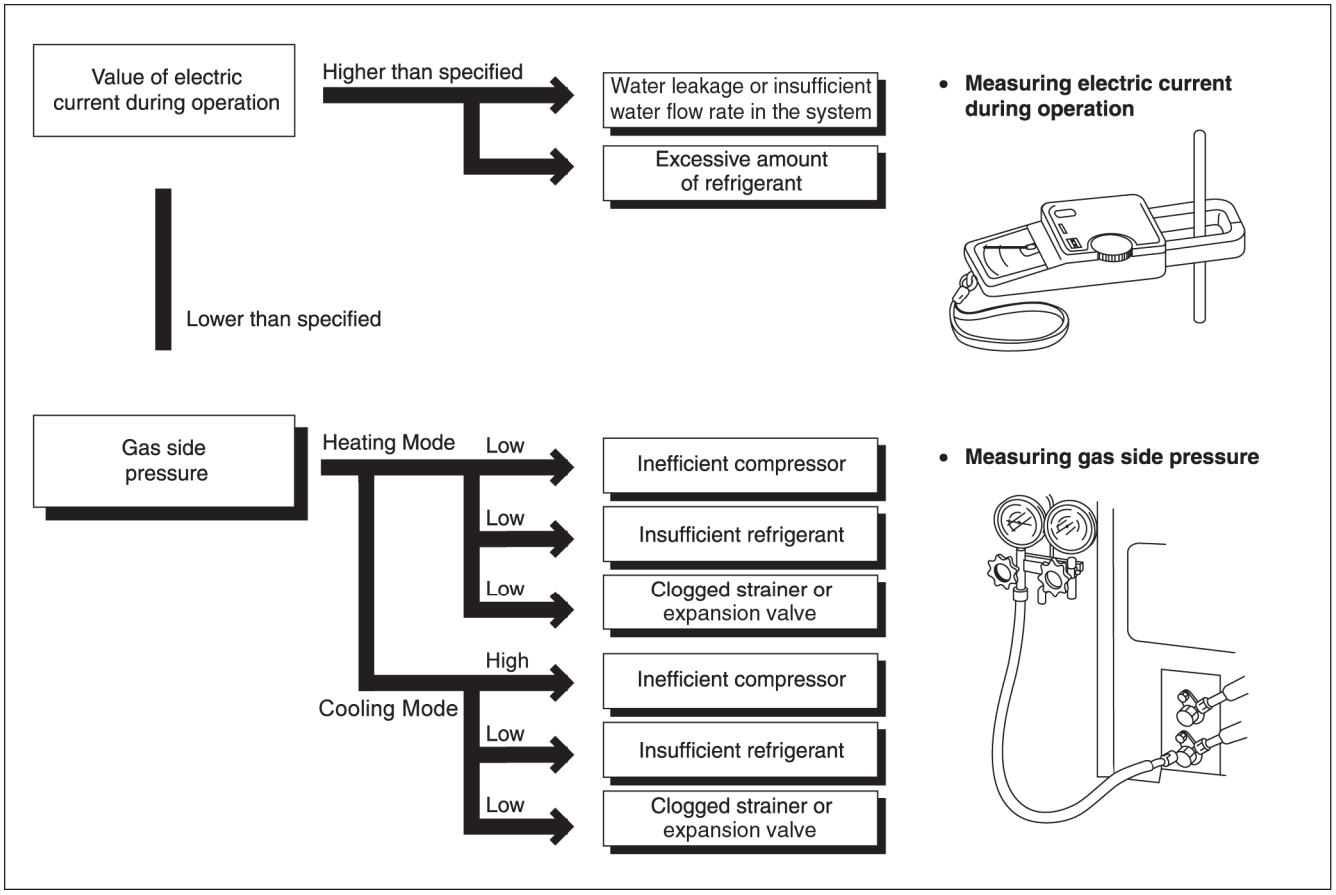
21.1 Refrigeration Cycle System

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle.
Such problems include insufficient insulation, problem with the power source, malfunction of a compressor and a fan.
The normal pressure of the refrigeration cycle depends on various conditions, the standard values for them are shown in the table on the right.

Normal Pressure (Standard)

	Gas pressure MPa (kg/cm^{2}G)
Heating Mode	2.3 ~ 2.9 (23 ~ 29)
Cooling Mode	0.9 ~ 1.2 (9 ~ 12)

★Condition: ● Outdoor temperature 7°C at heating mode and 35°C at cooling mode.
● Compressor operates at rated frequency.



21.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current

Condition of the Air-to-Water Heatpump indoor and outdoor units	Heating Mode			Cooling Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Water leakage or insufficient water flow rate in the system						
Excessive amount of refrigerant						
Inefficient compression						
Insufficient refrigerant (gas leakage)						
Outdoor heat exchange deficiency						
Clogged expansion valve or Strainer						

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is started.

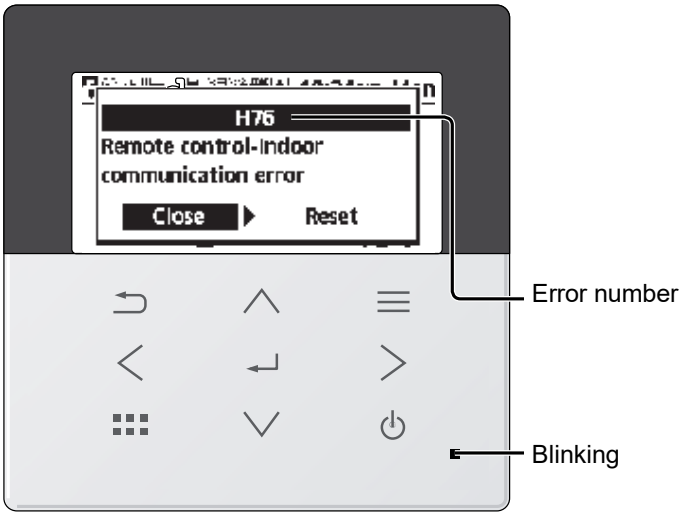
21.3 Breakdown Self Diagnosis Function

21.3.1 Self Diagnosis Function (Three Digits Alphanumeric Code)

- When abnormality occur during operation, the system will stop operation, and OFF/ON control panel LED will blink and error code will display on the control panel.
- Even error code is reset by turning OFF power supply or by selecting ERROR RESET, if the system abnormality is still unrepaired, system will again stop operation, and OFF/ON control panel LED will again blinks and error code will be display.
- The error code will store in IC memory.

- **To check the error code**
  1 When an abnormality occurs, system will stop operation and OFF/ON control panel LED will blink.
  2 Error code of the abnormality will be display on the control panel.
  3 To determine the abnormality description, the error code table needs to be referred.

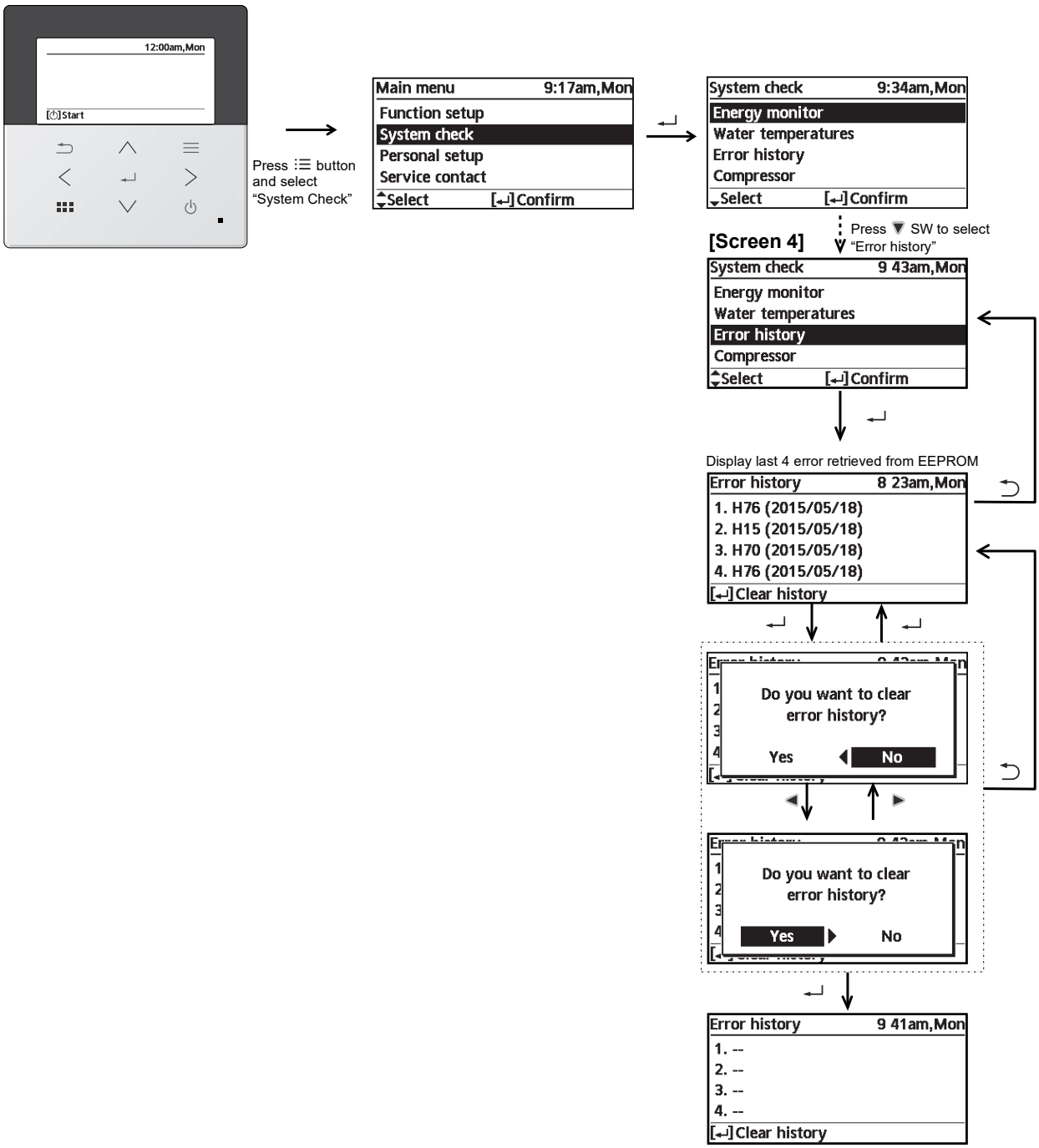
eg:



Press < > to select Close / Reset then press ↵

- **To display past/last error code**
  1 Turn ON power supply.
  2 Refer below procedure to retrieve the error code history.

- **To permanently delete error code from IC memory**
  1 Turn ON power supply.
  2 Refer below procedure to clear error history.


12:00am,Mon
Start
Press ☰ button and select "System Check"
Main menu 9:17am,Mon
Function setup
System check
Personal setup
Service contact
Select [↵]Confirm
System check 9:34am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
[Screen 4]
Press ▼ SW to select "Error history"
System check 9 43am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Display last 4 error retrieved from EEPROM
Error history 8 23am,Mon
1. H76 (2015/05/18)
2. H15 (2015/05/18)
3. H70 (2015/05/18)
4. H76 (2015/05/18)
[↵]Clear history
Do you want to clear error history?
Yes No
Do you want to clear error history?
Yes No
Error history 9 41am,Mon
1. --
2. --
3. --
4. --
[↵]Clear history

21.4 Error Codes Table

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
H00	No abnormality detected	—	—
H12	Indoor/Outdoor capacity unmatched	90s after power supply	• Indoor/outdoor connection wire • Indoor/outdoor PCB • Specification and combination table in catalogue
H15	Outdoor compressor temperature sensor abnormality	Continue for 5 sec.	• Compressor temperature sensor (defective or disconnected)
H20	Water pump abnormality	Continue for 10 sec.	• Indoor PCB • Water pump (malfunction)
H23	Indoor refrigerant liquid temperature sensor abnormality	Continue for 5 sec.	• Refrigerant liquid temperature sensor (defective or disconnected)
H27	Service valve error	Continue for 5 minutes	• High pressure sensor (defective or disconnected)
H28	Abnormal solar sensor	Continue for 5 sec.	• Solar temperature sensor (defective or disconnected)
H31	Abnormal swimming pool sensor	Continue for 5 sec.	• Pool temperature sensor (defective or disconnected)
H36	Abnormal buffer tank sensor	Continue for 5 sec.	• Buffer tank sensor (defective or disconnected)
H38	Brand code not match	When indoor and outdoor brand code not same	—
H42	Compressor low pressure abnormality	—	• Outdoor pipe temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
H43	Abnormal Zone 1 sensor	Continue for 5 sec.	• Water temperature Zone 1 sensor
H44	Abnormal Zone 2 sensor	Continue for 5 sec.	• Water temperature Zone 2 sensor
H62	Water flow switch abnormality	Continue for 1 min.	• Water flow switch
*H63	Refrigerant low pressure abnormality	Continue for 5 sec.	• Outdoor low pressure sensor (defective or disconnected)
H64	Refrigerant high pressure abnormality	Continue for 5 sec.	• Outdoor high pressure sensor (defective or disconnected)
*H65	Deice circulation error	Continue for 10 sec.	• Water flow switch sensor (defective or disconnected) • Water pump malfunction • Buffer tank (is used)
H67	Abnormal External Thermistor 1	Continue for 5 sec.	• Room temperature Zone 1 sensor
H68	Abnormal External Thermistor 2	Continue for 5 sec.	• Room temperature Zone 2 sensor
H70	Back-up heater OLP abnormality	Continue for 60 sec.	• Back-up heater OLP (Disconnection or activated)
H72	Tank sensor abnormal	Continue for 5 sec.	• Tank sensor
H74	PCB communication error	Communication or transfer error	• Indoor main PCB and Sub PCB
H75	Low water temperature control	Room heater disable and deice request to operate under low water temperature	• Heater operation must enable to increase water temperature
H76	Indoor - control panel communication abnormality	—	• Indoor - control panel (defective or disconnected)
H90	Indoor/outdoor abnormal communication	> 1 min after starting operation	• Internal/external cable connections • Indoor/Outdoor PCB
H91	Tank heater OLP abnormality	Continue for 60 sec.	• Tank heater OLP (Disconnection or activated)
H95	Indoor/Outdoor wrong connection	—	• Indoor/Outdoor supply voltage
H98 / F95	Outdoor high pressure overload protection Cooling high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB
H99	Indoor heat exchanger freeze prevention	—	• Indoor heat exchanger • Refrigerant shortage
F12	Pressure switch activate	4 times occurrence within 20 minutes	• Pressure switch
F14	Outdoor compressor abnormal revolution	4 times occurrence within 20 minutes	• Outdoor compressor
F15	Outdoor fan motor lock abnormality	2 times occurrence within 30 minutes	• Outdoor PCB • Outdoor fan motor
F16	Total running current protection	3 times occurrence within 20 minutes	• Excess refrigerant • Outdoor PCB

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
F20	Outdoor compressor overheating protection	4 times occurrence within 30 minutes	• Compressor tank temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
F22	IPM (power transistor) overheating protection	3 times occurrence within 30 minutes	• Improper heat exchange • IPM (Power transistor)
F23	Outdoor Direct Current (DC) peak detection	7 times occurrence continuously	• Outdoor PCB • Compressor
F24	Refrigeration cycle abnormality	2 times occurrence within 20 minutes	• Insufficient refrigerant • Outdoor PCB • Compressor low compression
F25	Cooling/Heating cycle changeover abnormality	4 times occurrence within 30 minutes	• 4-way valve • V-coil
F27	Pressure switch abnormality	Continue for 1 min.	• Pressure switch
F29	Low Discharge Superheat	1 times occurrence within 2550 minutes	• Discharge Temperature Sensor • Discharge Pressure Sensor • Pressure Switch • Outdoor PCB
F30	Water outlet sensor 2 abnormality	Continue for 5 sec.	• Water outlet sensor 2 (defective or disconnected)
F32	Abnormal Internal Thermostat	Continue for 5 sec.	• Control panel PCB thermostat
F36	Outdoor air temperature sensor abnormality	Continue for 5 sec.	• Outdoor air temperature sensor (defective or disconnected)
F37	Indoor water inlet temperature sensor abnormality	Continue for 5 sec.	• Water inlet temperature sensor (defective or disconnected)
F40	Outdoor discharge pipe temperature sensor abnormality	Continue for 5 sec.	• Outdoor discharge pipe temperature sensor (defective or disconnected)
F41	PFC control	4 times occurrence within 10 minutes	• Voltage at PFC
F42	Outdoor heat exchanger temperature sensor abnormality	Continue for 5 sec.	• Outdoor heat exchanger temperature sensor (defective or disconnected)
F43	Outdoor defrost sensor abnormality	Continue for 5 sec.	• Outdoor defrost sensor (defective or disconnected)
F45	Indoor water outlet temperature sensor abnormality	Continue for 5 sec.	• Water outlet temperature sensor (defective or disconnected)
F46	Outdoor Current Transformer open circuit	—	• Insufficient refrigerant • Outdoor PCB • Compressor low
F48	Outdoor EVA outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor EVA outlet temperature sensor (defective or disconnected)
F49	Outdoor bypass outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor bypass outlet temperature sensor (defective or disconnected)

Note: * This error code is not applicable for this system.

21.5 Self-diagnosis Method

21.5.1 Connection Capability Rank Abnormality (H12)

Malfunction Decision Conditions:

During startup operation of cooling and heating, the capability rank of indoor checked by the outdoor is used to determine connection capability rank abnormality.

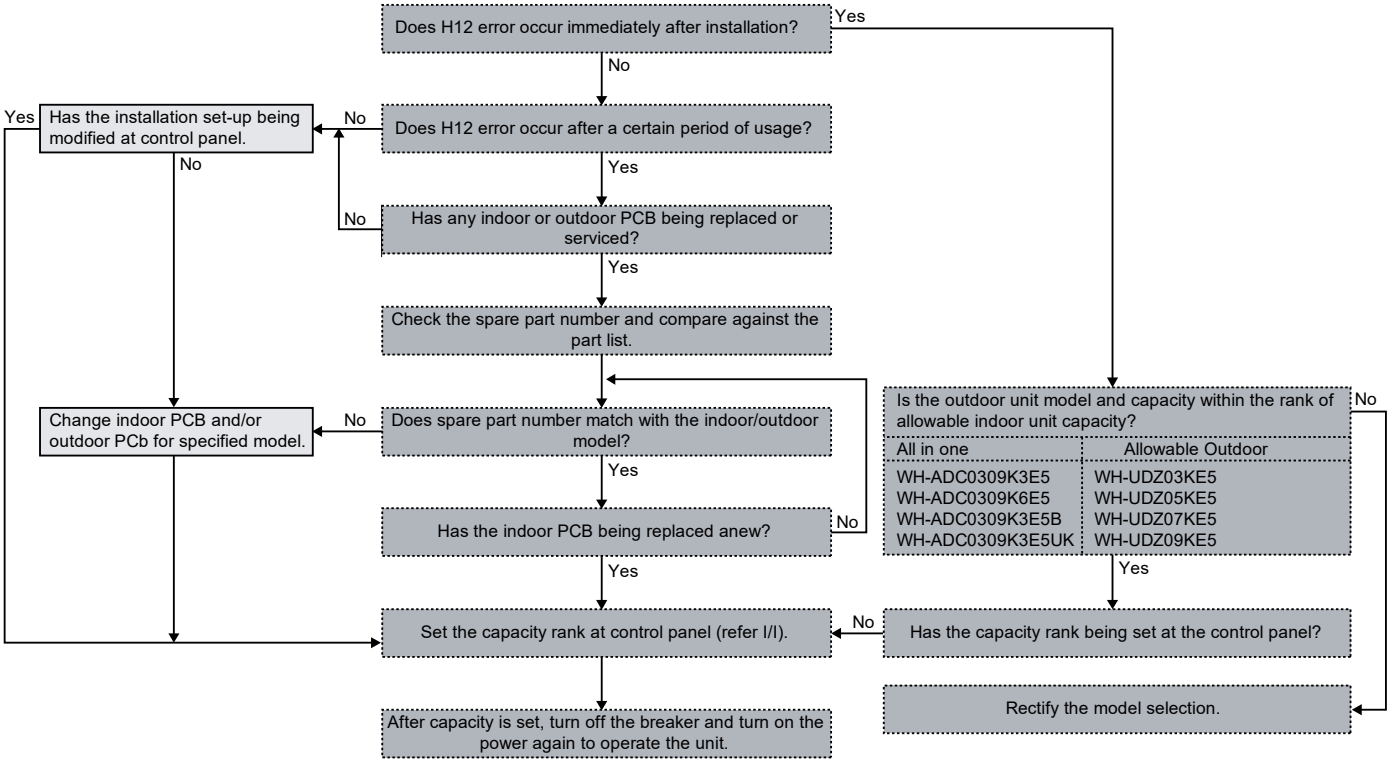
Malfunction Caused:

1 Wrong model interconnected.
2 Wrong indoor unit or outdoor unit PCB (main) used.
3 Faulty indoor unit or outdoor unit PCB (main).

Abnormality Judgment:

Continue for 90 seconds.

Troubleshooting:



21.5.2 Compressor Tank Temperature Sensor Abnormality (H15)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the compressor tank temperature sensor are used to determine sensor error.

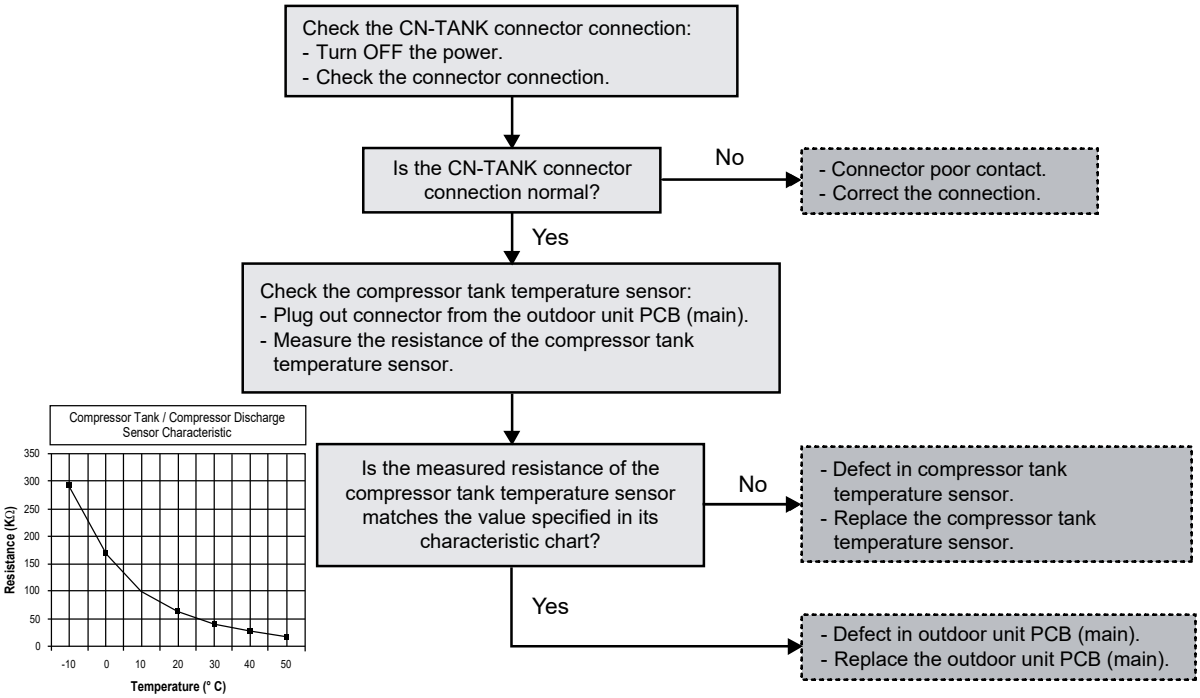
Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:



21.5.3 Water Pump Abnormality (H20)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the rotation speed detected by the IPM of water pump motor during water pump operation is used to determine abnormal water pump (feedback of rotation > 6,000rpm or < 1,000rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the water pump motor winding.
2 Operation stop due to breaking of wire inside the water pump motor.
3 Operation stop due to breaking of water pump lead wires.
4 Operation stop due to water pump motor IPM malfunction.
5 Operation error due to faulty indoor unit PCB.

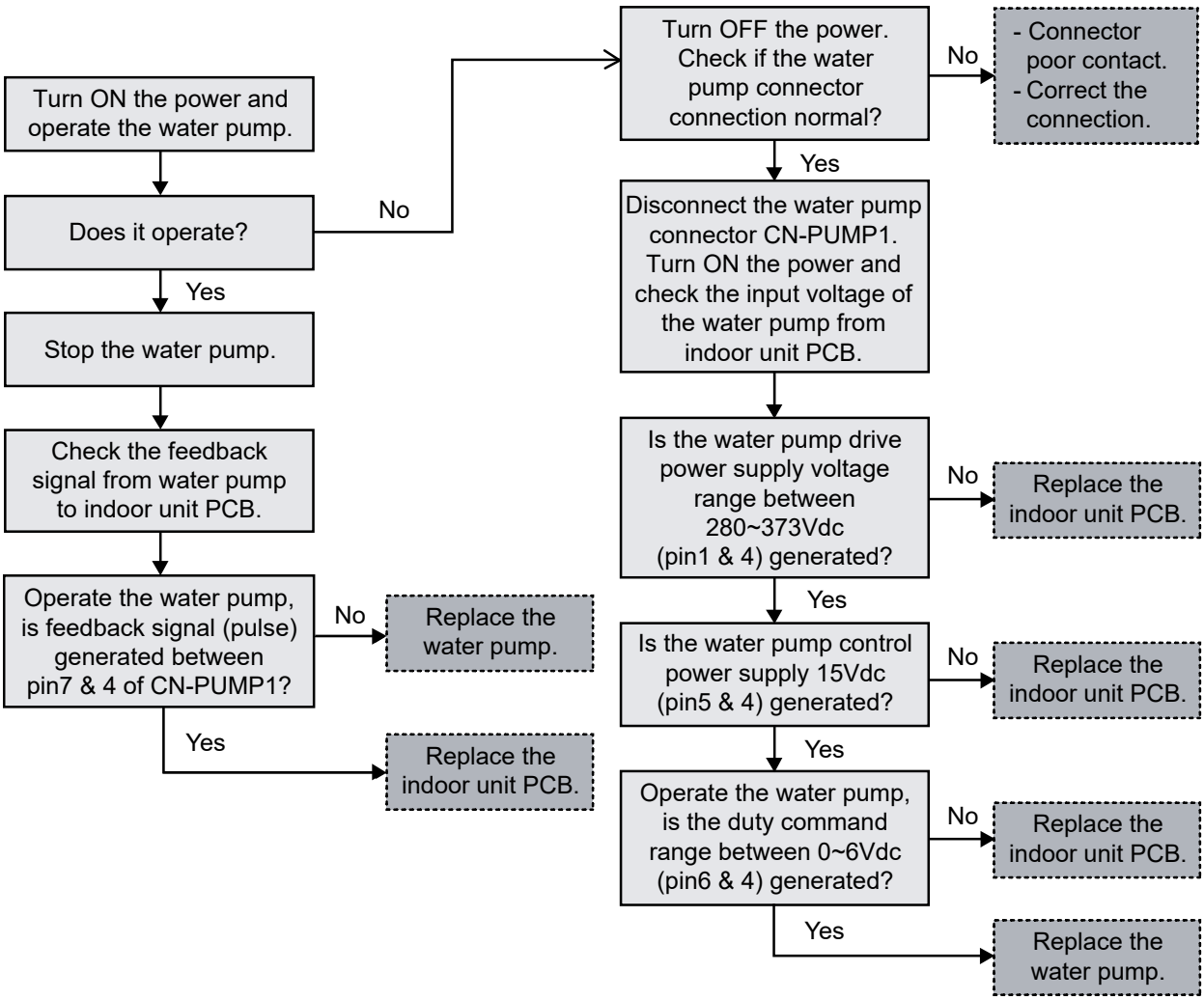
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.4 Indoor Refrigerant Liquid Temperature Sensor Abnormality (H23)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor refrigerant liquid temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

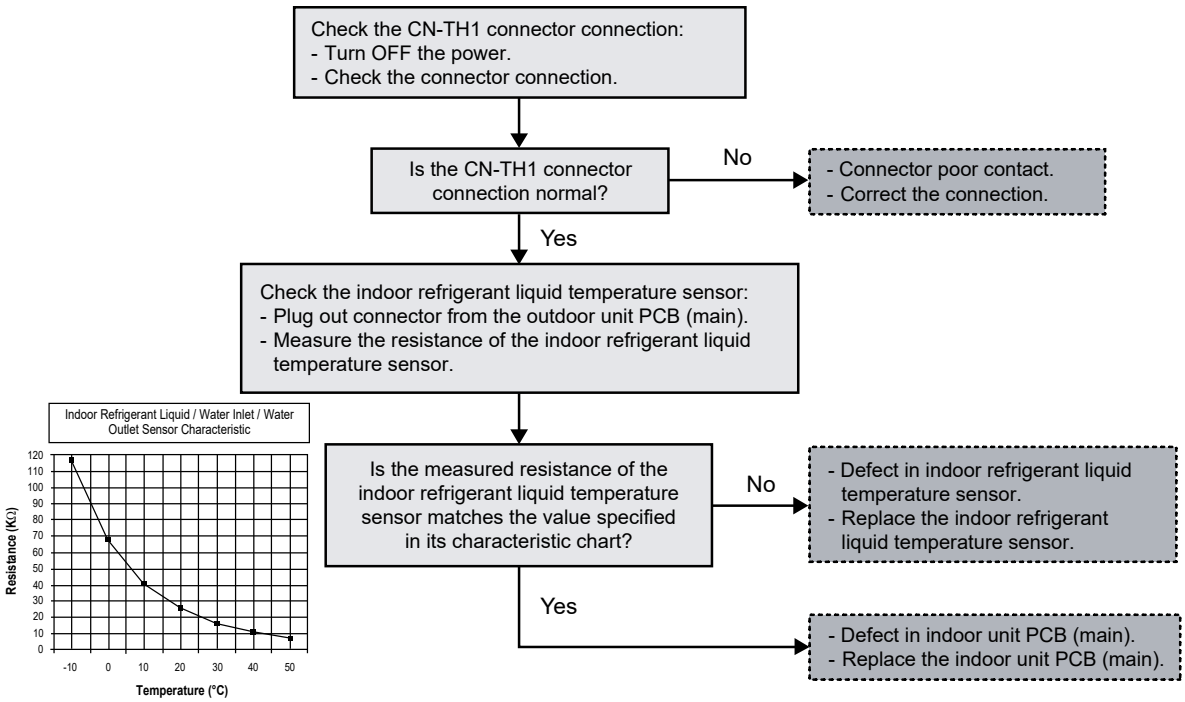
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.5 Service Valve Error (H27)

Malfunction Decision Conditions:

During cooling operation, when:-

[a] Indoor refrigerant pipe temperature at compressor startup - present indoor refrigerant pipe temperature < 2°C

[b] Present high pressure - high pressure at compressor startup < 5kg/cm^2

**Judgment only for first time cooling operation and not during pump down operation.

Malfunction Caused:

1 3 way valves closed.
2 Faulty high pressure sensor.
3 Faulty indoor refrigerant pipe temperature sensor
4 Faulty outdoor unit PCB (main).

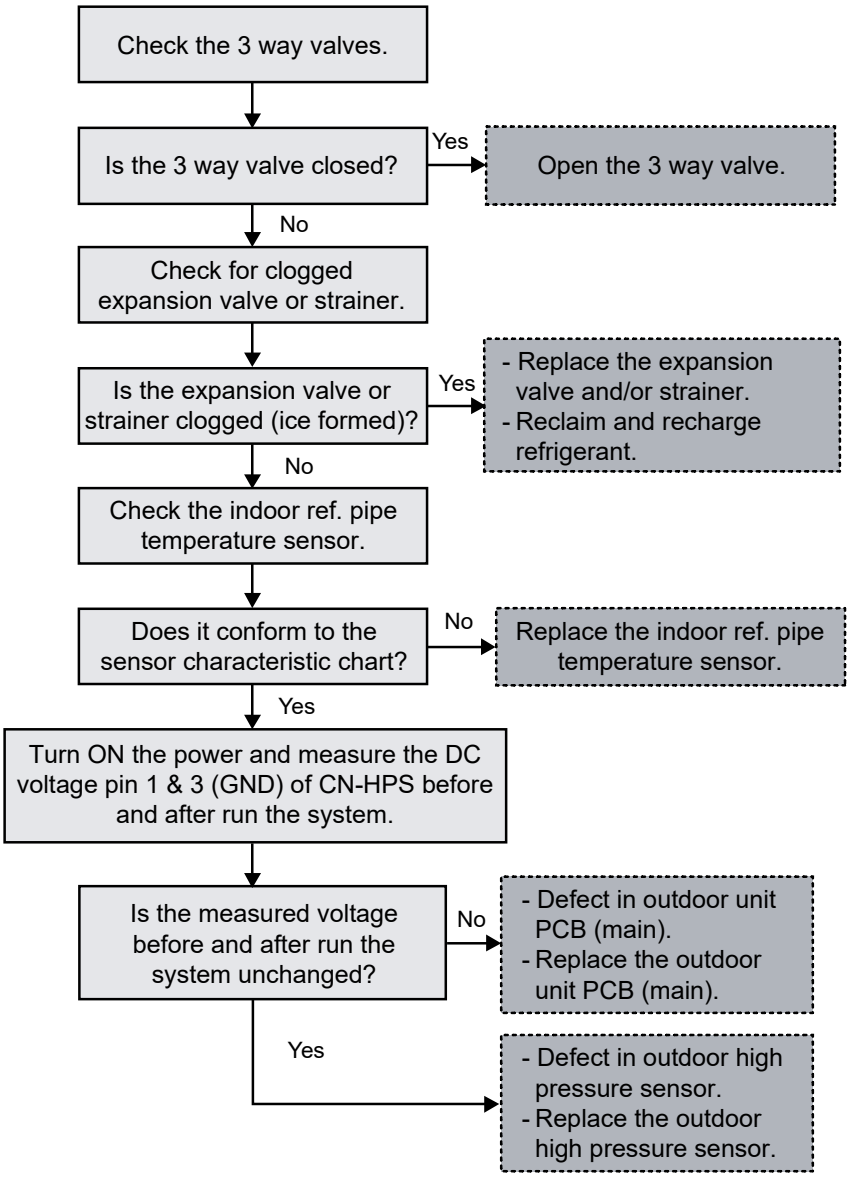
Abnormality Judgment:

Continue for 5 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



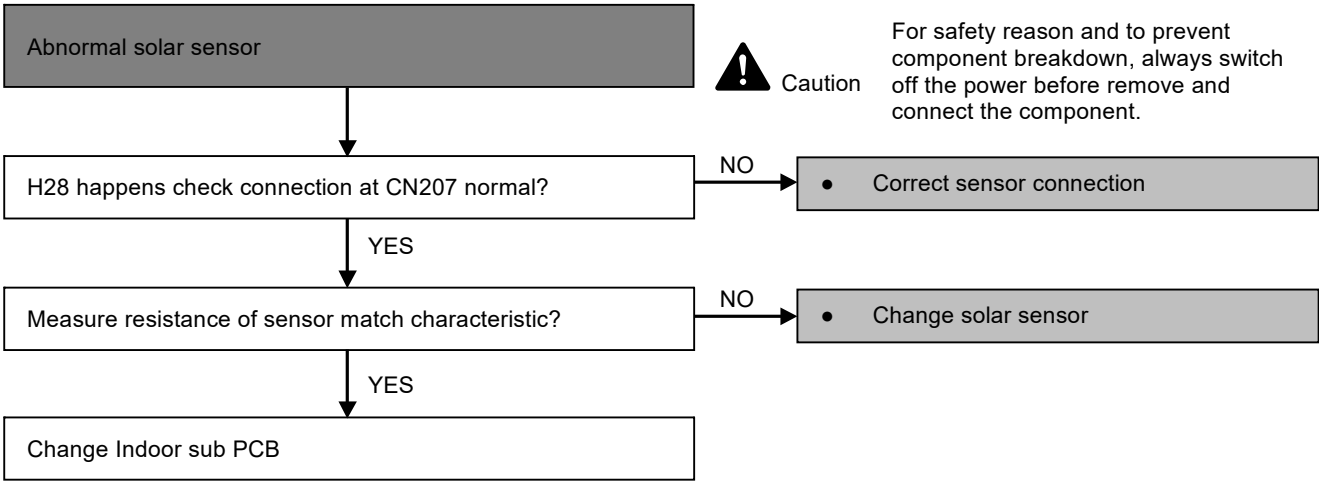
21.5.6 Abnormal Solar Sensor (H28)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty solar sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



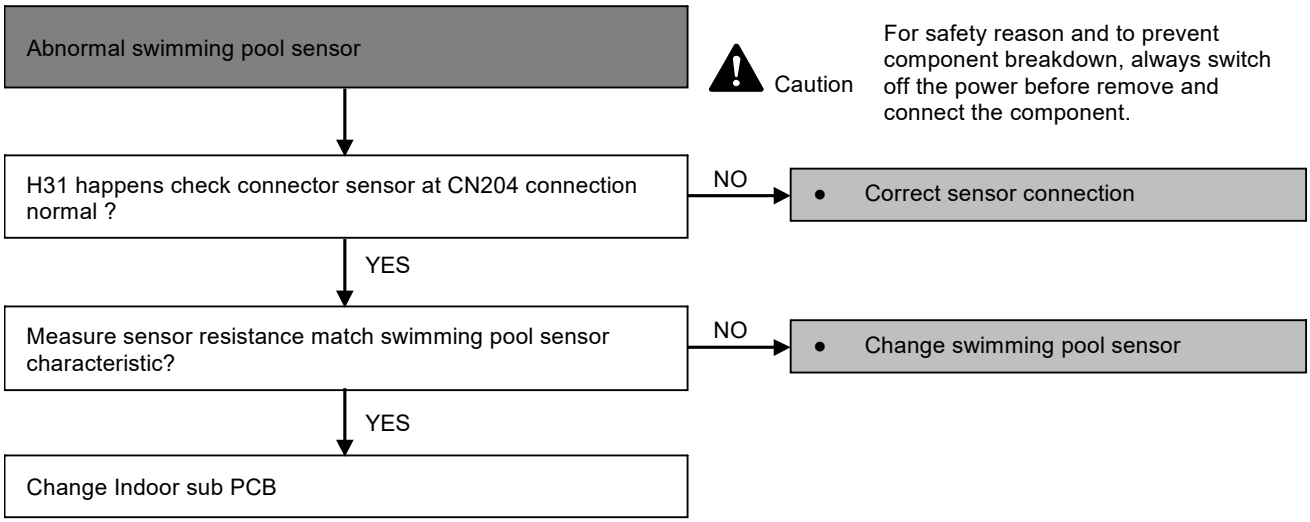
21.5.7 Abnormal Swimming Pool Sensor (H31)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty swimming pool sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



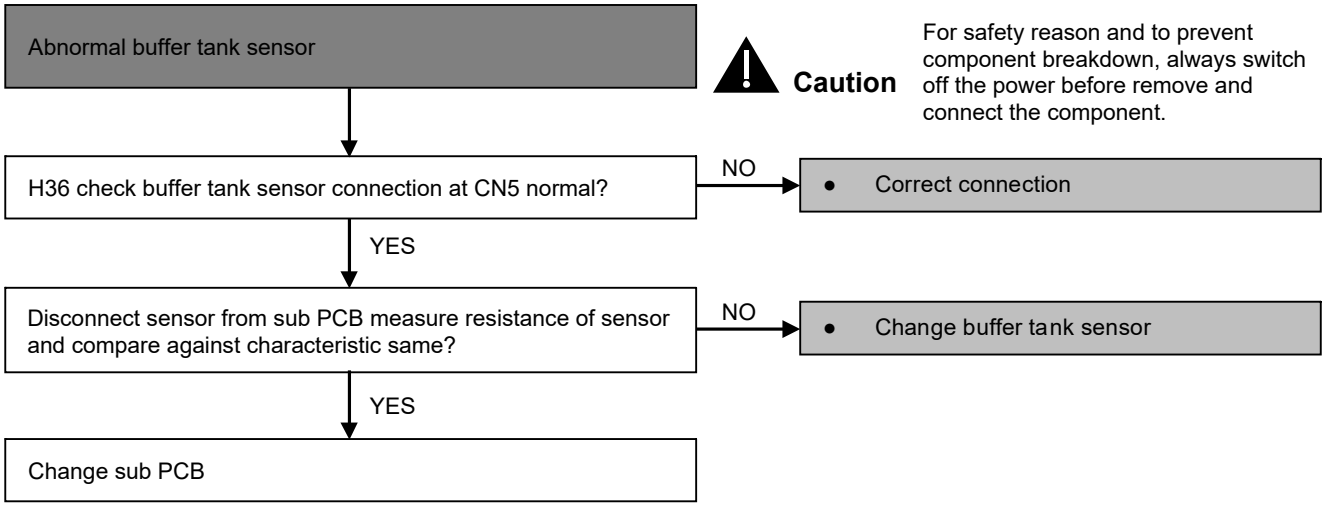
21.5.8 Abnormal Buffer Tank Sensor (H36)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

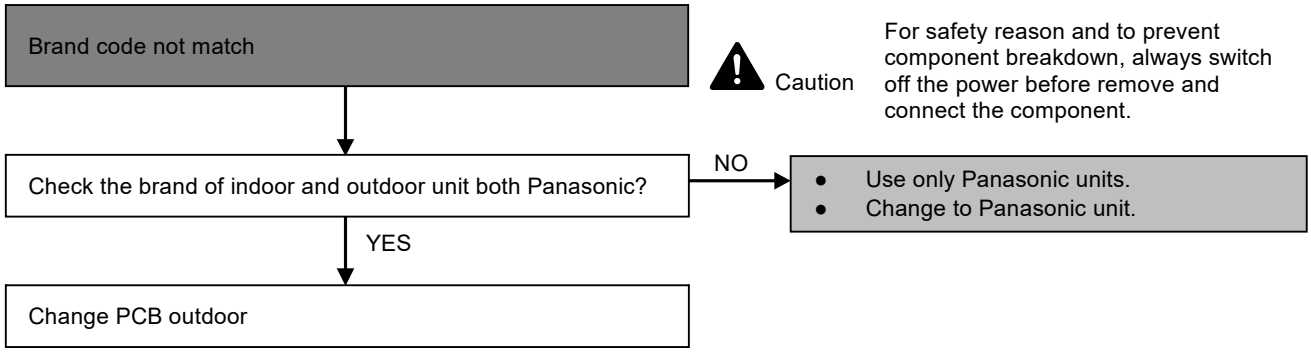
Continue for 5 seconds.



21.5.9 Brand Code Not Matching (H38)

Malfunction Caused:

1 Indoor and outdoor brand code not match.



21.5.10 Compressor Low Pressure Protection (H42)

Malfunction Decision Conditions:

During operation of heating and after 5 minutes compressor ON, when outdoor pipe temperature below -29°C or above 26°C is detected by the outdoor pipe temperature sensor.

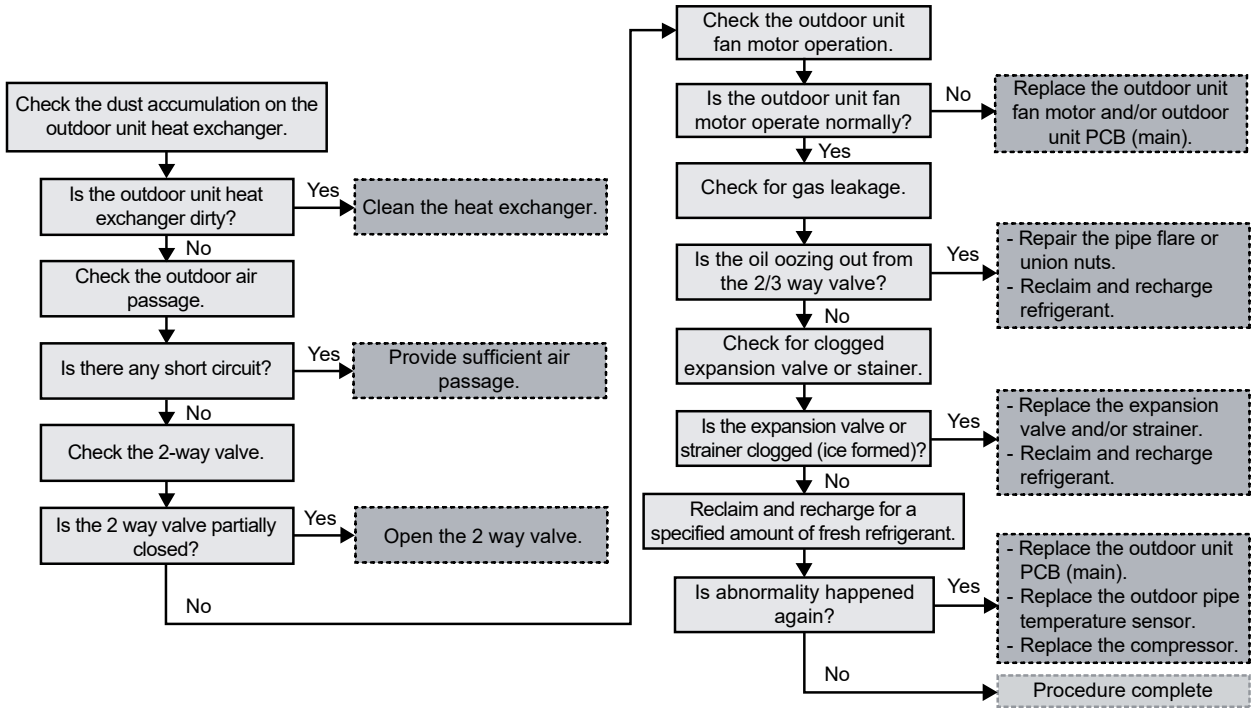
Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve partially closed.
4 Faulty outdoor unit fan motor.
5 Refrigerant shortage (refrigerant leakage).
6 Clogged expansion valve or strainer.
7 Faulty outdoor pipe temperature sensor.
8 Faulty outdoor unit main PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



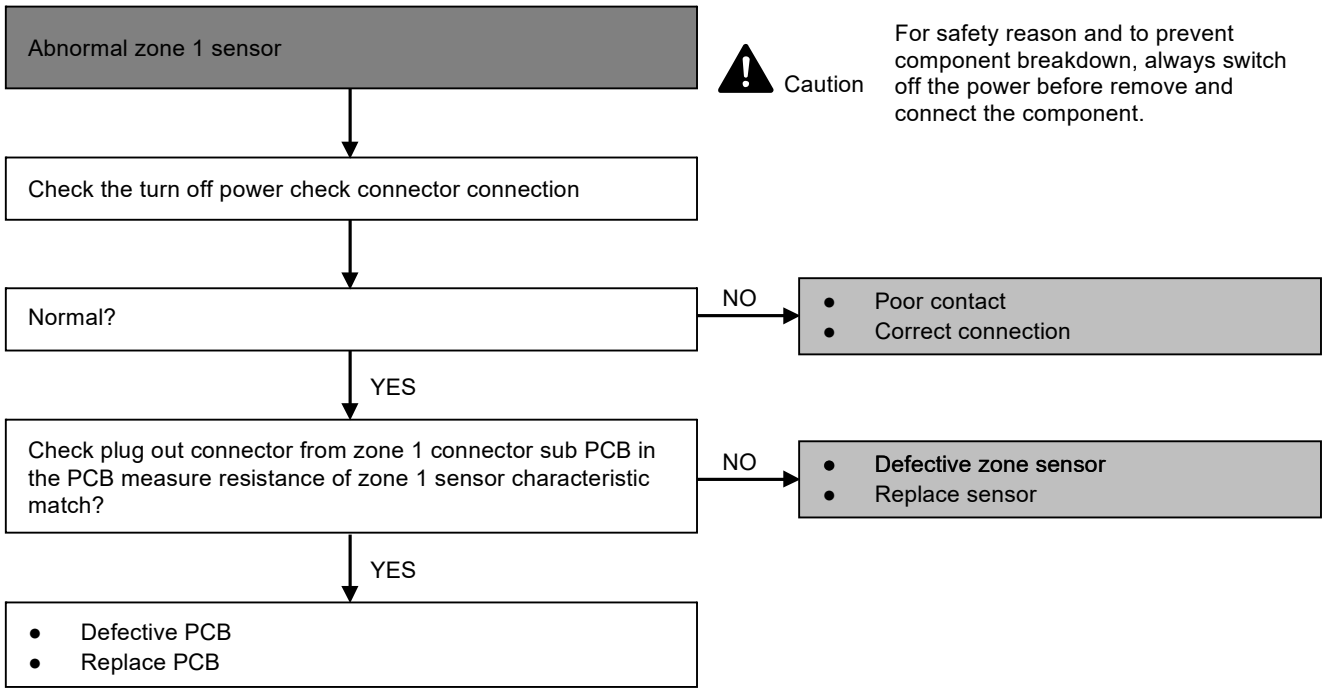
21.5.11 Abnormal Zone 1 Sensor (H43)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



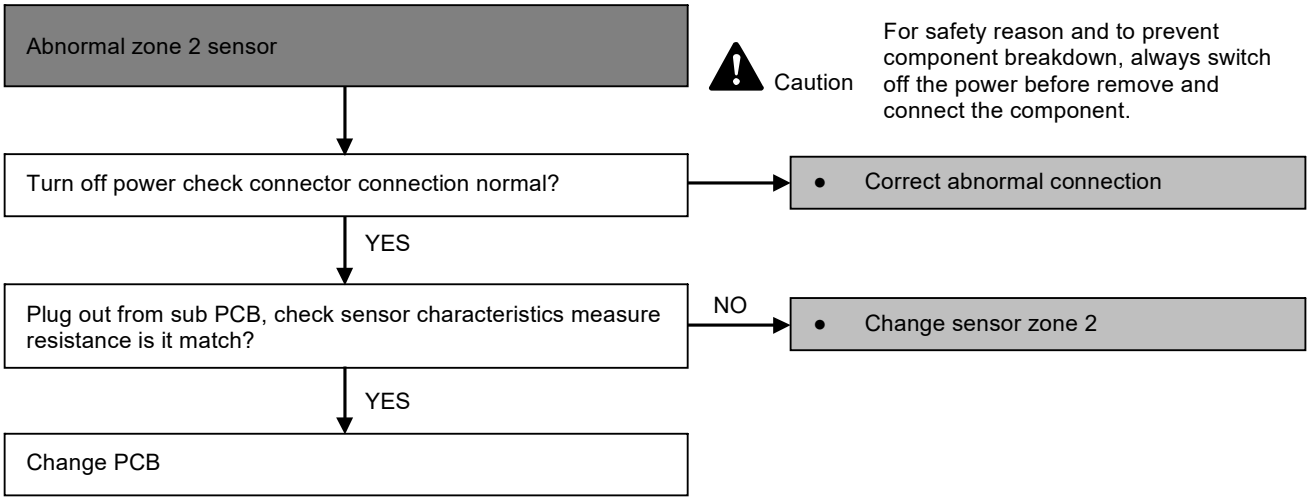
21.5.12 Abnormal Zone 2 Sensor (H44)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



21.5.13 Water Flow Switch Abnormality (H62)

Malfunction Decision Conditions:

During operation of cooling and heating, the water flow detected by the indoor water flow switch is used to determine water flow error.

Malfunction Caused:

1 Faulty water pump.
2 Water leak in system.
3 Faulty connector connection.
4 Faulty water flow switch.
5 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 10 seconds (but no judgment for 9 minutes after compressor startup/restart).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

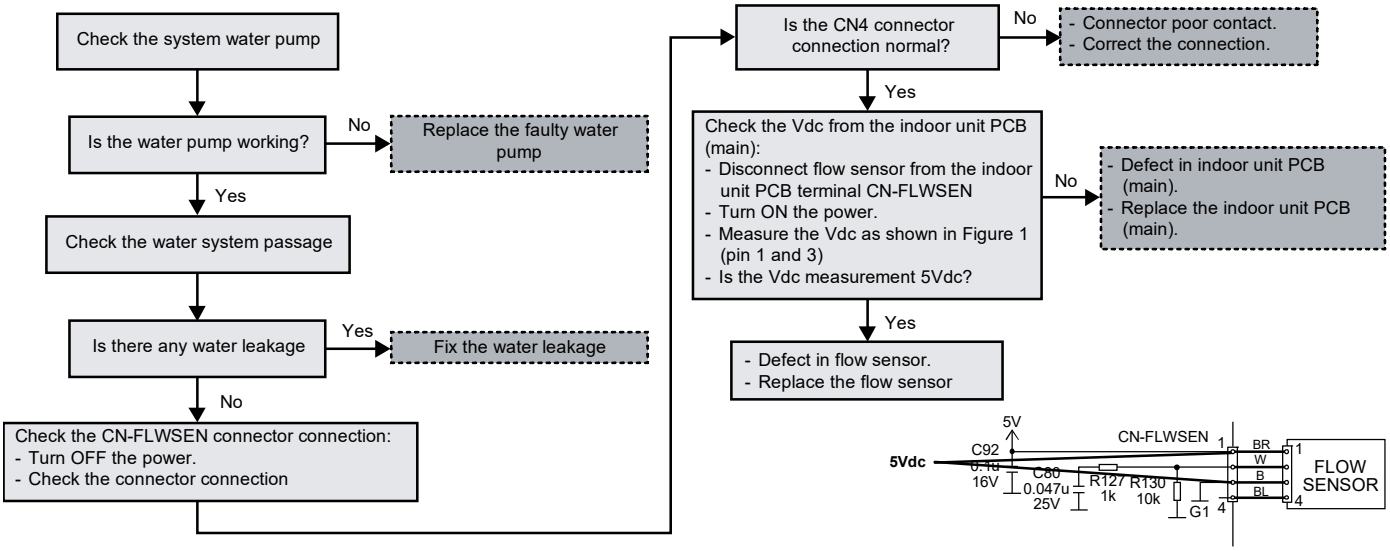


Figure 1

21.5.14 Outdoor High Pressure Abnormality (H64)

Malfunction Decision Conditions:

During operation of cooling and heating, when the outdoor high pressure sensor output signal is 0 Vdc or 5 Vdc.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

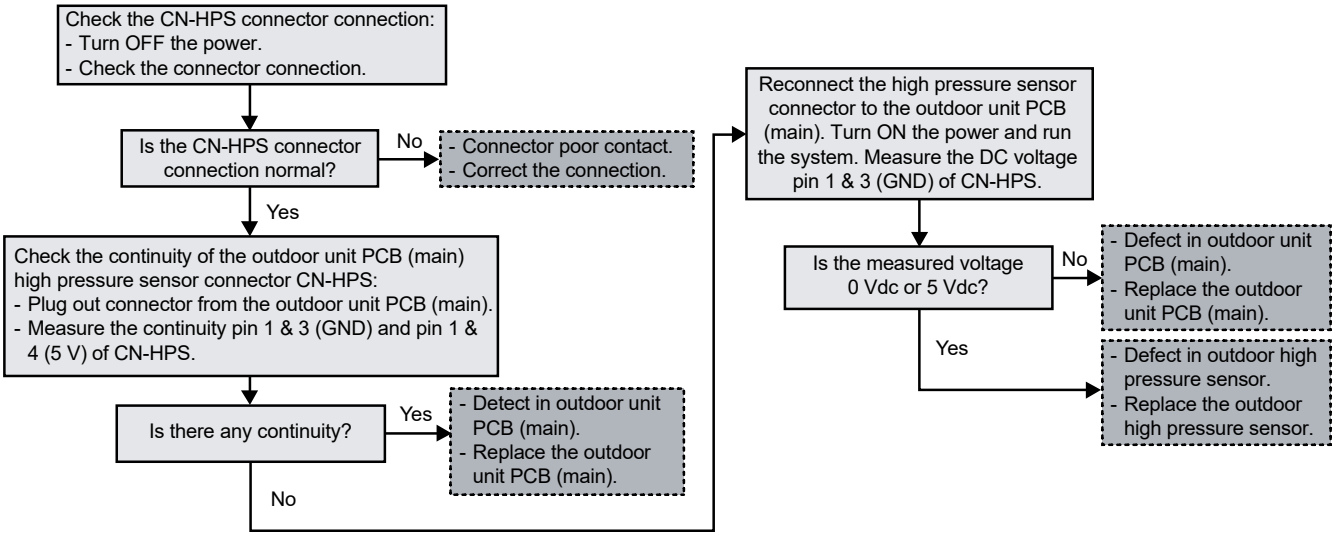
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.15 Deice Circulation Error (H65)

Malfunction Decision Conditions:

During startup and operation of deice (mode 2), the water flow (> 10l/min) detected by the water flow switch is used to determine deice circulation error.

Malfunction Caused:

1 Water flow in air-to-water heatpump indoor unit circuitry.
2 Faulty indoor unit water flow switch.
3 Faulty indoor unit water pump.
4 Faulty indoor unit PCB.

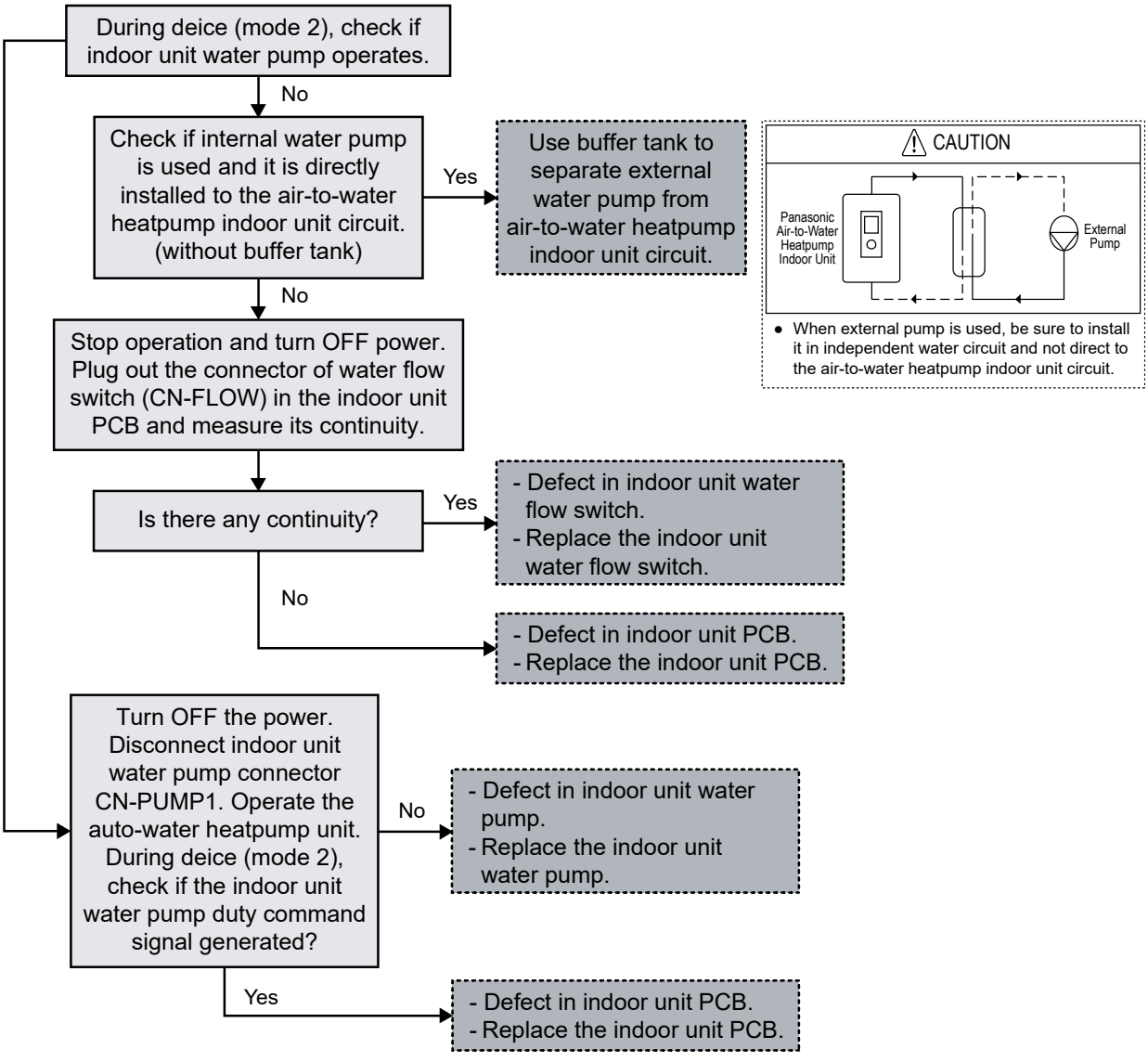
Abnormality Judgment:

Continue for 10 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



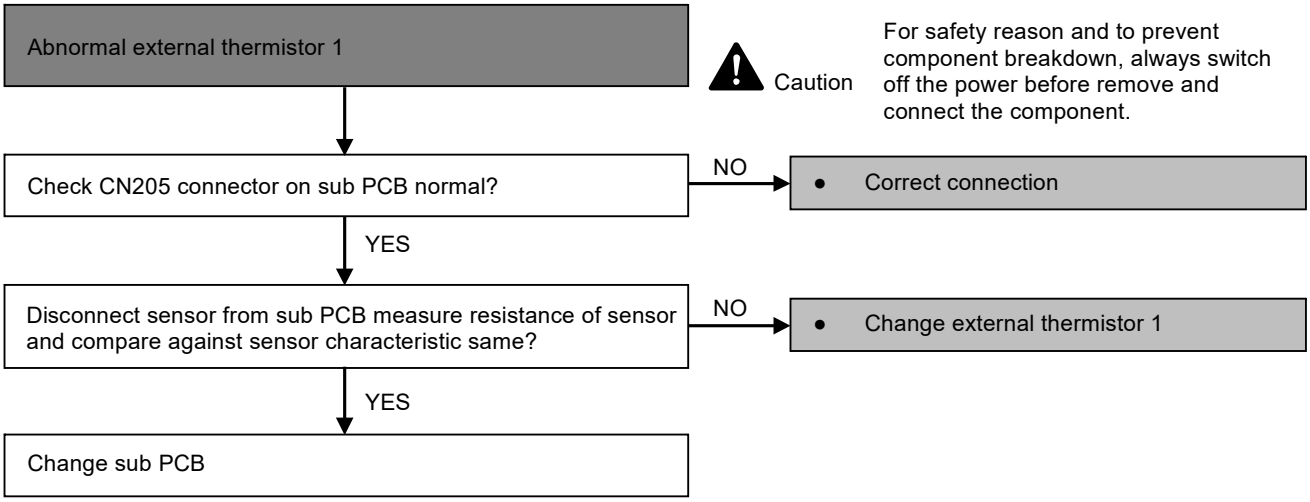
21.5.16 Abnormal External Thermistor 1 (H67)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 1 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



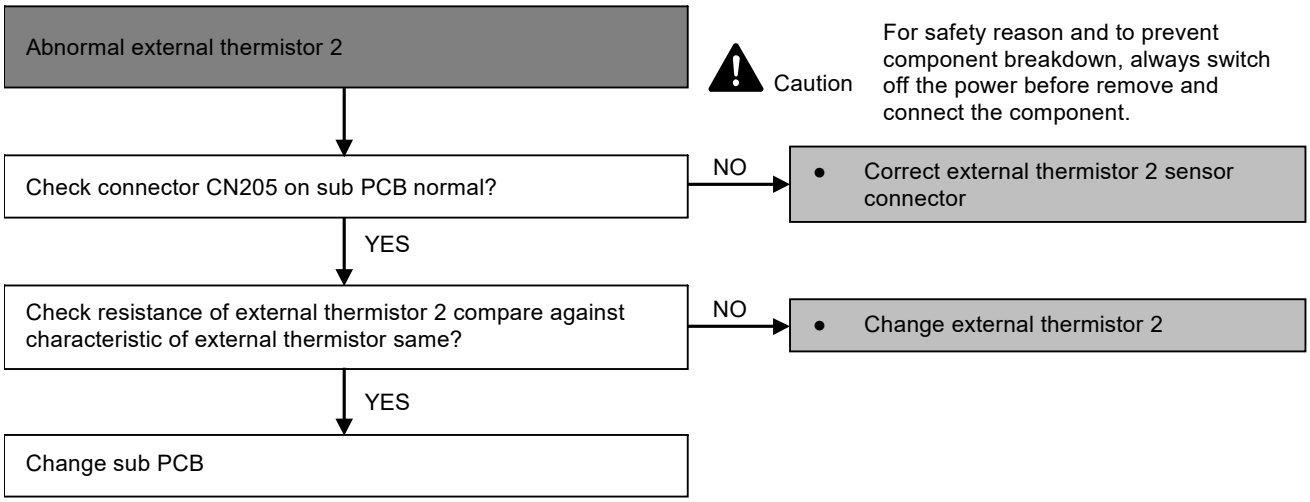
21.5.17 Abnormal External Thermistor 2 (H68)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 2 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



21.5.18 Indoor Backup Heater OLP Abnormality (H70)

Malfunction Decision Conditions:

During operation of indoor backup heater, when no power supplies to indoor backup heater or OLP open circuit.

Malfunction Caused:

1 Faulty power supply connector connection.
2 Faulty connector connection.
3 Faulty indoor backup heater overload protector (OLP).
4 Faulty indoor unit PCB (main).

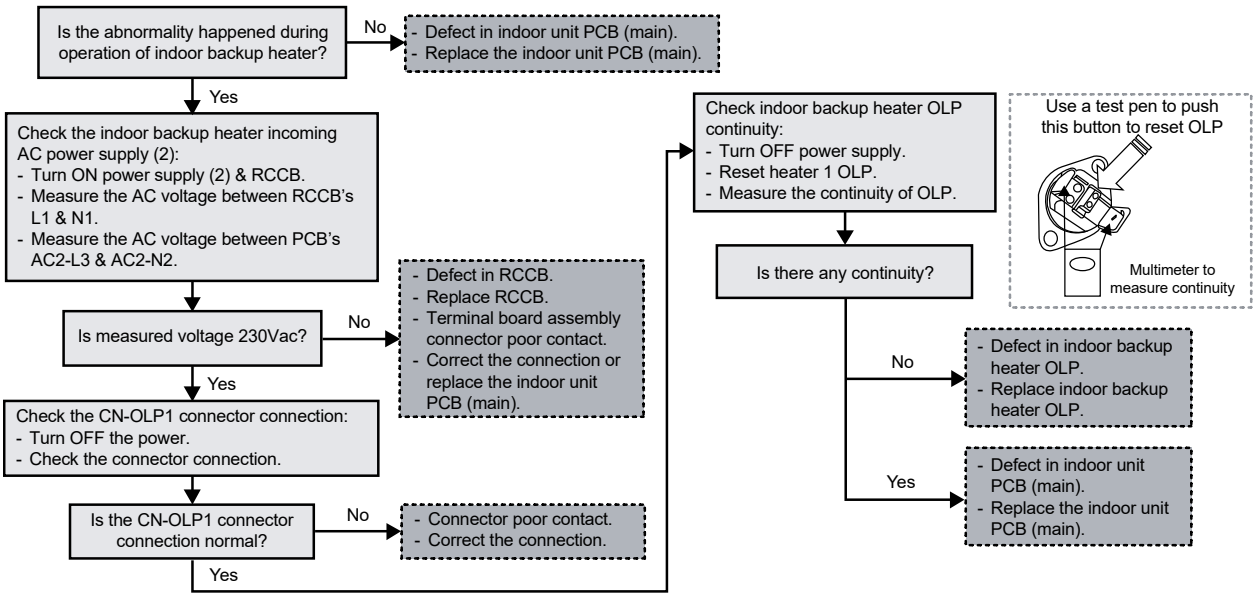
Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.19 Tank Temperature Sensor Abnormality (H72)

Malfunction Decision Conditions:

When tank connection is set to ON, the temperatures detected by the tank temperature sensor are used to determine sensor error.

Malfunction Caused:

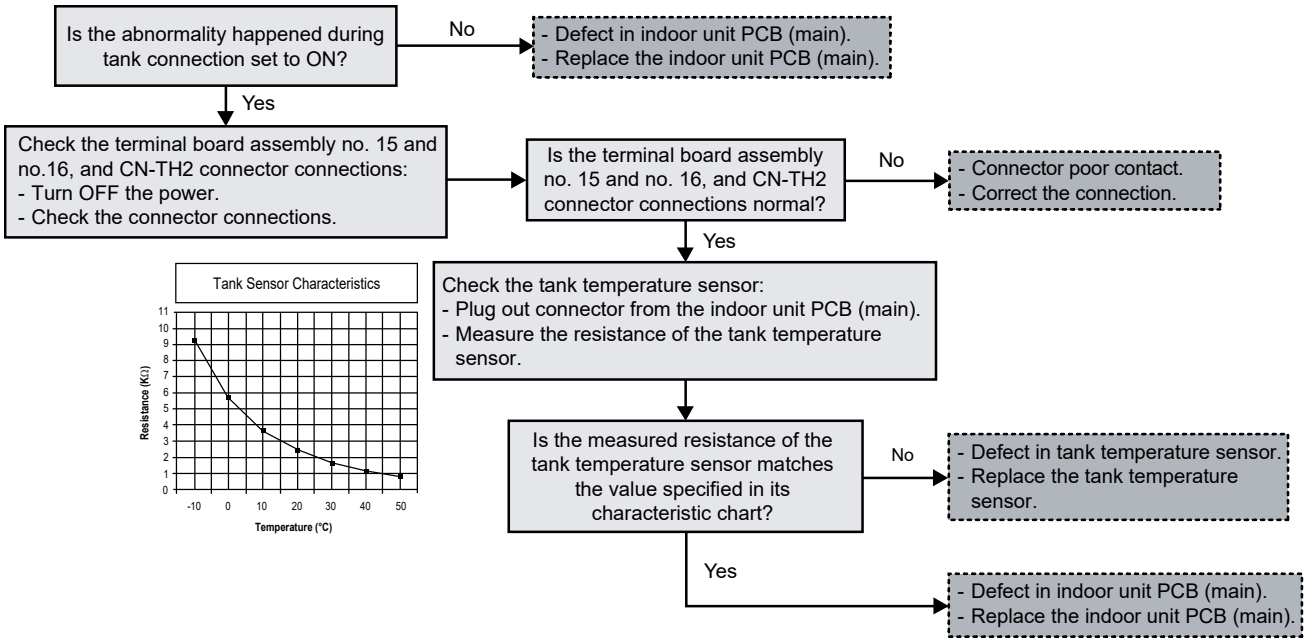
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.20 PCB Communication Error (H74)

Malfunction Decision Conditions:

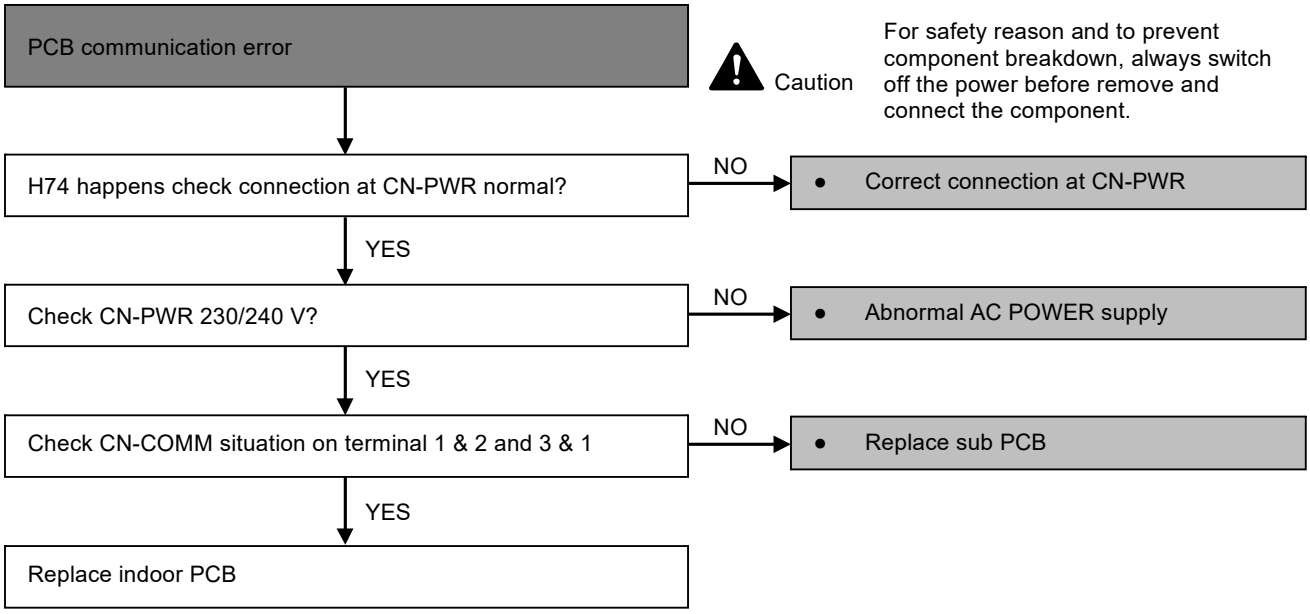
When External PCB connection is select “YES” and no communication with External PCB micon for 10 seconds and above.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty indoor PCB.
3 Faulty indoor sub PCB.

Abnormality Judgment:

After 1 minute operation started.



21.5.21 Indoor-Control Panel Communication Abnormality (H76)

Malfunction Decision Conditions:

During standby and operation of cooling and heating, indoor-control panel error occur.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty control panel.
3 Faulty indoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

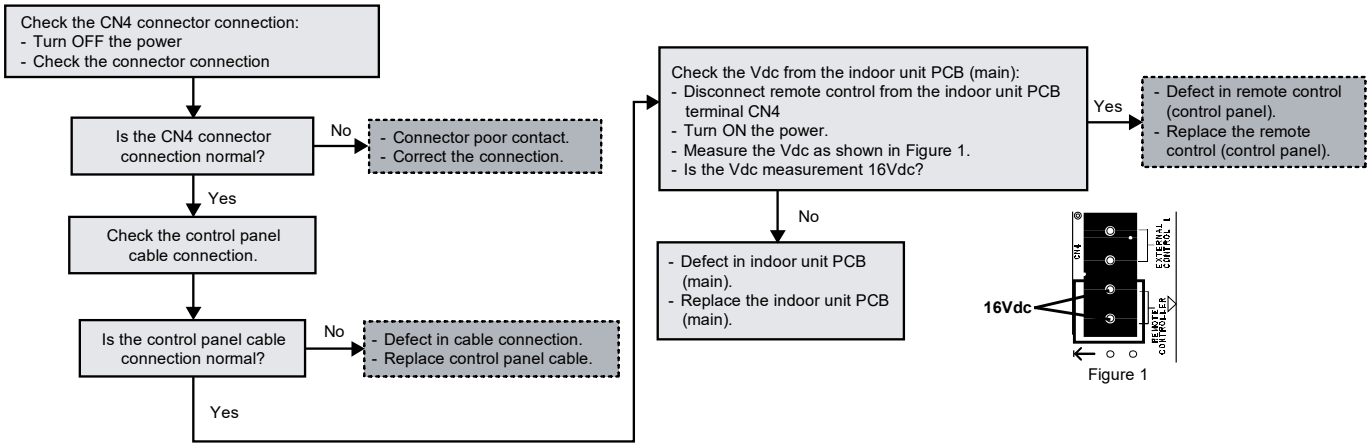


Figure 1

21.5.22 Indoor/Outdoor Abnormal Communication (H90)

Malfunction Decision Conditions:

During operation of cooling and heating, the data received from outdoor unit in indoor unit signal transmission is checked whether it is normal.

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty indoor unit PCB (main).
3 Indoor-outdoor signal transmission error due to wrong wiring.
4 Indoor-outdoor signal transmission error due to breaking of wire in the connection wires between the indoor and outdoor units.
5 Indoor-outdoor signal transmission error due to disturbed power supply waveform.

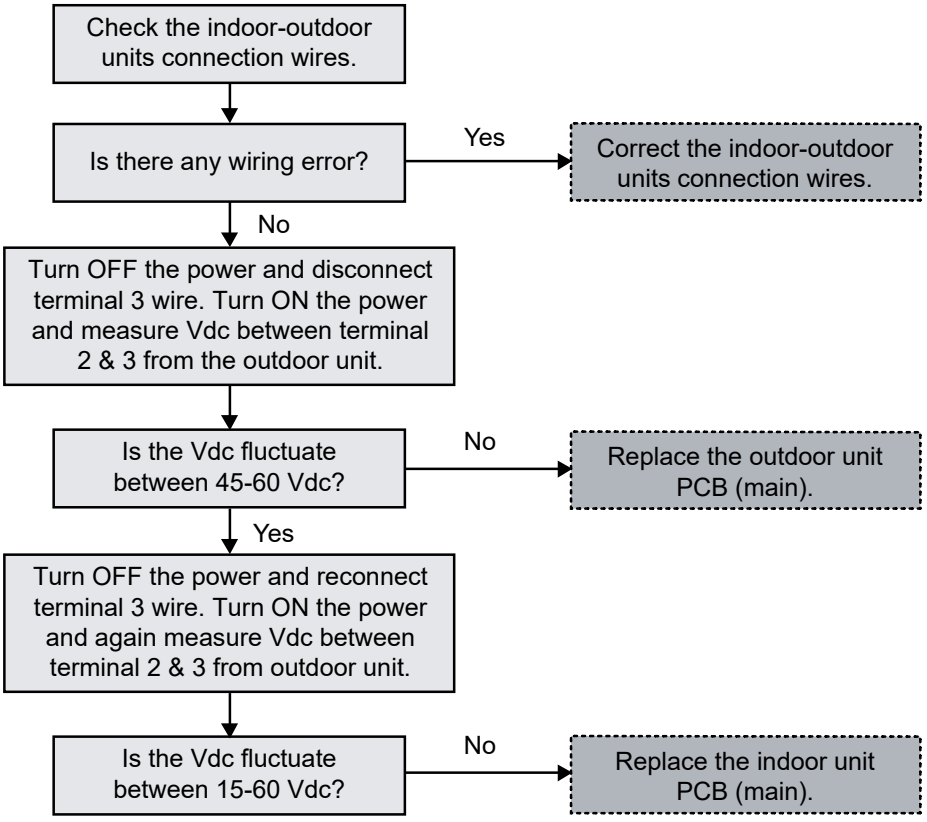
Abnormality Judgment:

Continue for 1 minute after operation.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.23 Tank Booster Heater OLP Abnormality (H91)

Malfunction Decision Conditions:

During operation of tank booster heater, and tank booster heater OLP open circuit.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty tank booster heater overload protector (OLP).
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

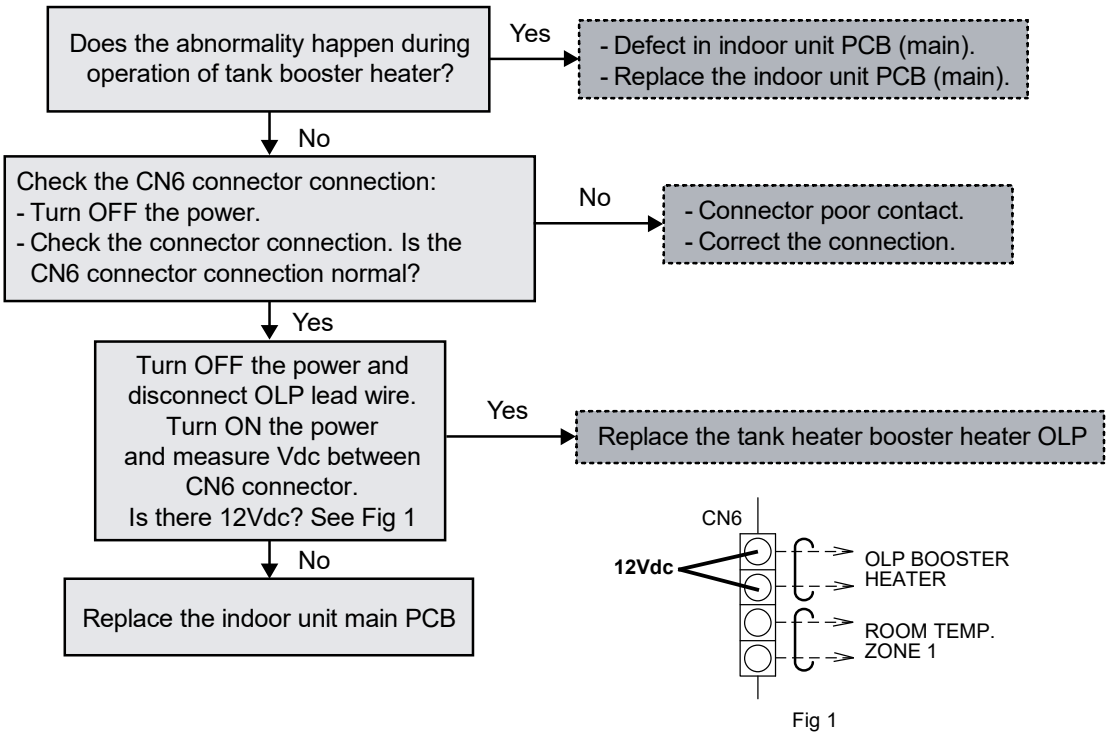


Fig 1

21.5.24 Unspecified Voltage between Indoor and Outdoor (H95)

Malfunction Decision Conditions:

The supply power is detected for its requirement by the indoor/outdoor transmission.

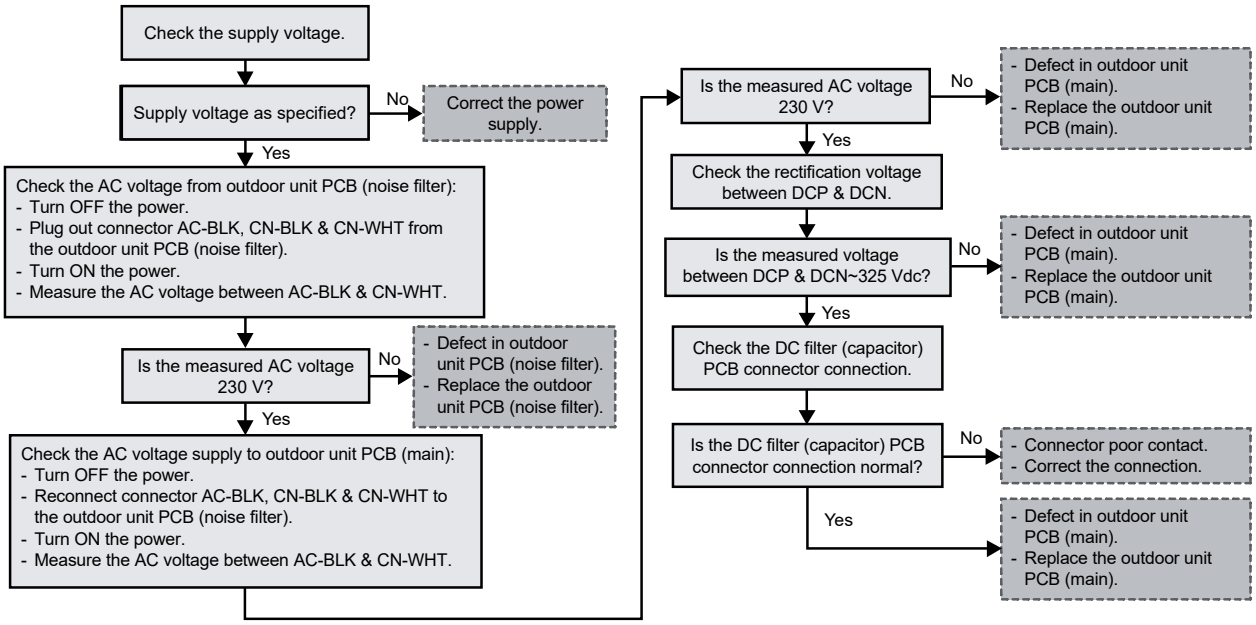
Malfunction Caused:

1 Insufficient power supply.
2 Faulty outdoor unit PCB (noise filter/main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.25 Outdoor High Pressure Protection (H98 / F95)

Malfunction Decision Conditions:

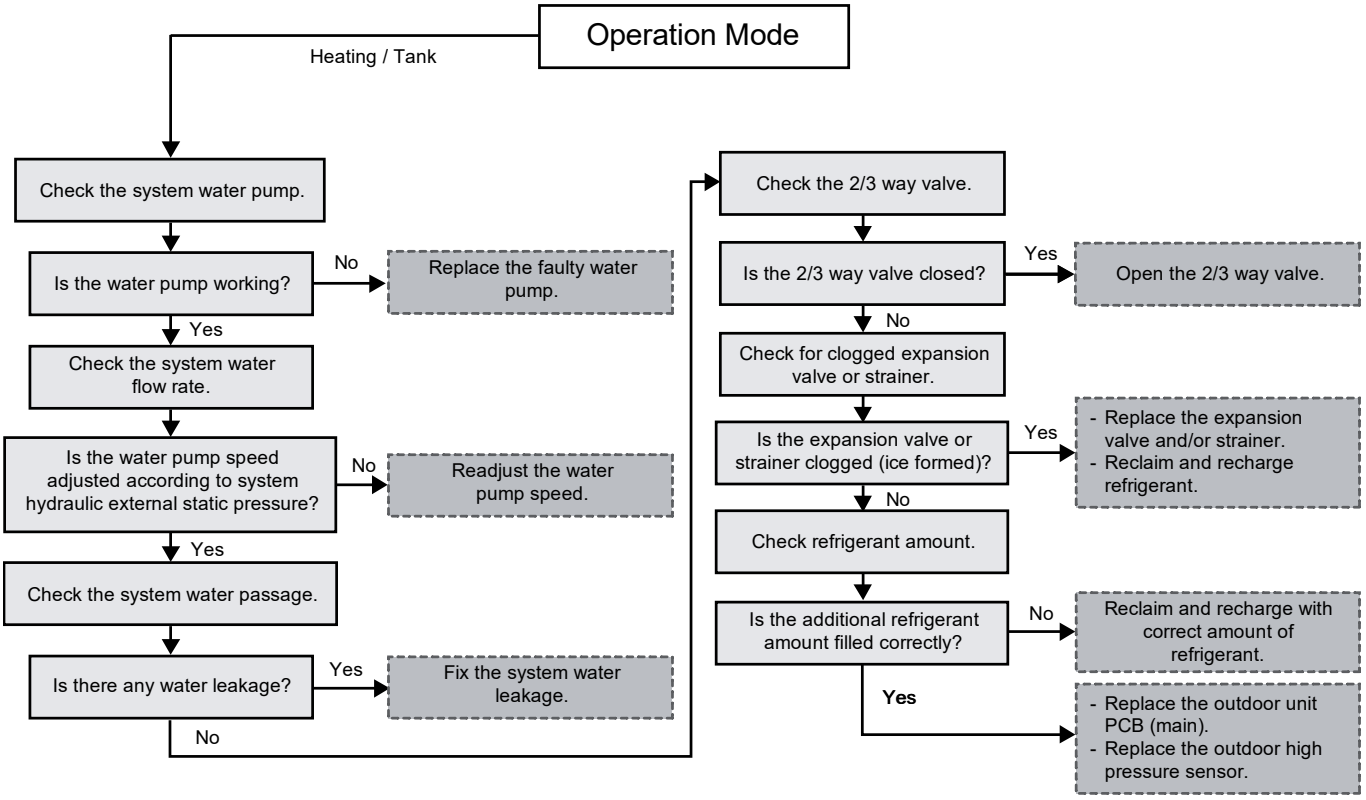
During operation of cooling / heating, when pressure 4.2 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 Dust accumulation in the outdoor unit heat exchanger.
5 Air short circuit at outdoor.
6 Faulty outdoor unit fan motor.
7 2/3 way closed.
8 Clogged expansion valve or strainer.
9 Excessive refrigerant.
10 Faulty outdoor high pressure sensor.
11 Faulty outdoor unit PCB (main).

Troubleshooting:

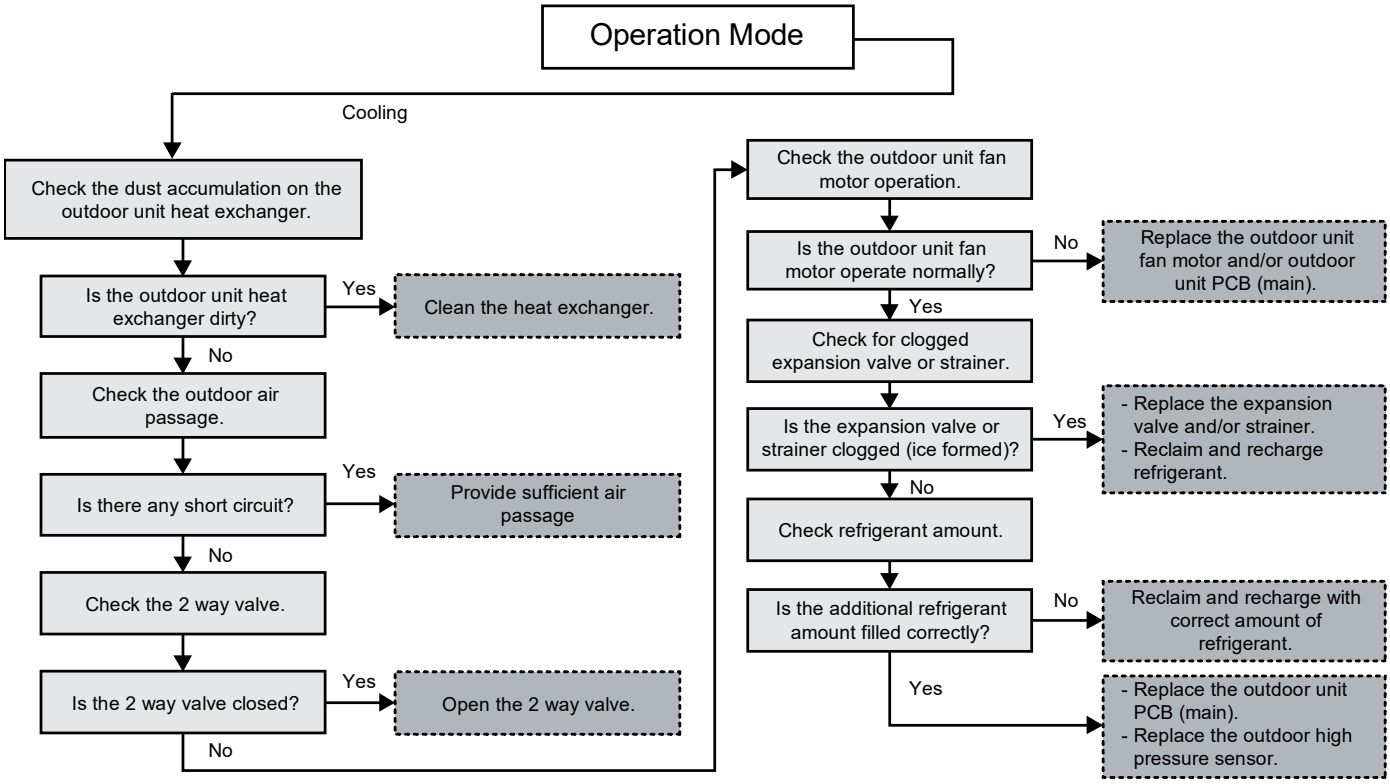
Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.26 Indoor Freeze-up Protection (H99)

Malfunction Decision Conditions:

During anti-freezing control in cooling operation, when the indoor refrigerant liquid temperature < 0°C.

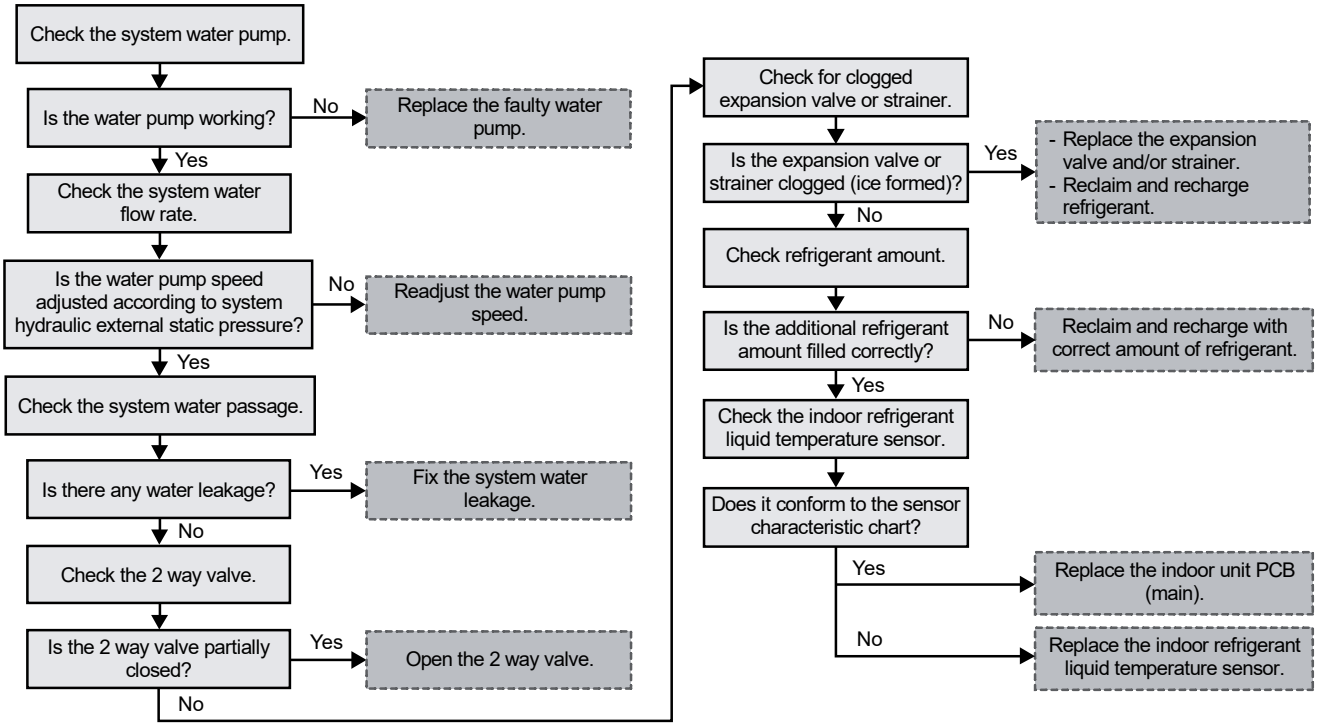
Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2 way valve partially closed.
5 Clogged expansion valve or strainer.
6 Refrigerant shortage (refrigerant leakage).
7 Faulty indoor refrigerant liquid temperature sensor.
8 Faulty indoor unit PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.27 Outdoor High Pressure Switch Activate (F12)

Malfunction Decision Conditions:

During operation of cooling and heating, when pressure 4.5 MPa and above is detected by outdoor high pressure switch.

Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 Faulty water pump.
4 Insufficient water flow rate in system.
5 Water leak in system.
6 2/3 way valve closed.
7 Clogged expansion valve or strainer.
8 Excessive refrigerant.
9 Faulty outdoor high pressure sensor and switch.
10 Faulty outdoor unit PCB.

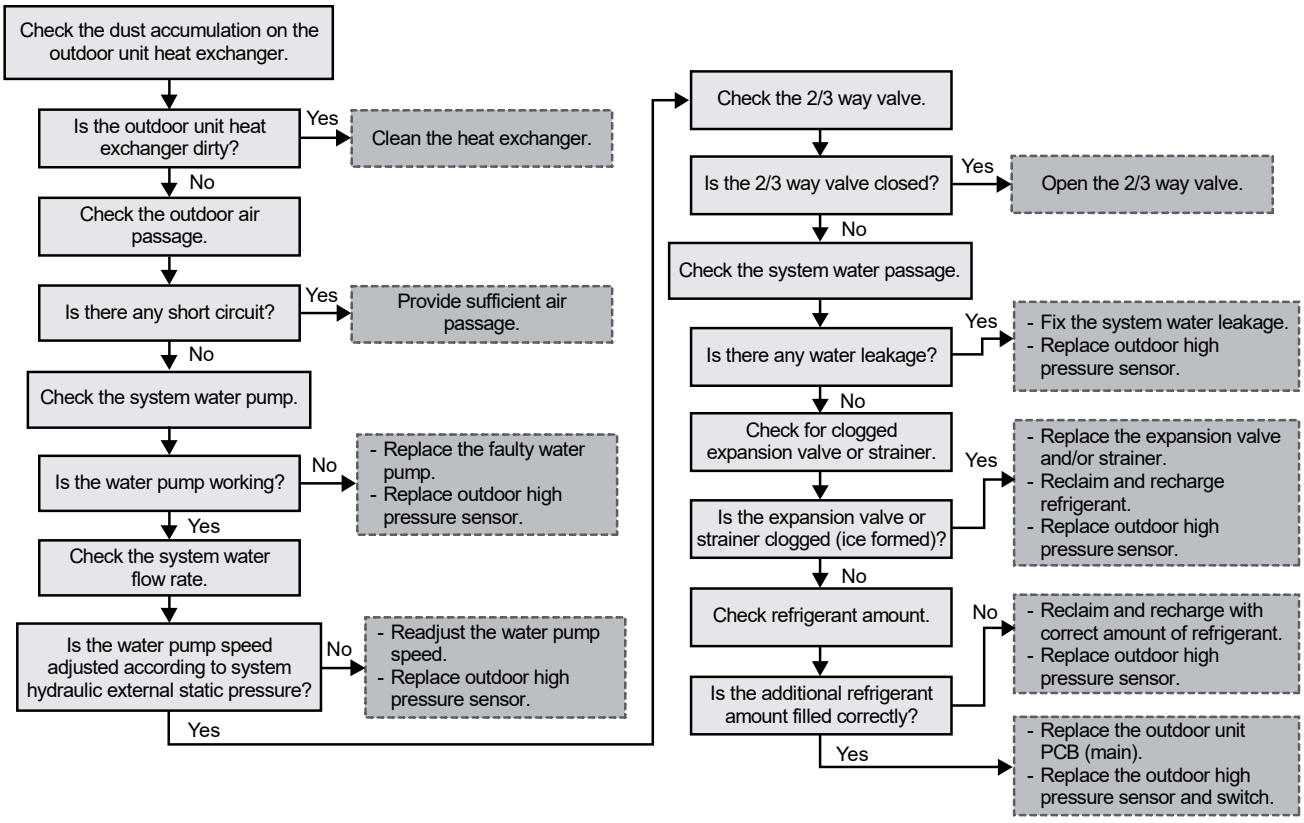
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.28 Compressor Rotation Failure (F14)

Malfunction Decision Conditions:

A compressor rotation failure is detected by checking the compressor running condition through the position detection circuit.

Malfunction Caused:

1 Compressor terminal disconnect.
2 Faulty outdoor unit PCB (main).
3 Faulty compressor.

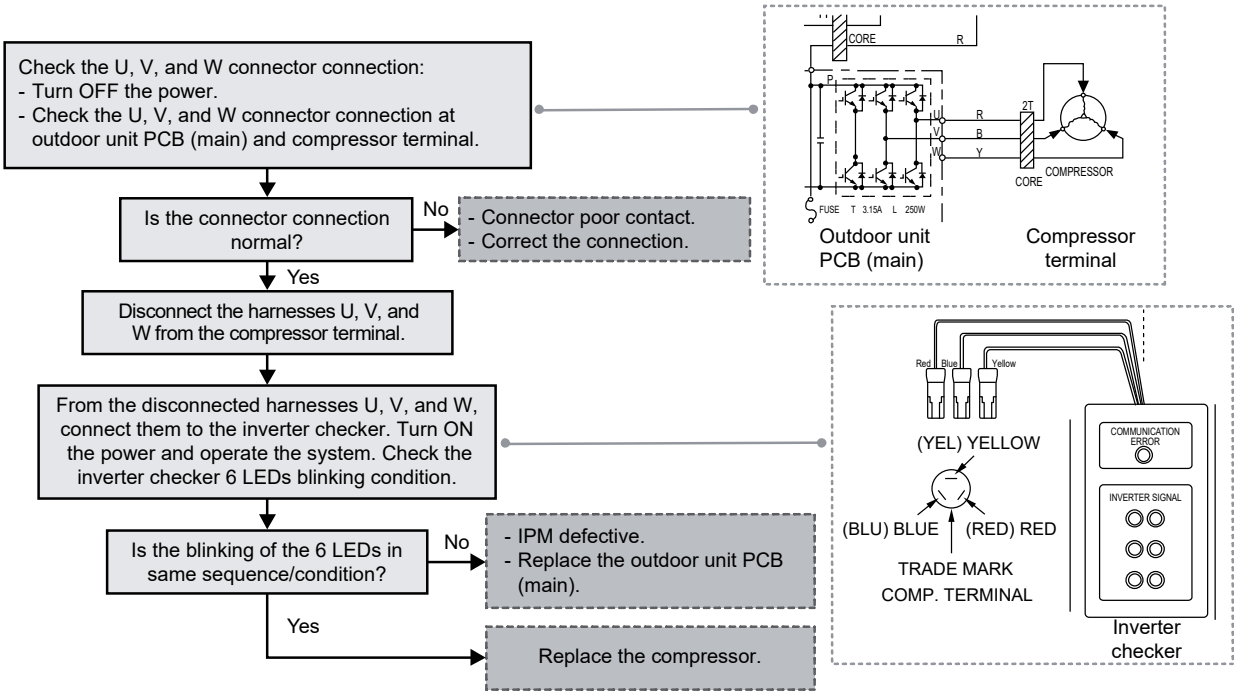
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.29 Outdoor Fan Motor (DC Motor) Mechanism Locked (F15)

Malfunction Decision Conditions:

The rotation speed detected by the Hall IC of the fan motor during fan motor operation is used to determine abnormal fan motor (feedback of rotation > 2550 rpm or < 50 rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the fan motor winding.
2 Operation stop due to breaking of wire inside the fan motor.
3 Operation stop due to breaking of fan motor lead wires.
4 Operation stop due to fan motor Hall IC malfunction.
5 Operation error due to faulty outdoor unit PCB.

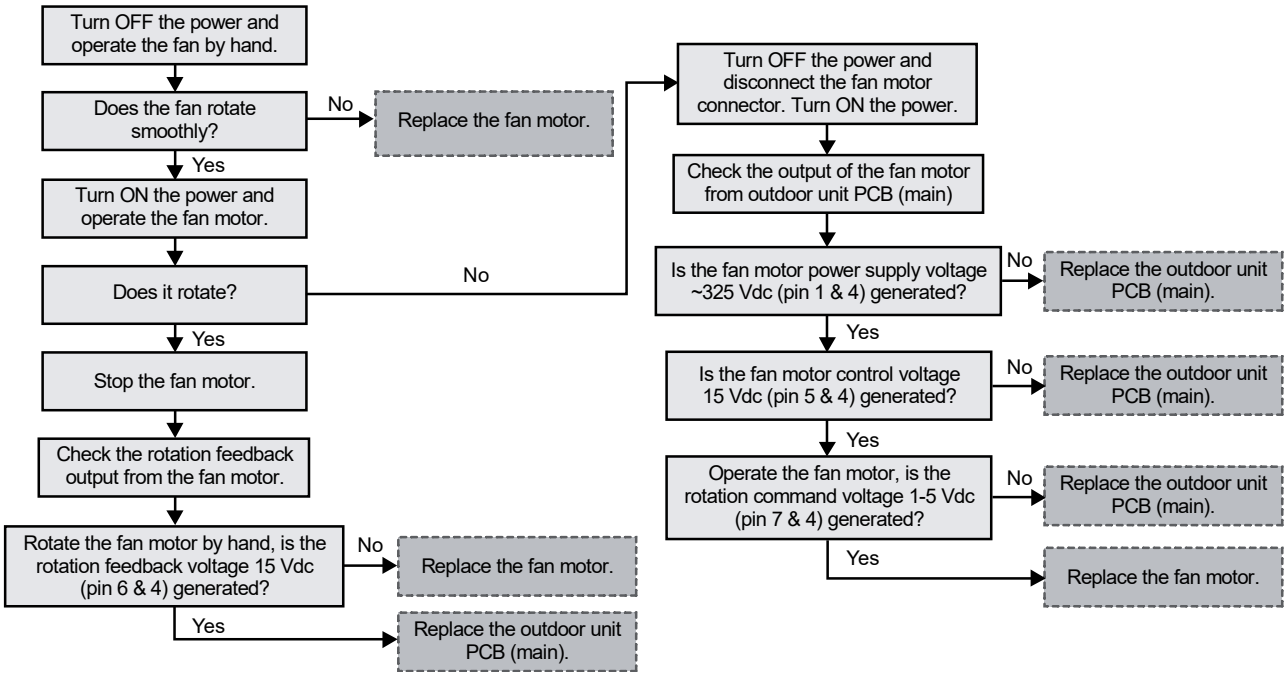
Abnormality Judgment:

Continue 2 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.30 Input Over Current Detection (F16)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor current above UDZ03 / 05KE5 14.6A & UDZ07 / 09KE5 16.9A is detected by the current transformer (CT) in the outdoor unit PCB.

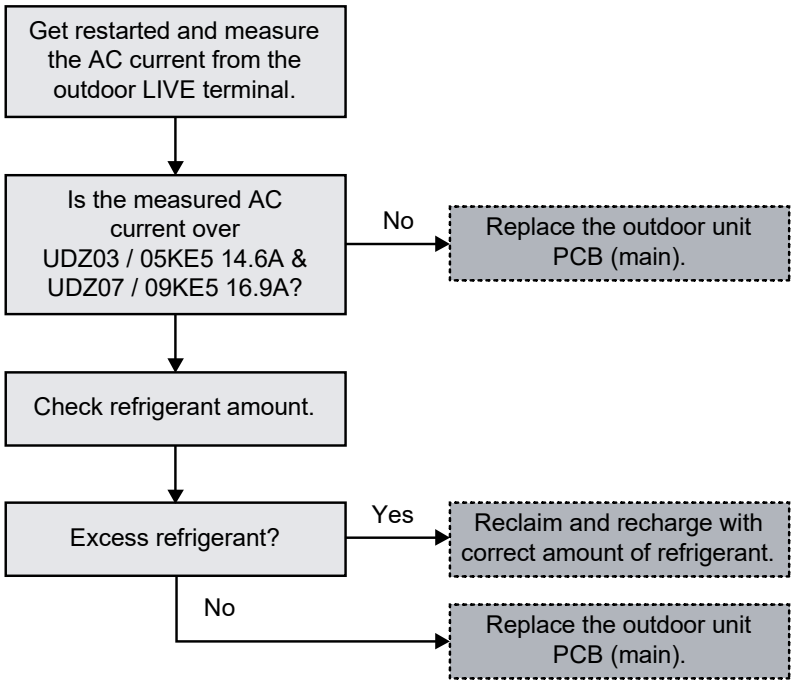
Malfunction Caused:

1 Excessive refrigerant.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting: **Caution** For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.31 Compressor Overheating (F20)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature above 112°C is detected by the compressor tank temperature sensor.

Malfunction Caused:

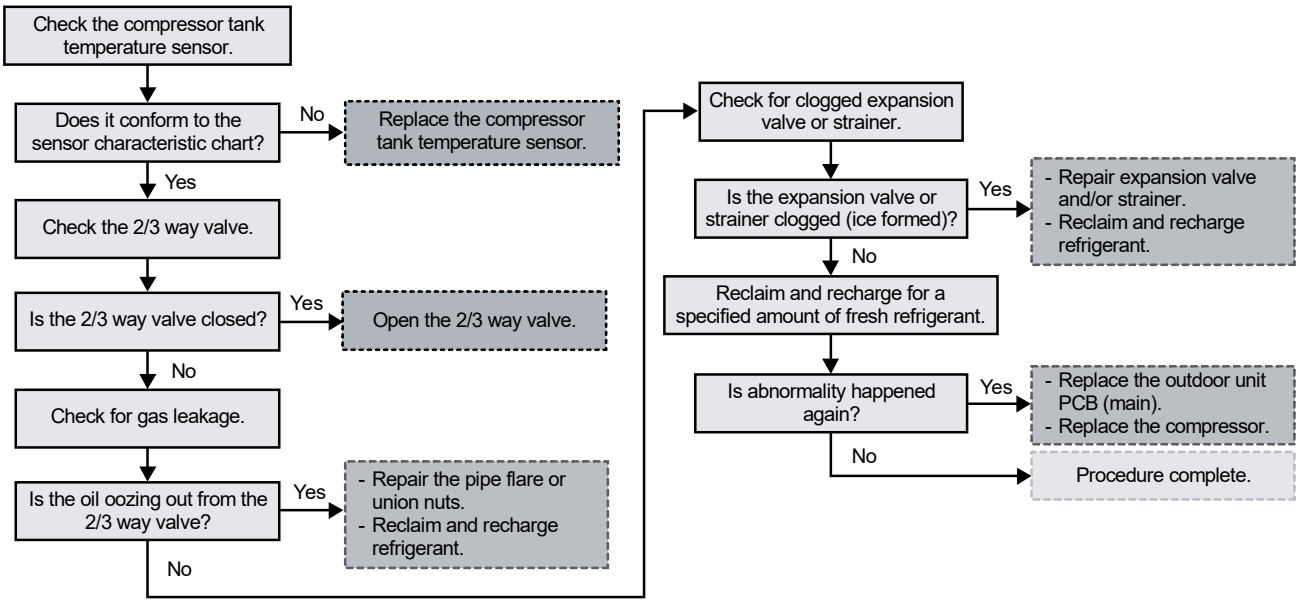
1 Faulty compressor tank temperature sensor.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Faulty compressor.

Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.32 IPM Overheating (F22)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature 95°C is detected by the outdoor IPM temperature sensor.

Malfunction Caused:

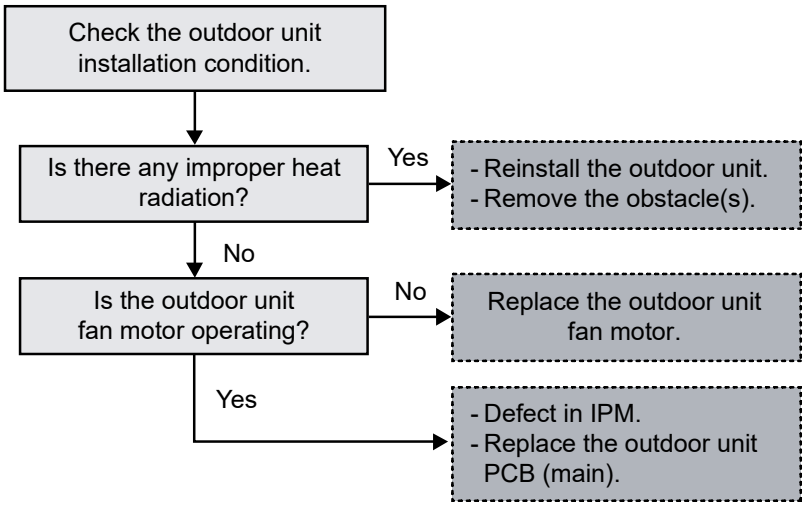
1 Faulty outdoor unit fan motor.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 30 minutes.

Troubleshooting: Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.33 Output Over Current Detection (F23)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor DC current is above set value is detected by the IPM DC Peak sensing circuitry in the outdoor unit PCB (main).

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty compressor.

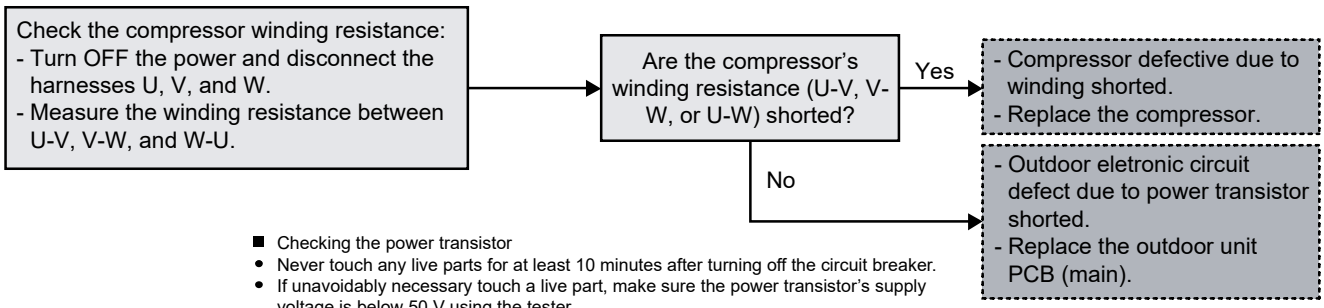
Abnormality Judgment:

Continue for 7 times.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



■ Checking the power transistor

- Never touch any live parts for at least 10 minutes after turning off the circuit breaker.
- If unavoidably necessary touch a live part, make sure the power transistor's supply voltage is below 50 V using the tester.
- For the UVW, make measurements at the Faston terminal on the board or the relay connector.

Tester's negative terminal	Power transistor (+)	UVW	Power transistor (-)	UVW
Tester's positive terminal	UVW	Power transistor (+)	UVW	Power transistor (-)
Normal resistance	Several kohms to several Mohms			
Abnormal resistance	0 or ∞			

21.5.34 Refrigeration Cycle Abnormality (F24)

Malfunction Decision Conditions:

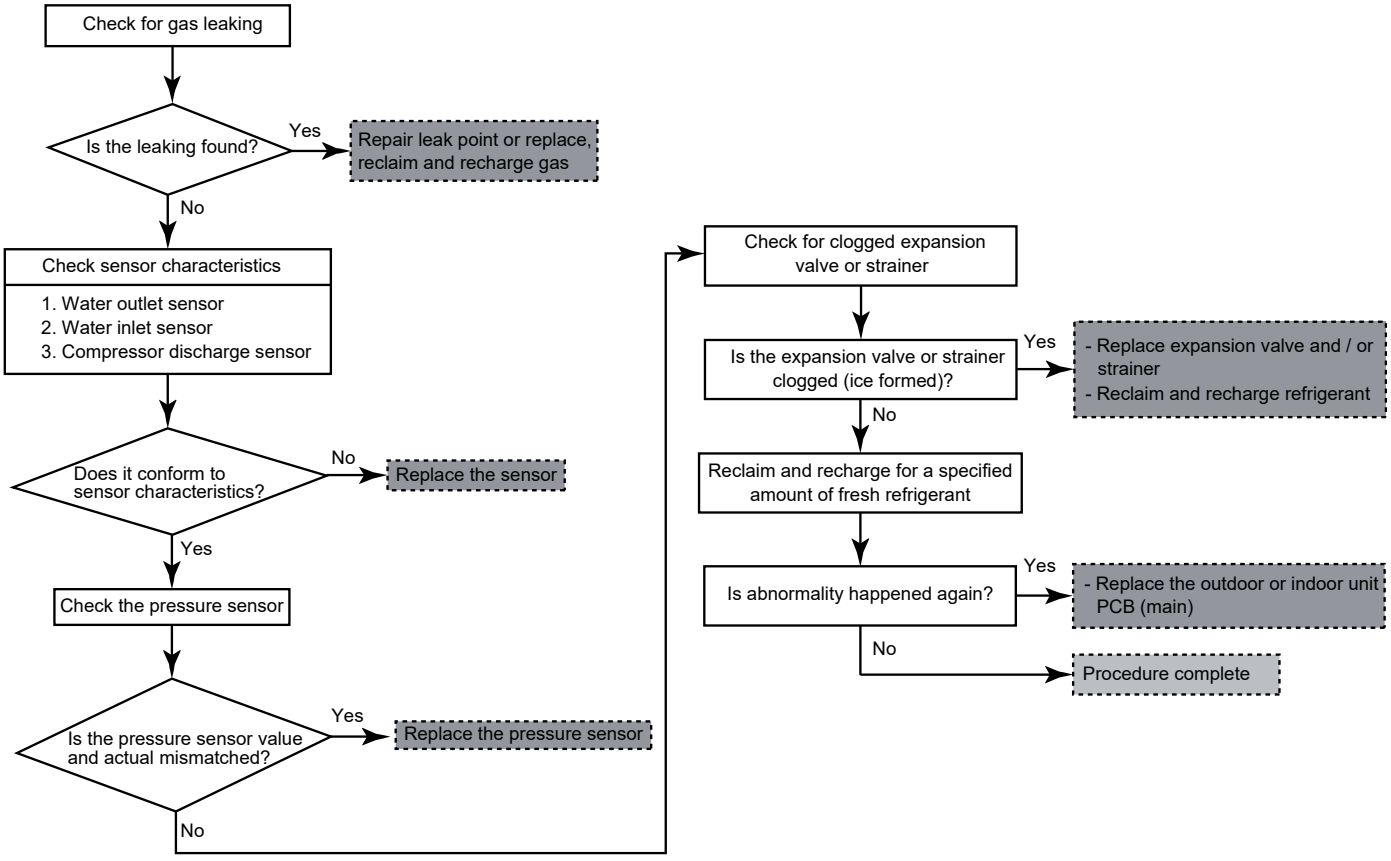
1 During compressor running (heating / cooling) for more than 10 minutes except deice, pumpdown and test mode.
2 During heating / cooling, water outlet and water inlet difference is less than 1°C.
3 During heating / cooling, high pressure < 1MPa (143 Psi) for more than 10 minutes or < 0.2 MPa (28 Psi) for more than 5 minutes.
4 During heating / cooling, discharge temperature - saturation temperature of high pressure ≥ 60°C.

Malfunction Caused:

1 Refrigerant shortage (refrigerant leakage).
2 Faulty indoor water inlet, indoor water outlet, compressor discharge temp sensor or high pressure sensor.
3 2/3 way valve closed.
4 Clogged expansion valve or strainer.
5 Faulty indoor or outdoor PCB (main).

Abnormality Judgment:

Continue 2 times in 30 minutes.



21.5.35 Four Way Valve Abnormality (F25)

Malfunction Decision Conditions:

1 During heating operation, when the indoor pipe temperature of thermostat ON indoor unit < 0°C.
2 During cooling operation, when the indoor pipe temperature of thermostat ON indoor unit > 45°C.

Malfunction Caused:

1 Faulty sensor.
2 Faulty connector connection.
3 Faulty outdoor unit PCB (noise filter/main).
4 Faulty four way valve.

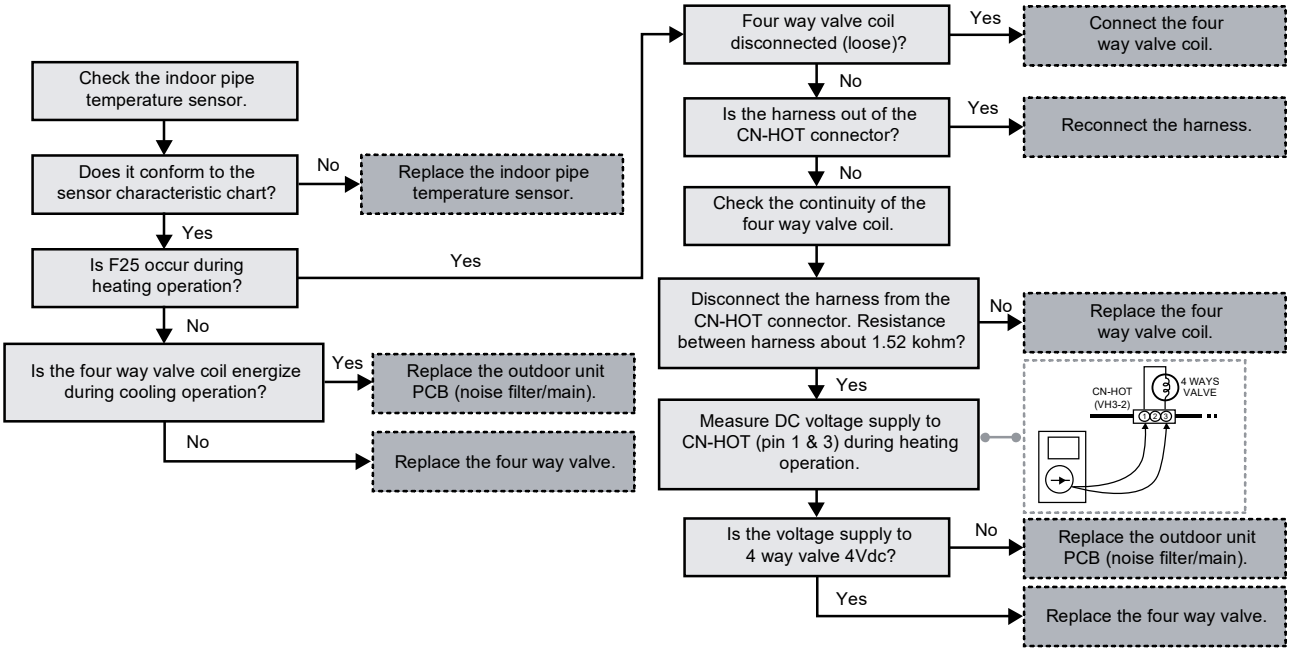
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.36 Outdoor High Pressure Switch Abnormal (F27)

Malfunction Decision Conditions:

During compressor stop, and outdoor high pressure switch is remain opened.

Malfunction Caused:

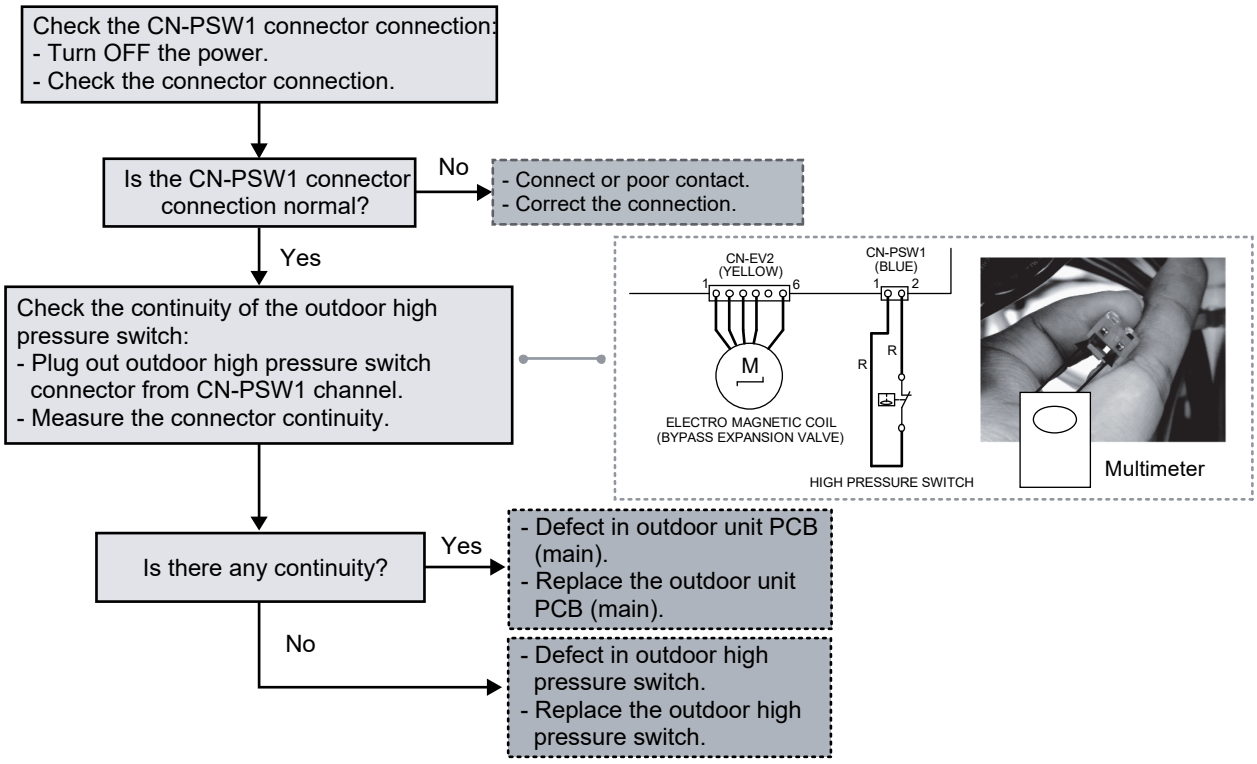
1 Faulty connector connection.
2 Faulty switch.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 1 minute.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component



21.5.37 Low Discharge Superheat (F29)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).
4 Faulty High Pressure Switch
5 Refrigerant shortage (refrigerant leakage).

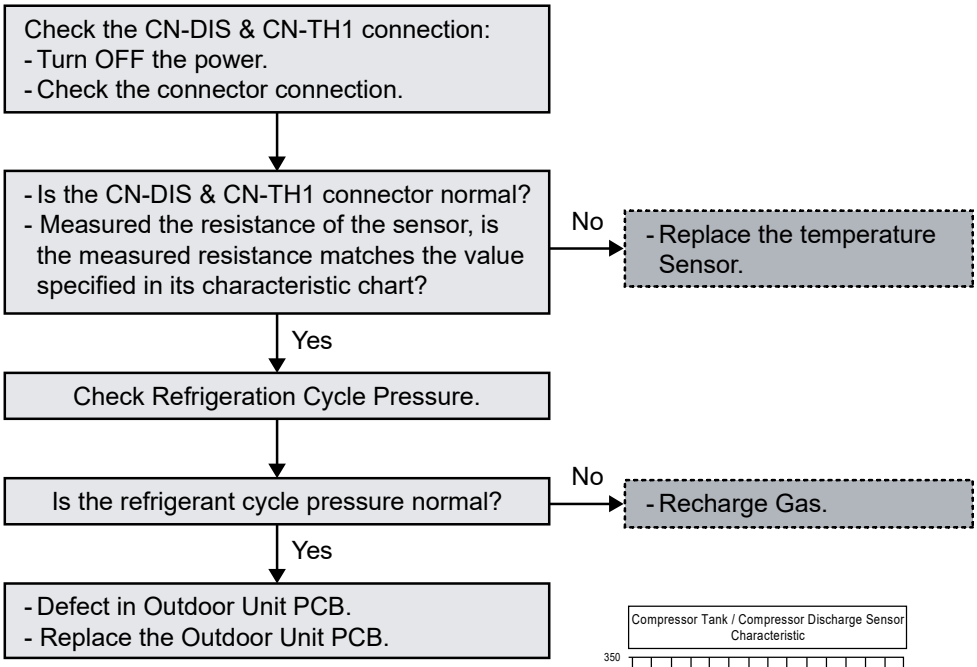
Abnormality Judgment:

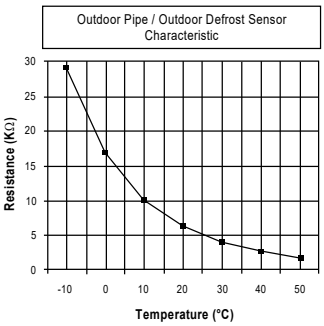
1 times occurrence within 2550 minutes.

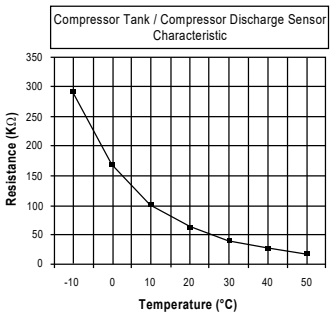
Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.







21.5.38 Indoor Water Outlet Temperature Sensor 2 Abnormality (F30)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor 2 are used to determine sensor error.

Malfunction Caused:

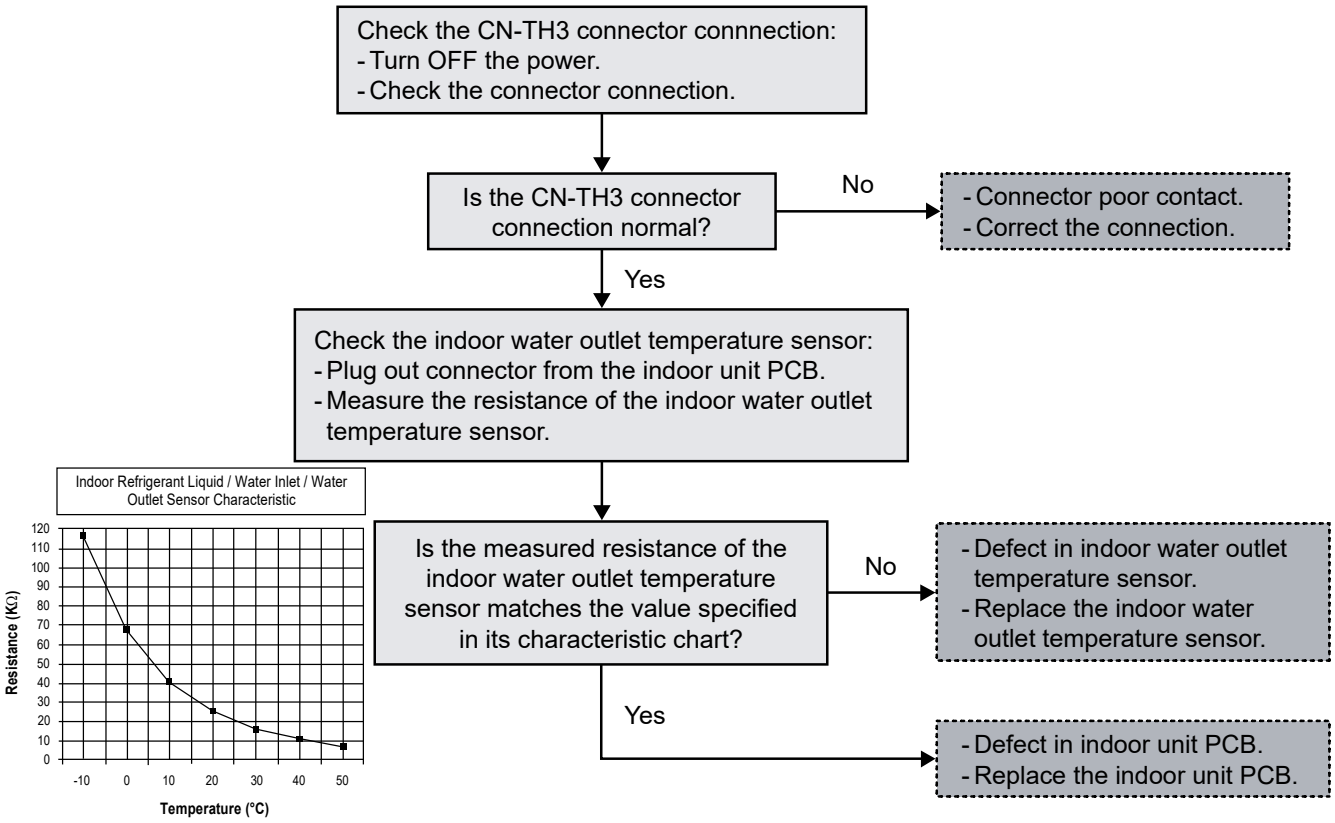
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.39 Outdoor Air Temperature Sensor Abnormality (F36)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor air temperature sensor are used to determine sensor error.

Malfunction Caused:

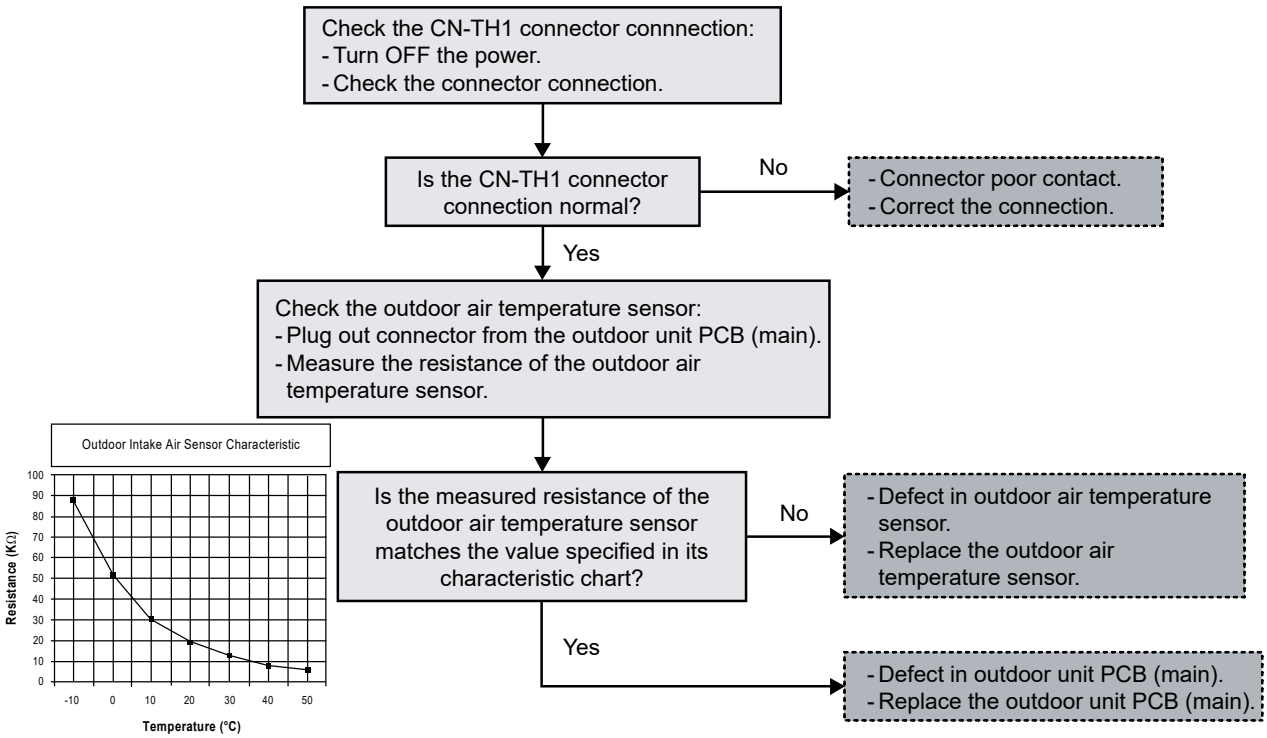
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.40 Indoor Water Inlet Temperature Sensor Abnormality (F37)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water inlet temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

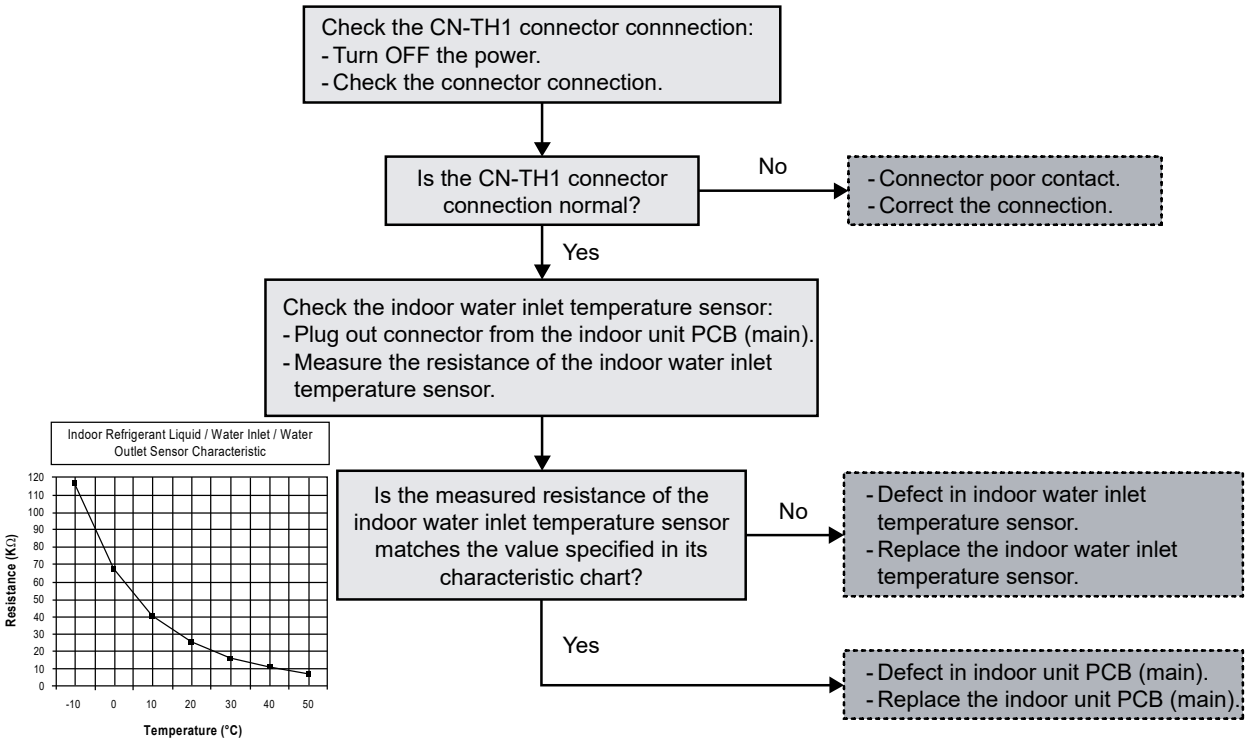
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.41 Outdoor Discharge Pipe Temperature Sensor Abnormality (F40)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor discharge pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

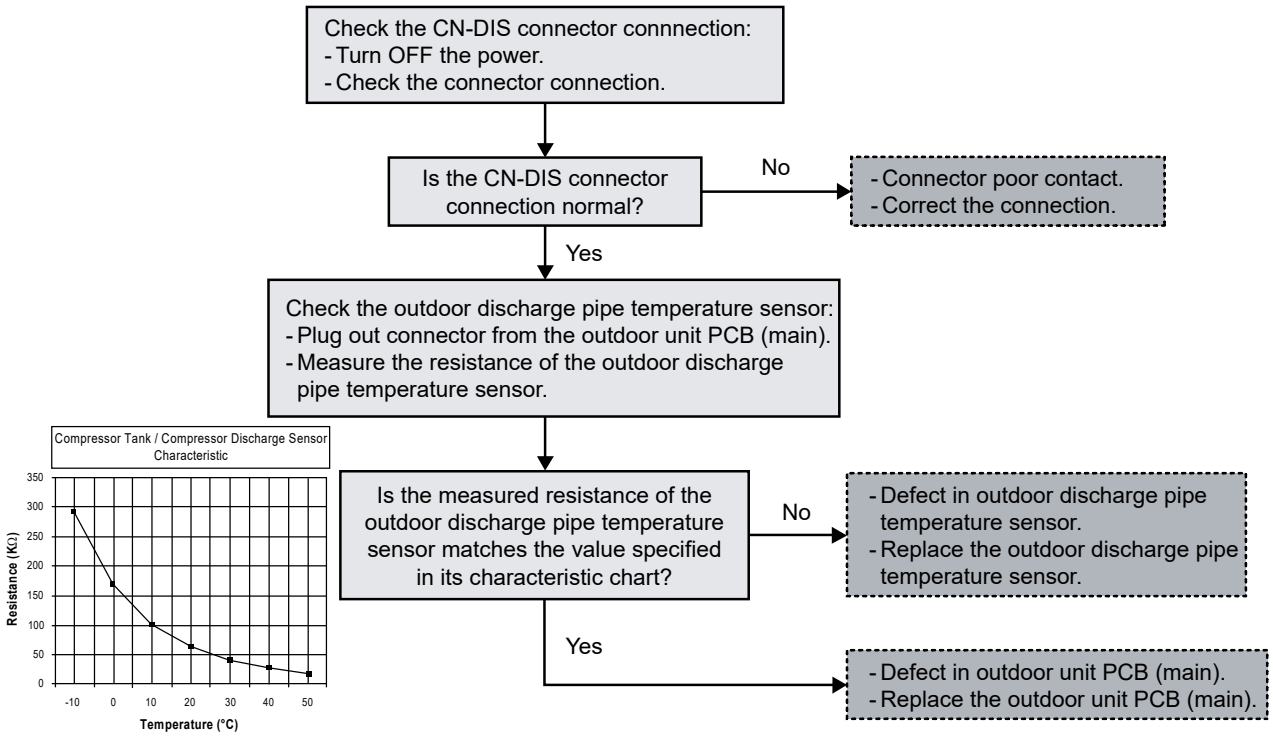
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.42 Power Factor Correction (PFC) Abnormality (F41)

Malfunction Decision Conditions:

During operation of cooling and heating, when the PFC protection circuitry in the outdoor unit PCB (main) senses abnormal high DC voltage level.

Malfunction Caused:

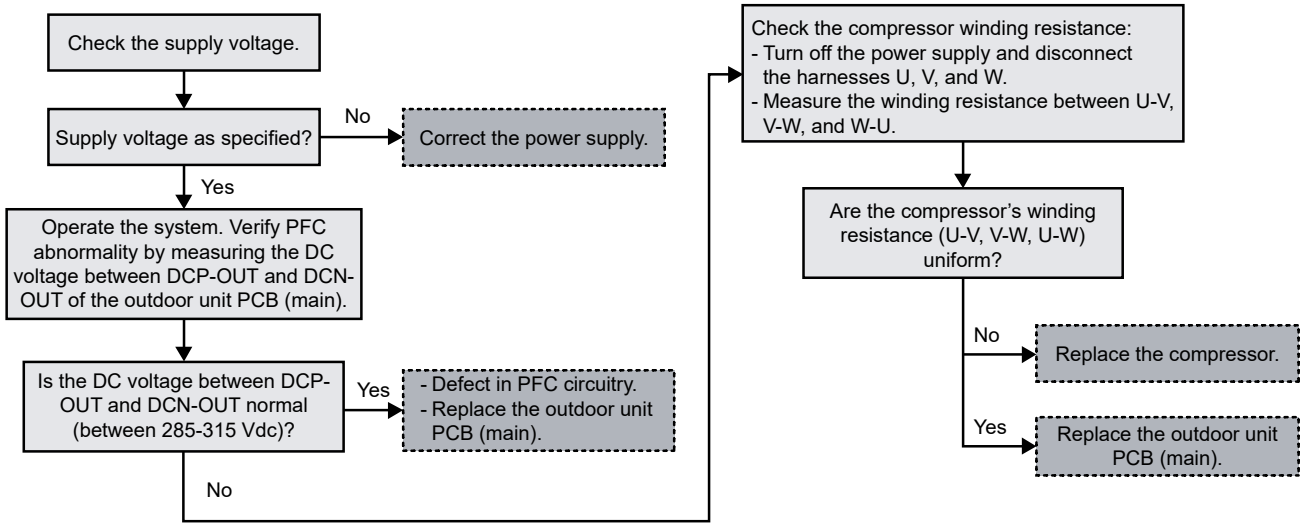
1 Power supply surge.
2 Compressor windings not uniform.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 4 times in 10 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.43 Outdoor Pipe Temperature Sensor Abnormality (F42)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

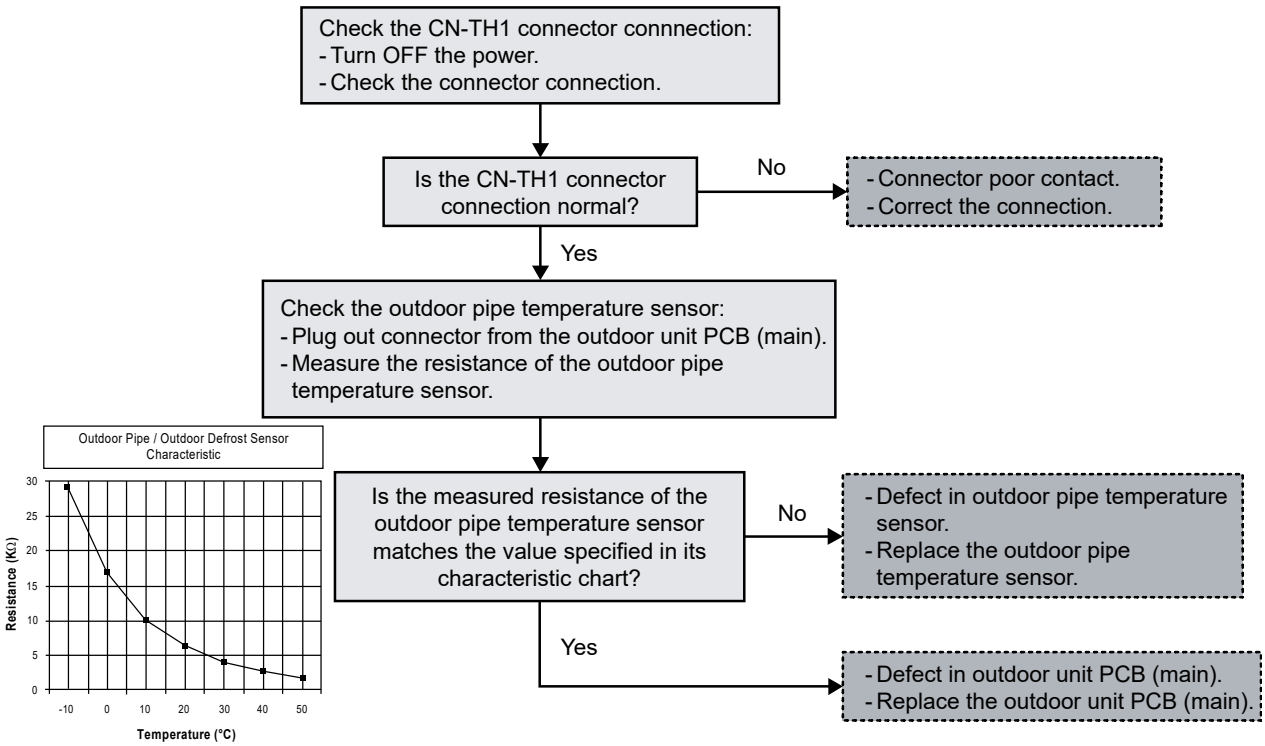
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



21.5.44 Outdoor Defrost Temperature Sensor Abnormality (F43)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor defrost temperature sensor are used to determine sensor error.

Malfunction Caused:

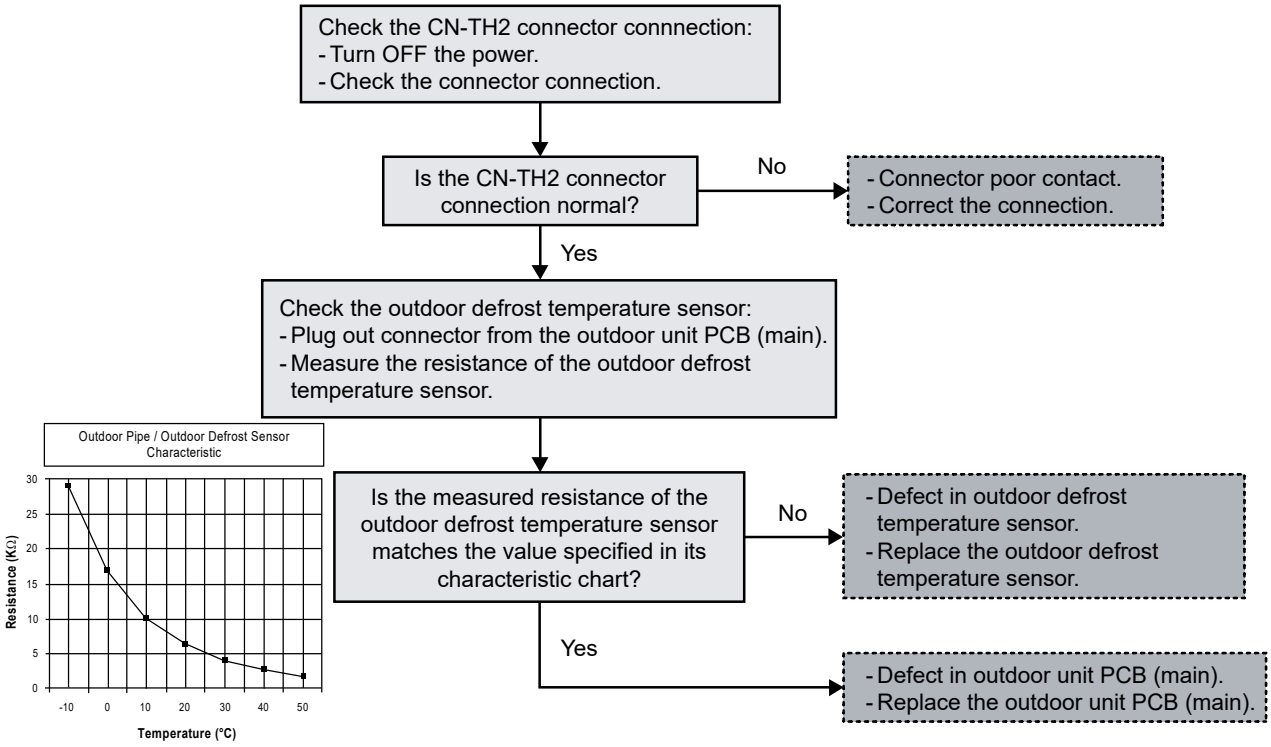
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.45 Indoor Water Outlet Temperature Sensor Abnormality (F45)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor are used to determine sensor errors.

Malfunction Caused:

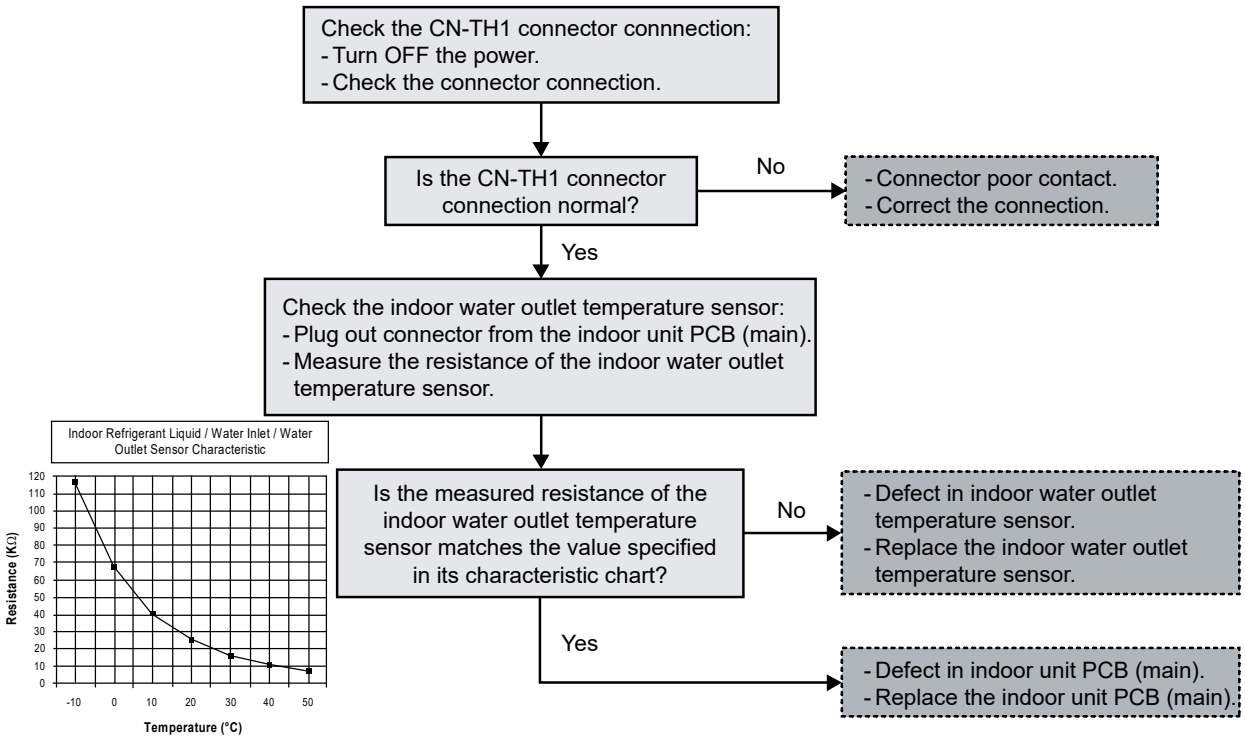
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.46 Outdoor Current Transformer Open Circuit (F46)

Malfunction Decision Conditions:

A current transformer (CT) open circuit is detected by checking the compressor running frequency (⩾ rated frequency) and CT detected input current (< 0.65 A) for continuously 20 seconds.

Malfunction Caused:

1 CT defective.
2 Faulty outdoor unit PCB (main).
3 Compressor defective (low compression).

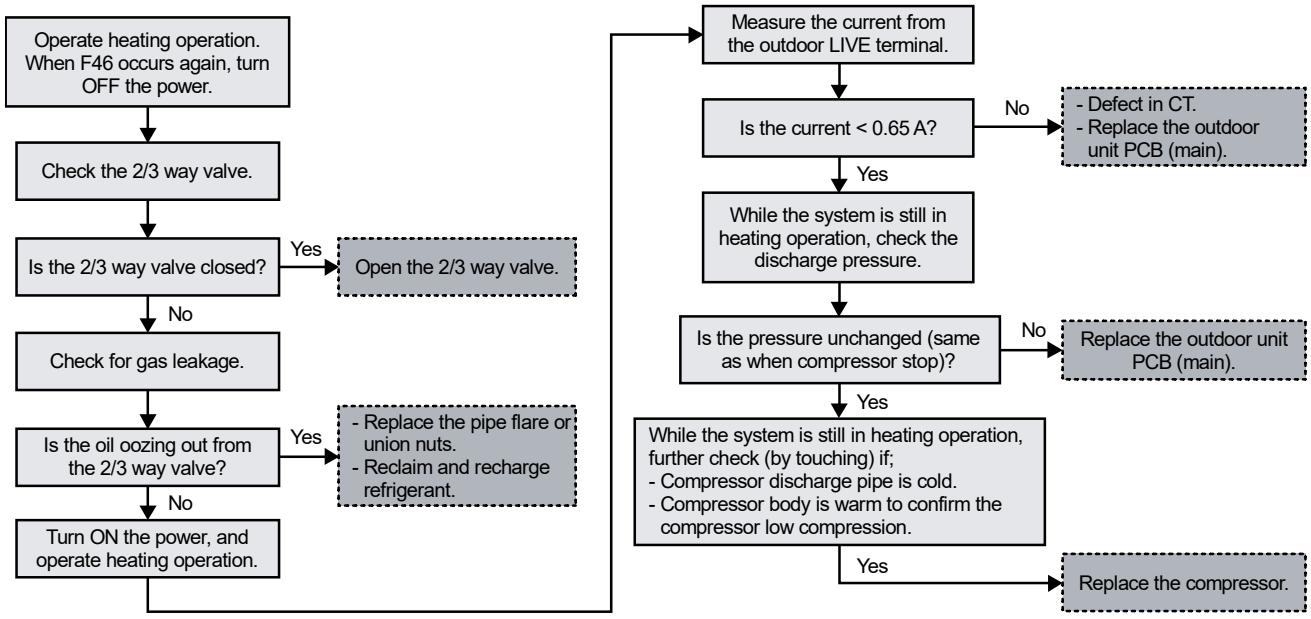
Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



21.5.47 Outdoor Bypass Outlet Temperature Sensor (F49)

Malfunction Decision Conditions:

During start up and operation of cooling and heating, the temperature detected by outdoor bypass sensor is used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection
2 Faulty sensor
3 Faulty outdoor unit PCB (main)

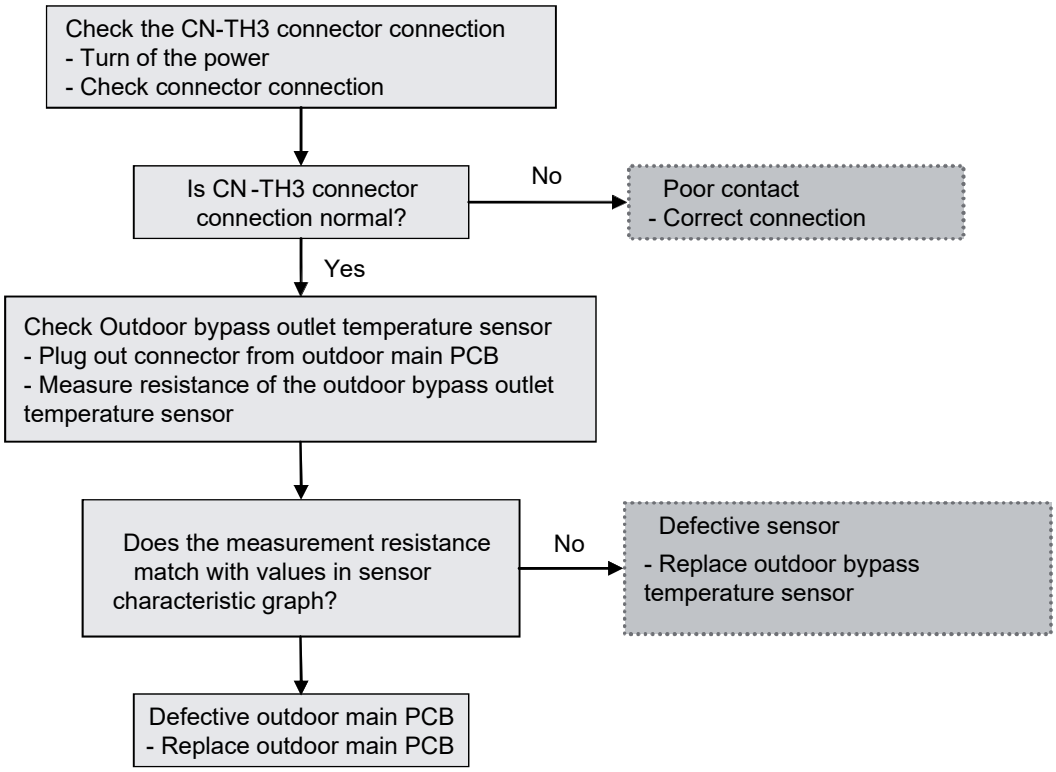
Abnormality Judgment:

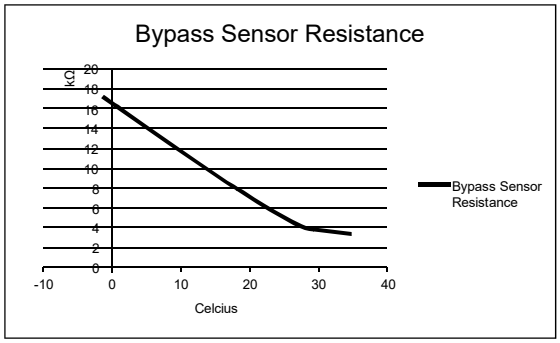
Continuous for 5 seconds

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



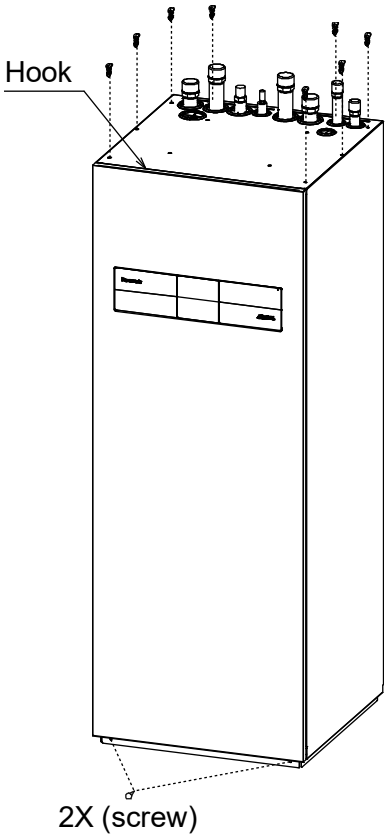


22. Disassembly and Assembly Instructions

⚠ WARNING

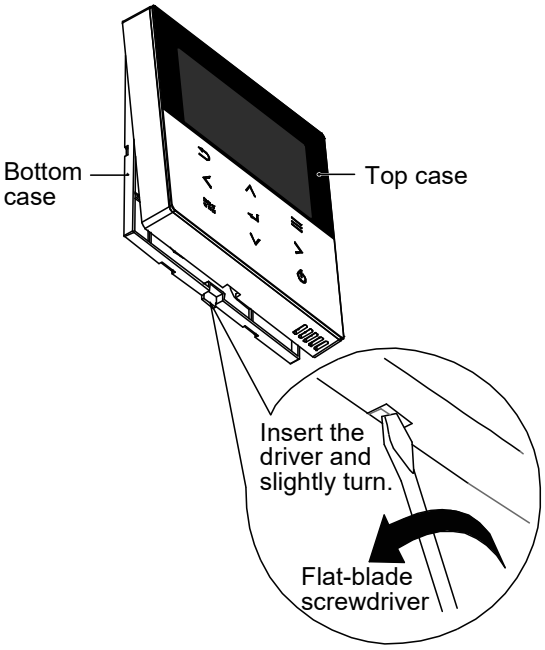
High Voltage are generated in the electrical parts area by the capacitor. Ensure that the capacitor has discharged sufficiently before proceeding with repair work. Failure to heed this caution may result in electric shocks.

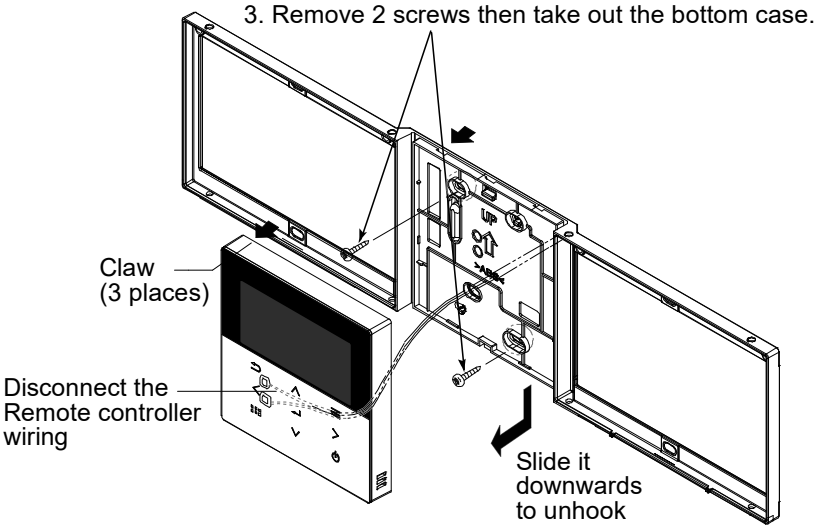
22.1 To Remove Front Plate and Top Plate



1 Remove 2 screws at the bottom to remove the Front Plate
2 Remove 8 screws at the top to remove the Top Plate

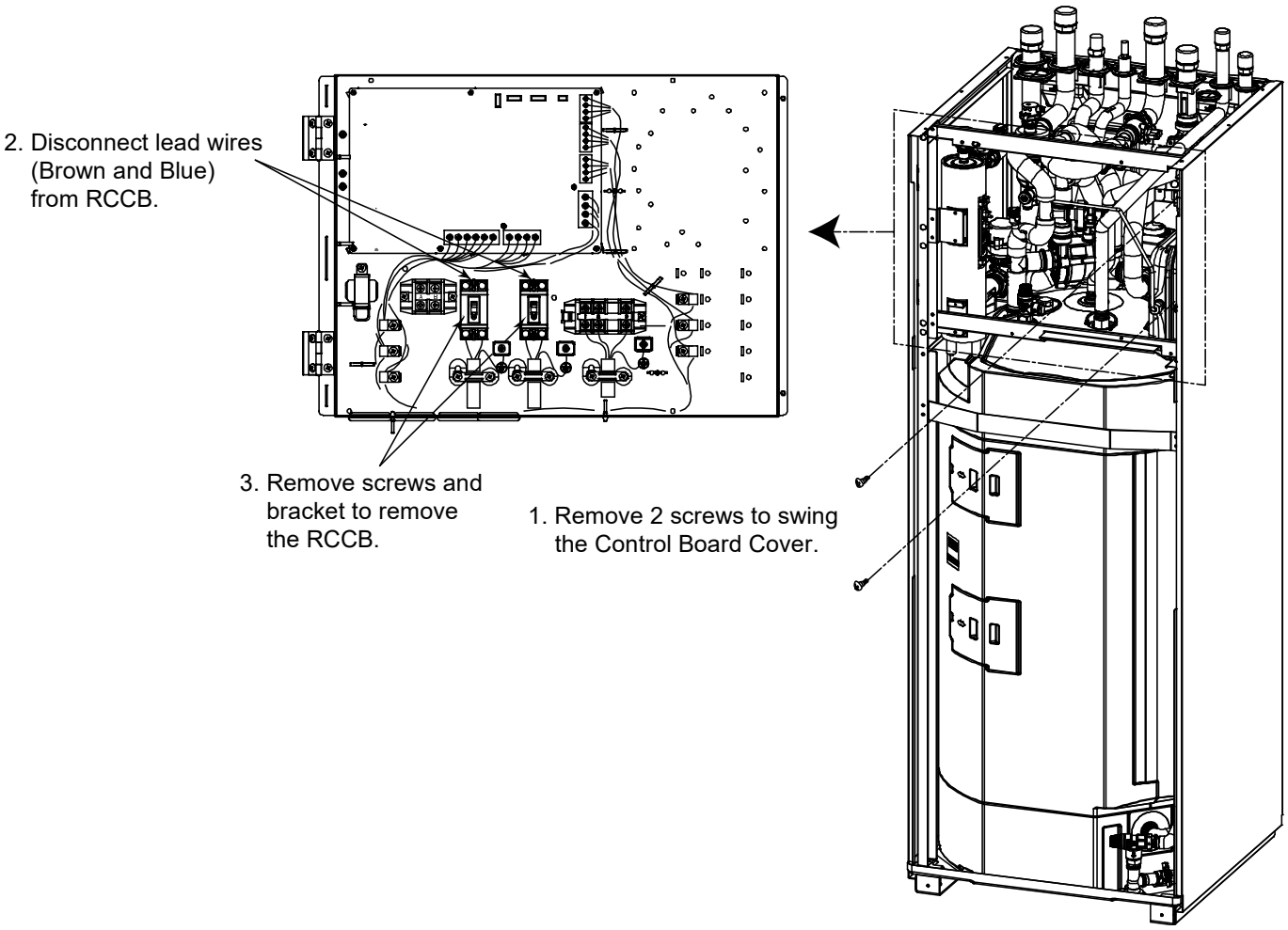
22.2 To Remove Remote Control





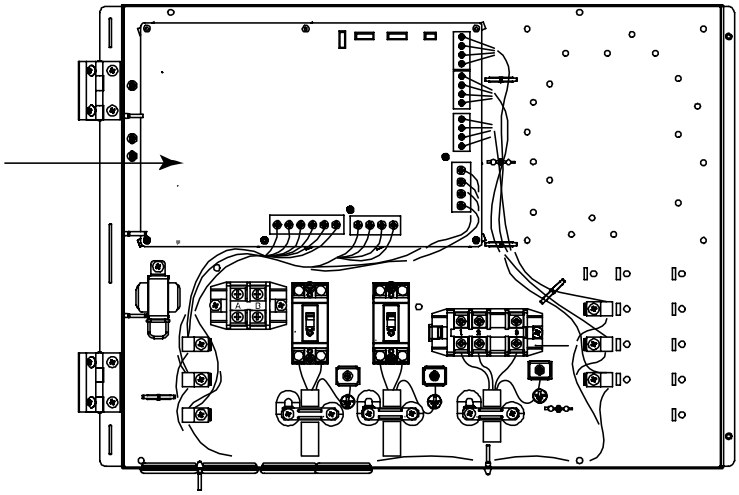
1. Remove the top case from the bottom case.
2. Disconnect the Remote Controller wiring.

22.3 To Remove RCCB

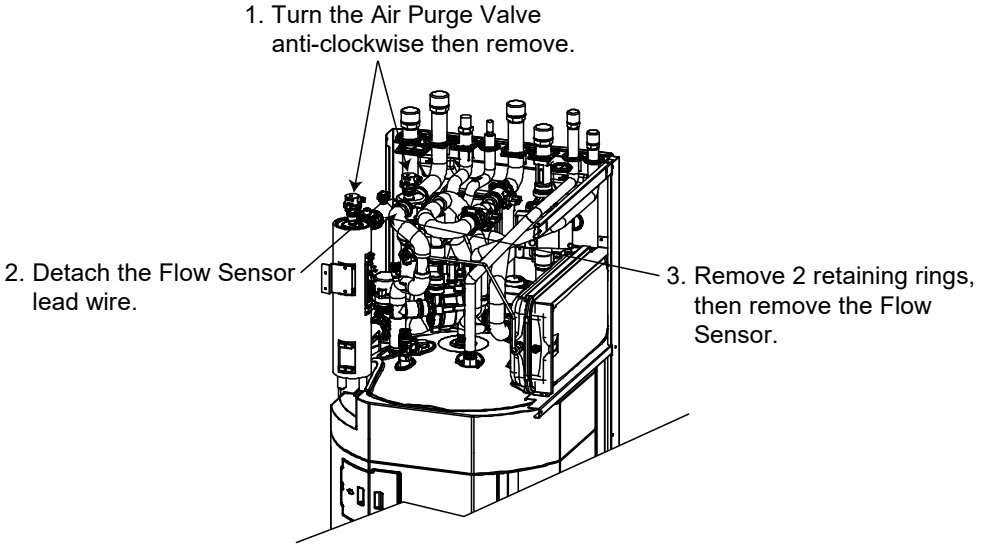


22.4 To Remove Electronic Controller

1. Disconnect all connectors from main PCB. (CN-FLWSEN, CN-PUMP1, CN-TH1, CN-TH2, CN-TH3, CN-OLP1, CN1, CN2, CN3, CN4, CN5, CN6, CN-CNT2, CN-DPS)
2. Detach remote controller, 3-way valve and all wires. (AC1-L3, AC1-N, G01, G05, DATA, G02, L2, L1, G03, AC2-L2, HT1-L2, HT1-L3, AC2-L3, AC2-N2, AC2-N3)

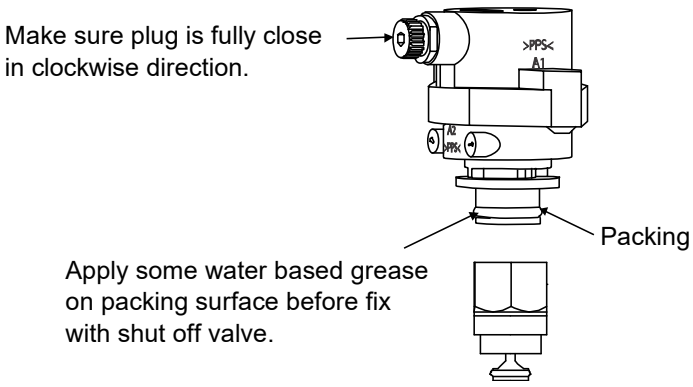


22.5 To Remove Flow Sensor and Air Purge Valve

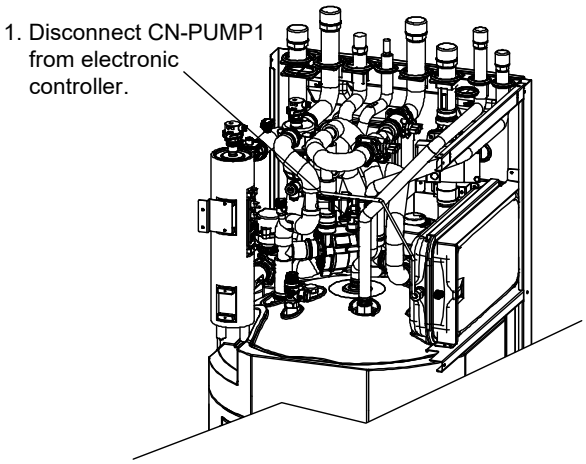


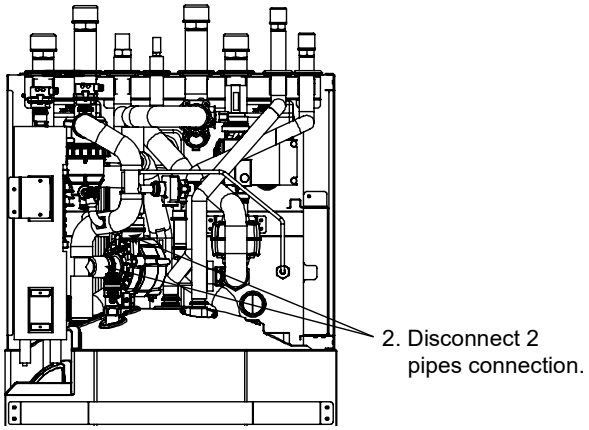
⚠ When reinstall Flow Switch, ensure the arrow on the flow switch is parallel with the pipe shaft and is facing in the direction of flow.

⚠ During reinstall Air Purge Valve.

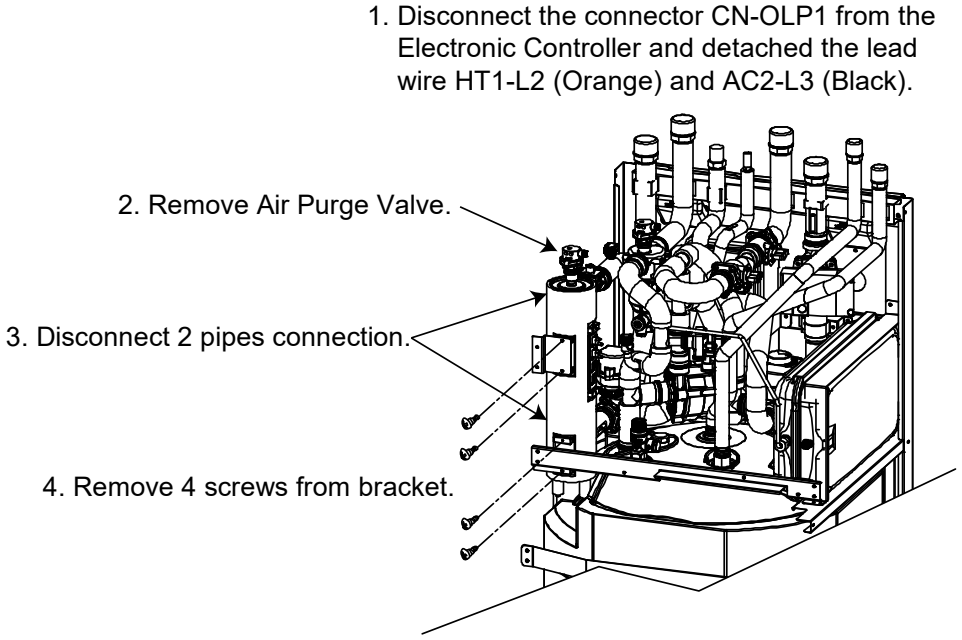


22.6 To Remove Water Pump

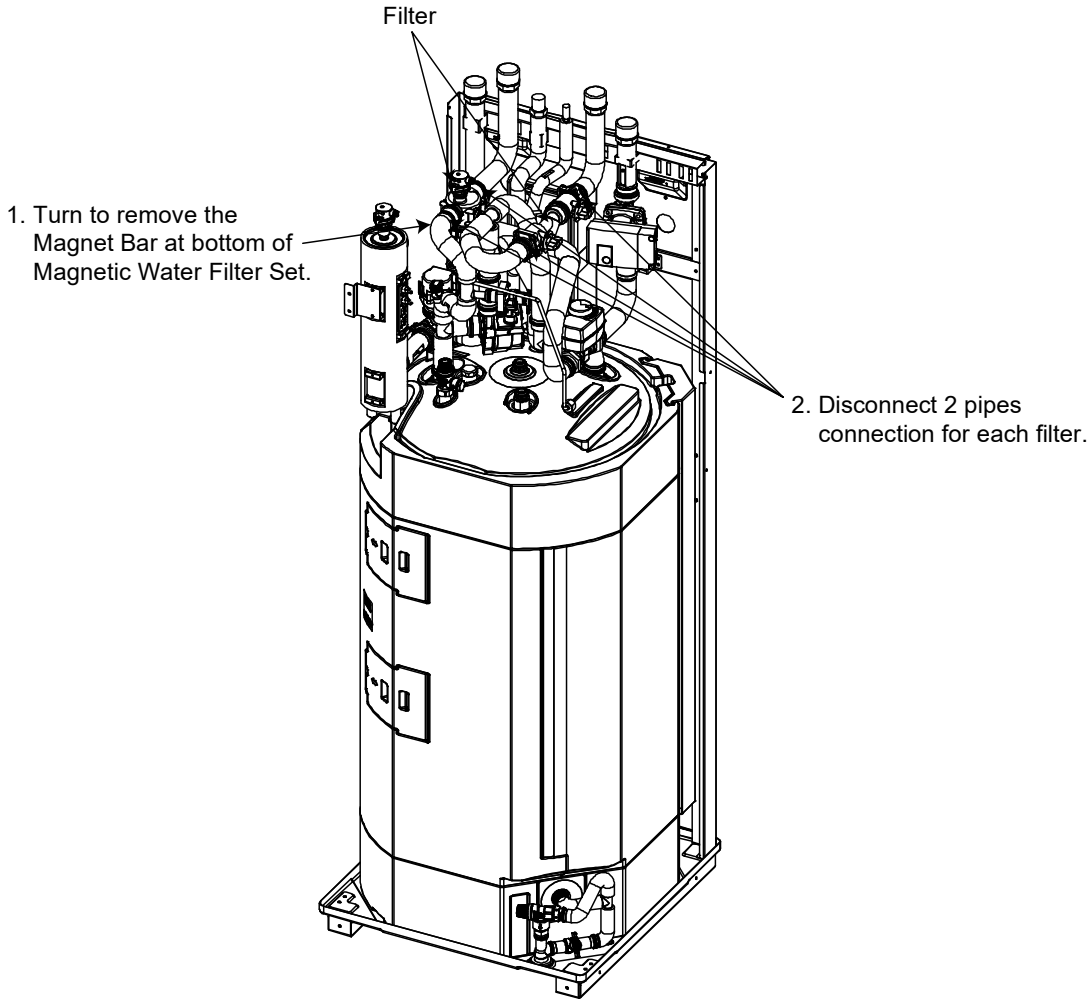




22.7 To Remove Bottle Complete



22.8 To Remove Water Filter



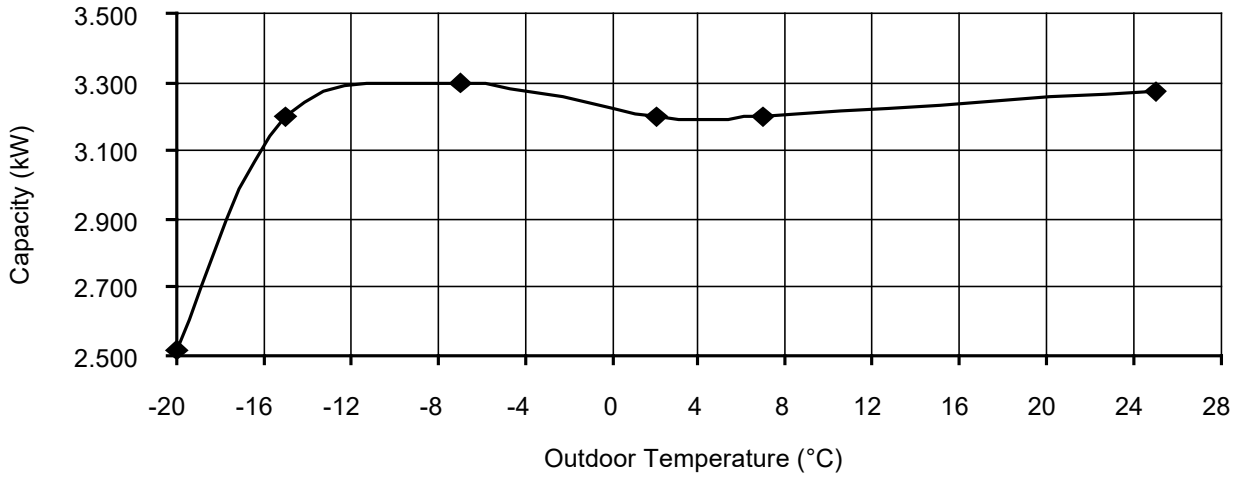
23. Technical Data

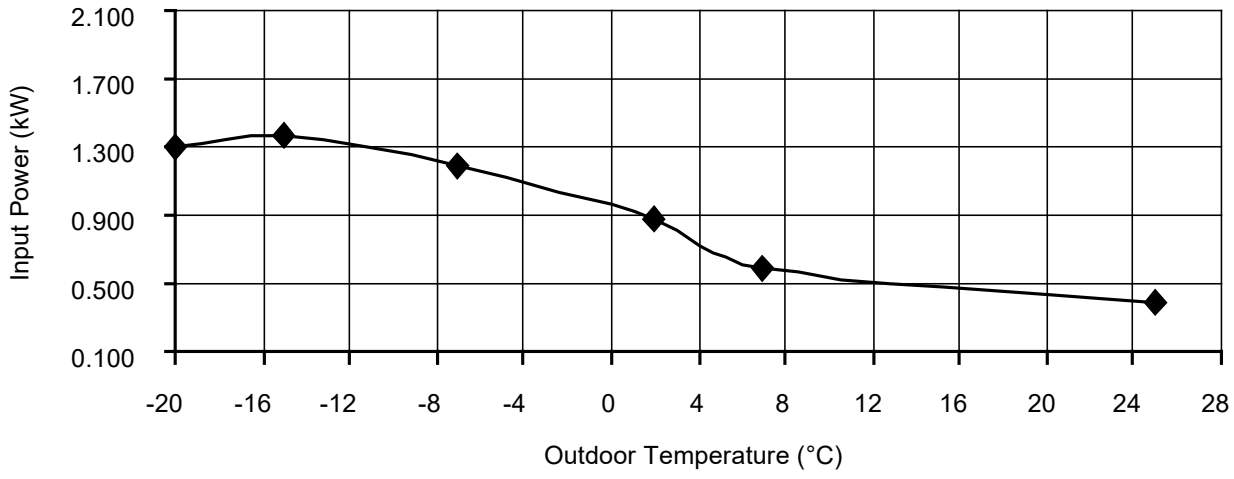
23.1 Operation Characteristics

23.1.1 WH-UDZ03KE5

Heating Characteristics at Different Outdoor Air Temperature

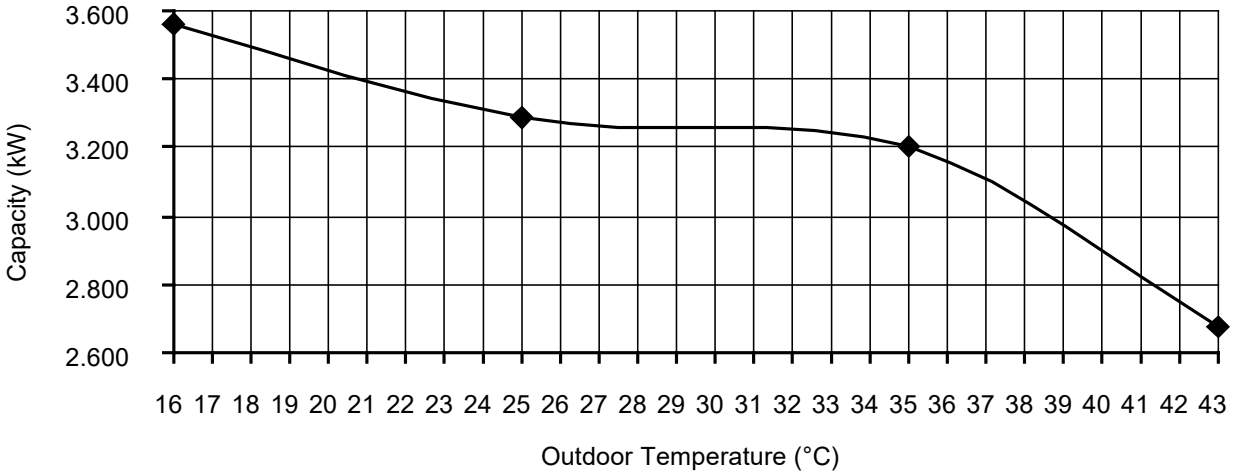
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

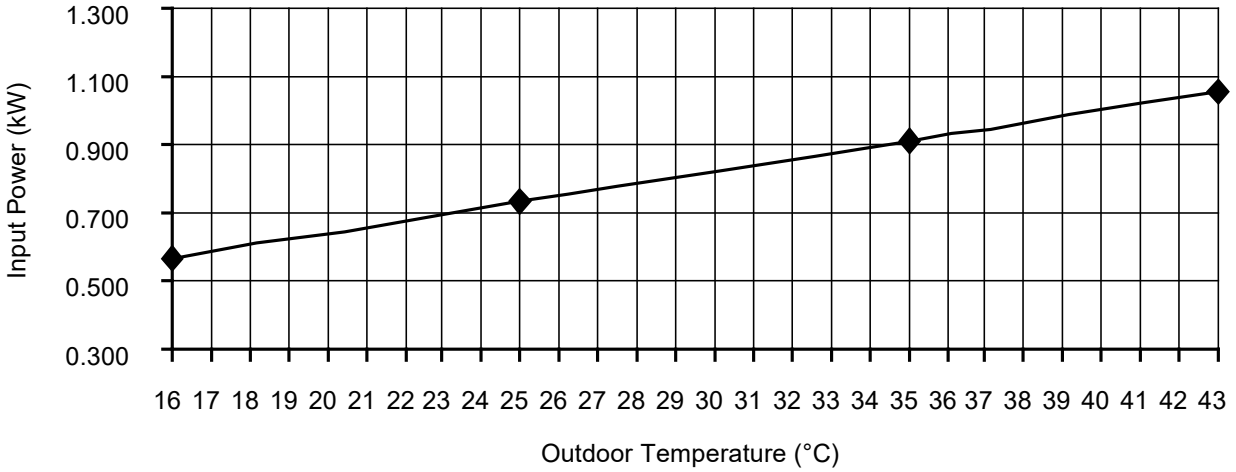




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

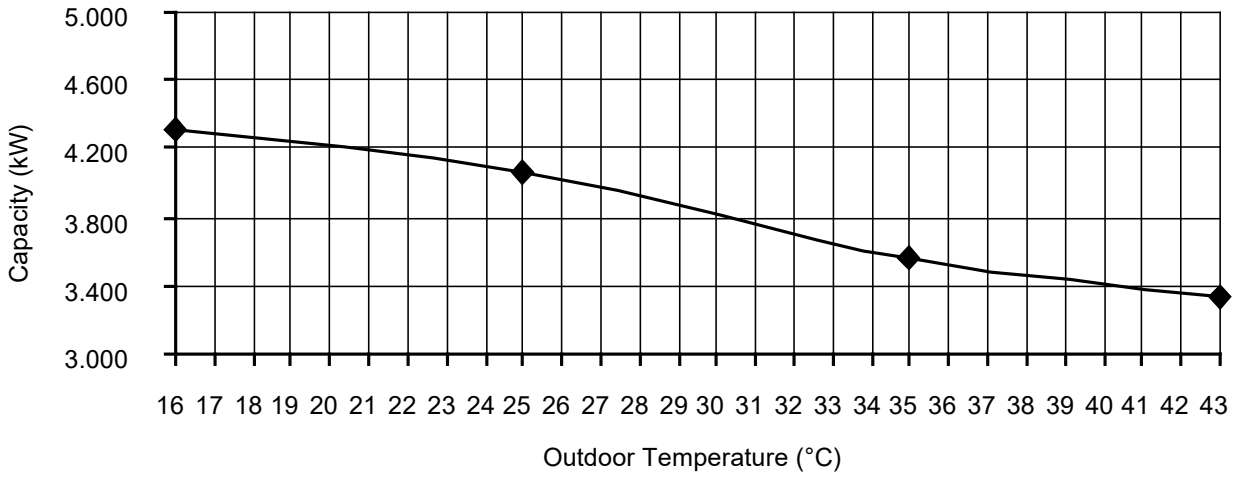
Condition

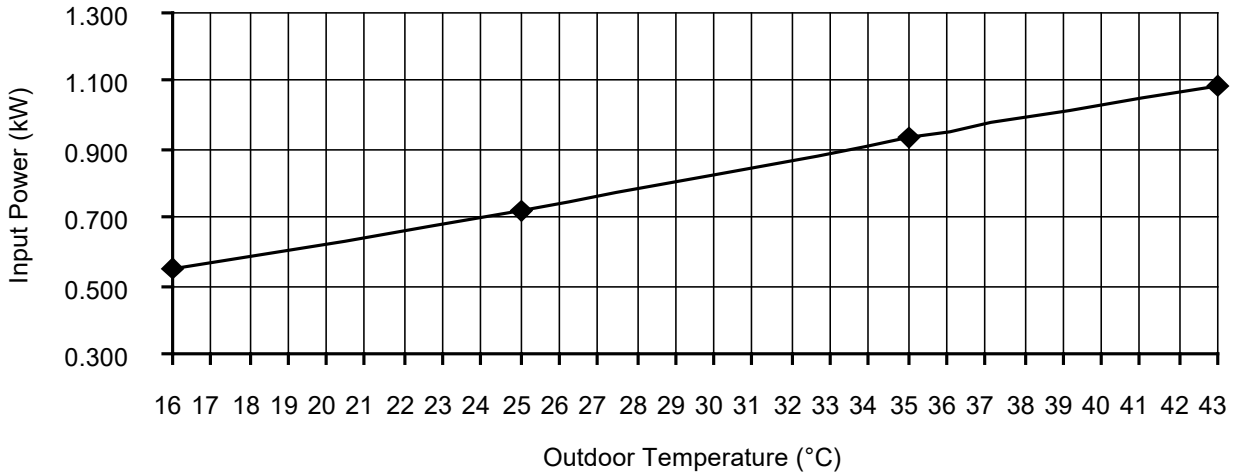
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

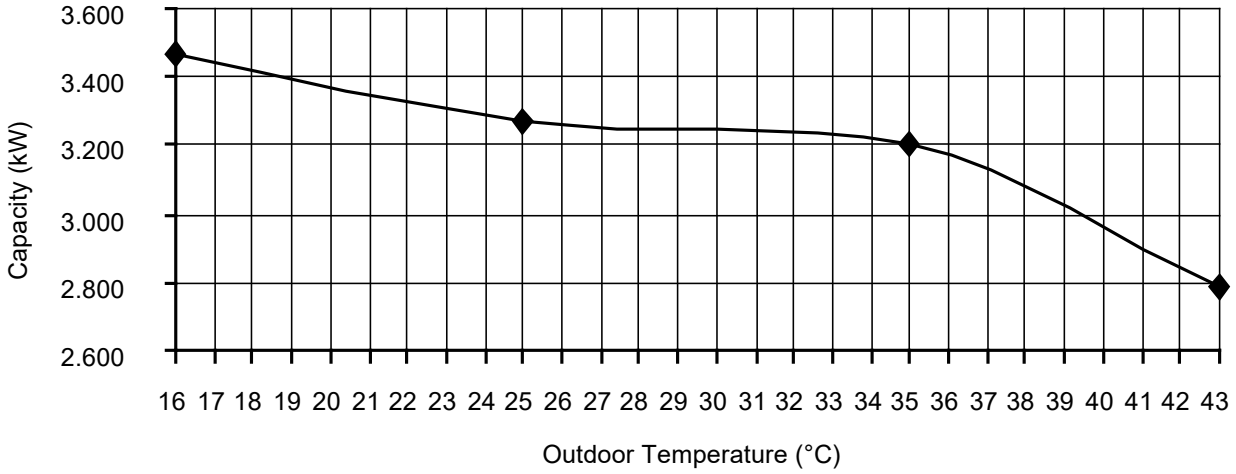
Piping length : 7 m

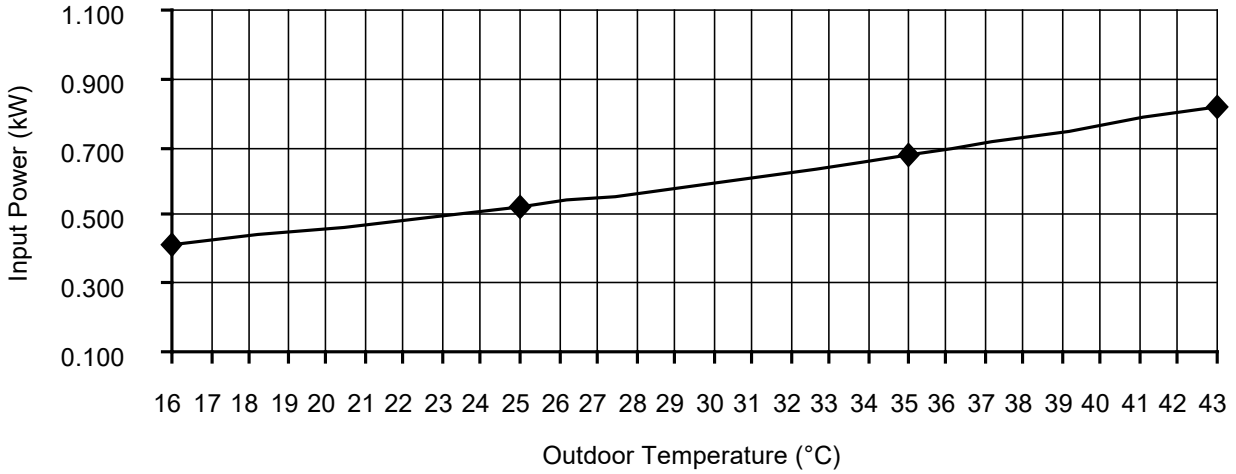




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 23°C
Indoor water outlet temperature : 18°C
Piping length : 7 m





Heating Characteristics at Different Piping Length

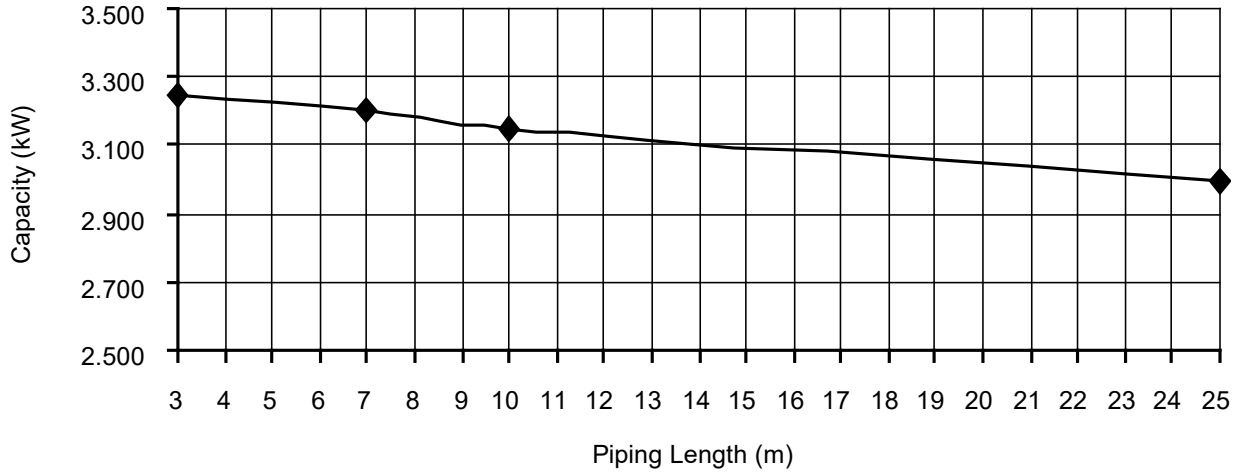
Condition

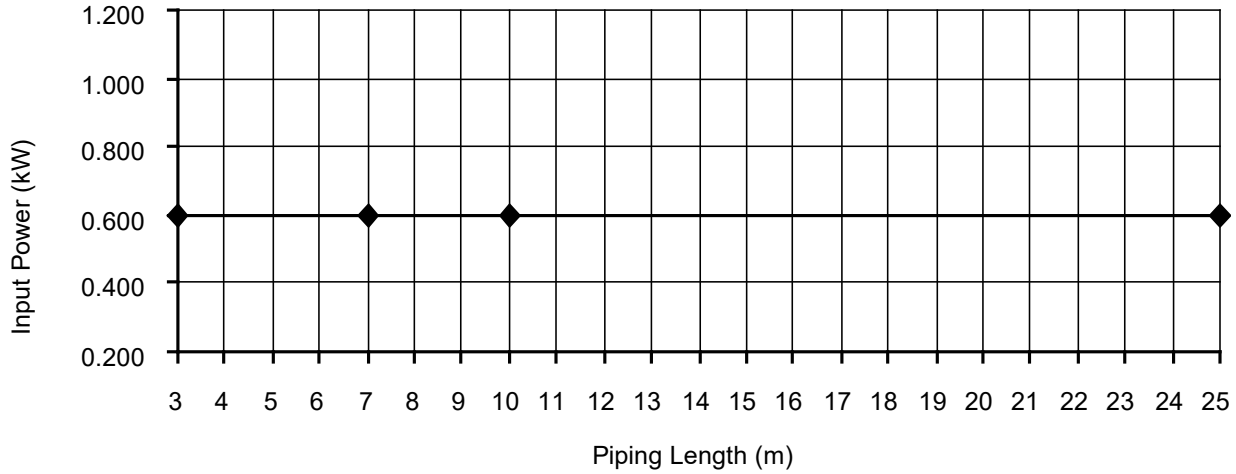
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 7 m





Cooling Characteristics at Different Piping Length

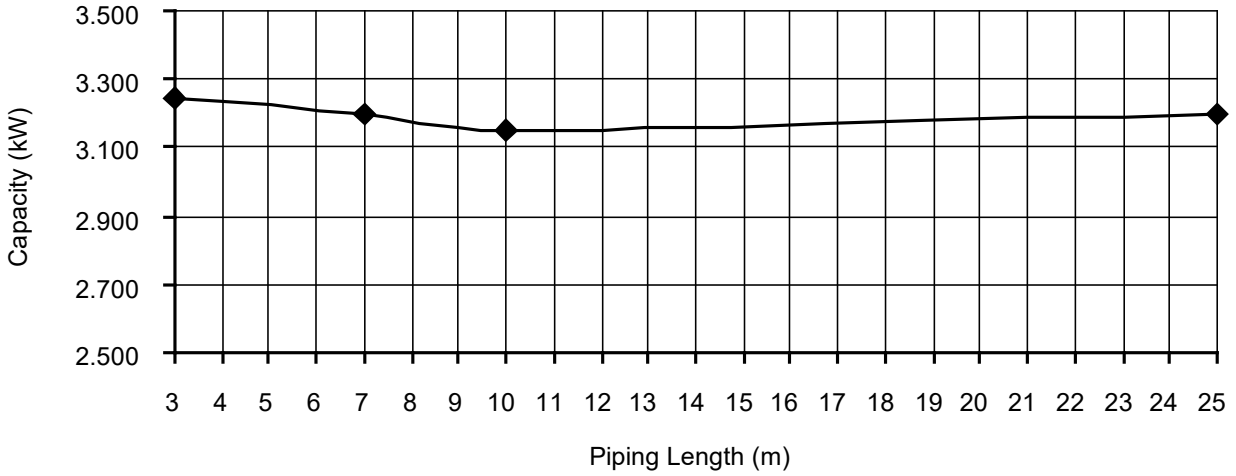
Condition

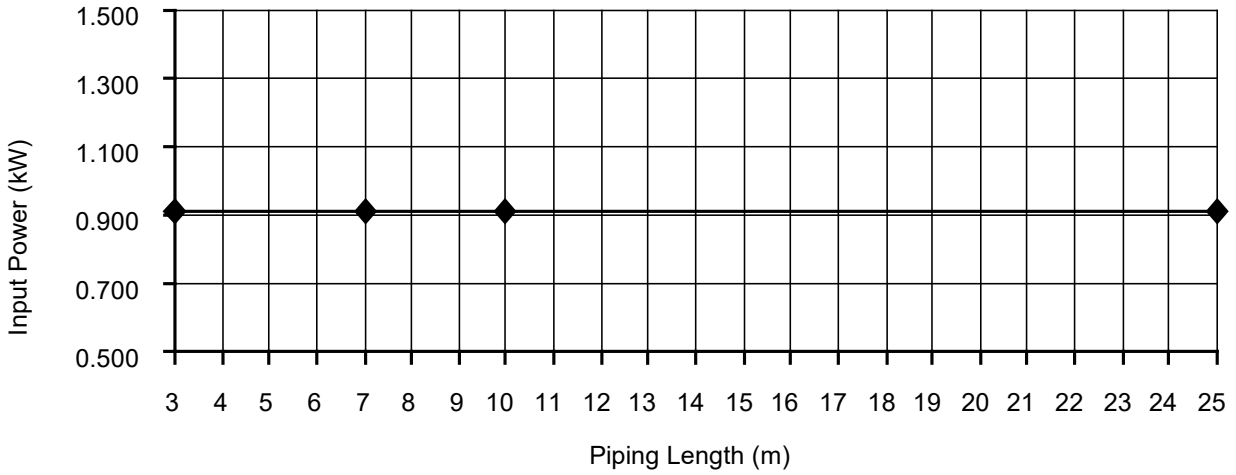
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 7 m

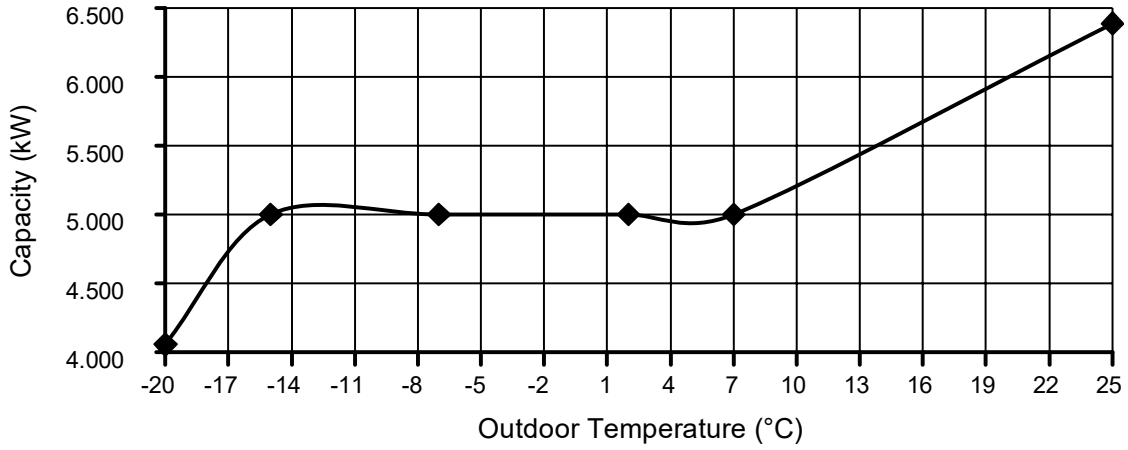


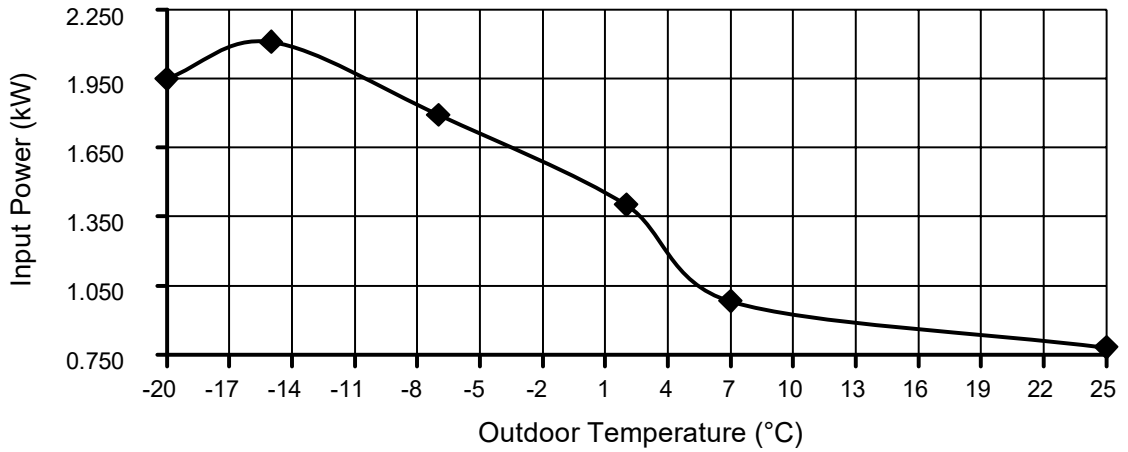


23.1.2 WH-UDZ05KE5

Heating Characteristics at Different Outdoor Air Temperature

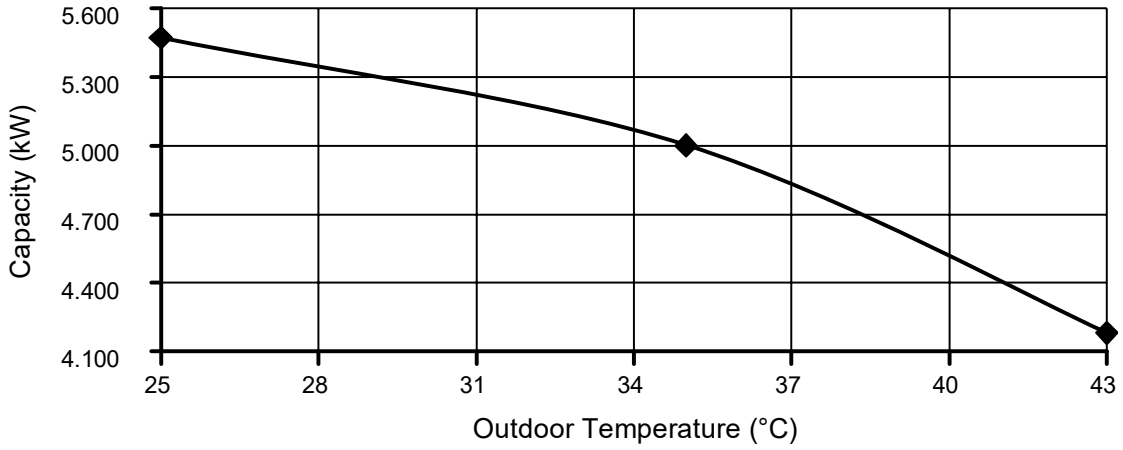
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

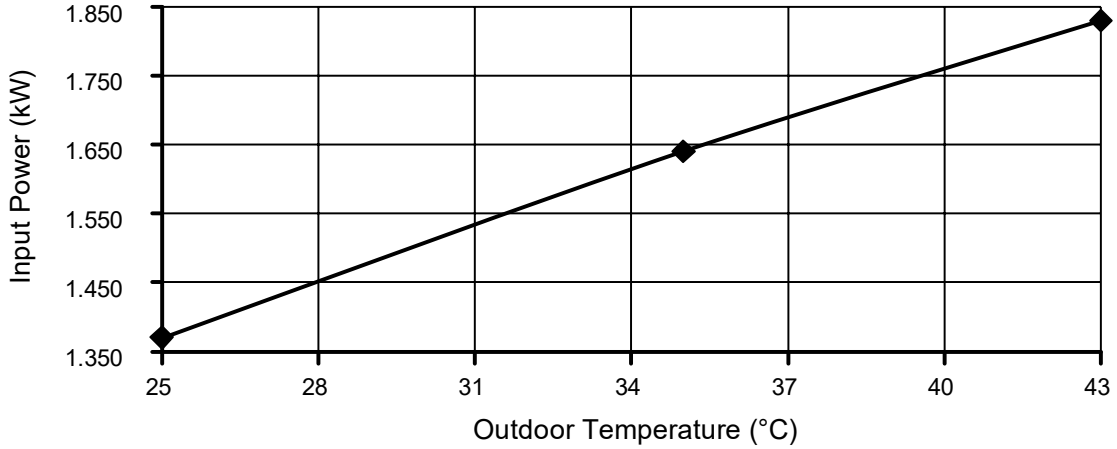




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

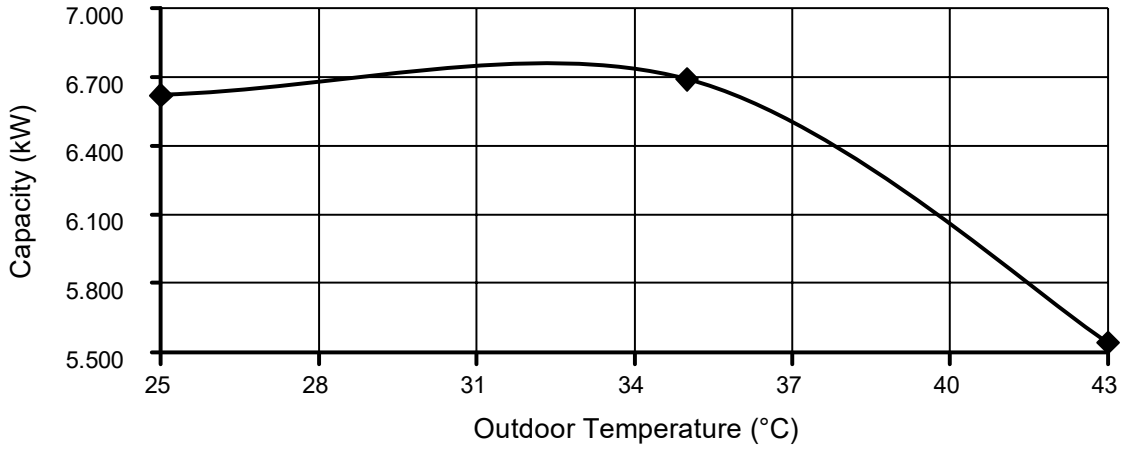
Condition

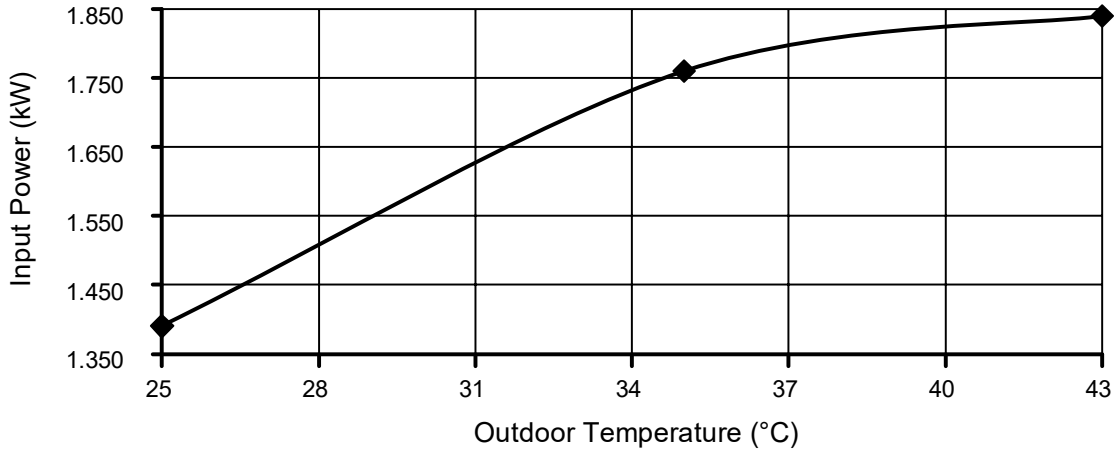
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

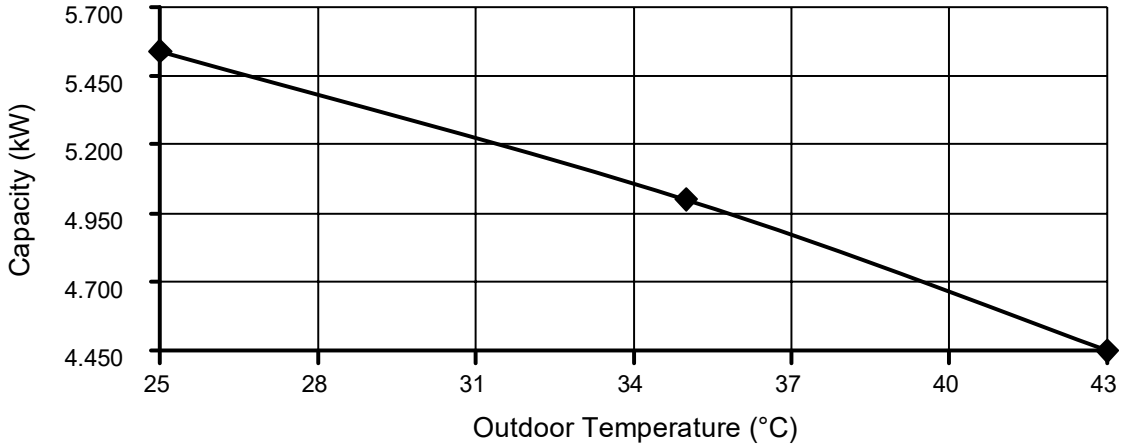
Condition

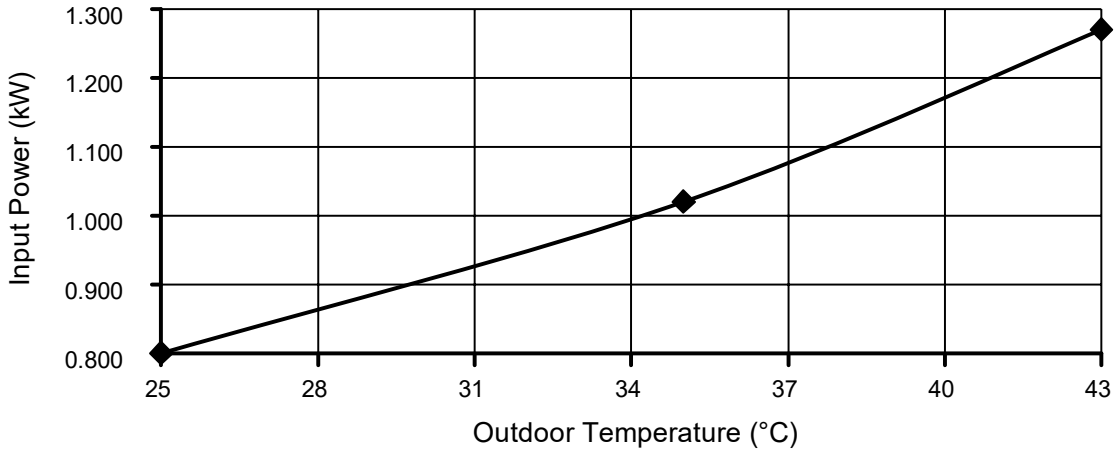
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m





Heating Characteristics at Different Piping Length

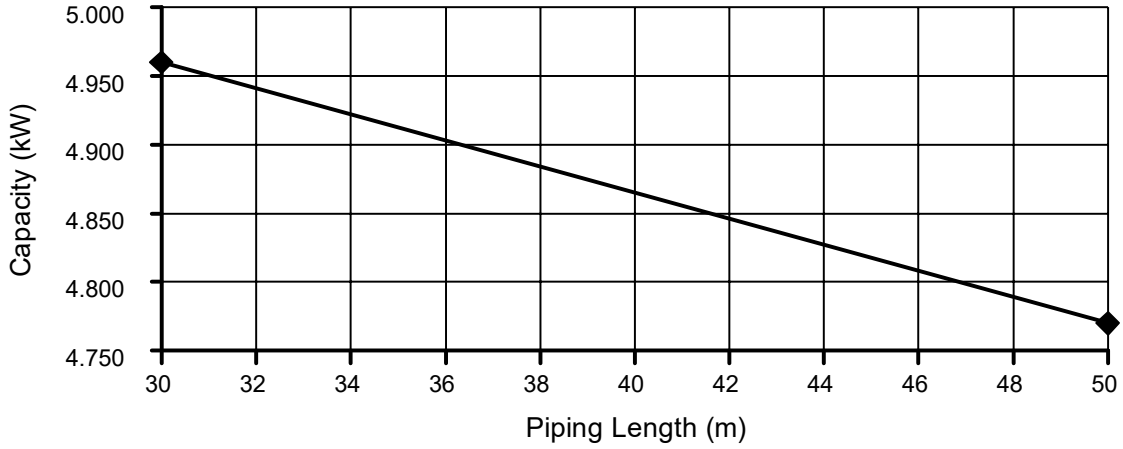
Condition

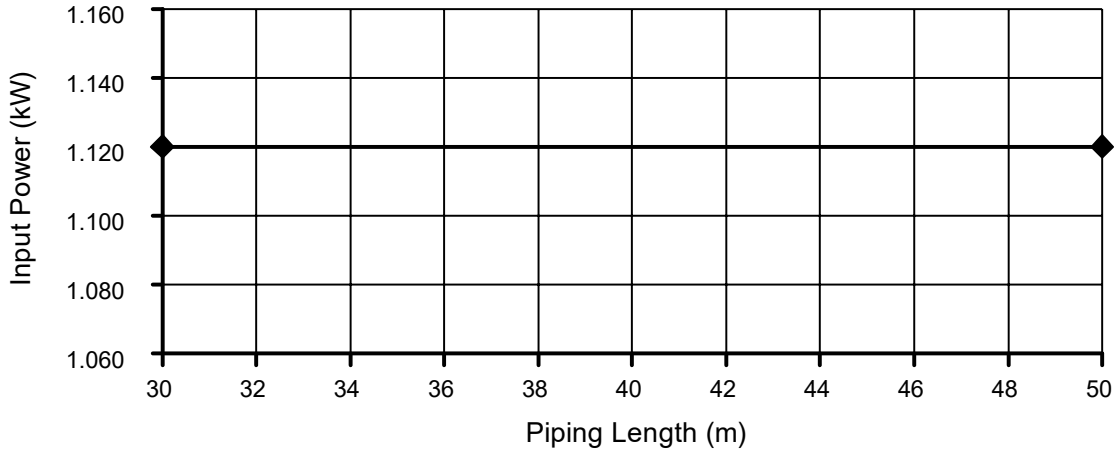
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

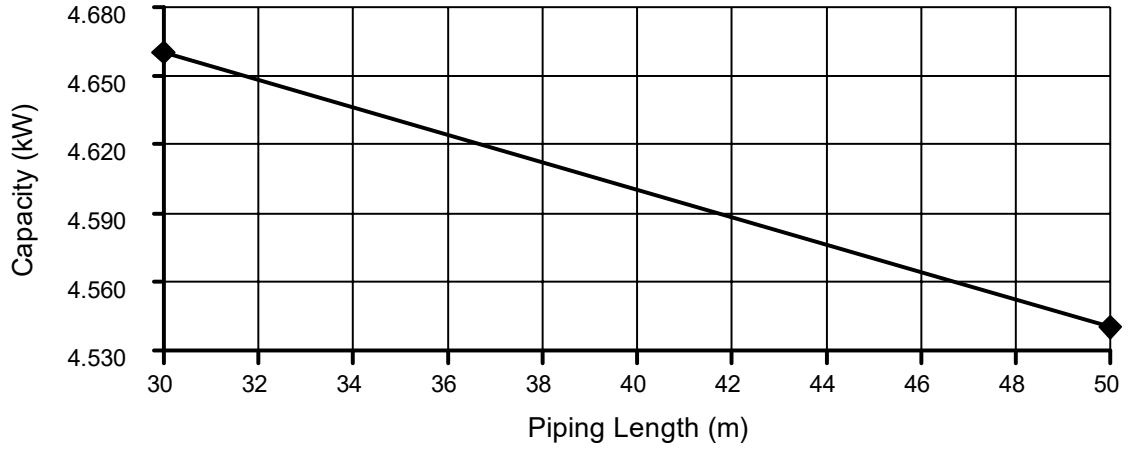
Condition

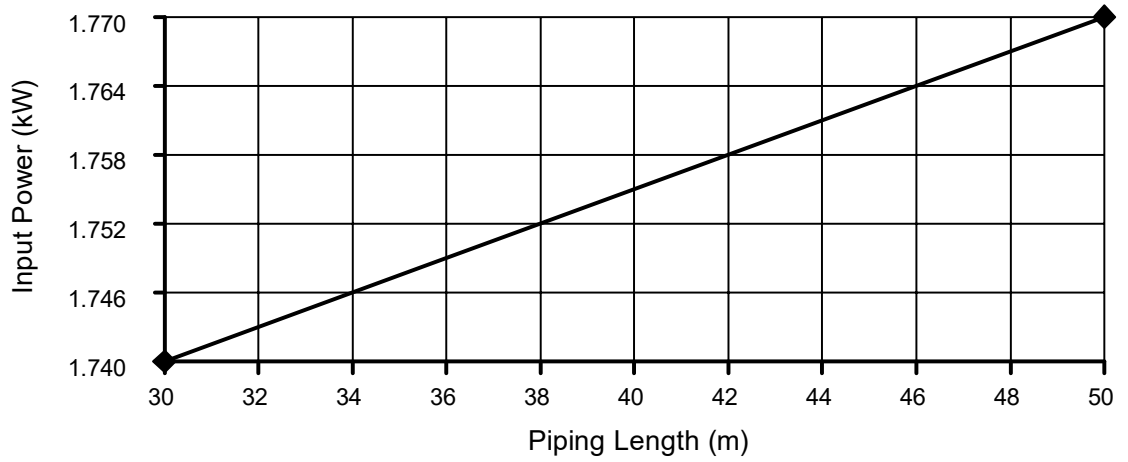
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m

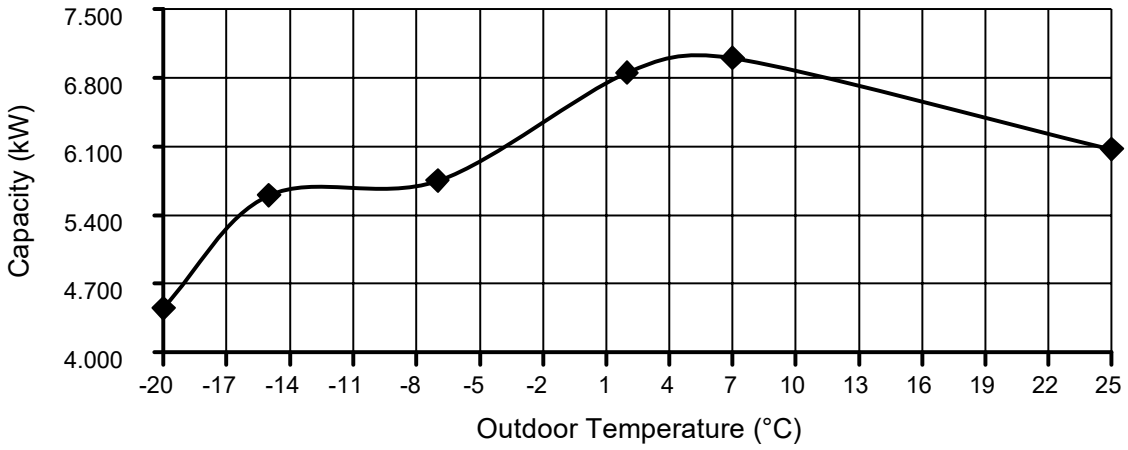


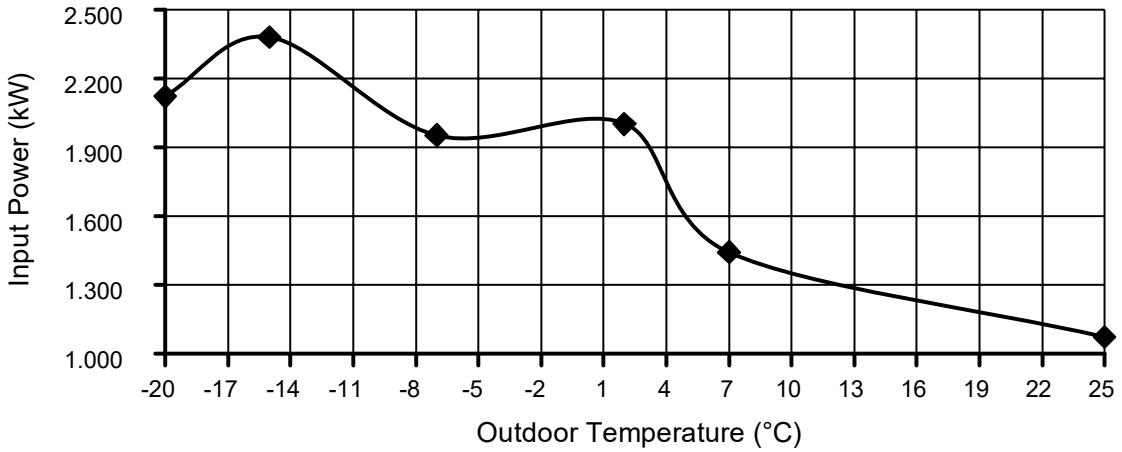


23.1.3 WH-UDZ07KE5

Heating Characteristics at Different Outdoor Air Temperature

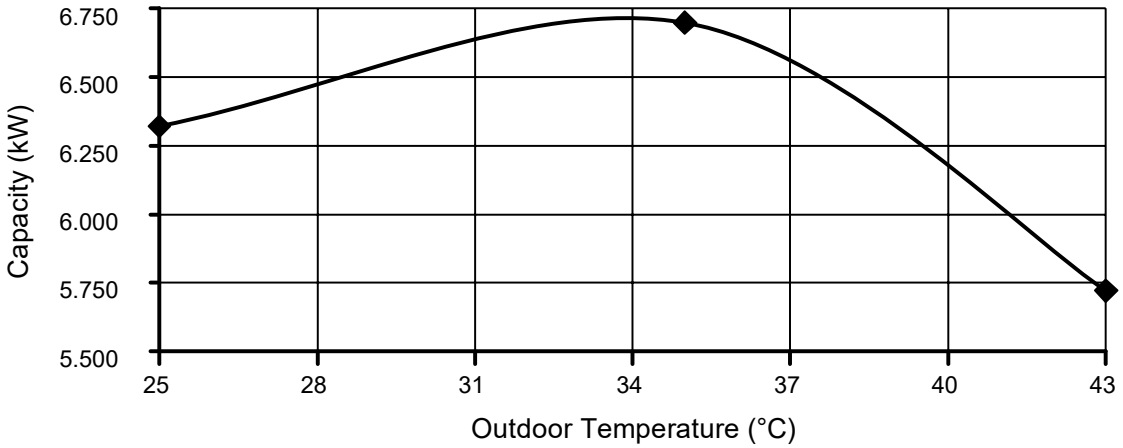
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

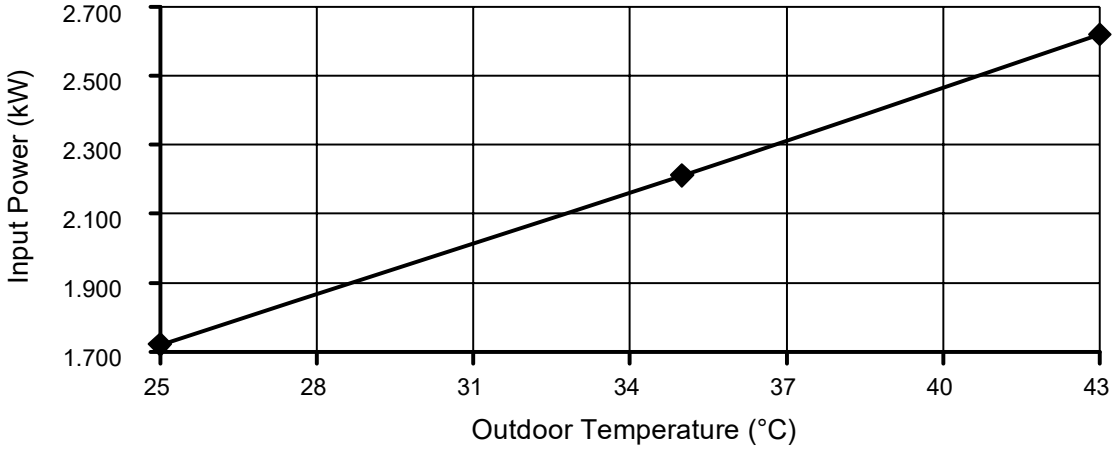




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

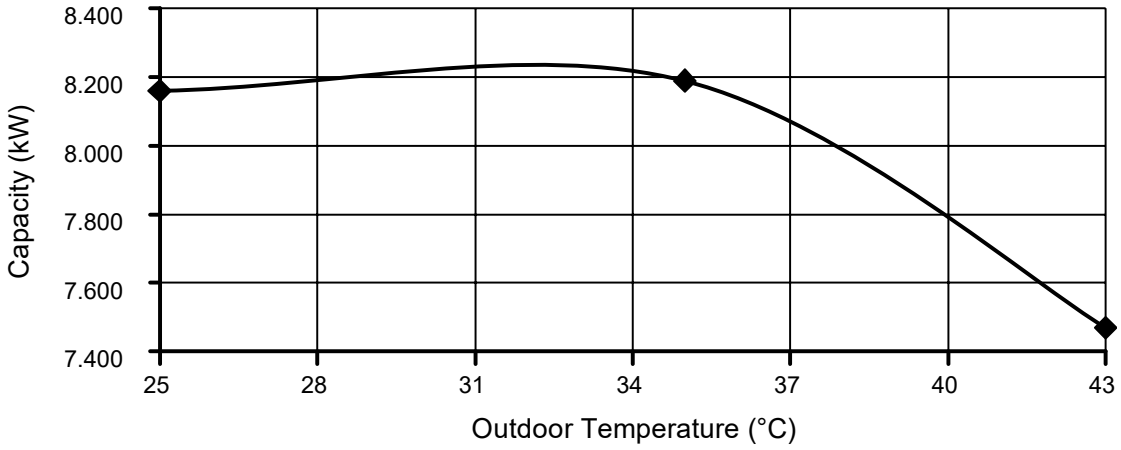
Condition

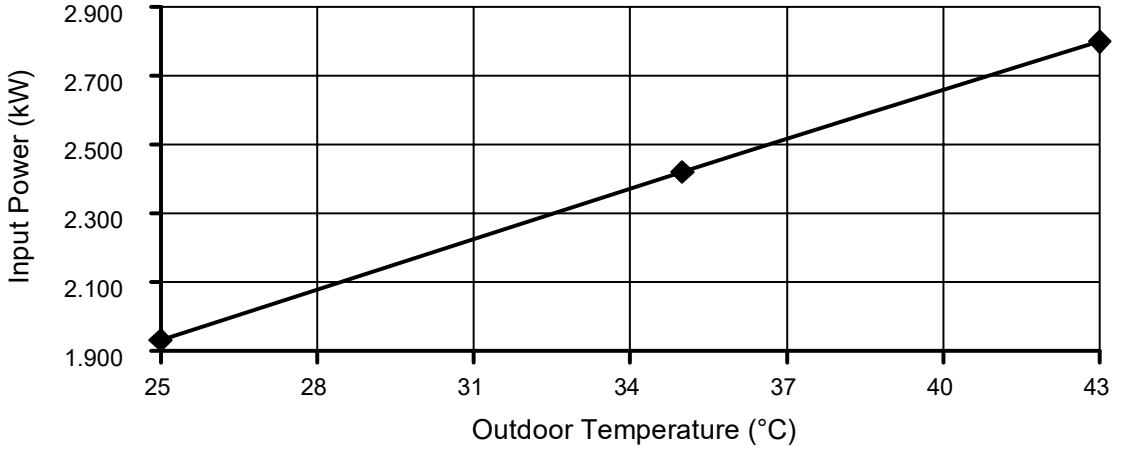
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

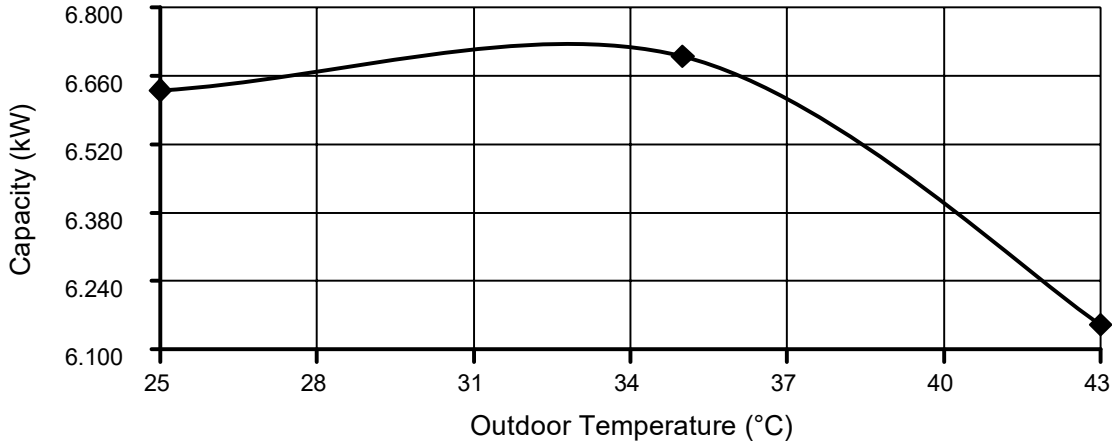
Condition

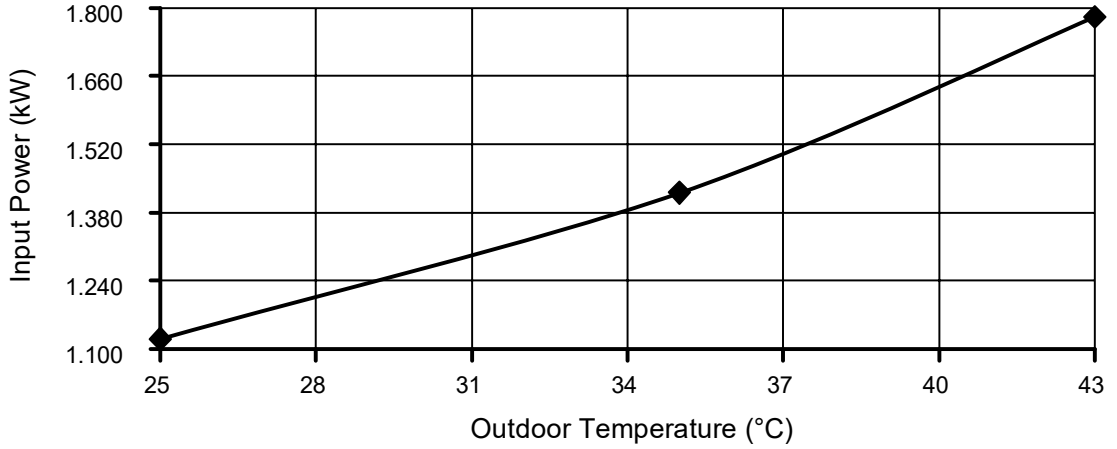
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m





Heating Characteristics at Different Piping Length

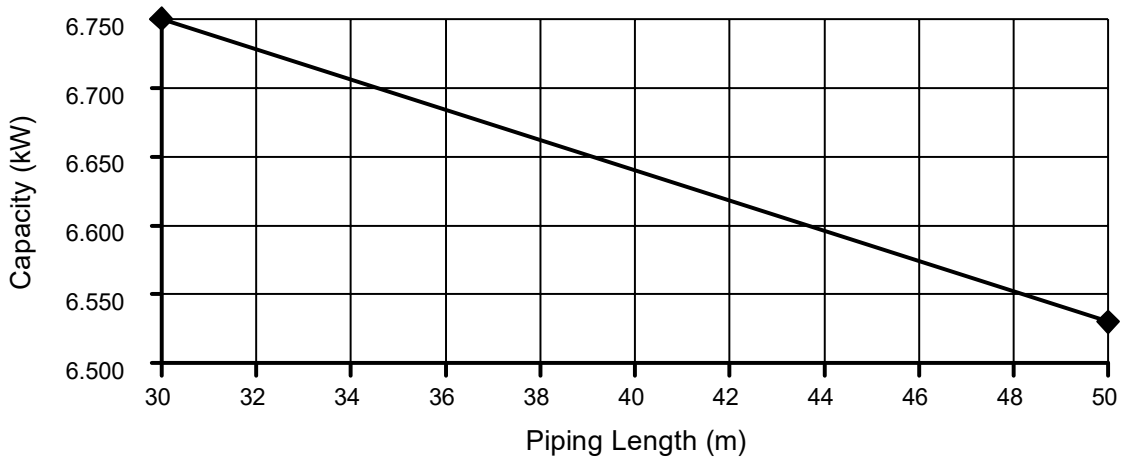
Condition

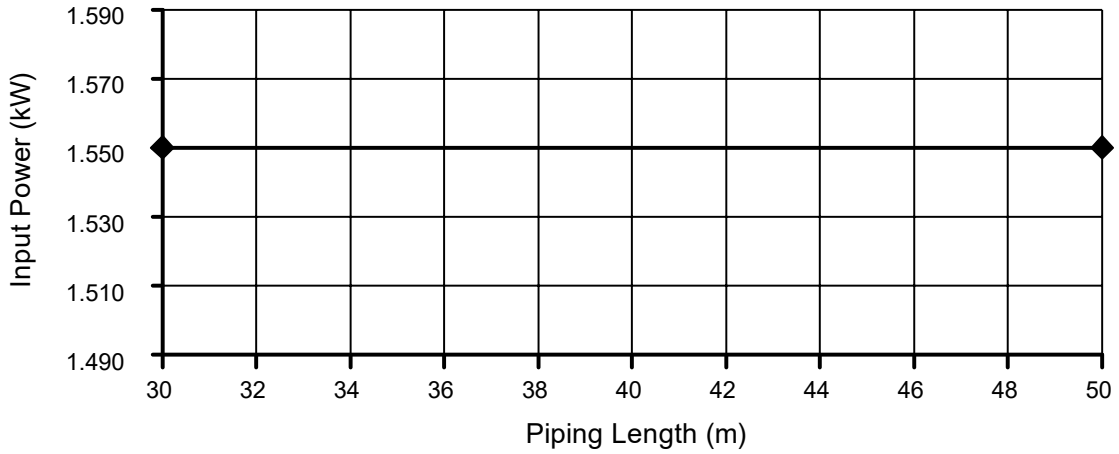
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

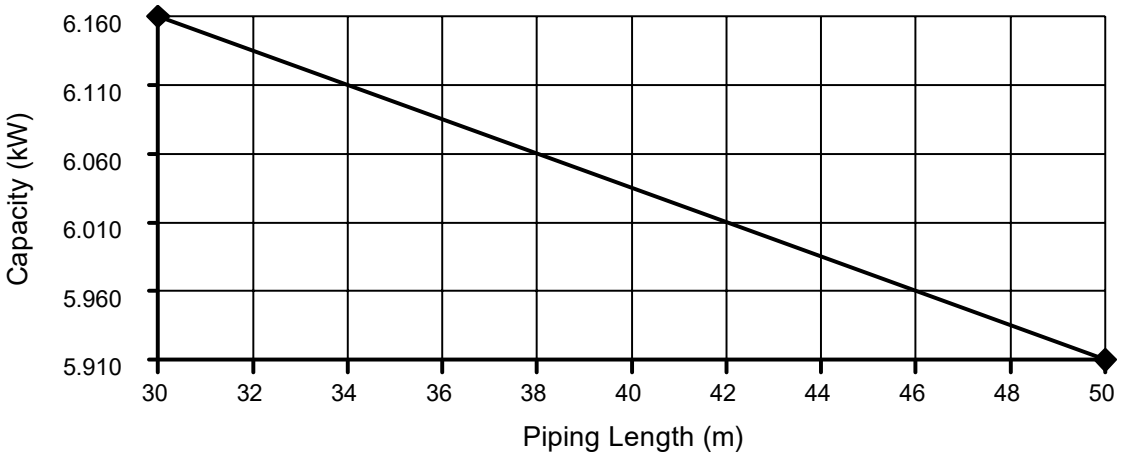
Condition

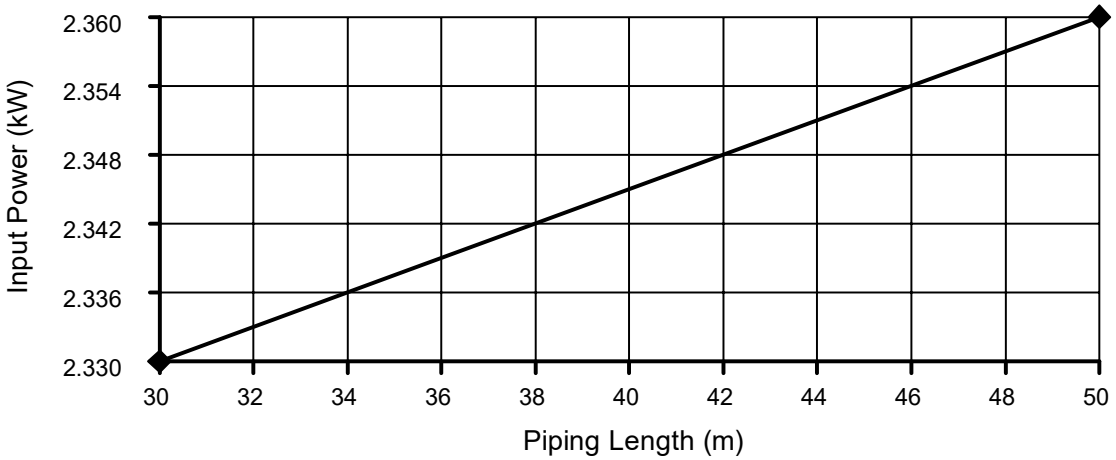
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m

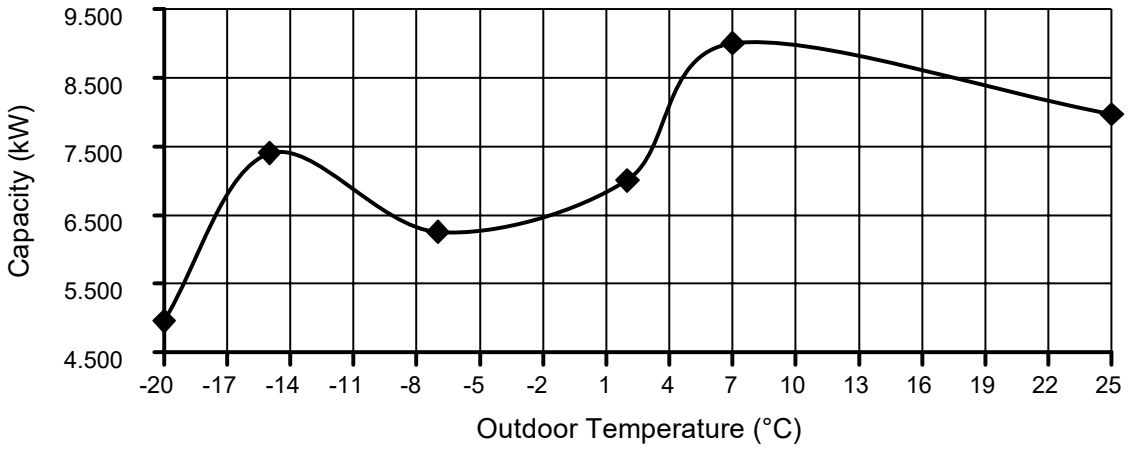


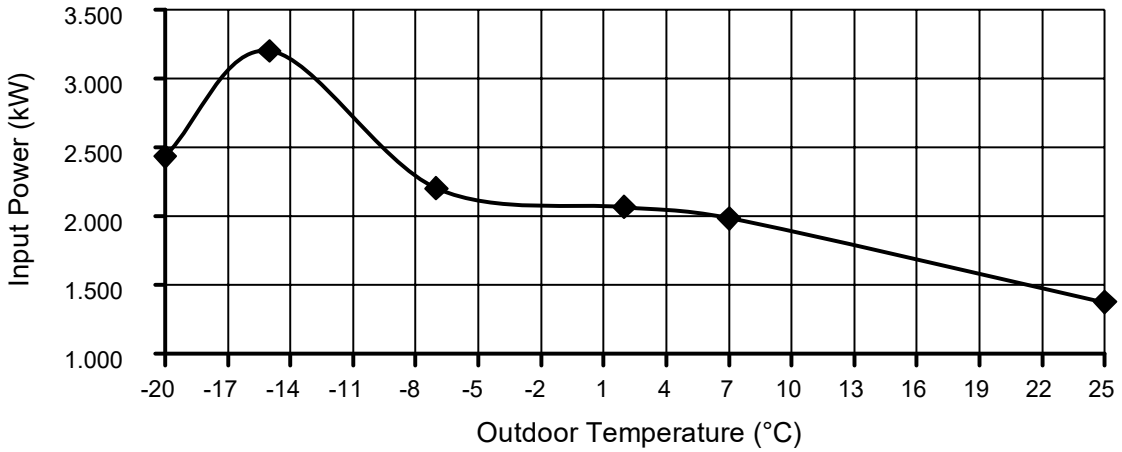


23.1.4 WH-UDZ09KE5

Heating Characteristics at Different Outdoor Air Temperature

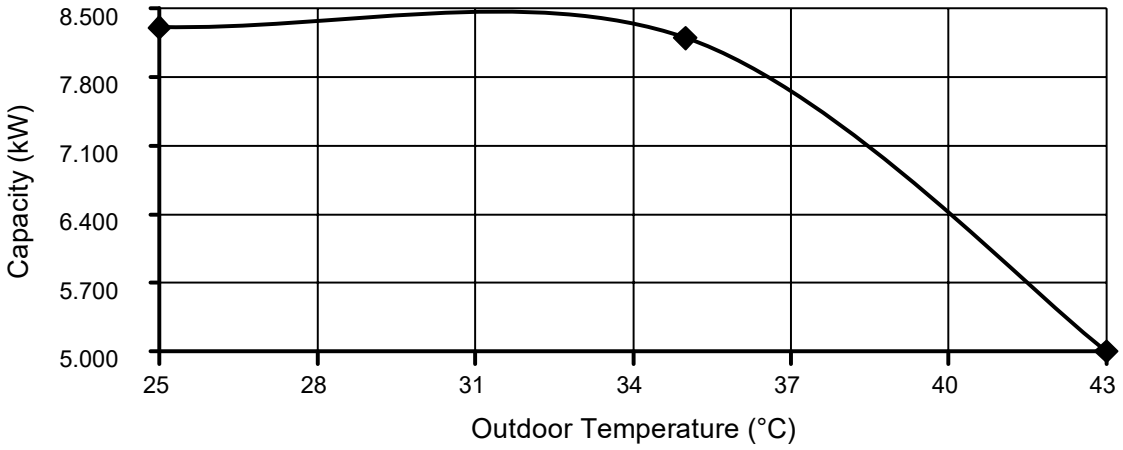
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

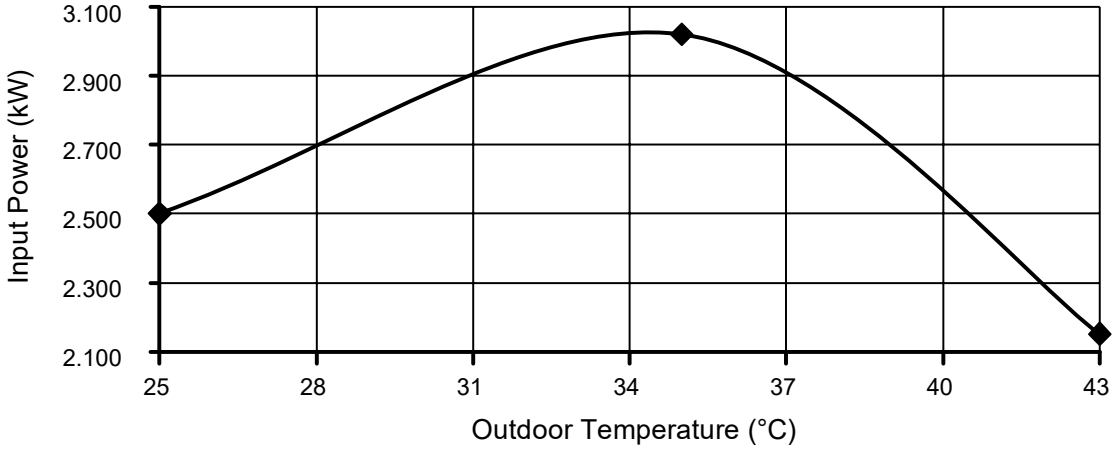




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

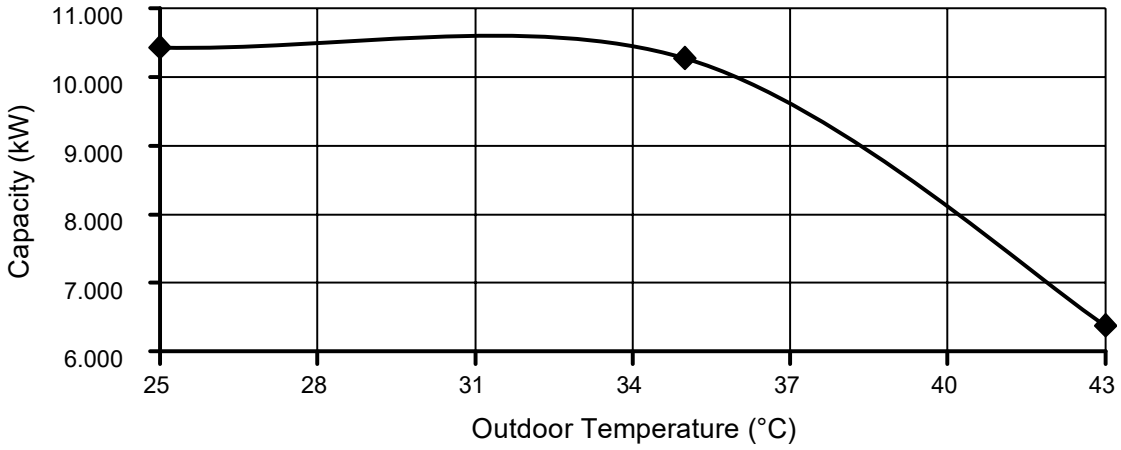
Condition

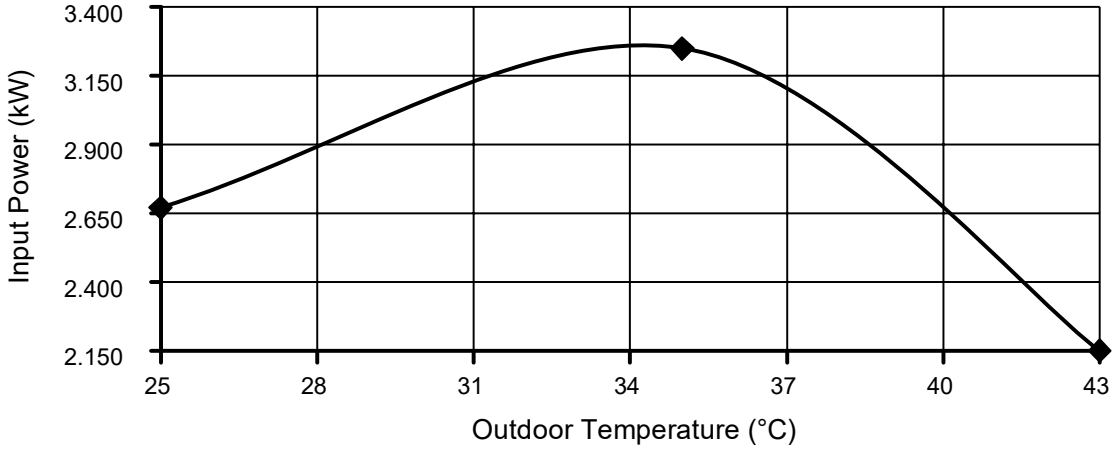
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

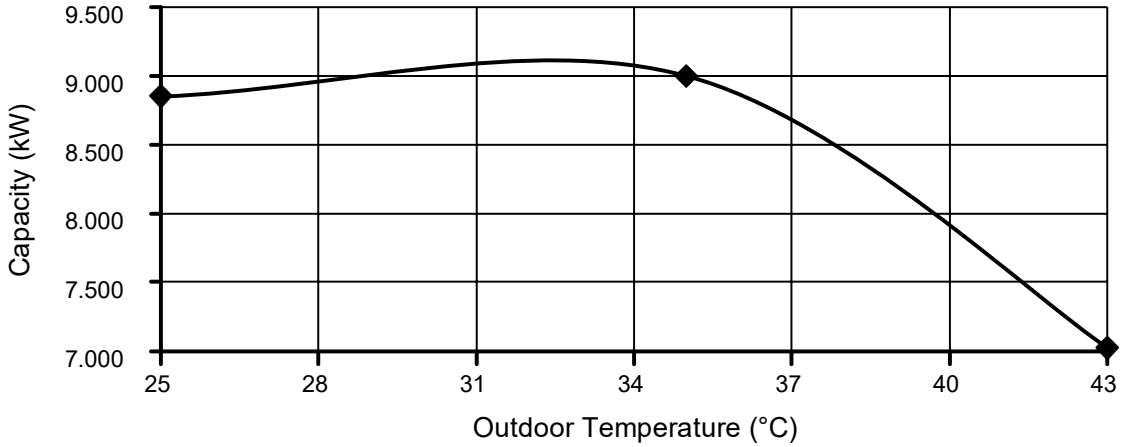
Piping length : 7 m

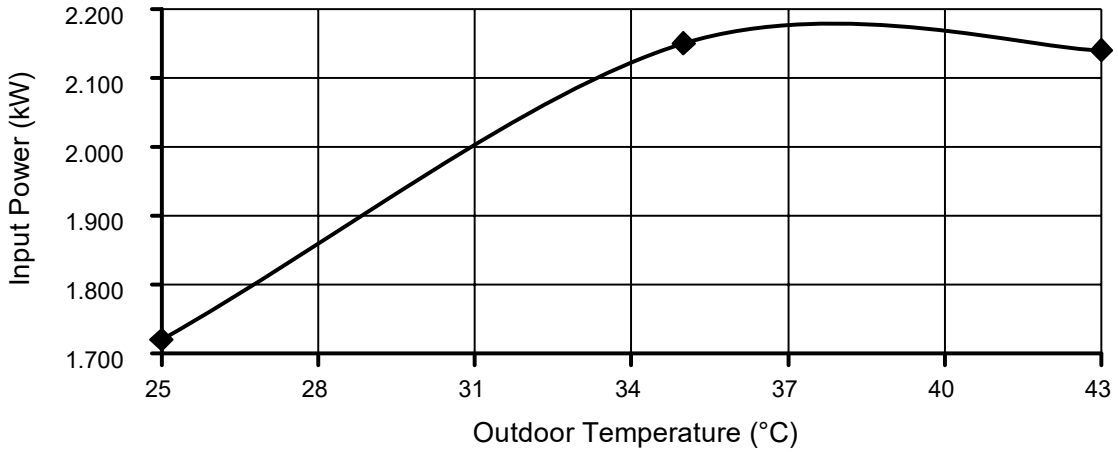




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 23°C
Indoor water outlet temperature : 18°C
Piping length : 7 m





Heating Characteristics at Different Piping Length

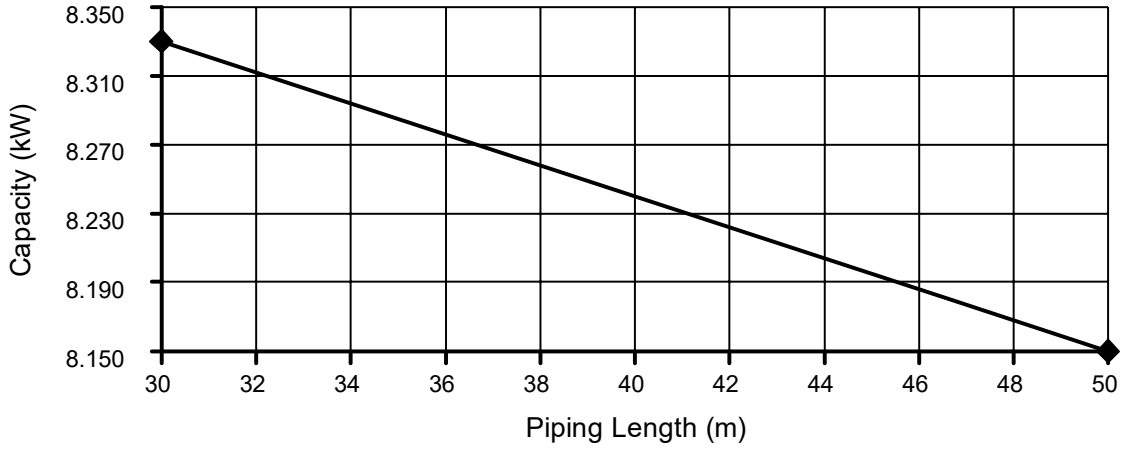
Condition

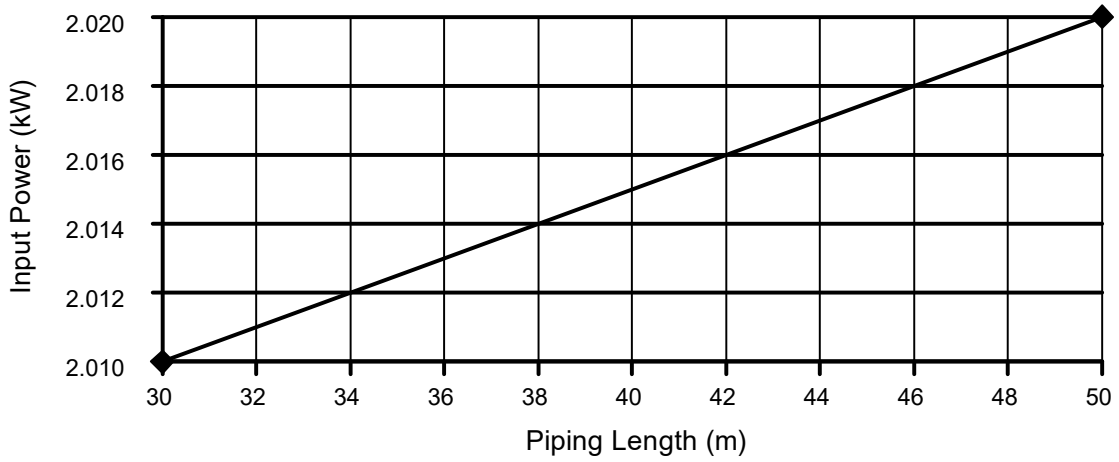
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

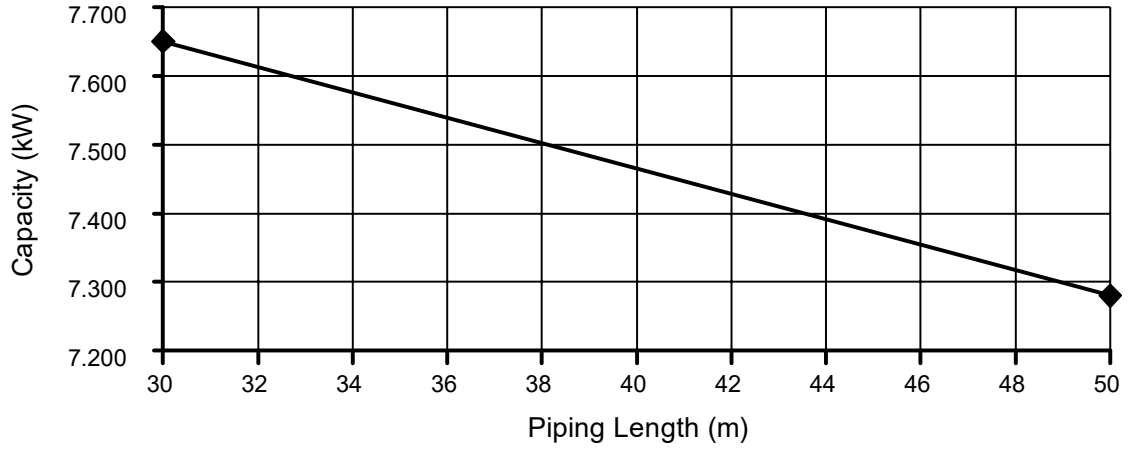
Condition

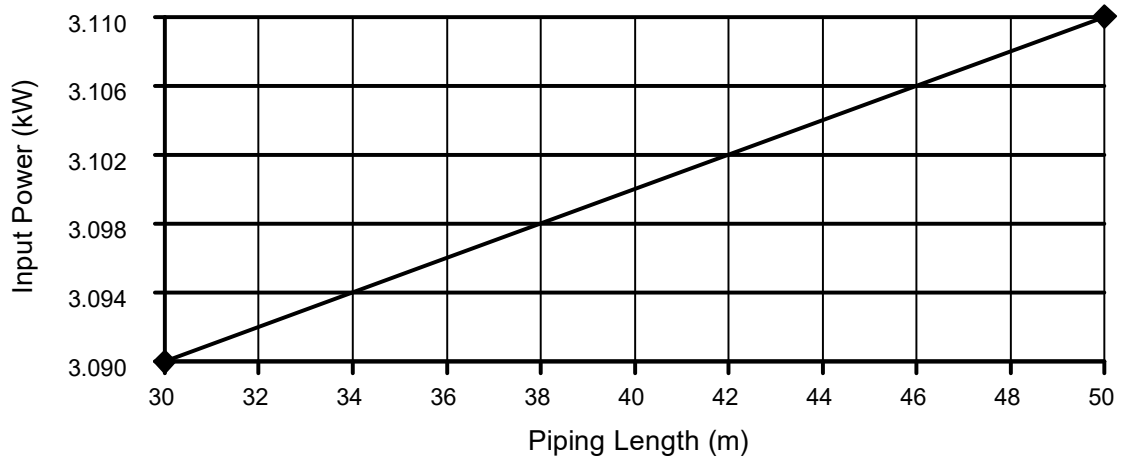
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m





23.2 Heating Capacity Table

23.2.1 WH-UDZ03KE5

Water Out (°C)	25		35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	2500	1110	2520	1310	2240	1590	2120	1800	-	-
-15	3000	1140	3200	1370	3000	1620	2750	1920	-	-
-7	2990	910	3300	1180	3250	1470	3200	1790	3000	1880
2	2920	690	3200	880	3200	1130	3200	1460	3150	1670
7	3090	490	3200	600	3200	840	3200	1140	2950	1220
25	3270	230	3270	380	3610	630	4060	1110	4030	1140

23.2.2 WH-UDZ05KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4050	1950	3760	2200	3390	2480	-	-
-15	5000	2110	4750	2490	4300	2610	-	-
-7	5000	1790	5000	2140	5000	2650	4680	2710
2	5000	1400	5000	1790	5000	2180	4800	2400
7	5000	980	5000	1310	5000	1650	4580	1900
25	6390	780	6170	1190	6530	1580	5150	1330

23.2.3 WH-UDZ07KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4450	2120	4230	2480	3900	2850	-	-
-15	5600	2380	5300	2780	5000	3200	-	-
-7	5750	1950	5650	2300	5350	2700	4980	2900
2	6850	2000	6750	2400	6250	2800	6180	2910
7	7000	1440	7000	1920	7000	2400	6860	2730
25	6070	1070	6670	1500	7150	1770	6880	1900

23.2.4 WH-UDZ09KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4950	2430	4580	2700	4040	3000	-	-
-15	7400	3200	6450	3280	5400	3420	-	-
-7	6250	2200	6100	2680	5900	3060	5650	3240
2	7000	2060	6850	2500	6300	2890	7260	3310
7	9000	1980	9000	2580	8900	3040	8600	3420
25	7970	1370	8750	1930	9160	2340	6080	1720

23.3 Cooling Capacity Table

23.3.1 WH-UDZ03KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
16	3560	570	4320	550	3470	410
25	3290	730	4060	720	3270	520
35	3200	910	3560	930	3200	680
43	2680	1060	3340	1090	2790	820

23.3.2 WH-UDZ05KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	5470	1370	6620	1390	5540	800
35	5000	1640	6690	1760	5000	1020
43	4180	1830	5540	1840	4450	1270

23.3.3 WH-UDZ07KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	6320	1720	8160	1930	6630	1120
35	6700	2210	8190	2420	6700	1420
43	5720	2620	7470	2800	6150	1780

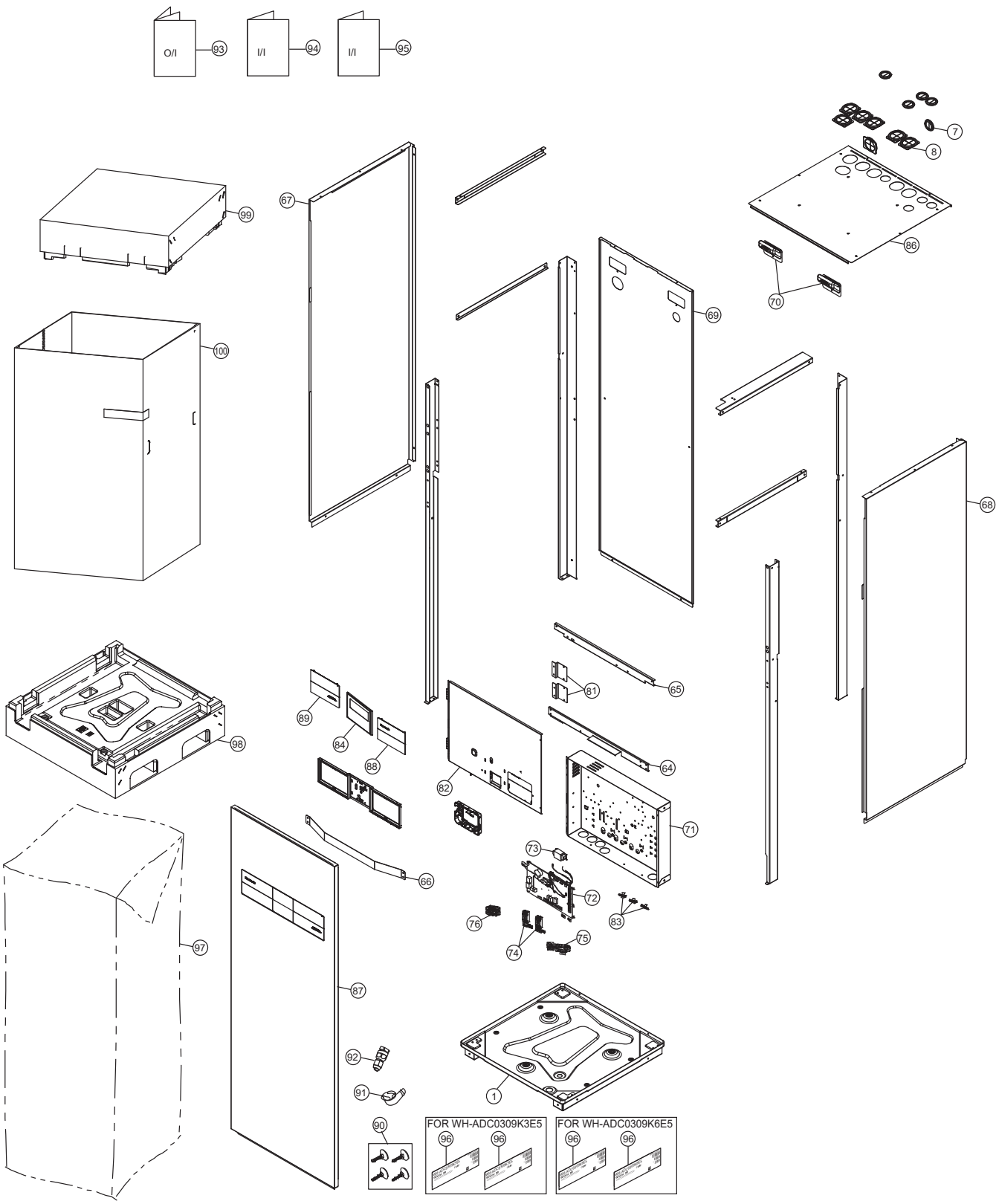
23.3.4 WH-UDZ09KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	8310	2500	10430	2670	8850	1720
35	8200	3020	10280	3250	9000	2150
43	5000	2150	6380	2150	7020	2140

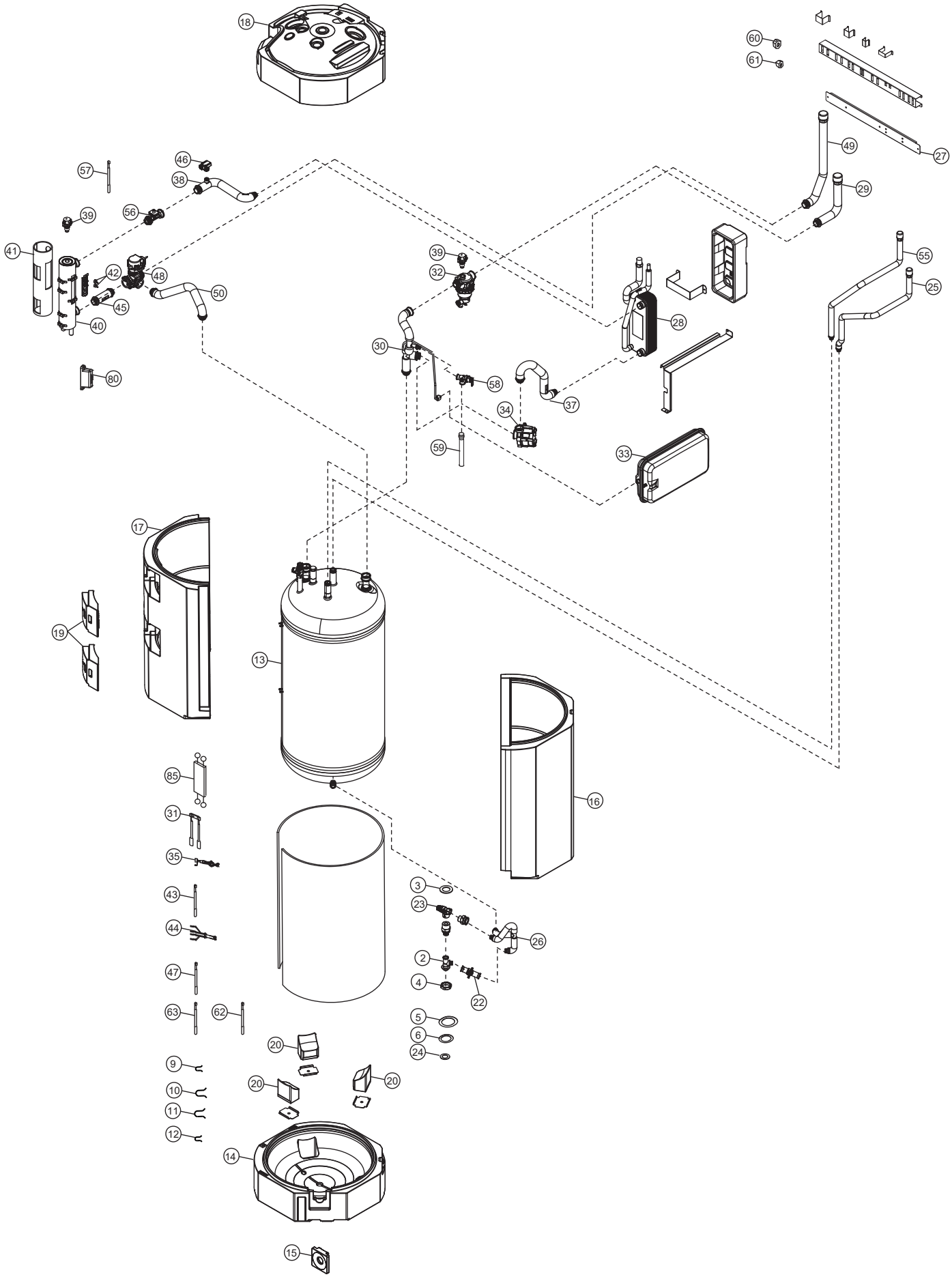
24. Exploded View and Replacement Parts List

24.1 Indoor Unit

24.1.1 WH-ADC0309K3E5 WH-ADC0309K6E5



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5	WH-ADC0309K6E5	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C04840	←	
	2	TUBE CONNECTER	1	ACXT29-02570	←	
	3	O-RING (ID: 27mm)	1	ACXB81-07070	←	
	4	NUT	1	ACXH56-01370	←	
	5	O-RING (ID: 25.70mm)	1	ACXB81-06910	←	
	6	O-RING (ID: 8.1mm)	1	CWB811179	←	
	7	O-RING (ID: 35mm)	1	ACXB81-00030	←	
	8	O-RING (ID: 54mm X 47mm)	1	ACXB81-00040	←	
	9	RETAINING RING	1	CWH581007	←	
	10	RETAINING RING	1	ACXH58-00370	←	
	11	RETAINING RING	1	CWH581012	←	
	12	RETAINING RING	1	CWH581038	←	
	13	TANK COMPLETE	1	ACXB56C01210	←	
	14	FOAMED POLYSTYRENE	1	ACXG07-07690	←	
	15	FOAMED POLYSTYRENE	1	ACXG07-07700	←	
	16	FOAMED POLYSTYRENE	1	ACXG07-07710	←	
	17	FOAMED POLYSTYRENE	1	ACXG07-07720	←	
	18	FOAMED POLYSTYRENE	1	ACXG07-07731	←	
	19	FOAMED POLYSTYRENE	2	ACXG07-07740	←	
	20	FOAMED POLYSTYRENE	3	ACXG07-07750	←	
	22	DRAIN VALVE	1	ACXB65C00111	←	
	23	PRESSURE AND TEMP RELIEF VALVE (TANK)	1	ACXB62-00111	←	
	24	O-RING (ID: 13.60mm)	1	ACXB81-00010	←	
	25	TUBE ASSY - COMPLETE	1	ACXT00C40900	←	
	26	TUBE ASSY	1	ACXT00-77551	←	
	27	CONNECTING BAR	1	ACXE26-02841	←	
	28	HOT WATER COIL - COMPLETE	1	ACXB90C01570	←	
	29	TUBE ASSY - COMPLETE	1	ACXT00C40920	←	
	30	TUBE ASSY - COMPLETE	1	ACXT00C40931	←	
⚠	31	SENSOR CO. (TANK TEMP CN-TH2)	1	ACXA50C18700	←	
⚠	32	FILTER COMPLETE	1	ACXB51C00110	←	
	33	RECEIVER	1	CWB141039	←	
⚠	34	WATER PUMP	1	ACXB53-00850	←	
⚠	35	LEAD WIRE FOR WATER PUMP	1	ACXA60C73630	←	
	37	TUBE ASSY - COMPLETE	1	ACXT00C41200	←	
	38	TUBE ASSY - COMPLETE	1	ACXT00C40961	←	
	39	AIR PURGE VALVE	2	ACXB62-00130	←	
⚠	40	HEATER ASS'Y	1	ACXA34K00460	ACXA34K00530	
	41	SOUND PROOF MATERIAL	1	ACXG30-08750	←	
⚠	42	THERMOSTAT	2/3	CWA151074	←	
⚠	43	LEAD WIRE - CO. (CN-OLP1)	1	ACXA60C93670	←	
⚠	44	LEAD WIRE - CO. (OLP & HEATER)	1	ACXA60C93680	ACXA60C84060	
	45	TUBE ASSY - COMPLETE	1	ACXT00C32732	←	

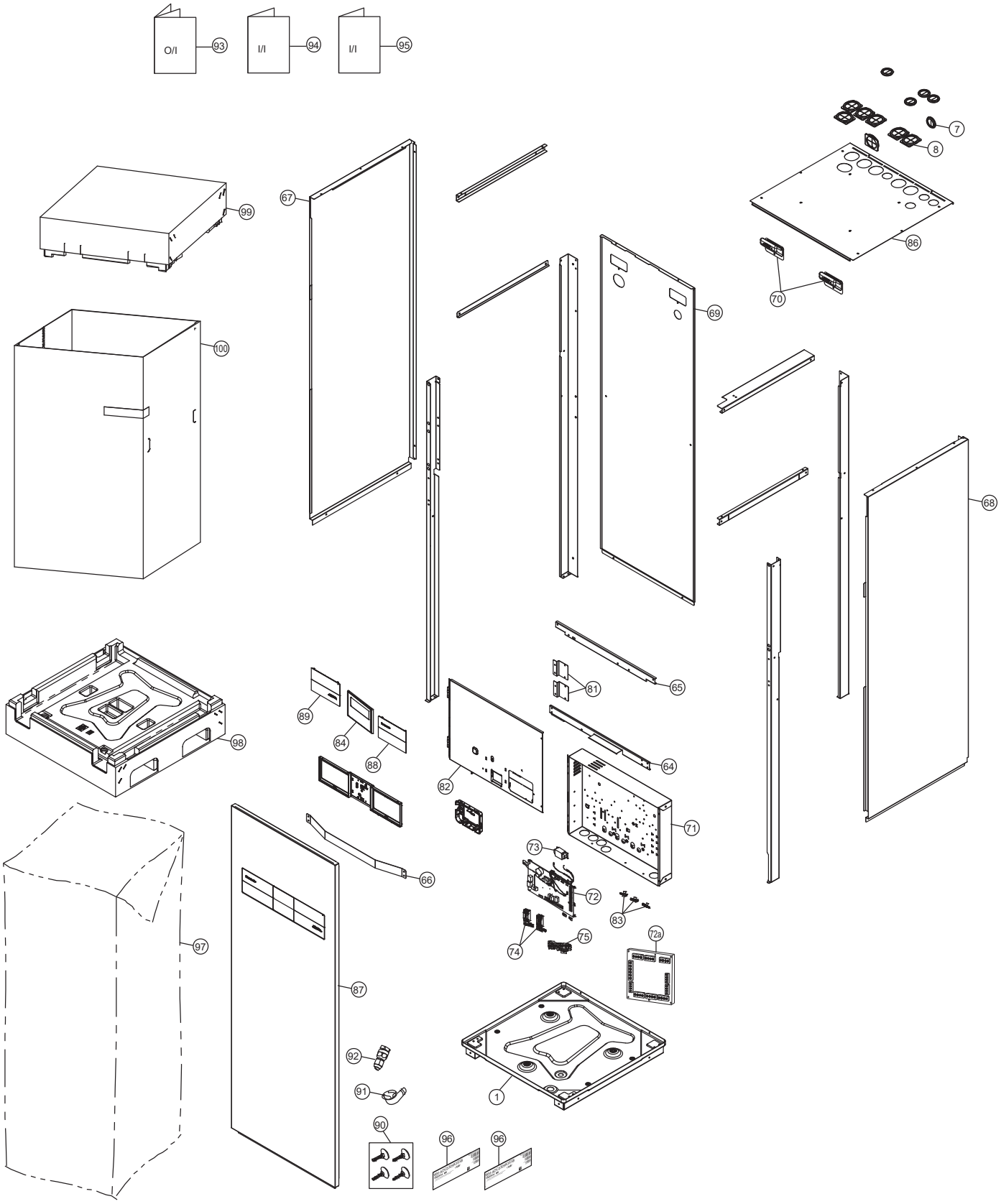
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5	WH-ADC0309K6E5	REMARK
	46	BOILER PRESSURE TRANSDUCER	1	ACXB62-01110	←	
	47	LEAD WIRE - CO. (CN-DPS)	1	ACXA60C93661	←	
	48	MOTORIZED 3-WAY VALVE	1	ACXB62-00092	←	
	49	TUBE ASSY - COMPLETE	1	ACXT00C41210	←	
	50	TUBE ASSY - COMPLETE	1	ACXT00C40980	←	
	55	TUBE ASSY - COMPLETE	1	ACXT00C41030	←	
	56	FLOW SWITCH	1	ACXB62-00932	←	
⚠	57	LEAD WIRE - FLOW SENSOR	1	ACXA60C82910	←	
	58	PRESSURE RELIEF VALVE	1	ACXB62-00100	←	
	59	TUBE ASSY	1	ACXT00-77260	←	
	60	FLARE NUT (5/8)	1	CWT251064	←	
	61	FLARE NUT (1/4)	1	CWT251063	←	
⚠	62	SENSOR - CO. (WATER IN OUT, REF TEMP SENSOR CN-TH1)	1	ACXA50C16380	←	
⚠	63	SENSOR - CO. (HEX WATER OUTLET TEMP CN-TH3)	1	ACXA50C16390	←	
	64	CONNECTING BAR	1	ACXE26-02860	←	
	65	CONNECTING BAR	1	ACXE26-02870	←	
	66	CONNECTING BAR	1	ACXE26-02880	←	
	67	CABINET SIDE PLATE (L)	1	ACXE04-13290A	←	
	68	CABINET SIDE PLATE (R)	1	ACXE04-13300A	←	
	69	CABINET REAR PLATE	1	ACXE02-03400	←	
	70	HANDLE	2	ACXE16-00300	←	
	71	CONTROL BOARD	1	ACXH10-09860	←	
⚠	72	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C88710	ACXA73C91070	
⚠	73	REACTOR	1	G0C103Z00003	←	
⚠	74	RESIDUAL CURRENT - CIRCUIT BREAKER	2	K5KYYAY00003	←	
⚠	75	TERMINAL BOARD ASSY (1,2,3)	1	CWA28K1217	←	
⚠	76	TERMINAL BOARD ASS'Y (A,B)	1	-	CWA28K1238	
	80	TERMINAL COVER	1	CWH171051	←	
	81	HINGE	2	CWH611006	←	
	82	CONTROL BOARD COVER	1	ACXH13-09290	←	
	83	HOLDER - P.S. CORD	3	CWH31103	←	
⚠	84	REMOTE CONTROL COMPLETE	1	ACXA75C23211	←	
⚠	85	LEAD WIRE FOR REMOTE CONTROL	1	ACXA60C93691	←	
	86	CABINET TOP PLATE	1	ACXE03-04880	←	
	87	CABINET FRONT PLATE	1	ACXE06-05161A	←	
	88	DECORATION BASE ASS'Y	1	ACXE35K03630	←	
	89	DECORATION BASE ASS'Y	1	ACXE35K03640	←	
	90	ACCESSORY - CO. (DECORATION BASE ASS'Y)	1	ACXH82C04160	←	
	91	ACCESSORY ADJUSTABLE FEET	1	ACXH82C21740	←	
	92	SOCKET COMPLETE	1	ACXT27C00020	←	
	93	OPERATING INSTRUCTION	1	ACXF55-34120	←	
	94	INSTALLATION INSTRUCTION	1	ACXF60-49960	←	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5	WH-ADC0309K6E5	REMARK
	95	INSTALLATION INSTRUCTION	1	ACXF60-49910	←	
	96	MODEL LABEL	2	ACXF85-95610	ACXF85-95630	
	97	BAG	1	ACXG86-04782	←	
	98	BASE BOARD - COMPLETE	1	ACXG62C02790	←	
	99	TOP BOARD COMPLETE	1	ACXG60C00300	←	
	100	C.C. CASE	1	ACXG50-61790	←	

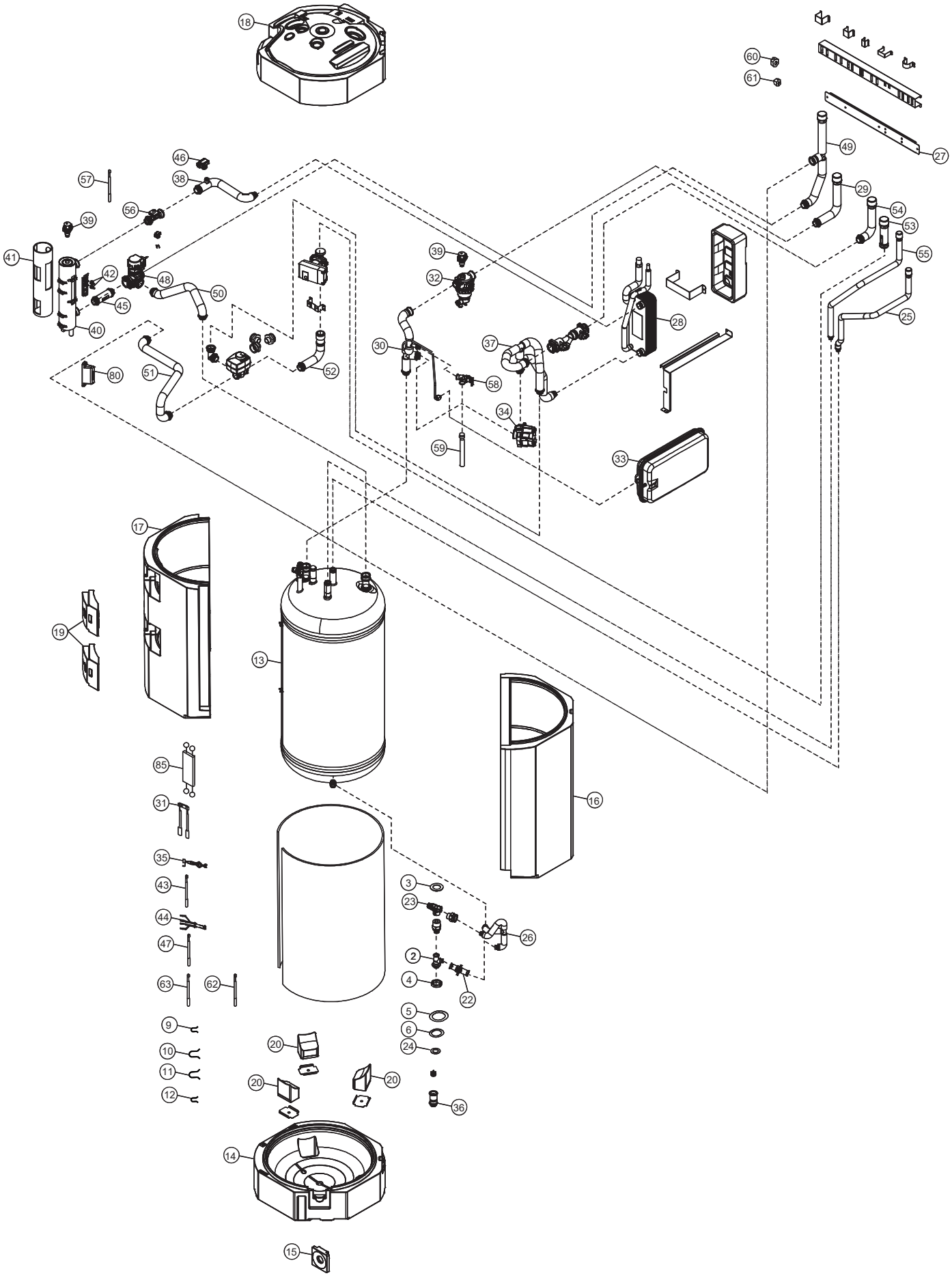
Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

24.1.2 WH-ADC0309K3E5B



Note:
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Note:
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SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5B	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C04840	
	2	TUBE CONNECTER	1	ACXT29-02570	
	3	O-RING (ID: 27mm)	1	ACXB81-07070	
	4	NUT	1	ACXH56-01370	
	5	O-RING (ID: 25.70mm)	1	ACXB81-06910	
	6	O-RING (ID: 8.1mm)	1	CWB811179	
	7	O-RING (ID: 35mm)	1	ACXB81-00030	
	8	O-RING (ID: 54mm X 47mm)	1	ACXB81-00040	
	9	RETAINING RING	1	CWH581007	
	10	RETAINING RING	1	ACXH58-00370	
	11	RETAINING RING	1	CWH581012	
	12	RETAINING RING	1	CWH581038	
	13	TANK COMPLETE	1	ACXB56C01210	
	14	FOAMED POLYSTYRENE	1	ACXG07-07690	
	15	FOAMED POLYSTYRENE	1	ACXG07-07700	
	16	FOAMED POLYSTYRENE	1	ACXG07-07710	
	17	FOAMED POLYSTYRENE	1	ACXG07-07720	
	18	FOAMED POLYSTYRENE	1	ACXG07-07731	
	19	FOAMED POLYSTYRENE	2	ACXG07-07740	
	20	FOAMED POLYSTYRENE	3	ACXG07-07750	
	22	DRAIN VALVE	1	ACXB65C00111	
	23	PRESSURE AND TEMP RELIEF VALVE (TANK)	1	ACXB62-00111	
	24	O-RING (ID: 13.60mm)	1	ACXB81-00010	
	25	TUBE ASSY - COMPLETE	1	ACXT00C40900	
	26	TUBE ASSY	1	ACXT00-77551	
	27	CONNECTING BAR	1	ACXE26-02841	
	28	HOT WATER COIL - COMPLETE	1	ACXB90C01570	
	29	TUBE ASSY - COMPLETE	1	ACXT00C40920	
	30	TUBE ASSY - COMPLETE	1	ACXT00C40931	
⚠	31	SENSOR CO. (TANK TEMP CN-TH2)	1	ACXA50C18700	
⚠	32	FILTER COMPLETE	1	ACXB51C00110	
	33	RECEIVER	1	CWB141039	
⚠	34	WATER PUMP	1	ACXB53-00850	
⚠	35	LEAD WIRE FOR WATER PUMP	1	ACXA60C73630	
	36	TUBE ASSY - COMPLETE	1	ACXT00C40940	
	37	TUBE ASSY - COMPLETE	1	ACXT00C40950	
	38	TUBE ASSY - COMPLETE	1	ACXT00C40961	
	39	AIR PURGE VALVE	2	ACXB62-00130	
⚠	40	HEATER ASS'Y	1	ACXA34K00460	
	41	SOUND PROOF MATERIAL	1	ACXG30-08750	
⚠	42	THERMOSTAT	2	CWA151074	
⚠	43	LEAD WIRE - CO. (CN-OLP1)	1	ACXA60C93670	
⚠	44	LEAD WIRE - CO. (OLP & HEATER)	1	ACXA60C93680	

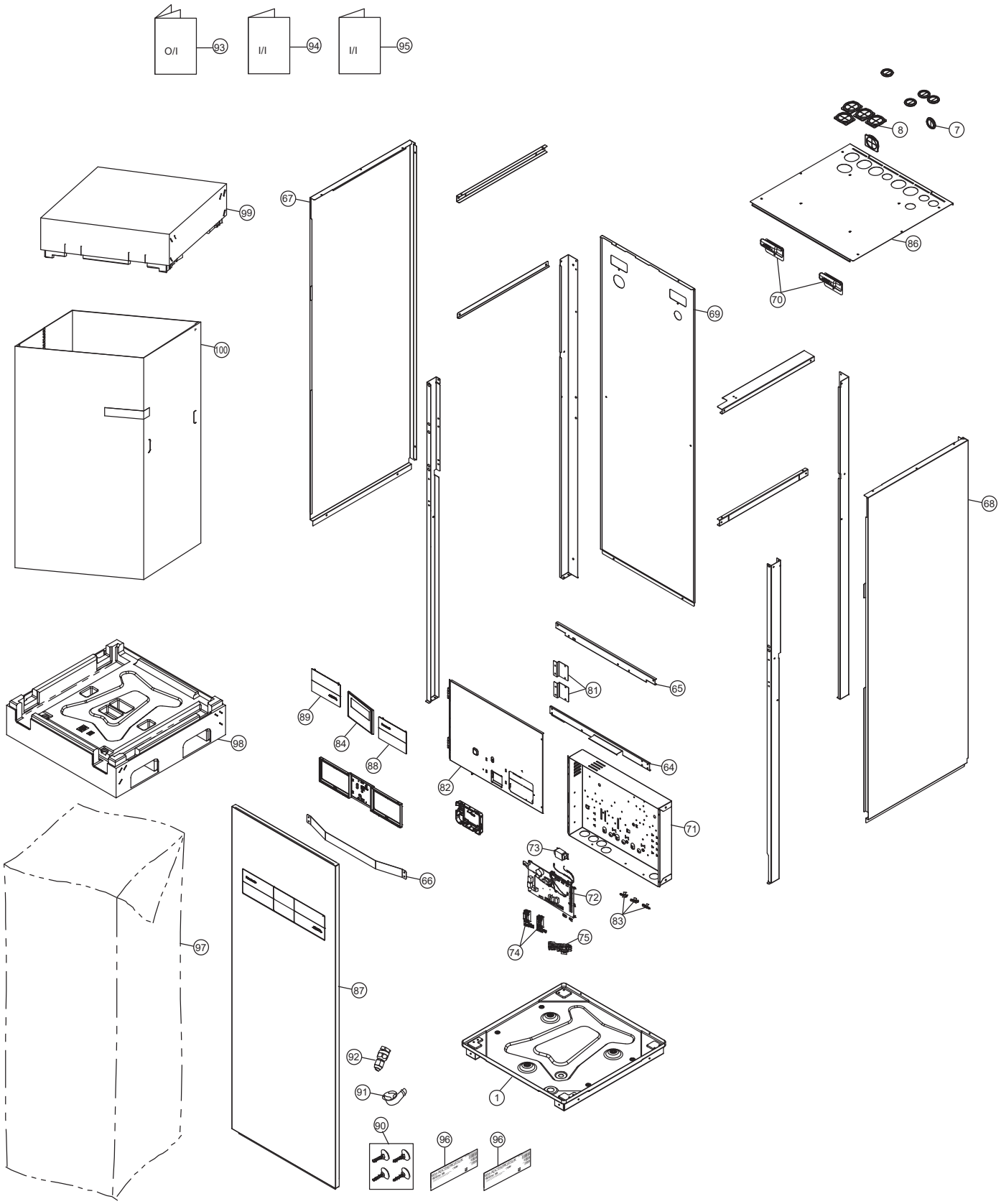
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5B	REMARK
	45	TUBE ASSY - COMPLETE	1	ACXT00C32732	
	46	BOILER PRESSURE TRANSDUCER	1	ACXB62-01110	
	47	LEAD WIRE - CO. (CN-DPS)	1	ACXA60C93661	
	48	MOTORIZED 3-WAY VALVE	1	ACXB62-00092	
	49	TUBE ASSY - COMPLETE	1	ACXT00C40970	
	50	TUBE ASSY - COMPLETE	1	ACXT00C40980	
	51	TUBE ASSY - COMPLETE	1	ACXT00C40990	
	52	TUBE ASSY - COMPLETE	1	ACXT00C41000	
	53	TUBE ASSY - COMPLETE	1	ACXT00C41010	
	54	TUBE ASSY - COMPLETE	1	ACXT00C41020	
	55	TUBE ASSY - COMPLETE	1	ACXT00C41030	
	56	FLOW SWITCH	1	ACXB62-00932	
⚠	57	LEAD WIRE - FLOW SENSOR	1	ACXA60C82910	
	58	PRESSURE RELIEF VALVE	1	ACXB62-00100	
	59	TUBE ASSY	1	ACXT00-77260	
	60	FLARE NUT (5/8)	1	CWT251064	
	61	FLARE NUT (1/4)	1	CWT251063	
⚠	62	SENSOR - CO. (WATER IN OUT, REF TEMP SENSOR CN-TH1)	1	ACXA50C16380	
⚠	63	SENSOR - CO. (HEX WATER OUTLET TEMP CN-TH3)	1	ACXA50C16390	
	64	CONNECTING BAR	1	ACXE26-02860	
	65	CONNECTING BAR	1	ACXE26-02870	
	66	CONNECTING BAR	1	ACXE26-02880	
	67	CABINET SIDE PLATE (L)	1	ACXE04-13290A	
	68	CABINET SIDE PLATE (R)	1	ACXE04-13300A	
	69	CABINET REAR PLATE	1	ACXE02-03400	
	70	HANDLE	2	ACXE16-00300	
	71	CONTROL BOARD	1	ACXH10-09860	
⚠	72	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C92180	
	72a	ELECTRONIC CONTROLLER - MAIN (SUB)	1	ACXA73-42460-W	
⚠	73	REACTOR	1	G0C103Z00003	
⚠	74	RESIDUAL CURRENT - CIRCUIT BREAKER	2	K5KYYAY00003	
⚠	75	TERMINAL BOARD ASSY (1,2,3)	1	CWA28K1217	
	80	TERMINAL COVER	1	CWH171051	
	81	HINGE	2	CWH611006	
	82	CONTROL BOARD COVER	1	ACXH13-09290	
	83	HOLDER - P.S. CORD	3	CWH31103	
⚠	84	REMOTE CONTROL COMPLETE	1	ACXA75C23211	
⚠	85	LEAD WIRE FOR REMOTE CONTROL	1	ACXA60C93691	
	86	CABINET TOP PLATE	1	ACXE03-04870	
	87	CABINET FRONT PLATE	1	ACXE06-05161A	
	88	DECORATION BASE ASS'Y	1	ACXE35K03630	
	89	DECORATION BASE ASS'Y	1	ACXE35K03640	
	90	ACCESSORY - CO. (DECORATION BASE ASS'Y)	1	ACXH82C04160	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5B	REMARK
	91	ACCESSORY ADJUSTABLE FEET	1	ACXH82C21740	
	92	SOCKET COMPLETE	1	ACXT27C00020	
	93	OPERATING INSTRUCTION	1	ACXF55-34180	
	94	INSTALLATION INSTRUCTION	1	ACXF60-49950	
	95	INSTALLATION INSTRUCTION	1	ACXF60-49910	
	96	MODEL LABEL	2	ACXF85-95600	
	97	BAG	1	ACXG86-04782	
	98	BASE BOARD - COMPLETE	1	ACXG62C02790	
	99	TOP BOARD COMPLETE	1	ACXG60C00300	
	100	C.C. CASE	1	ACXG50-61790	

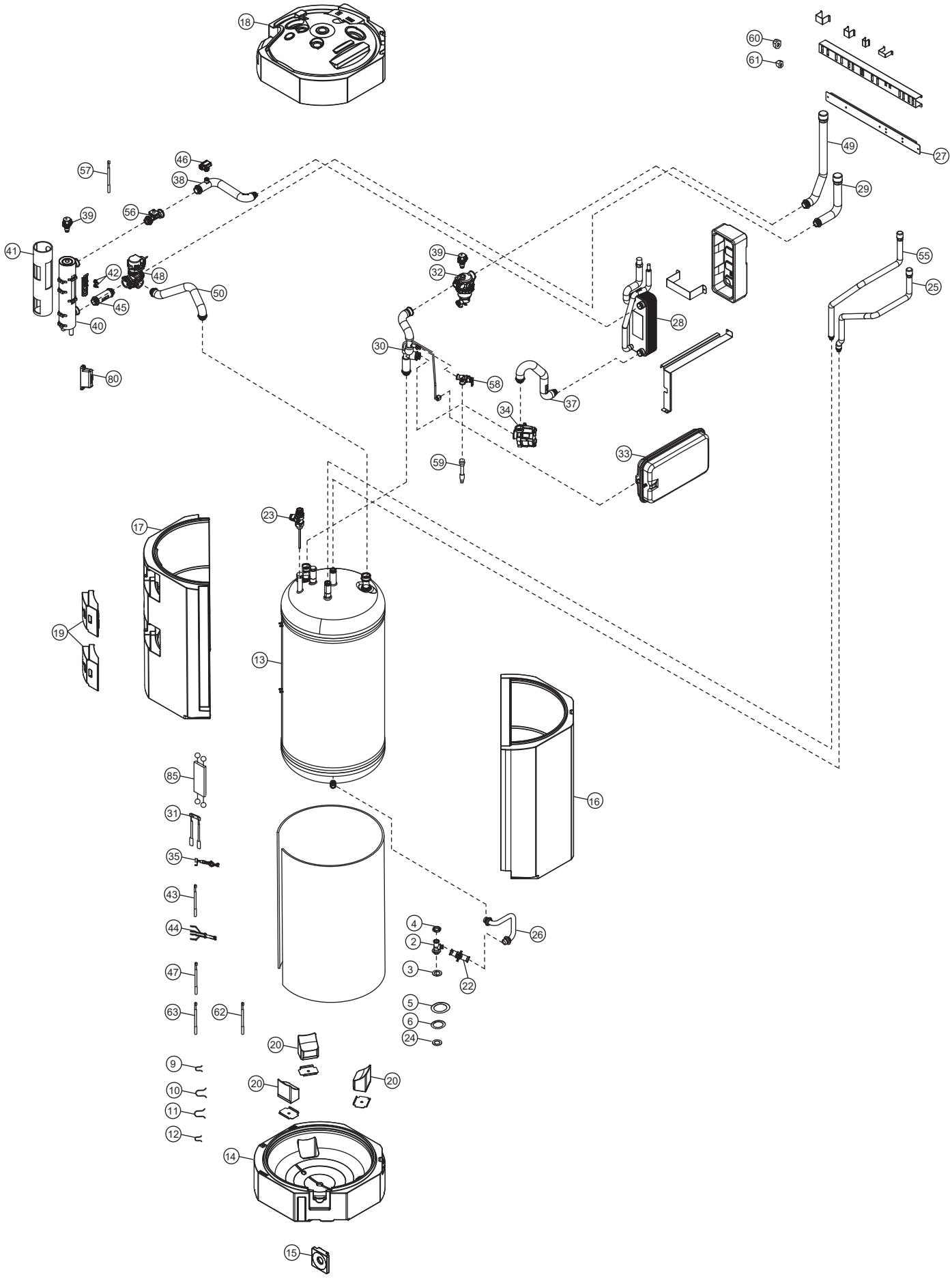
Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

24.1.3 WH-ADC0309K3E5UK



Note:
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Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5UK	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C04840	
	2	TUBE CONNECTER	1	ACXT29-02600	
	3	O-RING (ID: 27mm)	1	ACXB81-07070	
	4	NUT	1	ACXH56-01370	
	5	O-RING (ID: 25.70mm)	1	ACXB81-06910	
	6	O-RING (ID: 8.1mm)	1	CWB811179	
	7	O-RING (ID: 35mm)	1	ACXB81-00030	
	8	O-RING (ID: 54mm X 47mm)	1	ACXB81-00040	
	9	RETAINING RING	1	CWH581007	
	10	RETAINING RING	1	ACXH58-00370	
	11	RETAINING RING	1	CWH581012	
	12	RETAINING RING	1	CWH581038	
	13	TANK COMPLETE	1	ACXB56C01210	
	14	FOAMED POLYSTYRENE	1	ACXG07-07690	
	15	FOAMED POLYSTYRENE	1	ACXG07-07700	
	16	FOAMED POLYSTYRENE	1	ACXG07-07710	
	17	FOAMED POLYSTYRENE	1	ACXG07-07720	
	18	FOAMED POLYSTYRENE	1	ACXG07-07731	
	19	FOAMED POLYSTYRENE	2	ACXG07-07740	
	20	FOAMED POLYSTYRENE	3	ACXG07-07750	
	22	DRAIN VALVE	1	ACXB65C00111	O
	23	PRESSURE AND TEMP RELIEF VALVE (TANK)	1	CWB621135	
	24	O-RING (ID: 13.60mm)	1	ACXB81-00010	
	25	TUBE ASSY - COMPLETE	1	ACXT00C40900	
	26	TUBE ASSY	1	ACXT00-77930	
	27	CONNECTING BAR	1	ACXE26-02841	
	28	HOT WATER COIL - COMPLETE	1	ACXB90C01570	
	29	TUBE ASSY - COMPLETE	1	ACXT00C40920	
	30	TUBE ASSY - COMPLETE	1	ACXT00C40931	
⚠	31	SENSOR CO. (TANK TEMP CN-TH2)	1	ACXA50C18700	O
⚠	32	FILTER COMPLETE	1	ACXB51C00110	O
	33	RECEIVER	1	CWB141039	
⚠	34	WATER PUMP	1	ACXB53-00850	O
⚠	35	LEAD WIRE FOR WATER PUMP	1	ACXA60C73630	O
	37	TUBE ASSY - COMPLETE	1	ACXT00C41200	
	38	TUBE ASSY - COMPLETE	1	ACXT00C40961	
	39	AIR PURGE VALVE	2	ACXB62-00130	O
⚠	40	HEATER ASS'Y	1	ACXA34K00460	
	41	SOUND PROOF MATERIAL	1	ACXG30-08750	
⚠	42	THERMOSTAT	2	CWA151074	O
⚠	43	LEAD WIRE - CO. (CN-OLP1)	1	ACXA60C93670	O
⚠	44	LEAD WIRE - CO. (OLP & HEATER)	1	ACXA60C93680	O
	45	TUBE ASSY - COMPLETE	1	ACXT00C32732	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5UK	REMARK
	46	BOILER PRESSURE TRANSDUCER	1	ACXB62-01110	
	47	LEAD WIRE - CO. (CN-DPS)	1	ACXA60C93661	
	48	MOTORIZED 3-WAY VALVE	1	ACXB62-00092	
	49	TUBE ASSY - COMPLETE	1	ACXT00C41210	
	50	TUBE ASSY - COMPLETE	1	ACXT00C40980	
	55	TUBE ASSY - COMPLETE	1	ACXT00C41030	
	56	FLOW SWITCH	1	ACXB62-00932	O
⚠	57	LEAD WIRE - FLOW SENSOR	1	ACXA60C82910	O
	58	PRESSURE RELIEF VALVE	1	ACXB62-00100	O
	59	TUBE ASSY	1	ACXT00-56600	
	60	FLARE NUT (5/8)	1	CWT251064	
	61	FLARE NUT (1/4)	1	CWT251063	
⚠	62	SENSOR - CO. (WATER IN OUT, REF TEMP SENSOR CN-TH1)	1	ACXA50C16380	O
⚠	63	SENSOR - CO. (HEX WATER OUTLET TEMP CN-TH3)	1	ACXA50C16390	O
	64	CONNECTING BAR	1	ACXE26-02860	
	65	CONNECTING BAR	1	ACXE26-02870	
	66	CONNECTING BAR	1	ACXE26-02880	
	67	CABINET SIDE PLATE (L)	1	ACXE04-13290A	
	68	CABINET SIDE PLATE (R)	1	ACXE04-13300A	
	69	CABINET REAR PLATE	1	ACXE02-03440	
	70	HANDLE	2	ACXE16-00300	
	71	CONTROL BOARD	1	ACXH10-09860	
⚠	72	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C92190	O
⚠	73	REACTOR	1	G0C103Z00003	O
⚠	74	RESIDUAL CURRENT - CIRCUIT BREAKER	2	K5KYYAY00003	O
⚠	75	TERMINAL BOARD ASSY (1,2,3)	1	CWA28K1217	O
	80	TERMINAL COVER	1	CWH171051	
	81	HINGE	2	CWH611006	
	82	CONTROL BOARD COVER	1	ACXH13-09290	
	83	HOLDER - P.S. CORD	3	CWH31103	
⚠	84	REMOTE CONTROL COMPLETE	1	ACXA75C23211	O
⚠	85	LEAD WIRE FOR REMOTE CONTROL	1	ACXA60C93691	
	86	CABINET TOP PLATE	1	ACXE03-04880	
	87	CABINET FRONT PLATE	1	ACXE06-05161A	
	88	DECORATION BASE ASS'Y	1	ACXE35K03630	
	89	DECORATION BASE ASS'Y	1	ACXE35K03640	
	90	ACCESSORY - CO. (DECORATION BASE ASS'Y)	1	ACXH82C04160	
	91	ACCESSORY ADJUSTABLE FEET	1	ACXH82C21740	
	92	SOCKET COMPLETE	1	ACXT27C00020	
	93	OPERATING INSTRUCTION	1	ACXF55-34140	
	94	INSTALLATION INSTRUCTION	1	ACXF60-49970	
	95	INSTALLATION INSTRUCTION	1	ACXF60-50010	
	96	MODEL LABEL	2	ACXF85-95620	

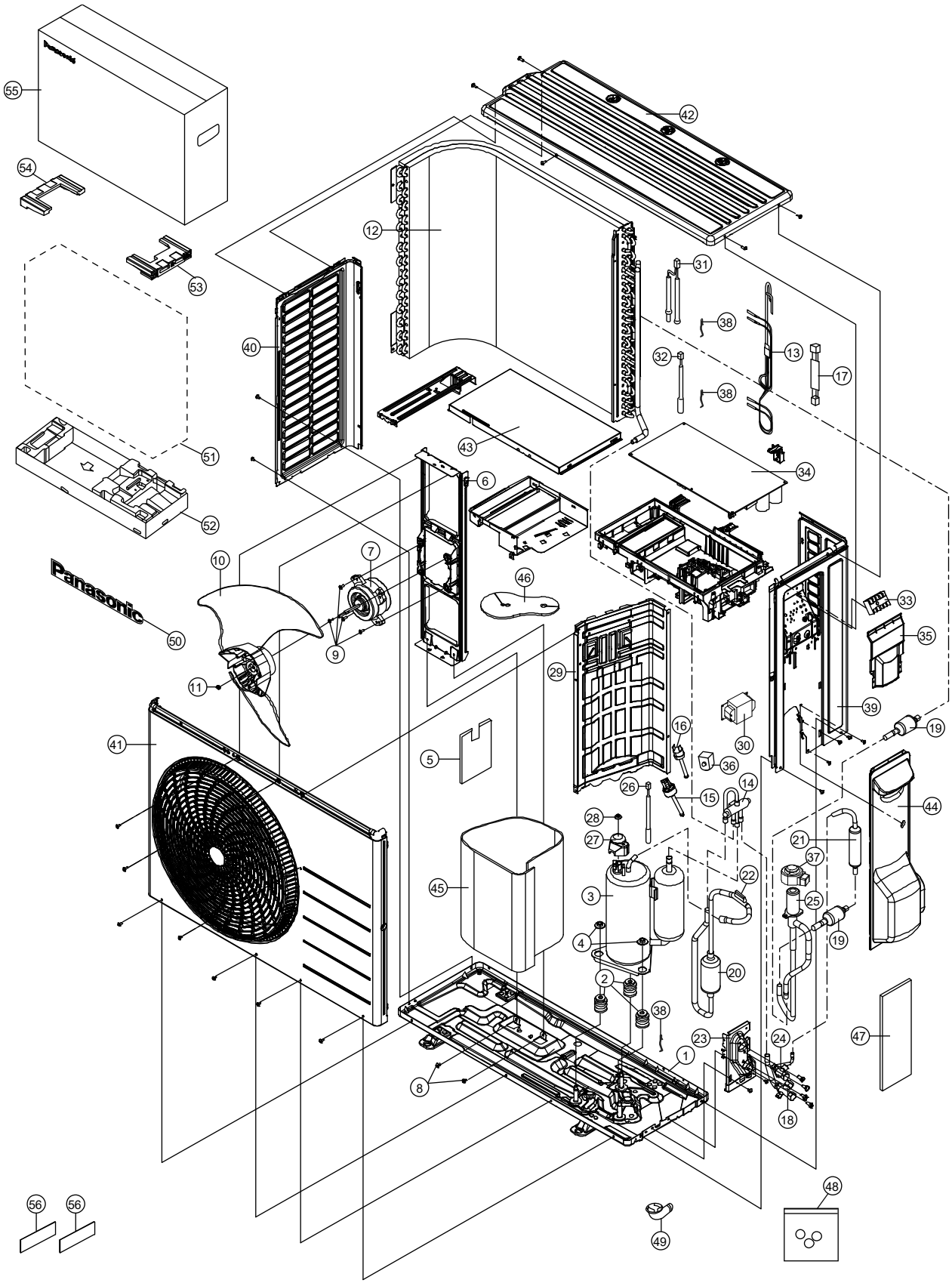
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K3E5UK	REMARK
	97	BAG	1	ACXG86-04782	
	98	BASE BOARD - COMPLETE	1	ACXG62C02790	
	99	TOP BOARD COMPLETE	1	ACXG60C00300	
	100	C.C. CASE	1	ACXG50-61790	

Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

24.2 Outdoor Unit

24.2.1 WH-UDZ03KE5



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

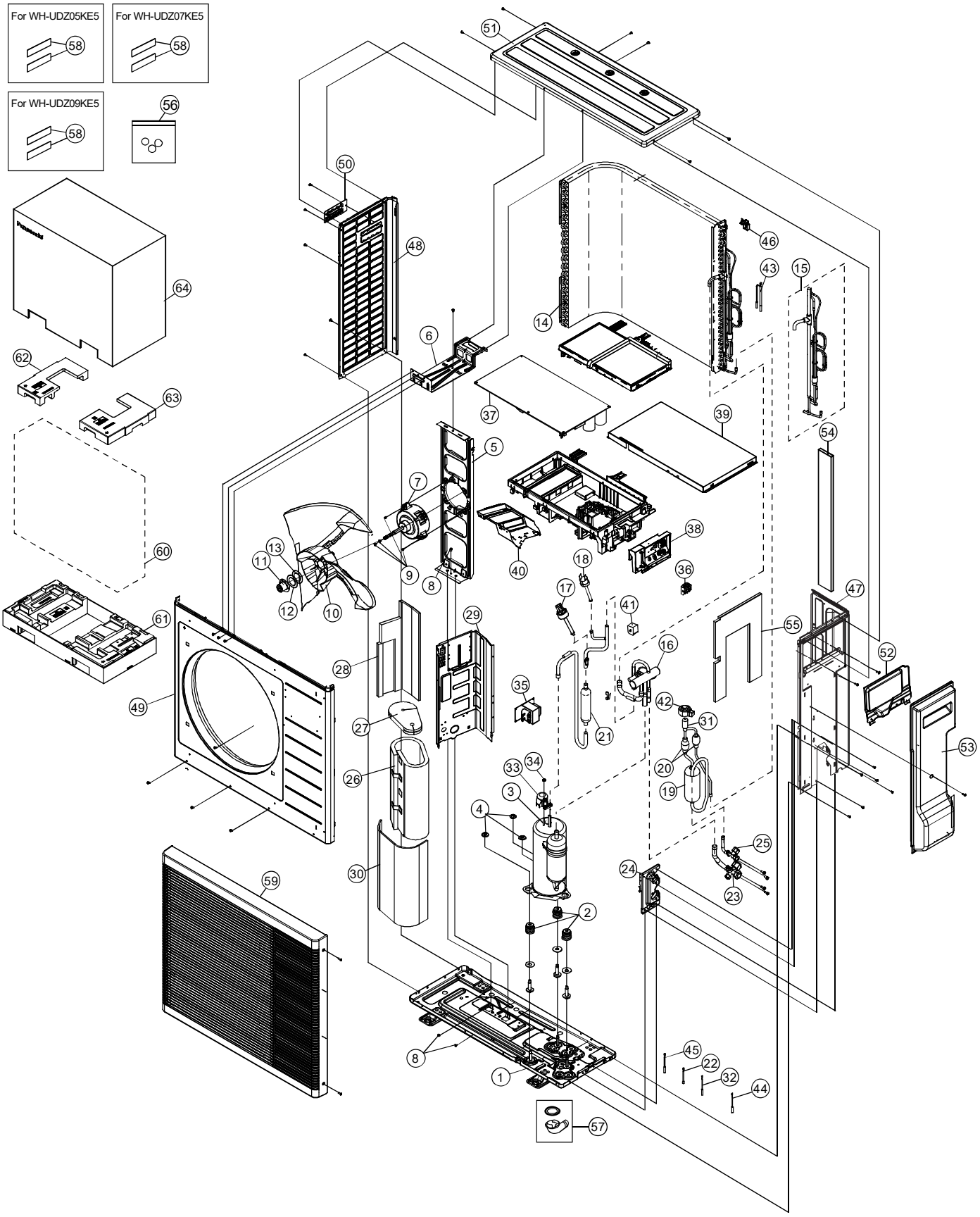
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ03KE5	REMARK
	1	CHASSY ASS'Y	1	CWD52K1317	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	
⚠	3	COMPRESSOR	1	9RD138ZAB21	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	
	5	SOUND PROOF MATERIAL	1	CWG302762	
	6	BRACKET FAN MOTOR	1	CWD541167	
⚠	7	FAN MOTOR, DC 40W 3PH	1	L6CAYYYL0064	O
	8	SCREW - BRACKET FAN MOTOR	2	ACXH55-07140	
	9	SCREW - FAN MOTOR MOUNT	4	CWH55252J	
	10	PROPELLER FAN ASSY	1	CWH03K1066	
	11	NUT - PROPELLER FAN	1	CWH56053J	
	12	CONDENSER	1	ACXB32C26820	
	13	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	CWT07K1831	
	14	4-WAYS VALVE	1	CWB001063	O
⚠	15	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	O
	16	PRESSURE SWITCH	1	ACXA10-00640	O
	17	LEAD WIRE - COMPLETE	1	CWA68C1224	O
	18	3-WAYS VALVE	1	ACXB01-04050	O
	19	STRAINER	2	CWB11094	
	20	DISCHARGE MUFFLER (1)	1	CWB121010	
	21	DISCHARGE MUFFLER (2)	1	CWB121063	
	22	HOLDER - SENSOR	1	CWH32075	
	23	HOLDER - COUPLING	1	CWH351233	
	24	2-WAYS VALVE	1	ACXB02-02680	O
	25	EXPANSION VALVE	1	CWB051029	O
⚠	26	SENSOR - CO. (DISCHARGE TEMP CN-DIS)	1	ACXA50C15550	O
	27	TERMINAL COVER	1	CWH171039A	
	28	NUT - TERMINAL COVER	1	CWH7080300J	
	29	SOUND - PROOF BOARD	1	CWH151345	
⚠	30	REACTOR	1	G0C392J00027	O
⚠	31	SENSOR - CO. (OUTDDOR AIR & PIPING TEMP CN-TH1)	1	CWA50C2893	O
⚠	32	SENSOR - CO. (EVA EXIT TEMP CN-TH3)	1	CWA50C3374	O
⚠	33	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1036J	O
⚠	34	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C92000R	O
	35	PLATE - C.B. COVER	1	CWH131470	
⚠	36	V-COIL COMP. (4 WAY VALVE)	1	CWA43C2431	O
⚠	37	V-COIL CO. (EXPANSION VALVE)	1	ACXA43C06020	O
	38	HOLDER - SENSOR	3	CWH32143	
	39	CABINET SIDE PLATE CO.	1	ACXE04C08670	
	40	CABINET SIDE PLATE (L)	1	ACXE04-13070G	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ03KE5	REMARK
	41	CABINET FRONT PLATE - CO.	1	ACXE06C04700	
	42	CABINET TOP PLATE	1	ACXE03-04830G	
	43	CONTROL BOARD COVER	1	CWH131473	
	44	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06420	
	45	SOUND PROOF MATERIAL	1	ACXG30-08570	
	46	SOUND PROOF MATERIAL	1	CWG302630	
	47	SOUND PROOF MATERIAL	1	CWG302788	
	48	ACCESSORY COMP. (CAP)	1	ACXH82C03830	
	49	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	
	50	PANASONIC BADGE	1	ACXE37-19200	
	51	BAG	1	CWG861078	
	52	BASE BOARD - COMPLETE	1	CWG62C1162	
	53	SHOCK ABSORBER	1	ACXG70-14440A	
	54	SHOCK ABSORBER	1	ACXG70-14450A	
	55	C.C. CASE	1	ACXG50-48673	
	56	MODEL LABEL	2	ACXF85-99600	

Note:

- All parts are supplied from PAPAMY, Malaysia (Vendor Code: 00029488).
- “O” marked parts are recommended to be kept in stock.

24.2.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ05KE5	WH-UDZ07KE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05210	←	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	←	
⚠	3	COMPRESSOR	1	9RD220XBC21	←	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	←	
	5	BRACKET FAN MOTOR	1	ACXD54-04490	←	
	6	PARTICULAR PLATE	1	CWD911802	←	
⚠	7	FAN MOTOR, DC 60W 3PH	1	L6CBYYYL0422	←	O
	8	SCREW - BRACKET FAN MOTOR	3	ACXH55-07140	←	
	9	SCREW - FAN MOTOR MOUNT	4	CWH551323	←	
	10	PROPELLER FAN ASSY	1	CWH00K1006	←	
	11	NUT - PROPELLER FAN	1	CWH561092	←	
	12	WASHER	1	CWH571067	←	
	13	WASHER	1	CWH571068	←	
	14	CONDENSER	1	ACXB32C26350	←	
	15	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	ACXT00C27650	←	
	16	4-WAYS VALVE	1	ACXB00-00140	←	O
⚠	17	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	←	O
	18	PRESSURE SWITCH	1	ACXA10-00640	←	O
	19	RECEIVER	1	CWB14017	←	
	20	STRAINER	2	CWB111063	←	
	21	DISCHARGE MUFFLER	1	CWB121013	←	
⚠	22	LEAD WIRE - CO. (CN-HPS)	1	ACXA60C94010	←	O
	23	3-WAYS VALVE	1	CWB011363	←	O
	24	HOLDER - COUPLING	1	CWH351228	←	
	25	2-WAYS VALVE	1	CWB021464	←	O
	26	SOUND PROOF MATERIAL	1	ACXG30-13230	←	
	27	SOUND PROOF MATERIAL	1	ACXG30-13240	←	
	28	SOUND PROOF MATERIAL	1	ACXG30-13310	←	
	29	SOUND - PROOF BOARD	1	ACXH15-03900	←	
	30	SOUND PROOF MATERIAL	1	ACXG30-13250	←	
	31	EXPANSION VALVE	1	CWB051029	←	O
⚠	32	SENSOR - CO. COMPRESSOR TEMP (CN-TANK)	1	ACXA50C17400	←	O
	33	TERMINAL COVER	1	CWH171039A	←	
	34	NUT - TERMINAL COVER	1	CWH7080300J	←	
⚠	35	REACTOR	1	G0C392J00027	←	O
⚠	36	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1076J	←	O
⚠	37	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C91120R	ACXA73C91130R	O
	38	PLATE - C.B. COVER	1	CWH102361	←	
	39	CONTROL BOARD COVER	1	ACXH13-09170	←	
	40	CONTROL BOARD COVER - COMPLETE	1	ACXH13K00830	←	
⚠	41	V-COIL COMP. (4 WAY VALVE)	1	ACXA43C00250	←	O
⚠	42	V-COIL CO. (EXPANSION VALVE)	1	A43C2342	←	O
⚠	43	SENSOR - CO. OUTDOOR AIR & PIPE TEMP (CN-TH1)	1	CWA50C2822	←	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ05KE5	WH-UDZ07KE5	REMARK
⚠	44	SENSOR - CO. DISCHARGE (CN-DIS)	1	CWA50C2722	←	O
⚠	45	SENSOR - CO. EVA EXIT (CN-TH3)	1	ACXA50C14960	←	O
	46	HOLDER - SENSOR	1	CWH321096G	←	
	47	CABINET SIDE PLATE CO.	1	ACXE04-13310G	←	
	48	CABINET SIDE PLATE (L)	1	CWE041585G	←	
	49	CABINET FRONT PLATE - CO.	1	ACXE06C04610	←	
	50	HANDLE	1	CWE161010G	←	
	51	CABINET TOP PLATE	1	CWE031083G	←	
	52	CONTROL BOARD COVER	1	CWH131332	←	
	53	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06350	←	
	54	SOUND PROOF MATERIAL	1	ACXG30-13260	←	
	55	SOUND PROOF MATERIAL	1	ACXG30-13270	←	
	56	ACCESSORY COMP. (CAP)	1	ACXH82C09831	←	
	57	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	←	
	58	MODEL LABEL	2	ACXF85-99610	ACXF85-99620	
	59	DISCHARGE GRILLE - COMPLETE	1	ACXE20C07970	←	
	60	BAG	1	CWG861154	←	
	61	BASE BOARD - COMPLETE	1	ACXG62C02810	←	
	62	SHOCK ABSORBER	1	ACXG70-15180	←	
	63	SHOCK ABSORBER	1	ACXG70-15190	←	
	64	C.C. CASE	1	ACXG50-61910	←	

Note:

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- “O” marked parts are recommended to be kept in stock.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ09KE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05210	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	
⚠	3	COMPRESSOR	1	9RD220XBC21	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	
	5	BRACKET FAN MOTOR	1	ACXD54-04490	
	6	PARTICULAR PLATE	1	CWD911802	
⚠	7	FAN MOTOR, DC 60W 3PH	1	L6CBYYYL0422	O
	8	SCREW - BRACKET FAN MOTOR	3	ACXH55-07140	
	9	SCREW - FAN MOTOR MOUNT	4	CWH551323	
	10	PROPELLER FAN ASSY	1	CWH00K1006	
	11	NUT - PROPELLER FAN	1	CWH561092	
	12	WASHER	1	CWH571067	
	13	WASHER	1	CWH571068	
	14	CONDENSER	1	ACXB32C26350	
	15	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	ACXT00C27650	
	16	4-WAYS VALVE	1	ACXB00-00140	O
⚠	17	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	O
	18	PRESSURE SWITCH	1	ACXA10-00640	O
	19	RECEIVER	1	CWB14017	
	20	STRAINER	2	CWB111063	
	21	DISCHARGE MUFFLER	1	CWB121013	
⚠	22	LEAD WIRE - CO. (CN-HPS)	1	ACXA60C94010	O
	23	3-WAYS VALVE	1	CWB011363	O
	24	HOLDER - COUPLING	1	CWH351228	
	25	2-WAYS VALVE	1	CWB021464	O
	26	SOUND PROOF MATERIAL	1	ACXG30-13230	
	27	SOUND PROOF MATERIAL	1	ACXG30-13240	
	28	SOUND PROOF MATERIAL	1	ACXG30-13310	
	29	SOUND - PROOF BOARD	1	ACXH15-03900	
	30	SOUND PROOF MATERIAL	1	ACXG30-13250	
	31	EXPANSION VALVE	1	CWB051029	O
⚠	32	SENSOR - CO. COMPRESSOR TEMP (CN-TANK)	1	ACXA50C17400	O
	33	TERMINAL COVER	1	CWH171039A	
	34	NUT - TERMINAL COVER	1	CWH7080300J	
⚠	35	REACTOR	1	G0C392J00027	O
⚠	36	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1076J	O
⚠	37	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C91140R	O
	38	PLATE - C.B. COVER	1	CWH102361	
	39	CONTROL BOARD COVER	1	ACXH13-09170	
	40	CONTROL BOARD COVER - COMPLETE	1	ACXH13K00830	
⚠	41	V-COIL COMP. (4 WAY VALVE)	1	ACXA43C00250	O
⚠	42	V-COIL CO. (EXPANSION VALVE)	1	A43C2342	O
⚠	43	SENSOR - CO. OUTDOOR AIR & PIPE TEMP (CN-TH1)	1	CWA50C2822	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ09KE5	REMARK
⚠	44	SENSOR - CO. DISCHARGE (CN-DIS)	1	CWA50C2722	O
⚠	45	SENSOR - CO. EVA EXIT (CN-TH3)	1	ACXA50C14960	O
	46	HOLDER - SENSOR	1	CWH321096G	
	47	CABINET SIDE PLATE CO.	1	ACXE04-13310G	
	48	CABINET SIDE PLATE (L)	1	CWE041585G	
	49	CABINET FRONT PLATE - CO.	1	ACXE06C04610	
	50	HANDLE	1	CWE161010G	
	51	CABINET TOP PLATE	1	CWE031083G	
	52	CONTROL BOARD COVER	1	CWH131332	
	53	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06350	
	54	SOUND PROOF MATERIAL	1	ACXG30-13260	
	55	SOUND PROOF MATERIAL	1	ACXG30-13270	
	56	ACCESSORY COMP. (CAP)	1	ACXH82C09831	
	57	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	
	58	MODEL LABEL	2	ACXF85-99630	
	59	DISCHARGE GRILLE - COMPLETE	1	ACXE20C07970	
	60	BAG	1	CWG861154	
	61	BASE BOARD - COMPLETE	1	ACXG62C02810	
	62	SHOCK ABSORBER	1	ACXG70-15180	
	63	SHOCK ABSORBER	1	ACXG70-15190	
	64	C.C. CASE	1	ACXG50-61910	

Note:

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